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同學可以透過scan右邊的QR code 到訪我們的免費網上資源站，即可見到不同按鈕選項以取得不同資源，以下是不同按鈕選項的介紹



I. 有關回報題目的問題

由於一般的by topic pastpaper內含三十年的AL，三十年的CE及十幾年的DSE題目，總共嘅題目數量都接近4000-10000題，我們的合作導師有時候都難免會有少許手民之誤，有可能會打錯字，又或者編排上面會有少許問題，小編本身每年亦會進行恆常校對，所以現時有問題嘅問題數量已經係極少。但係如果同學仍發現到題目或者答案有問題的話，歡迎在網上資源站 click **Link 1. 回報Past paper問題**，填寫表格向我們通知回報，為表謝意，每發現一個錯誤我們都會送出一份名校試卷俾同學。

發現錯誤之後我們會盡快更正，而其他已購買的同學可以click **Link 2. Past paper 更正修改記錄**以得知最新的更正。

II. 有關最新一年的試卷

而最新一年的試卷我們一般不會編排入去做by topic，因為有相當多的同學是希望將最新一年的past paper留嚟成份卷咁樣做，所以我們會提供試卷的原卷及這年的by topic試題分類表在此書的最後（book1或2或3），以滿足不同同學的需要。

每一年的DSE在5月完結後，官方的答案會在11月公布，由於5月-11月暫時並未有官方答案，所以最新一年的試題的官方答案我們會在11月中的時候上載到Google Drive，同學可以click **Link 3. 最新一年的Past paper答案**以查看並且下載。

III. 有關兩版一頁的問題

為了提供最相宜的價格，大部份By topic試卷我們都會以兩版一頁這樣印刷，我們明白同學有機會會覺得字體可能會太小，所以如果同學有需要的話，可以向我們申請加錢換購一版一頁的By topic pp，詳情可以whatsapp/instagram/carousel與我們聯絡，不便之處，敬請原諒。

IV. 有關詳細題解

絕大部份的科目一般只有長題目會有詳細題解，至於多項選擇題的話，一般只有答案件，詳細的答案解釋暫時只有少數科目有，其他科目的詳細題解我們會慢慢推出，如果有更新的話我們亦會放在**Link 4. MC 詳細題解**，另外其實嘅YouTube上面亦可以簡單搜尋到相關题目的題解，例如如果想要的題解，只需要把「chem 2005 CE Paper 1 Q5」這個題號放上YouTube搜尋即可找到。

BY TOPIC PP INTRO

六個秘訣 活用BY TOPIC PAST PAPER**I. 先做MC再做LQ**

編者建議同學先溫好一個topic的重點，然後做該topic的MC並對答案先，並從錯了的問題中了解自己冇咩concept未清，而從中做到清concept的效果，清晰了解過後，便可以開始做LQ，操LQ的重點是了解如何答得中晒point，用最精簡省時的方法取得滿分。

II. 做一條就對一條

操卷時，編者建議你做一條然後即刻對答案，以便即刻了解自己的答案有什麼缺陷，並有機會試做其他題目，以確保自己清楚明白。如果做晒一個課題嘅所有題目先對卷，你會一次過發現自己錯d咩，有機會畀自己再試下作答。相反，當你做一條對一條嘅時候，你可以即刻了解自己啱啱咩錯啱咩，學識咗之後點樣做嗰一類嘅問題。當你做下一年遇到同樣嘅題目嘅時候，你已經學識咗點樣作答，仲可以答得更準確。

III. 參考MARKING SCHEME

喺LQ，考評局marking scheme嘅寫法，往往係最低成本風險嘅滿分方法，係有很大的參考價值，尤其係Econ,bio，所以操by topic時，除了要歸納出題趨勢外，更要歸納每題marking嘅共通之處，建立你自己嘅答題框架、技巧。

IV. 整理複習筆記及錯題簿

為咗可以加深記憶，以及避免重複犯錯。編者建議大家要整理自己的複習筆記，寫低每一個課題嘅考核重點，以及常犯錯誤。你唔一定要自己由頭整啲 notes，你可以畀學校老師畀嘅 notes / 教科書上再額外補充，最緊要係方便到自己之後溫書。

V. 每一次做至少一個 BY TOPIC PP課題嘅一半

編者唔建議大家分開好多次去一個 topic 嘅 dse by topic past paper 題目，咁樣冇咗做 by topic 個意義。所以，每一次操卷嘅時候大約用兩至三個鐘，至少要做夠一個課題嘅一半 by topic 題目。

VI. 做至少 30 年 PAST PAPER

考評局出卷模式傾向抄歷屆試題，所以操卷時做30年係必須的。視乎翻個科目，好多科目主要做會考 CE 卷同 DSE 卷都足夠。有一少部分科目就要多少少高考 AL 卷。做夠30年，基本上你已經睇到晒嗰一個課題出咩問題同埋需要答嘅答案係啱咩。

此科編輯或內容上的特殊地方（如有）：

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CH 1 BASIC CONCEPTS

1.1 SCARCITY, ECONOMIC GOODS & FREE GOODS

Microeconomics by Topic
I. Basic Concepts

MARKING SCHEME

1991/CE/II/01 D	2005/CE/II/01 D	1993/CE/II/16 D	2001/CE/II/47 C	2008/CE/II/01 A (63%)
1994/CE/II/01 C	2009/CE/II/01 B	1994/CE/II/02 D	2002/CE/II/01 B (60%)	2009/CE/II/01 B (81%)
1995/CE/II/01 D	1990/CE/II/02 C	1995/CE/II/03 D	2002/CE/II/02 D (84%)	2014/DSE/I/03 C (59%)
1995/CE/II/02 B (deleted)	1990/CE/II/25 C	1996/CE/II/55 C	2003/CE/II/01 B (52%)	2016/DSE/I/01 B (74%)
1997/CE/II/01 D	1991/CE/II/03 D	1997/CE/II/12 C	2004/CE/II/03 B (70%)	2023/DSE/I/05 D
1998/CE/II/54 A	1991/CE/II/55 B	1998/CE/II/01 D	2005/CE/II/02 C (68%)	2024/DSE/I/02 D
1999/CE/II/01 B	1992/CE/II/01 D	1998/CE/II/02 B	2006/CE/II/03 D (72%)	2024/DSE/I/03 A
2004/CE/II/01 D	1992/CE/II/02 X	2000/CE/II/01 C	2007/CE/II/01 B (77%)	2025/DSE/I/03 A

Note: Figures in brackets indicate the percentages of candidates choosing the correct answers.

1991/CE/I/2(c)(iii)

The supply is so abundant that it satisfies all human wants for it.

OR

No one is willing to incur a cost getting it.

(2)

The case of a free good: e.g. oxygen for normal breathing

(2)

The case of an economic good: e.g. concentrated oxygen

(2)

1994/CE/I/1

Not a free good, because

resources used in providing the bus service have alternative uses; there are costs of using the resources in providing the bus service.

(1)

OR

more of the bus service is preferred to less.

(3)

2000/CE/I/1

E.g. The ice-field in the Polar regions is a free good, because

the quantity available is sufficient to satisfy all our wants for it / no sacrifice of resources of the economy is involved

(1)

(1)

E.g. Ice cubes in a glass of soft drink are economic goods, because

the quantity available is insufficient to satisfy all our wants for it / one has to pay a cost to obtain it

(1)

(1)

2002/CE/I/1

Not free goods because

the resources used to produce them have alternative uses / there is a cost to society of producing them / the quantity available is insufficient to satisfy all our wants for them

(1)

(2)

Microeconomics by Topic
I. Basic Concepts

2004/CE/I/1

No, because (1)
its quantity available is insufficient to satisfy our wants / more of it is preferred.

OR

resources used to provide the service had alternative uses / there is opportunity cost of providing the service. (2)

2012/DSE/II/1

No, because (1)
the good may be produced from scarce resource which have alternative uses.

OR

more of the good may still be preferred. (2)

2013/DSE/II/9(b)

No, because (1)
the provision of MTR service involves the use of scarce resources having alternative uses. (1)

2015/DSE/II/(a)(i)

No, because (1)
it has positive marginal cost of production;

OR

it is produced from scarce resources, which have alternative uses. (1)

2020/DSE/II/1

Yes. As the quantity available of free good is able to satisfy all human wants. No one is willing to pay a price to obtain it. (3 marks)

2023/DSE/II/1(c)

(c) No, because (1)
the resources involved in writing these analyses have alternative uses. (1)

OR

more of these analyses are preferred by some people.

2024/DSE/II/10(a)

(a) (i) No. More free flight tickets are preferred. The resources used to produce these services have alternative uses. (2)

(ii) No. The flight tickets are rival in consumption. One's consumption of the tickets reduces their availability to others. (3)

1.2 OPPORTUNITY COST

MARKING SCHEME

1990/CE/II/01 A	1996/CE/II/01 C	2000/CE/II/53 D	2006/CE/II/02 D (82%)	2013/DSE/I/02 C (74%)
1991/CE/II/02 A	1997/CE/II/02 C	2001/CE/II/01 D	2007/CE/II/02 C (78%)	2014/DSE/I/01 A (32%)
1992/CE/II/03 A	1997/CE/II/03 D	2002/CE/II/04 A (61%)	2008/CE/II/02 D (58%)	2015/DSE/I/02 C (63%)
1993/CE/II/02 D	1997/CE/II/49 C	2003/CE/II/02 C (56%)	2009/CE/II/02 A (63%)	2016/DSE/I/03 D (76%)
1994/CE/II/03 C	1998/CE/II/50 A	2003/CE/II/03 C (52%)	2010/CE/II/02 D (78%)	2017/DSE/I/04 D (44%)
1994/CE/II/14 C	1999/CE/II/50 D	2004/CE/II/02 C (67%)	PP/DSE/I/02 D	2018/DSE/I/03 B (80%)
1995/CE/II/05 C	2000/CE/II/02 C	2005/CE/II/03 C (68%)	2012/DSE/I/02 A (62%)	2019/DSE/I/03 C
2020/DSE/I/10 C	2021/DSE/I/03 A	2021/DSE/I/04 C	2022/DSE/I/2 C	2022/DSE/I/3 C
2023/DSE/I/1 C	2023/DSE/I/2 A	2024/DSE/I/01 C	2025/DSE/I/01 C	

1990/CE/I/(a)(i)

The value of employing a Filipino maid (best alternative use of Mrs Wong's time, e.g. income earned by a lawyer) is larger than the cost of so doing (e.g. payment to the maid). (5)

1990/CE/I/4/(c)(i)

No, because (1)
alternatively Mr Lee can sell these quotas in the market and make money, i.e. the opportunity cost is the sum of money thus forsaken. (3)

1990/CE/I/5/(c)(i)

The following items have not been included in the \$127 billion:

- pollution arising from construction
- loss incurred by private individuals adversely affected by construction
- permanent loss of a countryside view in Lantau
- strain on resources bidding up factor cost to other sectors
- the sacrifice of more valuable alternative projects

(Mark the **FIRST TWO** points only.)

(2@, max: 4)

1991/CE/I/2/(a)

- (i) Opportunity cost is the highest-valued option or the best alternative that has to be forgone. (2)
Occupational mobility is the ease with which a factor moves from one occupation / trade / industry to another. (2)
- (ii) In changing occupation, the opportunity cost of a doctor (in terms of income forgone) can be very high. (2)
The value of taking up a new occupation (i.e. a doctor's earnings in another job) is usually lower than the cost of so doing (i.e. income, satisfaction and social status etc. from being a doctor). (3)

1993/CE/I/5(a)

The cost of the second choice is to give up the first choice. (1)

The cost of the third choice is to give up the first choice. (1)

∴ Both had the same highest-valued option forgone. (2)

(Remark: mere mention of the definition of opportunity cost = max: 1 mark)

1995/CE/I/10(a)(i)

(I) Opportunity cost is the highest-valued option forgone. (2)

(II) Her services, time, or other resources used in the performance had alternative uses
e.g. income forgone from other jobs. (1)

(III) - prestige / increased no. of fans / more popular
- experience gained
- enjoyment in performing the shows
(Mark the **FIRST TWO** points only.) (1@, max: 2)

1996/CE/I/1

(a) The cost decreased / changed, because
the value of the concert (which is the highest-valued option forgone) is lowered / changed. (1)

(b) No change, because
enjoyment in the karaoke lounge is not an option forgone. (1)

OR

the value of the concert (which is the highest-valued option forgone) has not changed. (1)

1997/CE/I/11(b)

The value of the highest-valued option forgone increases. (1)

e.g. the market rental of the premises increases. (2)

1998/CE/I/1

(a) It is the highest-valued option forgone. (2)

(b) Since the science stream is less preferred than the business stream, a change from the option of the business stream to the option of the science stream would lower the opportunity cost. (3)

1999/CE/I/1

Cost decreases, because (1)

value of the noodle (the highest-valued alternative) ↓ (2)

2000/CE/I/10(b)(ii)

(I) Opportunity cost is the highest-valued option forgone. (2)

(II) No, because
the shares of Bank Y bought by Bank X is the chosen option, not the highest-valued option forgone. (1)

OR

the hidden debt only reduces the value of the take over. (2)

2001/CE/I/10(d)

The opportunity cost of gold coin purchase is the value of savings in banks. e.g. the interest return on savings in banks. (2)

A tax on interest reduces / changes the value of savings in banks, (1)

which means the opportunity cost of gold coin purchased is lowered / has changed. (1)

Microeconomics by Topic

I. Basic Concepts

2002/CE/I/10(c)

The cost of depositing money with one bank is the deposit interest that can be earned from other banks (2)

∴ to deposit money into those large (small) banks has a higher (lower) cost. (1)

∴ there will be a flow of deposits from those large banks to those small banks. (1)

(Remark : Mere mentioning of the definition of cost without other elaboration - **NO** marks)

2004/CE/I/2

(a) Opportunity cost is the highest-valued option forgone. (2)

(b) Not necessarily, because (1)
their highest-valued option forgone may not be the same. E.g., they may give up different jobs.

(c) Not necessarily, because (1)
though their monthly salary forgone (= \$30 000) are the same, their teaching-related non-pecuniary returns and other (2)
monetary returns may not be the same.

(In other words, the value of the same job to them may be different.)

2005/CE/I/9

(b) (ii) The opportunity cost increases, because (1)
need to spend more time which has alternative uses (i.e. time cost increases).

OR

fuel cost increases. (1)

(c) No, because (1)
the highest-valued alternative use of the time may be different for different tunnel users. (2)

2006/CE/I/2

(a) Not affected because (1)
it only affects the value of studying law at a university but not the opportunity cost which is the value of studying (1)
medicine at a university.

(b) Affected / decreased because (1)
the value of the highest-valued option forgone, i.e., the value of studying medicine at a university, decreases / changes. (1)

2007/CE/I/10(c)

(i) Yes, because (1)
their time cost of visiting Hong Kong would have changed (increased). (1)

(ii) No, because (1)
their time cost had not changed / it only affects the value of staying 24 more hours in Hong Kong. (1)

2008/CE/I/9(b)(i)

The opportunity cost is the value of depositing money with banks, i.e. the interest returns from bank deposits. (2)

(1) Yes because (1)
the interest return from bank deposits increases. (1)

(2) No because (1)
the tax affects the value of stock market investments but not the cost. (1)

2009/CE/I/1

(a) Opportunity cost is the highest-valued option forgone. (2)

(b) No, because (1)
their alternative use of time may not be the same. (1)

2010/CE/11(c)

- (i) The opportunity cost is the return from investment in property, e.g., rental income or increase in the value of property. If the rental income rises or the value of property increases, the opportunity cost increases. (3)
- (ii) No, because (1)
it only affects the value of investing in shares but not the cost. (1)

2013/DSE/II/9(c)

- No, because (1)
people have different alternative uses of time, their time costs will not be the same. (2)

2014/DSE/II/1

- (a) Yes, because (1)
the reduction in the starting salary of administrative officers would only decrease the value of Serine's chosen option without affecting her highest valued option forgone. (1)
- (b) No, because (1)
government provision of unemployment benefits may increase the value of Serine's 3rd option to a level higher than that of the 2nd option, (1)
the cost of choosing to work as an administrative officer may now change from the value of the 2nd option to that of the 3rd option. (1)

2015/DSE/II/1(b)

- No, because (1)
the time cost of a person taking MOOC may be higher than the time cost of another person taking on campus course plus the course fee; (1)
OR
the former person (MOOC student) may have to spend extra time to access / download the course materials and to communicate with the course instructor or teaching assistant; (2)
OR
the former person may also have to pay for internet connection in order to access the course materials. (2)

2016/DSE/II/1

The full cost of buying the new model includes the price of new model (monetary cost) plus the value of time spent logging in the website. (1)

The resale value of the new model is not a forgone option so the cost of buying the new model does not change when such value falls.

OR
The drop in resale value will lower the (speculators') demand - hence, excess demand - for the new model, reducing the time spent logging in the website and thus the time cost of buying the new model. The cost of buying the new model falls. (2)

2017/DSE/II/1

- (a) No, because (1)
the value of the highest-valued option forgone (i.e., the first option) would drop. (1)
- (b) No, because (1)
the working time in South Korea may be quite long, so that the first option would become less preferred. (1)

2018/DSE/II/1

Full cost = monetary cost + time cost (1)
During holiday periods, although the monetary cost of travelling abroad (including air ticket fares and hotel expenses) is usually higher than that during non-holiday periods, the time cost (e.g., income foregone) may be a lot lower. (1)
If the time-cost saved outweighs the extra monetary cost, then in terms of full cost travelling abroad during holiday periods would be less expensive. (1)

- (a) The opportunity cost will remain unchanged. It is because the height limitation only reduces the value of the reclaimed land but does not affect the highest valued option forgone.

Marks
(2)

- (b) The government needs to spend extra money/resources to compete for the raw materials in the case of shortage so monetary/non-monetary cost will increase and the full cost will increase.

(3)

<HKDSE 2022 Paper 2 – 10f>

- (f) The rise in the average income of professional athletes would increase the value of being a full-time athlete without affecting its opportunity cost (highest-value option forgone).

<HKDSE 2023 Paper II- 11(c)>

- (c) Full cost = time cost + money cost.
The housewife chose to buy at the official ticketing office because her time cost of queuing (i.e. her value of the best alternative use of the queuing time) was smaller than the additional money cost of buying from touts (i.e. the difference between touting price and official price).

(3)

2024/DSE/II/12(c)

- (c) No, because
the offer by Football Academy B makes the second option more attractive / raises the value of the second option.
If the scholarship raises the value of the second option above that of the first option, the opportunity cost of continuing to seek jobs will change (increase) from the value of enrolling full-time in Football Academy A to that of enrolling full-time in Football Academy B.

(1)

(1)

(1)

1.3 ECONOMIC SYSTEMS

MARKING SCHEME

1995/CE/II/20 C	2018/DSE/I/02 D (81%)	1997/CE/II/04 C	2003/CE/II/04 B (62%)	2010/CE/II/05 B (79%)
2006/CE/II/01 C (54%)	1990/CE/II/03 B	1998/CE/II/03 D	2004/CE/II/04 C (55%)	2015/DSE/I/03 C (75%)
2009/CE/II/03 B (56%)	1992/CE/II/05 D	1999/CE/II/03 C	2005/CE/II/08 A (46%)	2018/DSE/I/01 D (59%)
2010/CE/II/03 C (84%)	1993/CE/II/50 C	1999/CE/II/37 D	2006/CE/II/04 D (58%)	2019/DSE/I/02 A
2013/DSE/I/01 C (76%)	1994/CE/II/04 D	2000/CE/II/04 D	2007/CE/II/03 D (80%)	2020/DSE/I/01 B
2014/DSE/I/04 D (75%)	1994/CE/II/05 C	2001/CE/II/02 B	2008/CE/II/03 B (75%)	2021/DSE/I/02 B
2017/DSE/I/01 C (90%)	1996/CE/II/03 D	2002/CE/II/05 C (74%)	2009/CE/II/04 D (72%)	2022/DSE/I/1 D
2025/DSE/I/02 B	2025/DSE/I/04 D	2025/DSE/I/26 A		

Note: Figures in brackets indicate the percentages of candidates choosing the correct answers.

1996/CE/I/10(d)

In Country A, most resources are allocated according to government plans and commands; in Country B, the allocation of most resources is guided by the market mechanism. (2)

In Country A, most resources are privately owned by individuals; in Country B, most resources are owned by the government. (2)

2023/DSE/II/2(b)

- (b) 'How to produce', because (1)
replacing labour with capital is related to the production method / combination of factor (1)
inputs.

1.4 POSITIVE AND NORMATIVE STATEMENTS

Microeconomics by Topic
I. Basic Concepts

MARKING SCHEME

1986/AL/II/01 A	1987/AL/II/29 D	2004/AL/II/03 B (56%)	SP/DSE/I/03 B	2015/DSE/I/01 C (71%)
1986/AL/II/02 C	1998/AL/II/22 B (modified)	2011/AL/II/03 D (84%)	2012/DSE/I/01 D (67%)	2018/DSE/I/04 B (79%)
2021/DSE/I/01 C	2023/DSE/I/04 D	2025/DSE/I/25 A		

Note: Figures in brackets indicate the percentages of candidates choosing the correct answers.

LQ - ANS

<HKDSE 2022 Paper 2 – 5b>

- (b) Positive statement. It is refutable by facts-people can use property-price data before and after the introduction of STCGT to verify whether the statement is correct. (2)

1.5 COMPETITION AND DISCRIMINATION

Microeconomics by Topic
I. Basic Concepts

MARKING SCHEME

1986/AL/I/07 D	1990/AL/I/18 D	1994/AL/I/02 D	2005/AL/I/03 D (81%)	2012/AL/I/03 D (86%)
1987/AL/I/06 B	1992/AL/I/05 A	1995/AL/I/01 C	2007/AL/I/03 D (88%)	2025/DSE/I/24 A
1987/AL/I/10 C	1992/AL/I/09 C	2001/AL/I/06 D	2007/AL/I/25 C (62%)	
1987/AL/I/18 D	1993/AL/I/21 D	2004/AL/I/02 D (86%)	2009/AL/I/01 A (49%)	

Note: Figures in brackets indicate the percentages of candidates choosing the correct answers.

1991/AL/I/3

A good is said to be scarce if, to an individual, more of it is preferred to less of it.

When more than one individuals in society want more of the same good, competition is implied.

With competition, some criterion is needed to determine who wins and who loses. For whatever criterion chosen, it will necessary be discriminatory in nature because different individuals will have different advantages in winning under the same criterion.

2012/DSE/II/5(a)

No, because

when there is excess demand / shortage, non-price competition, such as queuing, will emerge, and the new competitive criteria does not necessarily depend on income.

(1)

(3)

1.6 PROPERTY RIGHTS

MARKING SCHEME

1988/AL/I/23 B	1994/AL/I/10 B	1995/AL/I/16 A	2000/AL/I/28 A	2011/AL/I/02 C (51%)
1990/AL/I/29 D	1994/AL/I/18 C	1997/AL/I/12 D	2004/AL/I/26 C (78%)	2011/AL/I/25 C (56%)
1992/AL/I/24 C	1994/AL/I/26 D	1997/AL/I/19 A	2010/AL/I/20 D	2016/DSE/I/05 C (88%)
2019/DSE/I/01 C	2023/DSE/I/03 B	2024/DSE/I/04 D		

Note: Figures in brackets indicate the percentages of candidates choosing the correct answers.

1993/AL/I/8

At the time of auction, the future is not certain. Events may happen which would affect the value of the bid. If the bid is transferable, a bid winner may transfer his rights to another operator if adverse events happen that are particularly harmful to him. Therefore, bidders are willing to offer higher bids in the case where the bid is transferable.

Even if no unexpected events were to follow, the bids may still be higher with transferable rights because the bidders must be aware at the time of the bidding that their decisions might be mistakes, given positive information costs.

2004/AL/I/7(b)

Yes, there are, property rights to the deer in the wild are delineated by the know-how or special skills of the less-than-10-percent superior hunters. This is just like a strong man occupying a piece of land without legal assignment will have an implicit claim to the land if no competitor can take it away or infringe upon his use rights.

2013/DSE/II/2(a)

Private property rights refer to the exclusive right to use, the exclusive right to receive income from, and the right to transfer a property owned by an individual. (3)

The private property rights over the flats under the scheme are weakened as the flat-owners' right of transferring their flats to non-residents is confined. (1)

1.7 INTEREST

MARKING SCHEME

1989/AL/I/20 C	1998/AL/I/26 B	2006/AL/I/11 D (78%)	2016/DSE/I/4 D (45%)
1993/AL/I/28 B	2003/AL/I/24 A	2012/DSE/I/3 D (90%)	2017/DSE/I/2 C (53%)

Note: Figures in brackets indicate the percentages of candidates choosing the correct answers.

1992/AL/I/2

Interest is a price because it is a sacrifice one must make to exchange for an economic good.

It is a price one pays for the earlier availability of goods or a price one pays for earlier over later consumption.

2004/AL/I/5

Disagree. Interest is the price of earlier availability of goods, and without money this price or premium still exists as long as a market (for borrowing and lending) exists. Hence it may be charged in a barter economy.

I loan you an apple, when you return it a year later, our agreement would be you return an apple plus a bite from another apple. That bite is interest. Money is not implied.

PP/DSE/II/1

To Peter: The bookmark was the price / premium he paid for the earlier availability of the book.

(2)

To Mary: The bookmark was the compensation she received for deferring consumption.

(2)

2025/DSE/II/4(Cross Topic4.1)

(a) Reasons:

- Not durable, as rice may decay within years and cannot be stored for a long time.
- Not homogeneous, as rice does not have uniform physical characteristics.
- Not portable, as people need to carry a large quantity of rice for high-value transactions.
- Any other relevant point

[Mark the FIRST TWO points only.]

} 2@
max: 4

(b) As a lender, Villager B receives the bag of rice as compensation for deferred consumption.

(2)

CH 2 PRODUCTION

Microeconomics by Topic
2. Production

2.1 TYPES OF PRODUCTION

MARKING SCHEME

1990/CE/II/27 C	1993/CE/II/04 B	1995/CE/II/15 C	2004/CE/II/19 B (43%)	2010/CE/II/15 C (49%)
1991/CE/II/04 B	1993/CE/II/05 C	1997/CE/II/28 A	2005/CE/II/14 C (81%)	2014/DSE/I/10 D (67%)
1991/CE/II/05 D	1994/CE/II/60 D	2002/CE/II/12 B (54%)	2007/CE/II/13 D (35%)	2025/DSE/I/05 D
1992/CE/II/04 D	1995/CE/II/04 D	2003/CE/II/15 D (26%)	2008/CE/II/13 C (19%)	

Note: Figures in brackets indicate the percentages of candidates choosing the correct answers.

1990/CE/I/1(a)(ii)
Tertiary production, because provision of service. (1)
(1)

1996/CE/I/2
Secondary production - food materials are cooked in the shop and turned into cooked food. (2)
Tertiary production - when having meals, customers are served by waiters in the shop. (2)

2004/CE/I/9(d)
Tertiary production, because they provide services. (1)
(1)

2015/DSE/II/(a)
Tertiary production, because it provides retail services. (1)
(1)

2017/DSE/II/13(a)(i)
Tertiary production. (1)
(1)

2022/DSE/II/9(ai)

(a) (i) Types of production: (4)

Primary production: As farmers, they directly extract resources from nature.

Tertiary production: As hostel owners, they provide catering and accommodations services to visitors.

2024/DSE/II/10(b)

(b) Tertiary production, because airlines provide (flight) services. (2)

CH 2 PRODUCTION

Microeconomics by Topic
2. Production

2.1 TYPES OF PRODUCTION

2025/DSE/II/5

(a) Unemployment rate $= 117\,800 / (3\,708\,600 + 117\,800) \times 100\%$
 $= 3.08\%$

(1)
(1)

(b) Losses to society caused by unemployment:

- loss of output
- loss of human capital
- political and social unrest
- any other relevant point

} 1@
max: 2

[Mark the FIRST TWO points only.]

2.2 CONSUMER GOODS AND PRODUCER GOODS

MARKING SCHEME

1992/CE/II/06 B	1996/CE/II/02 D	1999/CE/II/10 B	2009/CE/II/14 B (55%)	2025/DSE/I/40 C
1995/CE/II/14 D	1999/CE/II/02 C	2007/CE/II/12 C (56%)	2016/DSE/I/02 C (66%)	
2020/DSE/II/4 D	2022/DSE/I/04 D	2023/DSE/I/05 D	2024/DSE/I/06 A	

Note: Figures in brackets indicate the percentages of candidates choosing the correct answers.

2.3 FACTORS OF PRODUCTION

Microeconomics by Topic
2. Production

MARKING SCHEME

1990/CE/II/13 D	1999/CE/II/11 A	1994/CE/II/16 A	2007/CE/II/17 A (69%)	1995/CE/II/19 B
1991/CE/II/10 B	2000/CE/II/14 B	1995/CE/II/18 A (deleted)	2007/CE/II/18 B (28%)	1997/CE/II/57 A
1992/CE/II/07 D	2003/CE/II/16 B (20%)	1998/CE/II/16 D	2010/CE/II/17 D (82%)	2001/CE/II/12 D
1993/CE/II/08 B	2005/CE/II/15 C (80%)	2000/CE/II/13 A	2016/DSE/I/06 B (84%)	2004/CE/II/14 D (84%)
1996/CE/II/15 D	2007/CE/II/16 C (41%)	2001/CE/II/09 B	1990/CE/II/28 D	2006/CE/II/13 A (79%)
1997/CE/II/13 B	2008/CE/II/14 B (84%)	2001/CE/II/11 B	1991/CE/II/07 D	2019/DSE/I/08 B
1998/CE/II/12 D	2009/CE/II/15 A (83%)	2002/CE/II/15 D (50%)	1992/CE/II/08 B	2020/DSE/I/6 D
2025/DSE/I/36 A				

Note: Figures in brackets indicate the percentages of candidates choosing the correct answers.

1991/CE/I/4(a)(i)

Yes, because

he makes decisions and bears risks in a firm.

(1)

(2)

1995/CE/I/11(b)

The distinctive economic role is that Mr Ip belongs to entrepreneurship, because
he has to bear the risk of the business / formulate policies.

(1)

(1)

Mary does not belong to entrepreneurship / she belongs to labour, because
she does not have to bear risk / she provides human effort in production.

(1)

(1)

1996/CE/I/4(b)

A traffic light is a man-made resource used for production.

(2)

1997/CE/I/1

Entrepreneurship, because

(1)

he bears risk / makes decisions.

(1)

Labour, because

(1)

he provides human effort in production.

(1)

2000/CE/II/3

It is man-made and it is used for production.

(2)

2004/CE/I/9(e)

Capital, because

(1)

those cars are man made resources (used for production).

(1)

Microeconomics by Topic

2. Production

2005/CE/I/2

Entrepreneurship, because (1)

he bears risks / makes decisions. (1)

Labour, because (1)

he provides human effort in production. (1)

2014/DSE/II/2(a)

Entrepreneurship, because (1)

he bears business risk and makes business decisions. (1)

2016/DSE/II/10(b)

Land, because (1)

it is a natural resource which is a gift of nature. (1)

2018/DSE/II/2(b)

Entrepreneurship, because (1)

she bears business risk. (1)

2021/DSE/II/1b

(b) Capital goods, when the riders use them to help further production, i.e. delivery service.. (2)

OR

Consumption goods, when the riders use them for self-consumption, e.g. playing games.

2023/DSE/II/2(a)

(a) Capital, because (1)

it is a man-made resource used in production. (1)

MARKING SCHEME

2.4 LABOUR SUPPLY AND PRODUCTIVITY

1990/CE/II/20 D	1994/CE/II/17 A	1998/CE/II/14 B	2005/CE/II/16 D (51%)	2012/DSE/I/06 C (86%)
1991/CE/II/09 C	1995/CE/II/16 C	1998/CE/II/15 C	2005/CE/II/17 C (84%)	2012/DSE/I/10 D (38%)
1991/CE/II/56 D	1995/CE/II/59 D	1999/CE/II/14 A	2006/CE/II/14 C (72%)	2014/DSE/I/09 B (79%)
1992/CE/II/10 C	1996/CE/II/12 B	1999/CE/II/20 D	2007/CE/II/15 B (67%)	2015/DSE/I/06 B (73%)
1992/CE/II/14 D	1996/CE/II/13 B	2001/CE/II/13 B	2009/CE/II/16 C (81%)	2016/DSE/I/08 B (86%)
1993/CE/II/06 A	1996/CE/II/14 A	2001/CE/II/15 A	2010/CE/II/18 B (68%)	2017/DSE/I/11 C (75%)
1993/CE/II/19 C	1997/CE/II/55 B	2002/CE/II/14 D (29%)	2010/CE/II/20 A (84%)	2018/DSE/I/09 C (94%)
2019/DSE/I/06 D	2023/DSE/I/06 D	2024/DSE/I/08 D	2024/DSE/I/09 C	2025/DSE/I/35 A

Note: Figures in brackets indicate the percentages of candidates choosing the correct answers.

1990/CE/II/2(b)

(i) Average labour productivity in 1985: $\$39\,589 \text{ mn} \div 177\,000 = \$223\,666.7$

Average labour productivity in 1986: $\$47\,052 \text{ mn} \div 190\,000 = \$247\,642.1$ (3)

$\therefore$ Higher labour productivity in 1986 (1)

(ii) - better training and knowledge

- more investment in technology and machines

- better management, e.g. division of labour

(Mark the **FIRST TWO** points only.) (2@ max :4)

1994/CE/II/2

Average labour productivity is output per unit of labour, e.g. output per man-hour. (2)

Methods to raise labour productivity:

- division of labour

- increasing quantity / quality of capital goods

- more training of workers

- incentive schemes, e.g. a change from time rate to piece rate

(Mark the **FIRST TWO** points only.) (2@, max: 2)

2000/CE/II/1(c)

Output per unit of labour or output per man-hour

- better education / training

- better capital

- better technology

- better management, e.g. division of labour

- better working conditions

(Mark the **FIRST THREE** points only.) (1@, max:3)

2002/CE/I/3

Firm A: $100\,000 \div (100 \times 500) = 2$ units of output per man-hour

(1)

Firm B: $144\,000 \div (120 \times 800) = 1.5$ units of output per man-hour

(1)

- different education / training
- different capital
- different technology
- different management
- different working conditions

(1@, max : 2)

(Mark the **FIRST TWO** points only.)

2005/CE/I/10(a)(ii)

- different quality of factor input
- different education and training
- different degree of technological advancement
- different management efficiency / different degree of division of labour ,
- different working environment
- different labour welfare and incentives

(1@ , max: 3)

[Mark the **FIRST THREE** points only.]

2007/CE/I/2

(a) In year 2006: $[36\,000 / (40 \times 300)] = 3$ units / man-hour

(1)

In year 2007: $[40\,000 / (50 \times 200)] = 4$ units / man-hour

(1)

- (b)
- better capital / mechanization \
 - better training / education
 - better working condition
 - better management, e.g. the practice of division of labour
 - higher incentives, e.g. change from time rate to piece rate
 - improvement in technology

(1@, max: 2)

[Mark the **FIRST TWO** points only.]

2010/CE/I/2

(a) No, because

(1)

the labour supply of Firm A = $(20 \times 100) = 2\,000$ man hours is smaller than that of Firm B

(1)

= $(12 \times 200) = 2\,400$ man hours.

(1)

(b) No, because

(1)

the labour productivity of Firm B = $(120\,000 / 2\,400) = 50$ units of output per man hour is

(1)

the same as that of Firm A = $(100\,000 / 2\,000) = 50$ units of output per man hour.

(1)

2024/DSE/II/13d

(d) Marks for effective communication (EC: max 2 marks)

Marks	Performance
2	● Support arguments with the source/data and appropriate economic theories. ● Present relevant material. ● Present well-organised and coherent answers without repetition of ideas. ● Use language that expresses ideas clearly and fluently with appropriate use of words/terms/symbols.
1	● Present arguments with some support of the source/data and economic theories. ● Present some irrelevant material. ● Present answers in a less organised way with some repetition. ● Use language that conveys a clear message with some inappropriate use of words/terms/symbols.
0	● Present arguments with no support of the source/data and economic theories. ● Present material unrelated to the gist of the question. ● Present inconsistent arguments. ● Express limited ideas with inappropriate use of words/terms/symbols.

The maximum mark for content is 12 marks.

2.5 DIVISION OF LABOUR

MARKING SCHEME

1992/CE/II/11 A	1997/CE/II/48 C	2001/CE/II/16 D	2007/CE/II/14 C (92%)	2018/DSE/I/11 A (85%)
1993/CE/II/15 D	1999/CE/II/52 A	2003/CE/II/17 D (71%)	2009/CE/II/17 D (94%)	2020/DSE/I/3 B
1994/CE/II/59 A	2000/CE/II/19 B	2004/CE/II/17 D (69%)	2010/CE/II/16 D (55%)	2022/DSE/I/09 D

2025/DSE/I/09

D

Note: Figures in brackets indicate the percentages of candidates choosing the correct answers

1995/CE/II/2

Division of labour / specialization

- jobs assigned according to talents and skills
- more skillful at jobs due to continuous practice
- time saved from moving from one job to another job
- time saved from learning all the jobs
- encouraged the use of capital / more advanced capital

(Mark the **FIRST TWO** points only)

(1)

(2@, max: 4)

1997/CE/II/10(c)

Average labour productivity is output per unit of labour input, e.g. output per man-hour.

- getting talented or skilful people to do the job
- continuous practice makes workers even more skilful
- saves time because not need to shift from one task to another task

(Mark the **FIRST TWO** points only)

(2@, max: 4)

2001/CE/II/1

- stimulus to mechanization
- practice makes perfect
- saves time in learning the job
- saves time in moving around jobs
- choose the best person to do the job

(Mark the **FIRST THREE** points only)

(1@, max : 3)

2003/CE/II/10(a)

Division of labour

- choosing the most suitable persons for the job
- practice makes perfect / learning by doing
- saving time from training
- saving time from moving around tasks
- stimulus to mechanization

(Mark the **FIRST THREE** points only)

(1)

(1@, max: 3)

2013/DSE/II/10(a)

Division of labour

- choosing the most suitable person for the job
- practice makes perfect
- saving time in training
- stimulus to mechanization

[E points only.]

(1)

(1@; max: 3)

2016/DSE/II/13(a)

Division of labour / specialization

(1)

Reasons:

- practice makes perfect
- choosing the most suitable person for the job
- save time in training
- stimulus to mechanization

(1@; max: 3)

[Mark the **FIRST THREE** points only.]

2018/DSE/II/13(b)

(i) Labour productivity would rise as improved technology or higher quality of capital goods would result in an increase in output per unit labour. (2)

(ii) Uncertain. Labour productivity would increase if the imported workers are more productive on average than the existing local ones. (3)

2019/DSE/II/5

(a) Division of labour/ (Principle of) comparative advantage (1)
[Mark the **FIRST** point only.]

Benefits:

- Finding the most suitable person for the job: the parents in both families can use their expertises in teaching their children so the children can learn both languages better.
- Specialisation and exchange: Mr Wong can specialise teaching English, i.e. the production in which his opportunities cost is lower, then exchange for his children learning Japanese from Mrs. Chan. The children can learn both languages better with a lower cost. (Similar argument applies to Mrs Chan in her teaching of Japanese as well.)
- any other relevant point

(2)

(b) Economies of scale. (1)
[Mark the **FIRST** point only.]

Benefits:

- They can have more discounts when food ingredients are purchased in bulk which would reduce the average cost of producing the dinners.
- Cooking together can save the average cooking time and at the same time can have a larger variety of food choices.
- any other relevant point

(2)

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(c) Division of labour. (1)

Reasons:

- Practice makes perfect.
- Choose the most suitable workers to do the job.
- Save time in training
- Any other relevant point

} @1
max:2

[Mark the **FIRST TWO** points only.]

2.6 LABOUR MOBILITY

MARKING SCHEME

1990/CE/II/09 C	1995/CE/II/21 C	2004/CE/II/15 D (74%)	2014/DSE/II/11 B (83%)
1991/CE/II/06 B	1997/CE/II/14 D	2006/CE/II/15 D (85%)	2015/DSE/II/04 D (89%)
1992/CE/II/09 C	1999/CE/II/16 C	2008/CE/II/15 C (79%)	2016/DSE/II/09 D (49%)
1993/CE/II/14 C	2001/CE/II/14 D	2012/DSE/II/05 C (70%)	2017/DSE/II/07 B (65%)
2022/DSE/II/08 C	2023/DSE/II/07 B		

Note: Figures in brackets indicate the percentages of candidates choosing the correct answers.

1991/CE/II/2(a)

- (i) Opportunity cost is the highest-valued option or the best alternative that has to be forgone. (2)
Occupational mobility is the ease with which a factor moves from one occupation / trade / industry to another. (2)
- (ii) In changing occupation, the opportunity cost of a doctor (in terms of income forgone) can be very high. (2)
The value of taking up a new occupation (i.e. a doctor's earnings in another job) is usually lower than the cost of so doing (i.e. income, satisfaction and social status etc. from being a doctor). (3)

1993/CE/II/2/(b)(ii)

- retraining is required / lack of qualifications, experience and skills in the new jobs
- age problem
- loss in income earnings
- licensing requirements / entrance barriers
- lack of information about the job market
- adaptation to new working environment

(2@, max: 6)

(Mark the **FIRST THREE** points only.)

2013/DSE/II/11(b)(i)

- It increases the geographical mobility, because (2)
workers are more willing to travel to other districts for work. (1)

[Note: The lump-sum subsidy will not change the marginal cost of travelling. But some of the workers originally constrained by high travelling expenses may have the incentive to travel further for work (for a higher paid job).]

2015/DSE/II/10(c)

- Professional football players have low occupational mobility, because (2)
their skills are very specific and not easily transferable to other jobs. (1)

2017/DSE/II/12(a)

- The geographical mobility of low-skilled workers from the EU would be lowered because (2)
the visa requirement would increase the cost for those EU workers to go to work in the UK. (1)

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2. Supply-side factors:

- Supply of 'riders' is smaller, as driving a motorbike has a licence requirement.
- Supply of 'riders' is smaller, as 'riders' have a higher cost of production (e.g. gasoline cost, depreciation cost, etc).
- Any other relevant point.

} @2
max:2

[Mark the FIRST point only.]

Demand-side factors:

- Demand for 'riders' is larger, as riders can serve a larger area than 'walkers'.
- Demand for 'riders' is larger, as riders can deliver the food in a shorter time.
- Any other relevant point.

} @2
Max:2

[Mark the FIRST point only.]

<HKDSE 2022 Paper 2 – 9aii>

- (ii) There would be an increase in the occupational mobility of the villagers, as they engaged in a new occupation (as hostel owners). (3)

2.7 WAGE PAYMENT METHODS

MARKING SCHEME

1991/CE/II/08 D	1996/CE/II/16 A	2000/CE/II/15 D	2009/CE/II/18 A (77%)	2021/DSE/I/07 A
1991/CE/II/22 D	1998/CE/II/17 D	2002/CE/II/16 D (66%)	2012/DSE/I/07 A (76%)	2025/DSE/I/08 C
1994/CE/II/18 D	1999/CE/II/13 C	2003/CE/II/18 A (84%)	2013/DSE/I/07 D (83%)	
1995/CE/II/17 A	1999/CE/II/51 B	2006/CE/II/16 B (75%)	2016/DSE/I/10 A (94%)	

Note: Figures in brackets indicate the percentages of candidates choosing the correct answers.

1990/CE/I/1(a)(iii)

- output is difficult or costly to measure
- work not standardized
- easy to calculate wage payment

(Mark the **FIRST TWO** reasons only.)

(2@, max: 4)

1992/CE/I/1(c)

(As compared with the case when the driver is paid a fixed wage.)

Advantages:

- part of the risks can be shifted onto the driver
- the owner may earn a higher income as the driver has a higher working incentive

(3@, max: 3)

Disadvantages:

- under-report of the true revenue as the driver tends to cheat
- higher maintenance cost as the driver tends to over-utilize the taxi

(3@, max: 3)

(Mere listing of point elaboration -- max. 1 mark. Mark the **FIRST** point only.)

1993/CE/I/1(d)

Time rates

- the manager's output is difficult or costly to measure / work not standardized
- easy to calculate the wage payment

(2@, max: 2)

Profit-sharing

- the manager has higher incentive to manage the firm
- better control of production cost
- the manager is more loyal to the firm

(2@, max: 2)

(Mark the **FIRST** point only.)

1996/CE/I/12(b)

Advantages:

- higher working incentive
- lower monitoring

(2@, max: 2)

Disadvantage:

- less careful in using the machinery
- better quality control is required

(2@, max: 2)

(Mark the **FIRST** point only.)

Microeconomics by Topic

2. Production

1998/CE/1/11(a)

- basic monthly salary to maintain a team of staff
 - commission to increase the incentive to sell
 - to lower the monitoring cost because the amount of commission already reflects the output of the salesman
 - to share risk of operation with the salesman because commission payment varies with the sales revenue
- (2@, max: 4)
- (Mark the **FIRST TWO** points only.)

1999/CE/1/3

Company A is using time rate (1)
and Company B is using piece rate. (1)

Time rate:

- to avoid the disadvantage of using piece rate under which workers may forego quality for quantity
 - to maintain a steady supply of workers within a time period
 - no need to measure the output for determining the wage payment
- (2@, max: 2)

Piece rate:

- to improve the incentive to produce more or to increase productivity of workers
 - to save the monitoring cost of workers (e.g. quality control)
 - less rigid to adjust the total production cost: extra workers can be employed or dismissed according to the business volume
- (2@, max: 2)
- (Mark the **FIRST** point only.)

2001/CE/1/9(c)

Advantage:

- his workers would have a higher working incentive
 - lower cost of supervision
- (1@, max: 1)

Disadvantage:

- higher cost of monitoring the quality of the products
 - more costly to calculate the wage payment
- (1@, max: 1)
- (Mark the **FIRST** point only.)

2002/CE/1/12(b)(i)

Advantage:

- to maintain a team of staff
 - lower cost of measuring the contribution of the workers
- (2@, max: 2)

Disadvantage:

- lower working incentive
 - higher supervising cost
- (2@, max: 2)
- (Mark the **FIRST** point only.)

2007/CE/1/2

(a) Post A: time rate (1)
Post B: piece rate (1)

- (b) (i) - higher monitoring cost
- lower working incentive
 - lower productivity
- (2)

- (ii) - lower quality of product
- higher cost of measuring the output
- (2)
- [Mark the **FIRST** point only.]

2008/CE/1/9(c)

- to increase the working incentive
 - to lower the cost of monitoring the performance of the employees
 - to spread the business risk between the employer and the employees
- (Mark the **FIRST TWO** point only.)

(2@, max: 4)

2009/CE/1/4

- (a) Job A: piece rate
Job B: time rate

(1)

(1)

- (b) One advantage:

- lower cost of supervising their workers
- higher working incentive
- higher productivity

(2)

One disadvantage:

- higher cost in measuring the output
- higher cost of monitoring the quality of service

(2)

[Mark the **FIRST** point only.]

2010/CE/1/3

Advantage of this payment method to shop owners over the payment method of a basic salary only:

- to increase the incentive to sell more
- lower the monitoring cost of the work effort
- share risk of operation with the salespersons

(2)

Advantage of this payment method to shop owners over the payment method of a commission only:

- to maintain a team of staff / reduce turnover rate as salespersons won't leave easily
- easier to employ staff due to more stable income

(2)

[Mark the **FIRST** advantage only.]

2013/DSE/11/10(b)

- (i) It improves the work incentive of the manager.

(2)

- (ii) It gives a more stable income as the manager still receives a basic salary even if the business is poor.

(2)

2015/DSE/11/10(b)

The owner of the football team can share some business risks with Player B as the former would make less payment to the latter when no match is played and/or no goal is scored

(2)

The owner of the football team can have a lower cost of monitoring players' performance as Player B would have a higher work incentive in training and in matches

(2)

2017/DSE/11/2(a)

- (i) possibility of getting higher income with more effort and/or better service

(2)

- (ii) lower cost of monitoring the performance of the waiters and/or higher incentives for the waiters to work harder and provide higher-quality service to the customers

(2)

2018/DSE/11/10(a)

The cost of measuring the output of the foreign domestic helpers is too high / the work of domestic helpers is not standardised.

(2)

2019/DSE/II/9

- (c) **Advantage:** Cost of measuring output of the employees can be saved. (2)
Disadvantage: Work incentive may not be high.
[Mark the FIRST advantage and FIRST disadvantage only.]
- (d) **Advantage:** Higher output will yield higher income. (2)
Disadvantage: Income may be unstable.
[Mark the FIRST advantage and FIRST disadvantage only.]

2021/DSE/II/1

- (a) Disadvantages:
..... Less job/income stability
..... Self-provision of capital goods¹
..... Tendency to drive fast and ignore road safety
..... Any other relevant point
[Mark the FIRST TWO points only.]

2022/DSE/II/10d

- (d) It can reduce the monitoring cost as athletes can have a higher incentive to win games. (2)

2023/DSE/II/10(e)

- (e) It is prohibitively costly to measure the outputs of the Financial Secretary as his outputs are not standardised. (2)

2024/DSE/II/11a

- (a) Advantage:
possibility of earning a higher income by providing better services (1)
- Disadvantage:
higher risk of income fluctuations (1)

MARKING SCHEME

2.8 PRIVATE GOODS & PUBLIC GOODS

1985/AL/I/10 D	1996/AL/I/04 A	2000/AL/I/28 A	2007/AL/I/29 A (91%)	2011/AL/I/29 C (73%)
1986/AL/I/08 B	1996/AL/I/16 C	2001/AL/I/15 C	2008/AL/I/27 A (95%)	2012/DSE/I/04 B (72%)
1988/AL/I/03 B	1997/AL/I/05 C	2002/AL/I/28 C	2010/AL/I/23 B	2013/DSE/I/06 D (55%)
1989/AL/I/24 D	1998/AL/I/25 D	2003/AL/I/17 B	2010/AL/I/30 B	2017/DSE/I/03 D (82%)
1990/AL/I/29 D	1999/AL/I/21 B	2005/AL/I/28 B (82%)	2011/AL/I/24 D (63%)	2018/DSE/I/05 B (58%)
2020/DSE/I/2 B	<i>Note: Figures in brackets indicate the percentages of candidates choosing the correct answers.</i>			

2015/DSE/II/1(a)(ii)

Yes, because

a person can download and study the course materials without reducing the availability of those materials to others, i.e., these materials are non-rivalrous in consumption. (1)

(2)

[Note: The answer can be "No" if there exist network capacity constraints (or congestion problems) in downloading the course materials.]

CH 3 INPUT-OUTPUT RELATIONSHIP

3.1 THE LAW OF DIMINISHING RETURNS

Microeconomics by Topic
3. Input-Output Relationship

MARKING SCHEME

1990/CE/II/06 C	1996/CE/II/22 D	2002/CE/II/13 C (58%)	2006/CE/II/17 A (58%)	2014/DSE/II/05 C (79%)
1992/CE/II/17 C	1996/CE/II/23 B	2003/CE/II/20 C (62%)	2008/CE/II/16 D (53%)	2015/DSE/II/09 A (79%)
1994/CE/II/21 C	1999/CE/II/15 D	2004/CE/II/18 C (69%)	2010/CE/II/19 C (44%)	2018/DSE/II/10 C (65%)

2025/DSE/II/06

B

Note: Figures in brackets indicate the percentages of candidates choosing the correct answers.

1994/CE/II/11(a)

Law of diminishing returns:

When variable factors are added continuously to the fixed factors, the marginal product will eventually diminish. (3)

Fertilizers (Units)	Land (Units)	Labour (Units)	TP (Units)	MP (Units)
0	10	40	30	-
1	10	40	70	40
2	10	40	120	50
3	10	40	160	40
4	10	40	190	30

From the above table, after 2 units of the variable factor (fertilizers) are employed, an additional unit of fertilizers leads to a fall in MP $\Rightarrow$ the law applies to the farm. (3)

1997/CE/II/3

Assume that the factory has 100 workers. (The number is arbitrary.) After 95 workers are employed, the change in MP caused by adding more and more workers in production can be found by converting the table given in the question into the following table:

No. of workers coming to work	MP (units / day)
95	-
96	60
97	50
98	40
99	30
100	20

(2)

In other words, when the variable factor (labour) is continuously added to the fixed factors (land, capital and entrepreneurship), the marginal product will eventually diminish. This is the law of diminishing marginal returns. (3)

1998/CE/II/4

(a) When a variable factor is successively added to the fixed factor(s), (holding technology constant) the marginal product will eventually diminish. (2)

(b) No, because all factors are variable / long-run production. (2)

2000/CE/I/4

The law states that when variable factors are added continuously to the fixed factors, (holding technology constant), the marginal product will eventually decrease. (3)

The marginal product calculated is shown below: (3)

AP	MP
30	-
34	62
37	61
39	57
40	50

∴ The data illustrate the law of D.M.P.

2001/CE/I/3

(The data show that) machinery is a fixed factor and when the variable factor, labour, is added successively to machinery, the marginal product eventually diminishes (1)
(1)
(2)

Labour (units)	Machinery (units)	MP (units)
10	5	-
11	5	50
12	5	100
13	5	150
14	5	130

∴ the data illustrate the law of diminishing marginal returns. (1)

2003/CE/I/2

(a) The law states that as the variable factor is continuously added to the fixed factor, the marginal product will eventually diminish. (3)

(b)

Machine (Unit)	MP (Unit)
1	-
2	14
3	16
4	10

It illustrates the law in (a) since MP eventually diminishes from 16 units of output to 10 units of output when the 4th unit of machine is added to the production. (3)

2006/CE/I/4

The law states that when variable factors are added continuously to the fixed factors, [holding technology constant,] the marginal product will eventually decrease. (3)

Labour (Units)	4	5	6
Marginal Product (Units)	60	30	20

OR

Diminishing marginal product of labour sets in when labour increases from 4 units to 5 units as the marginal product falls from 60 to 30 units. (2)

∴ The data illustrate the law of diminishing marginal returns. (1)

2007/CE/I/3

(a) The law states that when variable factors are added continuously to the fixed factors, the marginal product will eventually decrease. (3)

(b)

Labour (units)	Marginal product (units)
5	-
6	220
7	210
8	170

(2)

Yes, because
the marginal product decreases. (1)
(1)

2009/CE/I/3

The law states that when variable factors are added continuously to the fixed factors, the marginal product will eventually decrease. (3)

Machine (units)	Marginal output of labour (units)
3	18
4	22
5	21

(2)

Yes, because
the marginal product decreases. (1)
(1)

2012/DSE/II/2

The law states that (holding technology constant,) when more units of a variable factor are added successively to a given quantity of fixed factors, the marginal product of this factor will eventually diminish. (3)

No, because
there is no fixed factor / long-run production. (1)
(1)

2016/DSE/II/3

The law states that (holding technology constant) when more units of a variable factor are added successively to a given quantity of fixed factors, the marginal products of this factor will eventually diminish. (3)

Yes, because
when labour increases from 3 to 4 units, the marginal product of labour drops from 18 to 14 units (, so the law of diminishing returns holds). (1)
(2)

2021/DSE/II/1c

(c) Law of diminishing returns. When a larger amount of variable factor are continuously added to a given amount of a fixed factor, the marginal product (of the variable factor) would eventually diminish, holding technology constant. In this case, the rider is the variable factor and the firm's operating system is the fixed factor. (5)

2023/DSE/II/3

Marks
(1)

(a) Its quantity does not vary with output.

(b) Yes, because (1)
the marginal product of labour drops from 48 to 41 units when the quantity of labour (a (2)
variable factor) increases from 12 to 13 units, given a fixed quantity (12 units) of
machinery (fixed factor).

2025/DSE/II/1

The law states that (holding technology constant), when more units of a variable factor are (2)
added successively to a given quantity of fixed factors, the marginal product of this factor
will eventually decrease.

Yes, because given a constant number of machines, a fixed factor, when labour increases (3)
from 3 units to 4 units, the marginal product of labour drops from 25 units to 23 units.

3.2 VARIABLE AND FIXED COSTS

MARKING SCHEME

1990/CE/II/11
D

2003/CE/II/10
A (90%)

2017/DSE/I/10
B (87%)

2025/DSE/I/41
B

Note: Figures in brackets indicate the percentages of candidates choosing the correct answers.

1991/CE/I/4(a)(ii)

License fee of the lorry / monthly parking fee of the lorry / registration fee of the firm is a fixed cost, because
it does not vary with the output level.

(1)
(2)

Cost of petroleum / depreciation of the lorry is a variable cost, because
it varies with the output level.

(1)
(2)

2014/DSE/II/2(b)

Cost of buying the accessories, or wages of part-time workers, because
expenses on these items would vary (increase) with the output of Ah Wing's shop.

(1)
(1)

[Mark the **FIRST** example only.]

3.3 ECONOMIES & DISECONOMIES OF SCALE

Microeconomics by Topic
3. Input-Output Relationship

MARKING SCHEME

1990/CE/II/22 B	1993/CE/II/21 C	2000/CE/II/17 B	2006/CE/II/18 D (72%)	2016/DSE/I/07 B (52%)
1991/CE/II/20 D	1995/CE/II/22 B	2004/CE/II/20 D (63%)	2008/CE/II/17 D (38%)	2018/DSE/I/12 A (51%)
1992/CE/II/20 D	1998/CE/II/13 A	2005/CE/II/18 B (58%)	2013/DSE/I/05 C (71%)	2020/DSE/I/7 B
2022/DSE/I/07 C	2025/DSE/I/11 B			

Note: Figures in brackets indicate the percentages of candidates choosing the correct answers.

1992/CE/II/1(b)

It enjoys economies of scale

- purchasing economy: bulk purchase of spare parts at a discount (1)
- financial economy: lower cost of borrowing due to goodwill / sufficient collateral
- technical economy: fully utilize machines and spare parts (i.e. less than proportionate of part and components are kept for maintenance purpose) (3@, max: 6)

[Mere listing of points without elaboration – max. 2 marks. Mark the **FIRST TWO** points only.]

1993/CE/II/1(c)

- (i) Long run, because all factors are variable / the scale of production is enlarged because the employment of all factors is increased. (2)
- (ii) - economies of scale (without explanation) (1)
- management economy: increased productivity due to greater scope of specialization (3)
- technical economy: fully utilize machines e.g. a van for delivery service (3)
- purchasing economy: bulk purchase of food materials at a greater discount (3)

(max: 6)

[Mere listing of points without elaboration – max: 2 marks. Mark the **FIRST TWO** points only.]

1995/CE/II/10(a)(ii)

- the set up cost required for two shows is the same as that for one (2)
- management economies: the expertise required for two shows is the same as that for one (i.e. the payment to such professionals as designers and accountants can be spread over a larger output.) (2)
- marketing economies: the advertising fee for marketing two shows is the same as that for one (i.e. the advertising fee can be spread over a larger output.) (2)
- purchasing economies: purchase of materials required at a bigger discount than that for one show (2)

(2@, max:4)

[Mark the **FIRST TWO** points only.]

1997/CE/II/9(a)

Elaboration of the following points in relation to KCRC:

- technical economies: fully utilize plants and equipment, such as maintenance depots
- financial economies of scale: borrow at a lower interest rate due to more collaterals
- managerial economies of scale: increased productivity due to greater scope of specialization
- purchasing economies of scale: bulk purchase of new trains, parts and consumables, etc. at a greater discount
- research and development economies: spreading R&D cost over a larger output (2@, max: 4)

[Mark the **FIRST TWO** points only.]

[Remark : Mere mention of the term 'economies of scale' without elaboration – max: 1 mark.]

Microeconomics by Topic
3. Input-Output Relationship

2000/CE/10(b)(i)

Horizontal integration, because (1)
they are doing the same or similar businesses (1)

Elaboration of 3 of the following:

- financial economies of scale: borrow at a lower interest rate due to more collaterals
- managerial economies of scale: increased productivity due to greater scope of specialization
- purchasing economies of scale: bulk purchase of stationary, printing service (leaflet), etc. at a greater discount
- research and development economies: spreading R&D cost over a larger output (2@, max: 6)

[Mark the **FIRST THREE** points only.]

2004/CE/11(c)(i)

Name 2 relevant economies of scale with elaboration.

- marketing economies of scale: bulk purchase at lower cost, etc
- technical economies of scale: a fuller use of the plants and equipment
- managerial economies of scale: a wider scope for specialization
- financial economies of scale: easier to raise capital and at a lower interest rate (2@, max: 4)

[Remark: Mark the **FIRST TWO** points only.]

2007/CE/11

- management: higher productivity due to a greater scope for specialization / due to benefits from management expertise
- technical: fuller use of resources and facilities / elimination of duplication of resources
- financing: more flexible to raise funds / borrow from banks at a lower interest rate
- purchasing: bulk purchase of raw materials with discount
- marketing: can spread the advertising cost over a larger output
- research and development: can spread the R&D cost over a larger output (2@, max: 6)

[Mark the **FIRST THREE** points only.]

2009/CE/19(a)

- Technical economies of scale:	Large firms can utilize more fully the machines and plant equipment and thus reduce the average cost.
- Managerial economies of scale:	A large firm with a wider scope of specialization can raise efficiency and reduce the average cost
- Financial economies of scale:	Large firms can obtain capital at a lower cost, e.g., borrow from a bank at a lower interest rate.
- Marketing economies of scale in purchasing materials:	Buying raw materials in bulk helps to obtain discounts in the purchase and results in a decrease in average cost.
- Marketing economies of scale in advertising:	As the advertising cost and the cost of providing extra service to customers spread over a larger output, the average cost also decreases.
- Risk diversification:	Large firms can reduce risk by diversification of markets / sources of raw materials.
- Research and development:	Large firms can afford to do research and development say, for new production techniques, to reduce the average cost.

(2@, max: 4)

[Mark the **FIRST TWO** points only.]

2010/CE/1/9(a)

When the firm is larger in scale, it may enjoy the following economies of scale:

- Financial economies:	Obtaining capital at a lower cost, e.g., borrow from a bank at a lower interest rate.
- Managerial economies:	A wider scope of specialization can raise efficiency and reduce the average cost.
- Technical economies:	When the output is larger, machines can be utilized more fully and efficiently and reduce the average cost.
- Marketing economies of scale in purchasing materials:	Buying raw materials in bulk helps to obtain discounts in the purchase and result in a decrease in average cost.
- Marketing economies of scale in advertising:	Spreading the advertising cost and the cost of providing extra service to customers over a larger output can reduce the average cost.
- Research and development:	Spreading research expenses over a larger output can reduce the average cost.
- Risk bearing:	With more and varied attractions, it can spread risk by diversification of markets / sources of raw materials.

(2@, max: 4)

[Mark the **FIRST TWO** points only.]

2015/DSE/II/2(b)

- reduce cost of marketing and promotion
- reduce transportation cost for suppliers
- more related businesses available such as accessories and repair

(2@, max: 4)

[Mark the **FIRST TWO** points only.]

2016/DSE/II/2(c)

- cost of advertising shared by a larger number of students
- bulk purchase of raw material in production
- more extensive division of labour

(1@, max: 3)

2017/DSE/II/2(b)

- may have become too large in scale, so managerial efficiency starts declining
- may have large outstanding loans, so the marginal cost (interest rate) of further borrowing increases
- market may have saturated, leading to a rise in the marketing cost

(2@, max: 4)

[Mark the **FIRST TWO** points only.]

2018/DSE/II/3

- reduction of marketing and promotion costs, because the region would attract more customers for high-technology products/services.
- reduction of transportation cost, because transport network and other infrastructures would develop more quickly, thus improving accessibility.
- reduction of average/marginal cost of recruiting more experienced and skilled workers, because more of these workers would be attracted to the region.

(2@, max: 4)

[Mark the **FIRST TWO** points only.]

2020/DSE/II/2

- c. Financial economies of scale
- Technical economies of scale
- Marketing economies of scale
- Management economies of scale
- Purchasing economies of scale (2 marks)

2022/DSE/II/1b

(b) Reasons:

- The firm may have large outstanding loans, so the marginal cost of further borrowing increases.
- The firm may have become too large in scale, so managerial efficiency starts declining.
- The market may have saturated, leading to a rise in marketing cost.
- Any other relevant point.

} @2
max:2

[Mark the FIRST point only.]

2023/DSE/II/11(d)

(d) Possible reasons:

- The cost of advertising can be spread over a larger quantity of outputs (number of shows).
- The cost of rehearsal can be spread over a larger quantity of outputs.
- any other relevant point

} 2@;
max: 4

[Mark the FIRST TWO points only.]

2024/DSE/II/11f

(f) Possible reasons:

- discount for bulk purchase of inputs, e.g., coffee beans
- spreading of advertising cost over a larger number of outputs
- lower cost of borrowing with more collateral
- Any other relevant point

} 1@
max: 2

[Mark the FIRST TWO points ONLY.]

3.4 SHORT RUN OR LONG RUN?

MARKING SCHEME

1994/CE/II/22 C	2000/CE/II/16 B	2006/CE/II/19 B (55%)	2009/CE/II/19 D (66%)	2019/DSE/I/05 D
1996/CE/II/24 A	2001/CE/II/17 C	2007/CE/II/19 A (42%)	2013/DSE/I/03 D (45%)	2020/DSE/I/ D
1997/CE/II/15 A	2005/CE/II/19 C	2008/CE/II/18 D (78%)	2013/DSE/I/04 B (82%)	

Note: Figures in brackets indicate the percentages of candidates choosing the correct answers.

1993/CE/I/1(b)

Law of diminishing (marginal) returns

(2)

The law states that as more variable factors are added (successively) to the fixed factors, marginal output will (eventually) diminish.

(3)

3.5 PROFIT MAXIMISATION

MARKING SCHEME

1987/AL/I/05 C	1995/AL/I/10 A	2004/AL/I/12 C (86%)	2015/DSE/I/05 B (73%)	2018/DSE/I/06 D (42%)
1987/AL/I/25 A	1998/AL/I/29 C	2012/DSE/I/09 B (39%)	2016/DSE/I/12 B (76%)	2019/DSE/I/04 D
1992/AL/I/25 D	2001/AL/I/29 B	2014/DSE/I/06 C (56%)	2017/DSE/I/08 D (55%)	2019/DSE/I/10 D
2020/DSE/I/9 B	2021/DSE/I/10 A	2021/DSE/I/12 C	2022/DSE/I/11 A(74%)	2023/DSE/I/09 C(67%)
2024/DSE/I/10 D	2025/DSE/I/07 C			

SP/DSE/II/9(c)

- | | |
|---|-----|
| (i) It is a variable cost because it increases with output. | (1) |
| | (1) |
| (ii) 600 units | (1) |
| (iii) 600 units | (1) |

2024/DSE/II/11b

- | | |
|--|-----|
| (b) To maximise profit, a firm under perfect competition produces up to the point where the marginal cost of the last unit equals its marginal revenue (i.e., the market price). | (1) |
| In choosing daily output, the monthly rental cost is viewed as a fixed cost as it does not vary with the quantity of output produced in a day. | (1) |
| As the increase in rental cost affects neither the marginal cost nor the market price, it does not affect the profit-maximising daily output quantity. | (1) |

CH 4 OWNERSHIP & EXPANSION OF FIRMS

Microeconomics by Topic
4. Ownership & Expansion of Firms

4.1 PUBLIC ENTREPRISES

MARKING SCHEME

1991/CE/II/13
D

1991/CE/II/14
A

1992/CE/II/19
A

1993/CE/II/12
A

1993/CE/II/17
D

2025/DSE/II/31

D

Note: Figures in brackets indicate the percentages of candidates choosing the correct answers.

1993/CE/II/3(a)

(i) The government

(2)

(ii) Advantages over a government department

- profit motive
- more flexible operation e.g. less red-tape
- higher incentive to lower production cost / efficient use of employees
- better control of employees
- more responsive to consumers' demand
- more innovative e.g. stored valued ticket

(2@, max: 4)

[Mark the **FIRST TWO** points only.]

Advantages over a private firm:

- sound financial structure (financially backed up by the government) / lower interest on loans borrowed (with government as guarantor)
- better overall planning / co-ordination of transportation services
- government may receive dividend from the corporation

(2@, max: 2)

[Mark the **FIRST** point only.]

2017/DSE/II/10(b)

Arguments supporting the suggestion:

- Government will have a better control of the train fare, and it can freely reduce the fare to relieve the burden of the passengers
- If MTR is fully owned and supported by the government, that would make it easier for MTR to get loans to fund its operation or investment.

(2@, max: 2)

Arguments opposing the suggestion:

- It may lower the efficiency of MTR's management and reduce its sensitivity to market (price) signals.
- The money involved in the purchase of MTR is substantial. The government may have better alternative uses of the money, such as education and medical services.

(2@, max: 2)

[Mark the **FIRST** point only.]

2019/DSE/II/3

Examples:

- HK Airport Authority
- Hong Kong Export Credit Insurance Corporation
- any other relevant example

[Mark the **FIRST** point only.]

Marks
(1)

Advantages:

- easier to get information about the general public from the government for decision-making
- can provide stable service with a lower price
- easier to get loan with the backup of the government
- any other relevant point

[Mark the **FIRST TWO** points only.]

} @1
max: 2

CH 4 OWNERSHIP & EXPANSION OF FIRMS

4.1 PUBLIC ENTREPRISES

Microeconomics by Topic
4. Ownership & Expansion of Firms

2025/DSE/II/4(Cross Topic1.7)

(a) Reasons:

- Not durable, as rice may decay within years and cannot be stored for a long time.
- Not homogeneous, as rice does not have uniform physical characteristics.
- Not portable, as people need to carry a large quantity of rice for high-value transactions.
- Any other relevant point

[Mark the FIRST TWO points only.]

} 2@
max: 4

(b) As a lender, Villager B receives the bag of rice as compensation for deferred consumption.

(2)

4.2 SOLE PROPRIETORSHIPS & PARTNERSHIPS

Microeconomics by Topic
4. Ownership & Expansion of Firms

MARKING SCHEME

1990/CE/II/12 C	1993/CE/II/09 C	1999/CE/II/18 C	2001/CE/II/22 A	2014/DSE/I/08 B (74%)
1990/CE/II/21 B	1993/CE/II/09 B	1999/CE/II/21 D	2005/CE/II/22 B (80%)	
1991/CE/II/15 C	1999/CE/II/17 C	2000/CE/II/20 C	2009/CE/II/21 C (32%)	2022/DSE/I/05 B

Note: Figures in brackets indicate the percentages of candidates choosing the correct answers.

1995/CE/II/1

(General) partnership

- 2 to 20 partners

- unlimited liability / each partner is fully liable for the firm's debts

- death and bankruptcy of a partner will lead to dissolution of the partnership

- act by one partner binds all other partners

- difficult to admit or withdraw partners

- financial statements need not be publicly disclosed

[Mark the **FIRST THREE** features only.]

(1)

(1@, max: 3)

1995/CE/II/11(c)

- low set up cost

- easy to set up and dissolve, so suitable to business with sudden ups and downs

- lower profits tax rates

- low accounting cost

[Mark the **FIRST THREE** points only.]

(2@, max: 6)

1998/CE/II/9(b)

- wider source of capital

- better division of labour among owners (wider scope of specialization)

- risk and cost can be shared among partners

[Mark the **FIRST TWO** points only.]

(2@, max: 4)

2018/DSE/II/2(a)

- Peter wants to invite more investors to get new funds for expansion of his business.

- Peter can make use of different expertise from Paul and Mary in the new markets.

(1)

(1)

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1. (a) Advantages:

— Ability to raise capital by issuing shares to the general public

— Easier transfer of its ownership

— Any other relevant point.

[Mark the **FIRST** point only.]

@1

Max: 1

4.3 LIMITED COMPANIES

MARKING SCHEME

1990/CE/II/07 A	1999/CE/II/22 A	2010/CE/II/22 B (78%)	1997/CE/II/18 A	2009/CE/II/20 D (29%)
1991/CE/II/16 C	2000/CE/II/21 D	2012/DSE/I/09 B (39%)	1998/CE/II/18 B	2009/CE/II/22 D (61%)
1993/CE/II/10 D	2001/CE/II/18 B	2015/DSE/I/07 B (74%)	2002/CE/II/17 D (59%)	2010/CE/II/23 C (55%)
1993/CE/II/13 B	2002/CE/II/21 C (82%)	1990/CE/II/14 D	2002/CE/II/18 C (76%)	2014/DSE/I/07 D (71%)
1994/CE/II/24 B	2003/CE/II/19 C (60%)	1992/CE/II/18 C	2003/CE/II/22 D (73%)	2017/DSE/I/06 A (84%)
1994/CE/II/25 A	2005/CE/II/20 A (67%)	1993/CE/II/03 D	2004/CE/II/23 A (65%)	2019/DSE/I/07 A
1995/CE/II/24 B	2006/CE/II/20 A (41%)	1993/CE/II/07 B	2005/CE/II/21 D (74%)	2020/DSE/I/4 D
1995/CE/II/25 C	2007/CE/II/22 A (29%)	1993/CE/II/19 C	2006/CE/II/22 B (51%)	2025/DSE/I/32 C
1996/CE/II/57 A	2008/CE/II/20 B (75%)	1996/CE/II/20 A	2007/CE/II/21 A (64%)	
1997/CE/II/16 C	2009/CE/II/23 D (70%)	1996/CE/II/21 D	2008/CE/II/21 D (69%)	

Note: Figures in brackets indicate the percentages of candidates choosing the correct answers.

1992/CE/I/3(c)(ii)

- shares are freely transferable
- no upper limit on the number of shareholders
- financial accounts needed to be disclosed to the public
- can apply for listing in the stock market
- can raise fund from the public by issuing debentures

[Mark the **FIRST THREE** points only.]

(2@, max: 6)

1996/CE/II/3(a)

Public limited company

(1)

Advantages:

- limited liability: loss of an owner of the firm is confined to his investment in the firm. / lower risk in investment
- legal entity: the firm can act as a legal person to make contracts, to own properties and be sued. / profit or loss belongs to the firm, not the owners.
- Wider source of capital: can issue shares to the public.
- Easy transfer of shares: an investor can convert his shares into cash with ease.
- Professionals to run the business (separation of ownership and management and economies of scale)
- Lasting continuity: the firm's existence not affected by any investor's death withdrawal, etc.

(2@, max: 4)

[Mark the **FIRST TWO** points only.]

2001/CE/I/11(c)

- wider source of capital because capital can be raised from the public
- shares are freely transferable, approval of the Board of Directors is not required
- wider scope of specialization

(2@, max: 4)

[Mark the **FIRST TWO** points only.]

2002/CE/I/4

The liability of an owner of a firm is confined to the amount of his investment in the firm.

(2)

2004/CE/I/4

- no upper limit of the number of shareholders
- free transfer of shares
- wider source of capital / can invite the public to be shareholders
- financial accounts are open to the public

[Remark: Mark the **FIRST THREE** points only.]

(1@, max: 3)

2005/CE/I/3(a)

Public limited company.

(1)

Advantages:

- wider source of capital
- limited liability
- legal entity
- shares freely transferable
- continuity of business

(1@, max: 2)

[Mark the **FIRST TWO** points only.]

Disadvantages:

- financial accounts disclosed to the public
- higher profits tax rate
- higher risk of being taken over
- complicated set-up procedure

(1@, max: 2)

[Mark the **FIRST TWO** points only.]

2009/CE/I/2

Public limited company.

(1)

- independent legal status
- limited liability
- separation of ownership and management
- lasting continuity
- shares are freely transferrable
- issue shares to the public
- no upper limit on the number of shareholders
- required by law to disclose its financial status

(2@, max: 4)

[Mark the **FIRST TWO** points only.]

2012/DSE/II/12(b)

Advantages:

- tighter control of ownership / more difficult to be taken over
- no need to disclose the accounting information to public

(1)

(1)

Disadvantages:

- cannot issue shares to general public to raise capital
- difficult to transfer its ownership

(1)

(1)

Microeconomics by Topic
4. Ownership & Expansion of Firms

2014/DSE/II/10

- (c) Public limited company / listed company. (1)
- (d) (i) In a partnership, any given owner cannot transfer his/her shares of ownership without the consent of other partners while in a public limited company, the shares are freely transferable. (2)
- (ii) A partnership does not have independent legal status while a public limited company has an independent legal status. (2)
- (e) People expected the birth rate to rise, resulting in an expected increase in future demand for the baby products. (1)
So Goodbaby International was expected to earn bigger profits and to distribute more dividends to its shareholders in the future. (2)
The expected increase in future dividends would raise the demand for the company's shares and thus its stock price. (1)

2015/DSE/II/11(a)

Advantage:

- lower risk of being taken over
- no need to disclose accounting information to the public (1@, max: 1)

Disadvantage:

- owners cannot freely transfer their stocks
 - the firm cannot raise capital by issuing stocks to the general public (1@, max: 1)
- [Mark the **FIRST** point only.]

2016/DSE/II/2(a)

Public limited company / listed company. (1)

2017/DSE/II/10(a)

- Shares of a listed company are freely transferrable while shares of a private limited company are not.
 - A listed company is required to disclose its accounting information to the public while a private limited company is not.
 - A listed company can issue shares and bonds to the general public while a private limited company cannot. (1@, max: 2)
- [Mark the **FIRST TWO** points only.]

1994/CE/I/4

	Time deposit holder	Ordinary shareholder	
Name of return:	interest	dividends	(2)
Rate of return:	fixed rate	variable	(2)
Whether the return depends on the bank's business performance:	no	yes	(2)

1999/CE/I/4

Advantage:

- no interest burden / no redemption obligation (2)

Disadvantage:

- easier to be taken over (2)

2001/CE/I/10(c)(ii)

- no interest burden (1)
- no redemption obligation (1)

2012/DSE/II/10(a)(ii)

He can get interest return even if the company does not make any profit that year. (2)
Higher priority of getting paid than shareholders if the company is liquidated. (2)

2025/DSE/II/2(b)(Related to this topic)

- (b) (i) Limited liability means that when a firm is liquidated, the liability of an owner of the firm is confined to the amount of his/her investment in the firm. (2)

(ii) Advantages of a private limited company over a public limited company:

- lower risk of being taken over as the transfer of ownership requires consent of other owners
- no need to disclose financial accounts to the public

[Mark the FIRST TWO points only.]

} 1@
max: 2

4.4 COMPARING DIFFERENT FORMS OF OWNERSHIP

Microeconomics by Topic
4. Ownership & Expansion of Firms

MARKING SCHEME

1995/CE/II/23 C	1997/CE/II/58 C	2005/CE/II/23 D (65%)	2008/CE/II/19 C (57%)	2018/DSE/I/07 D (69%)
1996/CE/II/17 C	1998/CE/II/20 D	2006/CE/II/21 C (58%)	2012/DSE/I/08 B (77%)	2020/DSE/I/5 C
1997/CE/II/19 B	2002/CE/II/19 D (59%)	2007/CE/II/20 B (77%)	2017/DSE/I/05 D (88%)	2023/DSE/I/08 C
2024/DSE/I/05 B				

Note: Figures in brackets indicate the percentages of candidates choosing the correct answers.

1999/CE/I/10(b)

Advantages:

- lower tax rate (1)
 - easier to set up / close down (1)
 - less legal constraints in business operation, so save cost in this aspect (1)
- [Mark the **FIRST TWO** points only.]

Disadvantages:

- unlimited liability (1)
 - narrower scope of raising capital (1)
 - lack of continuity (1)
- [Mark the **FIRST TWO** points only.]

2003/CE/I/3

General partnership.

- number of partners ranges from 2 to 20 (1)
 - unlimited liability / each partner is fully liable for the firm's debt
 - all partners are bound by other partners' business decisions
 - difficult to admit or withdraw partners
 - easy to set up and dissolve (1@, max: 2)
- [Mark the **FIRST TWO** points only.]

2008/CE/I/4

- (a) The liability of an owner of the firm is confined to the amount of his investment in the firm. (2)

(b)

	Public limited company	Partnership	
(i) Transfer of ownership	Does not need the consent of other shareholders	Need the consent of other partners	(2)
(ii) Source of capital	Can issue shares and debentures	Cannot issue shares and debentures / borrowing from friends, etc	(2)

4.4 COMPARING DIFFERENT FORMS OF OWNERSHIP

2024/DSE/II/1a

(a) (i) Public limited company / Listed company (1)

(ii) Advantages:

- no interest burden
 - no redemption obligation
- [Mark the FIRST point ONLY.]

} 1@
max: 1

Disadvantages:

- dilution of ownership and reduction in control power over the firm
 - higher risk of being taken over
- [Mark the FIRST point ONLY.]

} 1@
max: 1

2024/DSE/II/11c

(e) Possible reasons:

- Ann does not want to be legally bound by the business decisions of a partner.
 - Ann prefers making decisions promptly by herself.
 - Any other relevant point
- [Mark the FIRST TWO points ONLY.]

} 1@
max: 2

2025/DSE/II/6

(a) Reasons:

- Yoga studios provide heterogeneous services, e.g. different types of classes / different styles of teaching.
 - Market information is imperfect, e.g. information about the prices and quality is not readily available to consumers.
 - Sellers engage in non-price competition, e.g. advertising.
 - any other relevant point
- [Mark the FIRST TWO points only.]

} 2@
max: 4

(b) (i) Limited liability means that when a firm is liquidated, the liability of an owner of the firm is confined to the amount of his/her investment in the firm. (2)

(ii) Advantages of a private limited company over a public limited company:

- lower risk of being taken over as the transfer of ownership requires consent of other owners
 - no need to disclose financial accounts to the public
- [Mark the FIRST TWO points only.]

} 1@
max: 2

4.5 EXPANSION OF FIRMS

Microeconomics by Topic
4. Ownership & Expansion of Firms

MARKING SCHEME

1990/CE/II/18 B	1995/CE/II/27 B	2001/CE/II/23 A	2006/CE/II/23 C (75%)	2010/CE/II/24 C (96%)
1991/CE/II/21 A	1995/CE/II/28 A	2002/CE/II/22 B (33%)	2007/CE/II/23 D (73%)	2013/DSE/I/08 A (82%)
1993/CE/II/23 D	1997/CE/II/20 A	2003/CE/II/23 D (74%)	2007/CE/II/24 D (52%)	2015/DSE/I/08 A (88%)
1993/CE/II/23 A	1998/CE/II/19 D	2003/CE/II/24 D (75%)	2008/CE/II/22 A (72%)	2016/DSE/I/11 B (72%)
1994/CE/II/27 A	1998/CE/II/21 A	2004/CE/II/24 D (77%)	2008/CE/II/23 A (92%)	2017/DSE/I/09 C (57%)
1994/CE/II/28 C	2000/CE/II/24 A	2005/CE/II/24 A (80%)	2009/CE/II/24 C (69%)	2018/DSE/I/08 A (83%)
2019/DSE/I/09 B	2020/DSE/I/18 C	2022/DSE/I/06 D	2024/DSE/I/07 D	2025/DSE/I/10 A
2025/DSE/I/34 D				

Note: Figures in brackets indicate the percentages of candidates choosing the correct answers.

1990/CE/I/4(a)(i)

- to secure outlet of product
- easier to obtain information on consumers' preferences
- to enjoy economies of scale
- to build up brand name

(2@, max: 4)

[Mark the first 2 points only. Mere listing of points without elaboration - max. 1 mark.]

1991/CE/I/5(b)(iv)

Horizontal integration

(1)

Explanation of two advantages:

- economies of scale
- greater market power
- reduce duplication of facilities

(3@, max: 6)

[Mark the **FIRST TWO** points only.]

1992/CE/I/4(a)

(i) Conglomerate integration

- (ii) - more efficient use of the land site
- diversification of production/spread risk
- promotion of goodwill
- economy of scale with reasonable explanation

(3@, max: 6)

[Mere listing of points without elaboration – max. 2 marks. Mark the **FIRST TWO** points only.]

Microeconomics by Topic

4. Ownership & Expansion of Firms

1994/CE/I/9(a)

Lateral expansion

Motives:

- diversification of product / spread risk
- make use of brand name
- economies of scale

[Mark the **FIRST TWO** points only.]

(1)

(2@, max: 4)

1996/CE/I/11(b)

(i) Lateral expansion, because

newspapers and magazines are related products but not necessarily competing with each other.

(ii) - better utilization of resources

- benefits derived from brand name
- diversification of investment / spread risk
- economies of scale

[Mark the **FIRST TWO** points only.]

(1)

(2)

(1@, max: 2)

1998/CE/I/3

- guarantee stable source of films to be shown on the television
- the television company is able to direct or influence the kind of films to be shot by the film company
- economies of scale illustrated by one appropriate example. e.g. the expertise
- to diversify the television company's risk of investment

[Mark the **FIRST TWO** points only.]

(2@, max: 4)

1999/CE/I/11(b)

Conglomerate integration.

Elaboration of two of the following:

- diversification of products / spread risk
- financial economy of scale
- managerial economy of scale
- marketing economy of scale
- any other relevant point (except technical economy of scale)

[Mark the **FIRST TWO** points only.]

(1)

(2@, max: 4)

2002/CE/I/11(d)

- to make use of brand name
- to diversify the business
- to promote its goodwill
- to better utilize its resources / to enjoy economies of scale

(1)

(1)

(1)

(1)

2003/CE/I/9(b)(ii)

Vertical forward expansion because

the manufacturer expands to the next stage of production.

- obtaining market information more easily
- ensuring market outlets for its products
- economies of scale, e.g. financial economies of scale
- promote goodwill / make use of brandname

[Mark the **FIRST TWO** points only.]

(1)

(1)

(2@, max: 4)

2006/CE/I/3

Conglomerate expansion because (1)
movie production and hotel business are two different lines of production. (1)

OR

Lateral expansion because (1)
the products are related but not competitive. (1)

Possible motives of conglomerate expansion or lateral expansion:

- product diversification to spread risk
- making use of brand name to sell another product / promoting goodwill
- more flexible use of resources
- to benefit from economies of scale (1@, max: 2)

[Mark the **FIRST TWO** points only.]

2010/CE/II/10(b)

Lateral expansion, because (1)
it expands to include a related but not competitive business. (1)

[Remark: No mark will be awarded to this part if the explanation contradicts with the type of expansion.]

2012/DSE/II/3

(a) Conglomerate expansion, because (1)
the firm has expanded into unrelated industries. (1)

(b) Possible motives:

- to have product / income diversification so as to spread risk
- to make use of its brand name of one product in other products
- to use resources more efficiently as the two companies can share some of their resources
- to enjoy economies of scale (e.g. lower average cost of advertising as the total cost can be spread over a larger quantity of output.) (1@, max: 3)

[Mark the **FIRST THREE** points only.]

2014/DSE/II/4

(a) Vertical backward expansion, because (1)
by starting Fai Lok Institute, Tai Fai Lok Group has expanded its operation to the previous stage of production to control directly the supply of labour inputs (chefs and managers) to its original (food) business. (1)

(b) Advantages

- more steady supply of factor inputs such as chefs and managers
- better use of the brand name
- program can be tailor-made for those graduates from the Institute who are hired to work in the Group's restaurant, so as to shorten their training time. (2@, max: 4)

[Mark the **FIRST TWO** points only.]

2015/DSE/II/9(b)

Lateral expansion, because (1)
fast food and coffee are related but not directly competitive products. (1)

2019/DSE/II/9

(a) **Public good.** It is because the videos are non-rivalrous in consumption as people can concurrently watch the video without affecting one another. (3)

(b) **Horizontal expansion.** (1)

Reasons:

- enlarge the market share
- enjoy economies of scale
- make use of the brand name
- any other relevant point

[Mark the **FIRST TWO** points only.]

} @1
max: 2

2020/DSE/II/2

2 a) Lateral. The integration involve in related industry but not directly competing (2 marks)

b. Economies of scale.

Better use of resources

Diversify of risk

Brand name effect

. any two of the above. (2 marks)

2021/DSE/II/3

Vertical backward integration.

Motives:

- To ensure a steady supply of raw materials
- To diversify sources of income and spread risks
- To make use of its brand name in other products
- Any other relevant point

[Mark the **FIRST TWO** points only.]

2023/DSE/I/1(a-b)

- (a) Conglomerate integration, because
electric car manufacturing and social media operation are totally unrelated businesses.

Marks

(1)

(1)

- (b) Possible motives:
- to spread risk through product diversification
 - to make use of its brand name in other products/services
 - any other relevant point

[Mark the **FIRST TWO** points only.]

} 1 @;
max: 2

CH 5 MARKET STRUCTURE

5.1 PERFECT COMPETITION

Microeconomics by Topic
5. Market Structure

MARKING SCHEME

1994/CE/II/29
A

2000/CE/II/26
C

2002/CE/II/23
C

2008/CE/II/24
A (52%)

2024/DSE/II/18
A

Note: Figures in brackets indicate the percentages of candidates choosing the correct answers.

1992/CE/I/1(a)

- heterogeneous service, e.g. 4-seaters and 5-seaters
- entry is not free, no. of licences is regulated by the government
- drivers and customers may not have perfect knowledge of the market, e.g. while some customers are waiting for taxis, some taxi drivers are looking for customers somewhere else

(3@, max: 6)

[Mere listing of points without elaboration -- max. 2 marks. Mark the **FIRST TWO** points only.]

1995/CE/II/4

- different services or heterogeneous product, e.g. different discs in different shops
- imperfect information, e.g. customers may not know well the service of other shops
- existence of non-price competition, e.g. free gifts

(2@, max: 4)

[Mark the **FIRST TWO** points only.]

2003/CE/II/4

- many sellers
- all firms are price-takers
- homogeneous product
- perfect information
- free entry and exit

(1@, max: 2)

[Mark the **FIRST TWO** points only.]

2004/CE/II/10(d)

- differentiated products: different services for the 5-day package
- under competition there are different prices (fees)

(2)

(2)

[Remark: Mark the **FIRST TWO** reasons only.]

2006/CE/II/9(b)

- differentiated products, e.g. different service provided by different shops, ...
- different prices / price searching, e.g. different discounts to different customers, ...
- imperfect information, e.g. customers do not know the prices of different shops, ...
- non-price competition, e.g., gifts given by some shops to their customers, ...

(2@, max: 4)

[Mark the **FIRST TWO** points only.]

2007/CE/II/8(c)

- different prices, e.g. not the same tuition fee
- heterogeneous products, e.g. different quality of teaching
- non-price competition, e.g. advertising in supermarkets
- imperfect information, e.g. one may not know who can teach well

(2@, max: 4)

[Mark the **FIRST TWO** points only.]

Microeconomics by Topic
5. Market Structure

2013/DSE/II/1

- sellers provide heterogeneous services, e.g. different tour groups.
- market information is imperfect, e.g. information about routes and services.
- sellers engage in both price and non-price competition, e.g. advertisements.

(2@, max: 4)

[Mark the **FIRST TWO** points only.]

2018/DSE/II/12(a)

- sellers provide heterogeneous services, e.g., rooms with different size / features.
- market information is imperfect, e.g., information about the quality of services.
- price searcher / under competition there are different prices, e.g., different hotels charge different room rates for similar rooms.

(2@, max: 4)

[Mark the **FIRST TWO** points only.]

1998/CE/I/10(c)

- (i) An appropriate example to illustrate differentiated products or services / imperfect information / different costs of production e.g. different after-sale maintenance services

(2)

- (ii) Monopolistic competition

(1)

2001/CE/I/4

- different products: different services, e.g. some shops sell cans of Coca-Cola 24 hours a day
- imperfect information: e.g. about the prices of the product sold in other shops; about the availability of other nearby shops selling the product, etc.
- different costs: different rents paid by different shops; discounted cost due to bulk purchase at the wholesale level, etc.

(2@, max: 4)

[Mark the **FIRST TWO** points only.]

2009/CE/I/11(c)(ii)

- different services, e.g., only some shops provide the delivery service
- imperfect information, e.g., customers do not know which shops sell at a lower price
- different production cost, e.g., different rental payments

(2@, max: 4)

[Mark the **FIRST TWO** points only.]

2015/DSE/II/9(c)

- lower-quality coffee
- lower production cost
- less pleasant environment
- less attractive brand name

(2@, max: 4)

[Mark the **FIRST TWO** points only.]

2021/DSE/II/4

Reasons:

- Sellers provide heterogeneous goods, e.g. smartphones with different specifications
- There exists non-price competition, e.g. advertisements
- Market information is imperfect, e.g. information about the performance of the smartphones
- Any other relevant point

[Mark the **FIRST TWO** points only.]

(c) Reasons:

Heterogeneous goods - some stored-value facilities (SVFs) can be used in most forms of public transportation, but some cannot. (2)

Non-price competition - different stored-value facilities (SVFs) companies engage in different kinds of promotions to attract customers. (2)

[Mark the FIRST TWO points only.]

5.2 IMPERFECT COMPETITION

MARKING SCHEME

1990/CE/II/16 B	1997/CE/II/59 D	2009/CE/II/25 B (81%)	1992/CE/II/56 A	2005/CE/II/25 C (39%)
1991/CE/II/27 C	1998/CE/II/22 C	2010/CE/II/25 D (65%)	1993/CE/II/22 D	2007/CE/II/25 D (83%)
1992/CE/II/29 C	1999/CE/II/23 B	2014/DSE/I/17 B (80%)	1994/CE/II/30 C	2008/CE/II/25 C (55%)
1993/CE/II/30 D	2003/CE/II/49 D (39%)	2015/DSE/I/20 D (88%)	1996/CE/II/58 D	2012/DSE/I/18 A (75%)
1995/CE/II/29 D	2004/CE/II/25 A (58%)	2017/DSE/I/19 D (70%)	2000/CE/II/25 B	2013/DSE/I/17 D (40%)
1996/CE/II/25 D	2006/CE/II/24 B (71%)	2018/DSE/I/21 B (86%)	2001/CE/II/24 B	2016/DSE/I/20 B (26%)
2019/DSE/I/18 D	2021/DSE/I/09 D	2022/DSE/I/19 A	2023/DSE/I/18 B	2025/DSE/I/19 C
2025/DSE/I/33 B	2025/DSE/I/39 B			

1990/CE/II/2(c)

- a few dominating banks in the market
- high degree of interdependence
- restricted entry
- differentiated services
- non-price competition e.g. advertising

(3@, max: 6)

[Mark the first 2 points only. Mere mention of points without elaboration on Hong Kong's situation – max. 2 marks]

1991/CE/II/2(c)

(i) Monopoly

(1)

(ii) Explanation of two sources of monopoly power:

- natural monopoly (i.e. high set-up cost and economies of scale)
- potential entrants find it difficult to capture a large market share
- limited market size

(3@, max : 6)

[Mark the **FIRST TWO** points only.]

1993/CE/II/3(b)

- natural monopoly, e.g. economies of scale
- insufficient demand to support more than one MTR
- duplication of tunnels etc. means wastage

(3@, max: 6)

[Mere listing of points without elaboration – max: 2 marks. Mark the **FIRST TWO** points only.]

1996/CE/II/5

Monopolistic competition

(1)

Any three kinds of price or non-price competition:

- price competition (i.e. discounts)
- gifts given by some shops to their customers
- advertising and branding
- differentiated service (e.g. accepting different membership cards)

(1@, max: 3)

[Mark the **FIRST THREE** points only.]

Microeconomics by Topic

5. Market Structure

1997/CE/I/5

(a) Oligopoly / Monopolistic competition (1)

- (b) - price competition (1)
- differentiated products (1)

(c) Monopolistic competition:

- many sellers
- other forms of non-price competition
- free entry
- imperfect information

(1@, max: 2)

OR

Oligopoly:

- a few dominating firms
- interdependency among competing firms or rivalry
- other forms of non-price competition
- imperfect information
- entry barriers

[Mark the **FIRST TWO** points only.]

[Either monopolistic competition or oligopoly which must match with (a)]

1998/CE/I/10(c)(ii)

Monopolistic competition (1)

1999/CE/I/9(d)

- a number of sellers / a few dominating firms
- interdependency
- free entry and exit
- differentiated products
- imperfect information
- non-price competition

(1@, max: 2)

[Mark the **FIRST TWO** points only.]

2000/CE/I/9(c)

Oligopoly. (1)

- a few dominating firms
- interdependence
- non-price competition
- price leadership

(1@, max: 2)

2002/CE/I/5

- many sellers
- differentiated products
- easy entry / exit
- imperfect information
- price competition
- non-price competition

(1@, max: 3)

[Mark the **FIRST THREE** points only.]

2005/CE/I/3(b)

Oligopoly

(1)

- a few dominating sellers
- interdependency
- non-price competition
- heterogeneous product
- entry barriers
- imperfect information
- price competition

(1@, max: 3)

[Mark the **FIRST THREE** features only.]

2008/CE/I/3

Oligopoly

(1)

- a few dominating firms
- price searcher
- interdependence of policies
- product can be heterogeneous
- non-price competition
- not easy to enter the industry
- imperfect information

(1@, max: 3)*

[Mark the **FIRST THREE** points only.]

[*Remark: Do not mark this part of the answer if the candidate gave a wrong answer such as monopolistic competition to the first part of the answer]

2010/CE/I/4

Monopolistic competition

(1)

- many sellers
- free / easy entry (and exit)
- heterogeneous / differentiated products
- imperfect market information
- non-price competition
- price searcher

(1@, max: 3)*

[Mark the **FIRST THREE** features only.]

[*Remark: No mark will be awarded to this part if the type of market structure is wrongly identified]

2012/DSE/II/12(a)

Oligopoly

(1)

- a few dominating firms
- price searcher
- interdependence of pricing policies
- product can be heterogeneous
- non-price competition
- not easy to enter the industry
- imperfect information

(1@, max: 3)

[Mark the **FIRST TWO** points only.]

2014/DSE/II/2(c)

Monopolistic competition

(1)

Reasons:

- many sellers / no dominant sellers
- sellers selling heterogeneous products
- free entry into the market

(1@, max: 2)

[Mark the **FIRST TWO** points only.]

Microeconomics by Topic
5. Market Structure

2016/DSE/II/10(a)

Monopolistic competition

Features:

- many sellers
- price searchers
- (slightly) differentiated products
- non-price competition
- imperfect market information

[Mark the **FIRST TWO** points only.]

(1)

(1@, max: 2)

2017/DSE/II/3

Oligopoly

Features:

- several dominant firms
- entry barrier
- interdependent pricing strategies
- imperfect market information

[Mark the **FIRST TWO** features only.]

(1)

(1@, max: 2)

2023/DSE/II/4

Possible reasons:

- Imperfect market information. Some customers may buy from shops charging higher prices because they do not know about shops selling at lower prices. (This may be due to high costs of acquiring such information, viz., search costs.)
 - Heterogeneous products/services. Some customers may be willing to pay a premium for better services, e.g. better after-sales services such as longer warranty period / better product descriptions by salespersons.
 - Any other reasonable answer
- [Mark the **FIRST TWO** points only.]

} 2@;
max: 4

2024/DSE/II/1(b-c)

(b) Oligopoly

(1)

Features:

- a few dominant sellers
- sellers are interdependent in pricing strategy
- price rigidity/stickiness

[Mark the **FIRST TWO** points ONLY.]

} 1@
max: 2

(c) Possible sources of monopoly power:

- government franchise
- high set-up costs and natural monopoly
- government ownership

[Mark the **FIRST** point ONLY.]

} 1@
max: 1

2025/DSE/II/2(a)

(a) Reasons:

- Yoga studios provide heterogeneous services, e.g. different types of classes / different styles of teaching.
- Market information is imperfect, e.g. information about the prices and quality is not readily available to consumers.
- Sellers engage in non-price competition, e.g. advertising.
- any other relevant point

[Mark the FIRST TWO points only.]

} 2@
max: 4

CH 6 DEMAND & SUPPLY (I)

Microeconomics by Topic
6. Demand & Supply (I)

6.1 THE LAW OF DEMAND

MARKING SCHEME

1995/CE/II/06 C	2004/CE/II/08 B (55%)	2008/CE/II/06 B (24%)	2019/DSE/I/12 B	2020/DSE/I/17 D
1997/CE/II/07 B	2006/CE/II/49 A (70%)	2000/AL/I/08 B	2020/DSE/I/12 D	2023/DSE/I/10 A
2023/DSE/I/11 A	2024/DSE/I/11 A	2024/DSE/I/12 A	2025/DSE/I/27 C	2025/DSE/I/37 C

Note: Figures in brackets indicate the percentages of candidates choosing the correct answers.

1995/CE/II/3

- (a) If the price of a good rises, its quantity demanded will decrease, (and vice versa,) ceteris paribus. (2)
- (b) 'Quantity demanded' refers to consumers' plan, and does not imply what can be actually bought (in what quantity and at what price) in the market. (2)

1996/AL/I/7(a)

The time cost of shopping is reduced by the supermarket. With sharply rising wages of domestic servants, this reduction of shopping time worth more.

1996/AL/I/8(a)

The cost of raising children has risen significantly in Hong Kong because of the rising costs of domestic servants, the rising employment opportunity, for women, a general increase in the cost of time, and a sharp rise in housing rent. Hence the quantity of children demanded falls.

2000/AL/I/6

- (a) No. For a given household size, the new charge will be a lump-sum fee, which will not vary with the amount of garbage disposed of. In other words, the marginal cost of disposing garbage remains unchanged.
- (b) No, if all the garbage is filled into one big bag before and after the change.

However, if more than one bag are originally used, the number of garbage bags will drop, with each bag squeezed more tightly. Because there are limits as to how tight the bag can be squeezed, the amount of garbage will also drop.

2001/AL/I/3

The cost of time falls for customers taking haircuts during holidays. The price of a haircut is the price charged by a barber plus the time cost, so during holidays the full price of a haircut falls.

(Therefore, according to the law of demand, the number of customers going to barber shops during holidays is larger.)

2023/DSE/II/12(a)(i)

- | | |
|---|---------------------|
| (a) (i) According to Source B, crude oil is an input for producing gasoline. As the prices of crude oil increase, the cost of producing gasoline increases. | Marks
(1) |
| Supply of gasoline decreases. | (1) |
| Price increases. | (1) |

CH 6 DEMAND & SUPPLY (I)

Microeconomics by Topic
6. Demand & Supply (I)

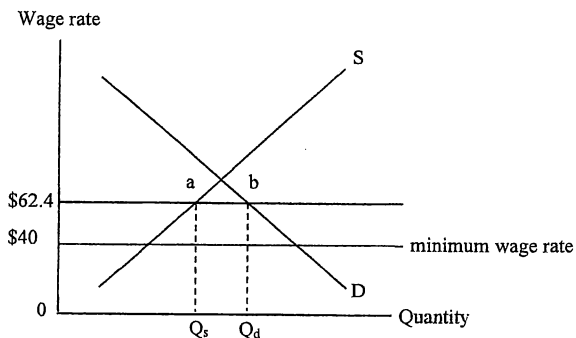
6.1 THE LAW OF DEMAND

2024/DSE/II/13(c)

- (c) (i) Indicate in the diagram:
wage rate (\$62.4) below equilibrium (1)
labour shortage = ab (1)
- (ii) Verbal elaboration:
– As a labour shortage exists at the prevailing wage rate (\$62.4), it can be concluded (1)
that the prevailing wage rate is below the equilibrium. As the minimum wage rate (\$40) is below the prevailing wage rate, it must be below the equilibrium wage rate as well.
– Thus, the minimum wage is ineffective in the labour market of waiters in 2023. (1)

Indicate in the diagram:

- minimum wage rate (\$40) < \$62.4 < equilibrium wage rate (1)



MARKING SCHEME

6.2 RELATIVE PRICE AND THE LAW OF DEMAND

1985/AL/I/06 C	1996/AL/I/12 D (deleted)	2002/AL/I/15 B	2009/AL/I/11 B (54%)	2015/DSE/I/18 C (53%)
1985/AL/I/29 C	1997/AL/I/08 D	2004/AL/I/21 A (78%)	2010/AL/I/09 B (64%)	2016/DSE/I/18 D (44%)
1986/AL/I/26 D	1997/AL/I/11 B	2006/AL/I/08 C (74%)	2011/AL/I/09 B (62%)	2017/DSE/I/13 A (46%)
1989/AL/I/14 B	2000/AL/I/22 D	2007/AL/I/08 C (43%)	2012/AL/I/08 A (55%)	2018/DSE/I/19 B (69%)
1992/AL/I/08 A	2001/AL/I/08 C	2008/AL/I/09 D (70%)	2012/AL/I/09 B (50%)	2019/DSE/I/20 C
1994/AL/I/07 B	2001/AL/I/20 A	2008/AL/I/16 A (80%)	2013/DSE/I/16 A (46%)	2022/DSE/I/18 C
2025/DSE/I/28 D	2025/DSE/I/29 A	2025/DSE/I/30 A		

Note: Figures in brackets indicate the percentages of candidates choosing the correct answers.

1993/AL/I/9

The uniform unit tax will lead to the import of higher quality automobiles because the price of a higher quality automobile will fall relative to the price of a lower quality automobile. (With the ad valorem tax, however, the relative prices of high vs. low quality automobiles will remain the same.)

By the law of demand, the quantity of higher quality automobiles rises relative to that of lower quality automobiles.

1995/AL/I/6(c)

The average quality of vehicles operating on the roads will rise.

This is because with a positive price of quota, the prices of higher quality vehicles imported will fall relative to lower quality vehicles.

By the law of demand, the quantity of higher quality vehicles rises relative to that of lower quality vehicles.

1998/AL/I/6

- (a) Because one unit of quota must be used for one unit of garment regardless of garment quality, and because the quota has a positive value, the price of higher quality garments fell relative to lower quality garments under the quota system. Hence according to the law of demand the quantity of higher quality garments demanded by the United States rose relative to that of lower quality garments.
- (b) Because Hong Kong had long been the only place in Asia to allow a free market for quota transactions, the value of the quota was higher and therefore the relative price of higher quality garments fell further. (This explains why Hong Kong became the leading exporter of higher quality garments and why the quota system, designed to protect the garment industry in the U.S., ended up hurting that industry far more than without the quota system -- Hong Kong garments before the quota were generally of such low qualities that they competed with those produced in Mexico or the Philippines, but hardly competed with those produced in the U.S.)

2010/AL/I/8(b)

Exactly the same as shipping the good apples out with transportation cost added, because the unit import tax added will reduce the price of higher quality tyres relative to lower quality tyres.

2014/DSE/II/9(c)

Law of demand: when the price of a good increases, the quantity demanded of the good decreases, vice versa, *ceteris paribus*.

Here the unit of measurement to which the tax applies is 'bottle', rather than volume (e.g., in litres), of soft drinks. When the same amount of tax is added to each unit (bottle), then -- per volume of liquid contained -- larger-bottle soft drinks would become cheaper than the small-bottle ones.

The relative price of larger-bottle soft drinks (in terms of small-bottle ones) thus drops. The law of demand then implies that people would buy more larger-bottle soft drinks relative to small-bottle ones, so the proportion of larger-bottle soft drinks sold in the market would increase. (5)

2016/DSE/II/13(b)(iii)

A lump-sum fee charged on both long-distance and short distance travellers will result in a drop of the price of long-distance flight relative to short-distance flight. A higher proportion of travellers will thus take long-distance flights, increasing the average flight distance. (4)

2021/DSE/II/2

2. When the surcharge is applied equally to long-distance trips and short-distance trips, the price of the former trips relative to the latter would fall. This would lead to a rise in the proportion of passengers travelling long distance. As a result, the average travel distance of late-night taxi passengers would increase.

2023/DSE/II/11(e)

- (e) For fans from overseas, the same amount of travelling expenses is added to both higher-priced and lower-priced tickets. (1)
Inclusion of these travelling expenses would lower the relative price of higher-priced tickets in terms of lower-priced tickets. (1)

2024/DSE/II/10(d)

- (d) As the same amount of lump-sum fuel surcharge is subtracted from the prices of both long- and short-distance flights, the percentage decrease in the nominal price of long-distance flights would be smaller than that in short-distance flights. (1)
The relative price of long-distance flights in terms of short-distance flights increases, resulting in a decrease in the ratio of long-distance flights relative to short-distance flights. (1)
Average flight distance decreases. (1)

MARKING SCHEME

6.3 DETERMINATION OF MARKET PRICE

1995/CE/II/08 A	1994/CE/II/07 B	1996/CE/II/04 D	1991/CE/II/12 C	2007/CE/II/04 C (79%)
2001/CE/II/04 D	1994/CE/II/58 A	1996/CE/II/56 A	1993/CE/II/26 B	2008/CE/II/05 A (56%)
2013/DSE/I/10 D (60%)	1996/CE/II/06 B	1998/CE/II/09 B	1994/CE/II/06 C	2009/CE/II/07 A (83%)
1992/CE/II/21 C	2002/CE/II/07 D (82%)	2000/CE/II/12 D	1995/CE/II/13 A	2010/CE/II/08 D (78%)
1999/CE/II/05 C	2003/CE/II/08 C (58%)	2001/CE/II/03 C	1996/CE/II/05 A	2012/DSE/I/13 D (78%)
2000/CE/II/09 C	2004/CE/II/05 D (60%)	2001/CE/II/05 A	1997/CE/II/05 B	2012/DSE/I/15 B (80%)
2009/CE/II/05 B (47%)	2005/CE/II/04 D (89%)	2003/CE/II/14 C (43%)	1998/CE/II/08 C	2013/DSE/I/11 B (87%)
2014/DSE/I/12 A (53%)	2006/CE/II/07 B (45%)	2004/CE/II/06 D (31%)	1998/CE/II/10 D	2014/DSE/I/14 C (84%)
2016/DSE/I/14 A (55%)	2007/CE/II/10 B (54%)	2006/CE/II/06 C (55%)	1999/CE/II/12 C	2016/DSE/I/16 B (71%)
2016/DSE/I/17 A (67%)	2008/CE/II/04 D (63%)	2006/CE/II/08 C (65%)	2000/CE/II/07 B	2018/DSE/I/16 A (82%)
2018/DSE/I/13 C (74%)	2010/CE/II/06 B (78%)	2007/CE/II/05 D (59%)	2001/CE/II/10 C	1992/CE/II/23 A
1990/CE/II/17 D	2015/DSE/I/15 A (58%)	2009/CE/II/06 D (60%)	2002/CE/II/09 C (72%)	1997/CE/II/06 A
1991/CE/II/24 A	2016/DSE/I/13 D (64%)	2013/DSE/I/12 A (25%)	2002/CE/II/50 D (51%)	2005/CE/II/06 B (77%)
1992/CE/II/22 A	2017/DSE/I/14 C (59%)	2015/DSE/I/14 A (69%)	2004/CE/II/07 C (79%)	2005/CE/II/07 D (70%)
1992/CE/II/42 B	1990/CE/II/17 D	2017/DSE/II/17 D (62%)	2005/CE/II/05 C (84%)	2018/DSE/I/15 B (47%)
1993/CE/II/18 A	1993/CE/II/57 B	2018/DSE/I/20 D (82%)	2006/CE/II/05 A (61%)	2019/DSE/I/13 B
2019/DSE/I/17 B	2020/DSE/I/14 A	2021/DSE/I/13 D	2021/DSE/I/14 B	2022/DSE/I/13 B
2022/DSE/I/14 C	2022/DSE/I/17 B	2023/DSE/I/12 B	2023/DSE/I/19 D	2025/DSE/I/13 A
2025/DSE/I/14 D	2025/DSE/I/16 C	2025/DSE/I/38 D		

Note: Figures in brackets indicate the percentages of candidates choosing the correct answer

1991/CE/1/2(b)(i)

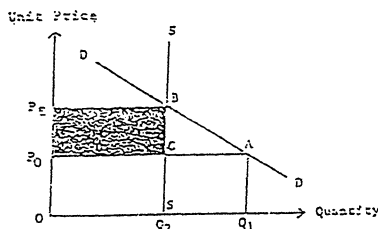


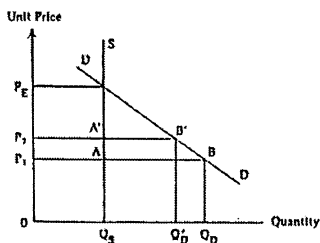
Diagram:

- $P_0 \uparrow$ to P_E (equilibrium price)
- $Q_1 \downarrow$ to Q_2
- shaded area = additional revenue

(i) According to the law of demand,
the quantity demanded decrease.

(ii) The total revenue increase (from area $0Q_2CP_0$ to area $0Q_2BPE$) because the fee is raised while the quantity remains unchanged.

1993/CE/1/3(c)



- P_E : equilibrium price
- P_1 : original price
- P_2 : new price

Correct graph:

- Vertical supply curve
- AB = excess demand at P_1
- $A'B'$ = excess demand at P_2

Correct verbal explanation:

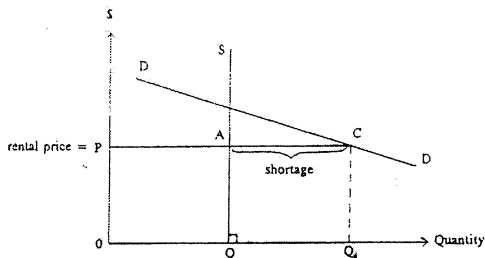
The supply of MTR service is perfectly inelastic (at maximum capacity).

The initial fare (P_1) is set below the equilibrium level, so there is an excess demand.

The higher fare (P_2) is still below the equilibrium level, but the excess demand will become smaller.

1995/CE/I/9(a)

(i)



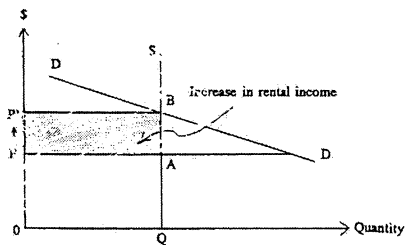
Candidates should indicate on the graph:

- a vertical supply curve
- the rental price P was below the equilibrium price / shortage of AC

(1)

(1)

(ii)



Candidates should indicate on the graph:

Correct indication of the rental income increase on the graph (i.e. area $ABP'P$).

(2)

Verbal elaboration:

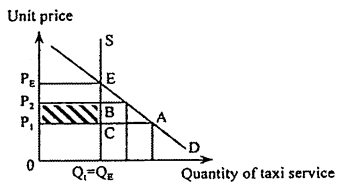
- The price increases to the equilibrium price.
- There is no change in the quantity transacted.
- The rental income increases.

(1)

(1)

(1)

1999/CE/I/11(a)



Graph showing:

- $P_1 < P_E$ (1)
- $P_1 < P_2 < P_E$ (1)
- no change in Q_t (1)
- total revenue increases from $P_1 \times Q_t$ to $P_2 \times Q_t$ / additional revenue is $P_1 P_2 BC$ (1)

Verbal explanation:

At P_1 , there is an excess demand for taxi service. (1)

After a rise in taxi fares from P_1 to P_2 (still below equilibrium price), there is still an excess demand. (1)

The quantity transacted remains unchanged. (1)

The total revenue increases from $P_1 \times Q_t$ to $P_2 \times Q_t$. (1)

1990/CE/I/4(b)

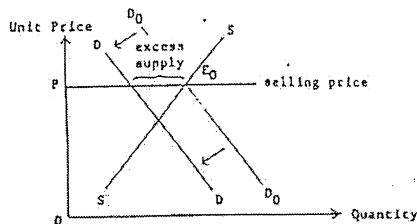


Diagram:

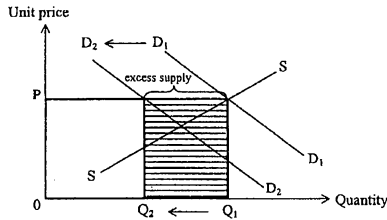
- demand curve shifts to the left (1)
- P unchanged (1)
- excess supply (1)

Elaboration:

The demand decreases (because of the fall in desire for consumption). (1)

There will be an excess supply at the original selling price P . (2)

2004/CE/10(a)



Indicate in the diagram:

- demand curve shifted to the left (1)
- P remained constant (1)
- Q decreased from Q_1 to Q_2 (1)
- excess supply (1)
- $TR (= P \times Q)$ decreased from PQ_1 to PQ_2 / loss in revenue (1)

[Remark: The above analysis starts from an equilibrium position. One can also start from a situation with excess supply or excess demand where similar analysis still applies.]

Verbal explanation:

- (Due to the outbreak of SARS,) there is a decrease in demand for airlines services. (1)
The quantity transacted decreased at the constant price. (1)
Therefore, total revenue decreased. (1)

1990/CE/1/5(c)(ii)

- (1) The demand would probably increase, because
the quality of living in Kowloon City will be improved (nuisances disturbances and danger caused by the old airport do not exist). (3)

OR

The demand would probably decrease, because
new airport spreads the population to Lantau and Western part of Kowloon and the N.T., the convenience of Kowloon City decreases. (3)

- (II) The supply would probably increase, because
removal of height limit / reconstruction of buildings / increase in supply of land released from the old airport (3)

1991/CE/1/4(c)

(i)

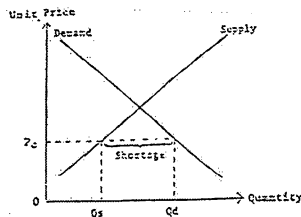


Diagram:

- P (lower than equilibrium) (1)
- shortage (1)

A shortage or excess demand arises when the price is below the equilibrium level. (1)

At such a price, the quantity demanded is larger than the quantity supplied and the difference between Q_d and Q_s , i.e., $Q_d - Q_s$, is a shortage. (2)

(ii) (I)

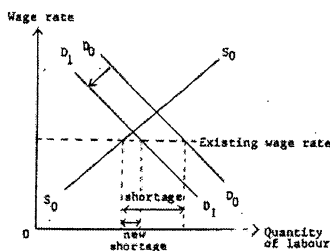


Diagram:

- demand curve shifts to the left
- P unchanged
- smaller shortage

(1)
(1)
(1)

The (derived) demand for labour will fall if there is a recession.
The excess demand will become smaller.

(1)
(1)

(II)

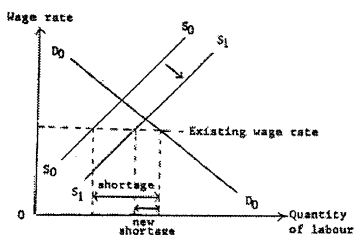


Diagram:

- supply curve shifts to the right
- P unchanged
- smaller shortage

(1)
(1)
(1)

The supply of labour will increase if foreign workers are imported.
The excess demand will become smaller.

(1)
(1)

1992/CE/I/4(d)

(i) The demand increases, because

(Construction of the new airport will bring forth the following:)

more business and investment in Tsuen Wan $\Rightarrow$ demand for shopping centres $\uparrow$

OR

more confidence in the future of Hong Kong $\Rightarrow$ people expect the value of these properties $\uparrow$

(1)

(2)

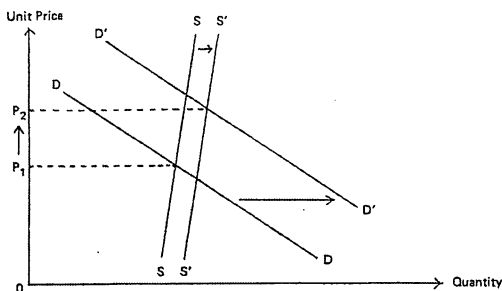
The supply increase, because

more and more factories there will be redeveloped into hotels and shopping centers

(1)

(2)

(ii)



Indicate in the diagram:

- supply curve shifts to the right
- demand curve shifts to the right
- larger shift of the demand curve and a rise in price

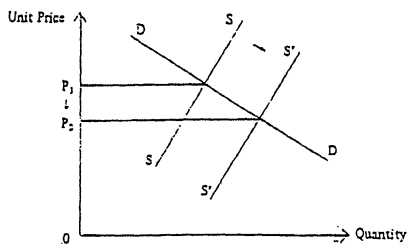
(1)
(1)
(2)

Verbal elaboration:

The increase in demand is larger than the increase in supply.

(2)

1993/CE/1/4(b)(ii)



Indicate in the diagram:

- supply curve shifts to the right
- $P \downarrow$

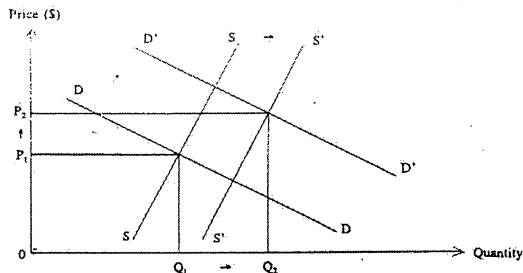
(1)
(1)

Verbal elaboration:

As the supply of TV sets in China increases (because of the smuggling), the price will decrease.

(1)
(1)

1995/CE/I/11(a)



Candidates should show on the graph:

- demand curve shifts to the right
- supply curve shifts to the right
- a greater shift of the demand curve

(1)
(1)
(1)

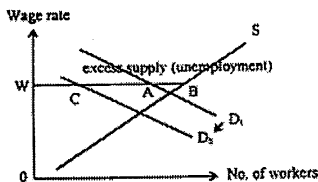
Verbal elaboration:

- The demand increases (because of easy traffic).
- The supply increases (because more land is available).
- The increase in demand is larger than the increase in supply.

(1)
(1)
(2)

1996/CE/I/12(a)

(i)



Indicate in the diagram:

- at W (higher than equilibrium), excess supply = AB
- demand curve shifts to the left
- new excess supply = BC

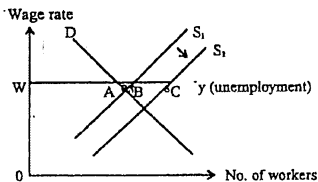
(1)
(1)
(1)

Elaboration:

- The (derived) demand for manufacturing workers decreases.
- At the going wage rate, there will be a greater excess supply.

(1)
(1)

(ii)



Indicate in the diagram:

- at W (higher than equilibrium), excess supply = AB
- demand curve shifts to the left
- new excess supply = AC

(1)
(1)
(1)

Elaboration:

The supply of manufacturing workers increases.
At the going wage rate, there will be a greater excess supply.

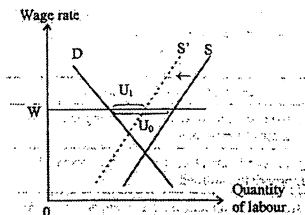
(1)
(1)

1997/CE/1/2

Initial unemployment problem exists when the wage rate is higher than the equilibrium wage rate.

(1)

Answer 1



Indicate in the diagram:

- wage rate $W >$ equilibrium wage rate
- supply curve shifts to the left
- unemployment drops from U_0 to U_1

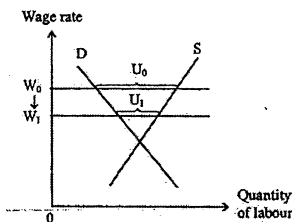
(1)
(1)
(1)

Elaboration:

If some workers have to leave the industry and find jobs in other industries,
the supply of labour will decrease (, and reduce unemployment).

(1)
(1)

Answer 2



Indicate in the diagram:

- wage rate $W_0 >$ equilibrium wage rate
- W_0 falls to W_1
- unemployment drops from U_0 to U_1

(1)

(1)

(1)

Elaboration:

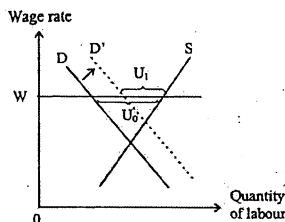
Workers of the industry have to accept lower wage rates.

(1)

The excess supply of labour (i.e. quantity supplied in excess of quantity demanded) will fall.

(1)

Answer 3



Indicate in the diagram:

- wage rate $W >$ equilibrium wage rate
- demand curve shifts to the right
- unemployment drops from U_0 to U_1

(1)

(1)

(1)

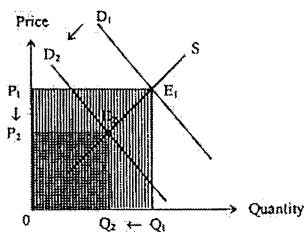
Elaboration:

If there is more business, or labour productivity of the industry increases, the derived demand for labour will increase (, and reduce unemployment).

(1)

(1)

1997/CE/I/9(b)



Candidates should show on the graph:

- demand curve shifts to the left

- $P \downarrow$ & $Q \downarrow$

- loss in total revenue = area $E_1Q_1Q_2E_2P_2P_1$

(1)

(1)

(1)

Verbal explanation:

The demand for minibus service will decrease, because

- the new railway service is a substitute of minibus service.

Total revenue will decrease (from P_1Q_1 to P_2Q_2).

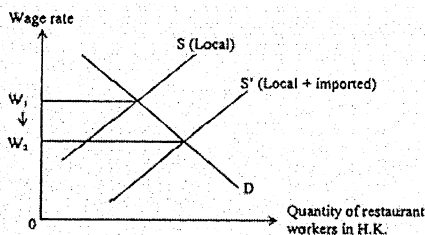
(1)

(1)

(1)

1998/CE/I/9(c)

(i)



Candidates should show on the graph:

- supply curve shifts to the right

- $W \downarrow$

(1)

(1)

Verbal explanation:

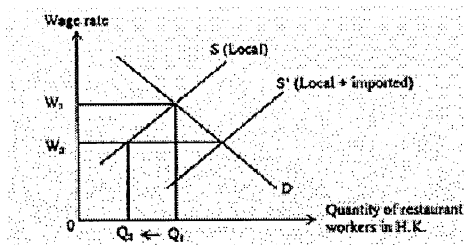
There is an increase in the supply of restaurant workers.

The equilibrium wage rate will fall.

(1)

(1)

(ii)



Verbal explanation:

Q_1 is the original employment level of local restaurant workers.

Q_2 is the new employment level of local restaurant workers.

The employment level decreases. (or diagram showing $Q_1 \Rightarrow Q_2$)

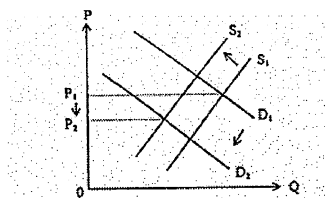
(1)

(1)

(1)

1998/CE/1/10(b)

(i)



Candidates should show on the graph:

- demand curve shifts to the left

- supply curve shifts to the left

(1)

(1)

Verbal explanation:

The demand for illegal computer software decreases.

The supply decreases.

The market price is uncertain,

depending on the extent of decrease in both demand and supply.

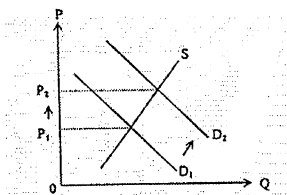
(1)

(1)

(1)

(1)

(ii)



Candidates should show on the graph:

- demand curve shifts to the right

- $P \uparrow$

(1)

(1)

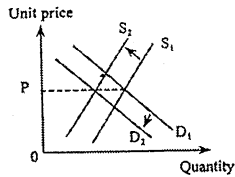
Verbal explanation:

The demand for legal computer software increases (because it is a substitute of illegal software).

As a result, the market price increases.

(1)

(1)



Graph showing:

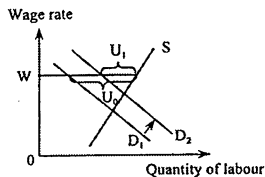
- demand curve shifts to the left (1)
- supply curve shifts to the left (1)
- P (determined by D_2 and S_2) remains unchanged (1)

Verbal explanation:

- There is a fall in demand (because people were afraid of eating poisonous seafood). (1)
- There is also a fall in supply (because the import volume of seafood decreased). (1)
- As the fall in demand was equal to the fall in supply, the price remained the same. (1)

1999/CE/I/10(a)

- (i) Firms' profits after tax will rise, (1)
- leading to a higher incentive to invest (or an increase in investment). (1)
- The (derived) demand for labour is likely to increase. (1)
- (ii)



Graph showing:

- demand curve shifts to the right (1)
- a fixed wage rate (1)
- a larger excess supply (from U_0 to U_1) (1)

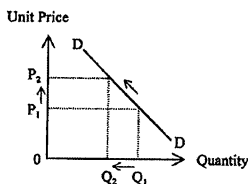
Verbal explanation:

- Unemployment exists when the fixed wage rate is higher than the equilibrium wage rate. (1)
- As the derived demand for labour increases, (1)
- (given the unchanged wage rate,) excess supply or unemployment decreases. (1)

2000/CE/1/2

David refers to the law of demand / a movement along a downward sloping demand curve.

(1)



Indicate in the graph:

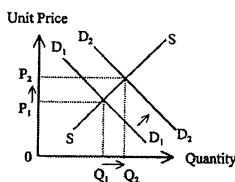
- a downward sloping demand curve
- a rise in price and a corresponding decrease in the quantity demanded

(1)

(1)

Amy refers to (a change in the equilibrium position due to) an increase in demand / a shift of the demand curve.

(1)



Indicate in the graph:

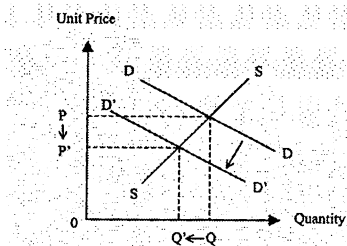
- rightward shift of demand curve
- an increase in equilibrium price
- an increase in equilibrium quantity

(1)

(1)

(1)

2000/CE/1/9(b)(i)



Indicate in the graph:

- demand curve shifts to the left
- $PQ \downarrow$ to $P'Q'$

(1)

(1)

Verbal explanation:

The demand for legal petroleum decreases, because

a substitute is available.

(1)

Both the price and the quantity transacted will fall.

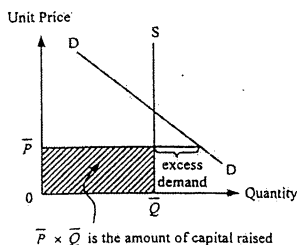
(1)

The total revenue will decrease.

(1)

(1)

2001/CE/I/10(c)(i)



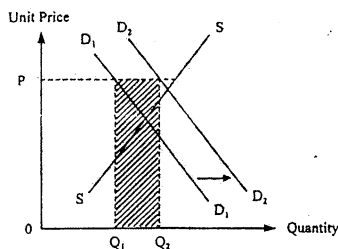
Indicate in the graph:

- a vertical supply curve (1)
- the price is set below the equilibrium price (1)
- excess demand (1)
- amount of capital raised (2)

Verbal explanation:

The selling price is fixed at a level below the equilibrium price, at which the quantity demanded is larger than the quantity supplied / there is an excess demand. (1)

2001/CE/I/11(d)



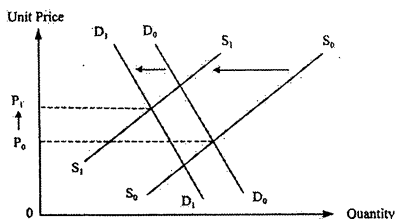
Indicate in the graph:

- fixed price P is set above the equilibrium price (1)
- demand curve shifts to the right (1)
- Shaded area = increase in sales revenue (1)

Verbal explanation:

The demand for private housing will increase, because (1)
public housing is a substitute. (1)
(Given constant P_s) the quantity transacted will increase, and the total revenue will increase. (2)

2002/CE/1/2



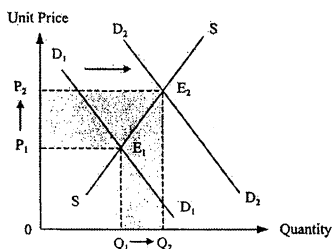
Indicate in the graph:

- supply curve shifts to the left (1)
- demand curve shifts to the left (1)
- correct indication of the increase in equilibrium price with the leftward shift of the supply curve > the leftward shift of the demand curve (1)

Verbal explanation:

- There is a decrease in the supply of gasoline (because gasoline is an oil product). (1)
- There is a decrease in the demand for gasoline, because gasoline and automobiles are complementary goods. (1)
- The change in supply is greater than the change in demand. (1)

2002/CE/1/10(b)



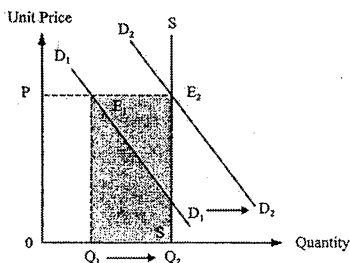
Indicate in the graph:

- demand curve shifts to the right (1)
- price $\uparrow$ and quantity transacted $\uparrow$ (1)
- change in total expenditure = $P_2Q_2 - P_1Q_1$ (1)

Verbal explanation:

- The demand increases (because the cost of borrowing is lowered). (1)
- Both the price and the quantity transacted increase. (1)
- Total expenditure increases. (1)

2002/CE/1/11(b)



Indicate in the graph:

- tuition fee level $P >$ equilibrium price (1)
- rightward shift of the demand curve (1)
- D_1 shifts to D_2 or the right of D_2 (1)
- increase in total revenue = area $E_1E_2Q_2Q_1$ (1)

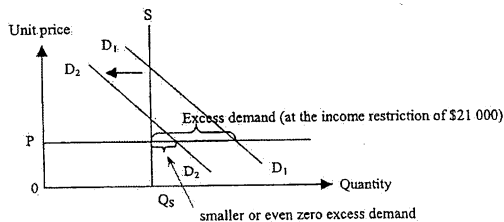
Verbal explanation:

The demand increases (because more students are attracted by the advertisement). (1)

The quantity transacted increases. (1)

The excess supply decreases from Q_1Q_2 to zero (i.e. disappeared). (1)

2003/CE/1/1



(a) Indicate in the diagram:

- a constant price (P) below the equilibrium price (1)
- excess demand at the income restriction of \$21 000 (demand is D_1) (1)

Verbal explanation:

The fixed price is below the equilibrium price. (1)

There is an excess demand at the beginning. (1)

(b) Indicate in the diagram:

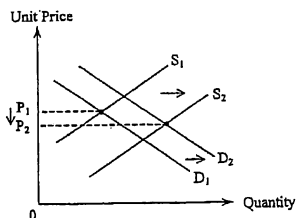
- demand curve shifts to the left (1)
- resulting in a smaller excess demand or even no more excess demand (1)

Verbal explanation:

The demand falls due to further income restriction. (1)

(The excess demand will be smaller or even eliminated). (1)

2003/CE/I/10(b)



- (i) The supply increases, because the cost of production (i.e. input price) decreases.
The demand increases, because there is a larger number of buyers

(1)
(1)
(1)

(ii) Indicate in the diagram:

- supply curve shifts to the right
- demand curve shifts to the right
- larger shift of the supply curve and a fall in price

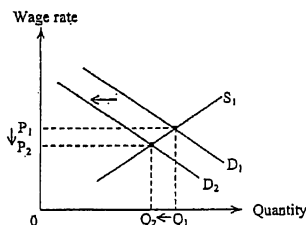
(1)
(1)
(1)

Verbal elaboration:

The increase in supply is larger than the increase in demand.

(1)

2003/CE/I/11(d)



Indicate in the diagram:

- demand curve shifts to the left
- $P \downarrow$
- $Q \downarrow$

(1)
(1)
(1)

Verbal elaboration:

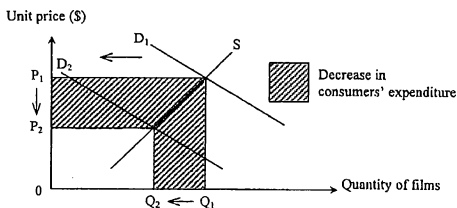
The (derived) demand for labour decreases.

(1)

The wage rate decreases, and employment decreases

(1)
(1)

2004/CE/I/3



Indicate in the diagram:

- D_1 shifts to D_2 (1)
- $P \downarrow$ & $Q \downarrow$ (1)
- Total expenditure $\downarrow$ (1)

Verbal elaboration:

The demand for films decreases, because (1)

traditional cameras and films are complements.

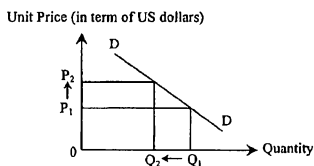
OR

The demand for traditional cameras decreases, because digital cameras are substitutes of traditional cameras. (1)

Consumers' total expenditure on films decreases (1)

2005/CE/I/1

(i)



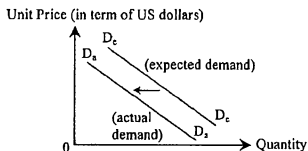
Indicate on the diagram:

- $P \uparrow$ (1)
- $Q \downarrow$ (1)

Verbal elaboration:

[A rise in exchange rate of the euro dollar would lead to a rise in price (in terms of US dollars) of the tourism service,] the quantity demanded of the tottrism service would decrease. (1)

(ii)



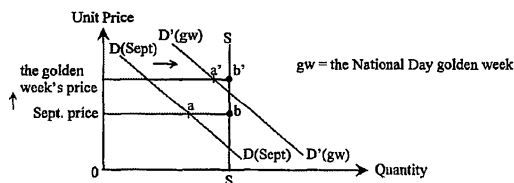
Indicate on the diagram:

- leftward shift of the demand curve (1)

Verbal elaboration:

[If tourists were afraid that they might be attacked by the terrorists,] their demand for the tourism service would decrease. (1)

2006/CE/I/9(a)



(i) Diagram:

- Sept. price > equilibrium price
- ab = excess supply in Sept.

(1)

(1)

Verbal Elaboration :

- initial excess supply in Sept.

(1)

(ii) Diagram:

- demand curve shifts to the right to position $D'D'$ in the golden week.
- price raised to a new level (there is an excess supply at the new price)
- excess supply in the golden week = $a'b'$
- $a'b' < ab$

(1)

(1)

(1)

(1)

Verbal Elaboration:

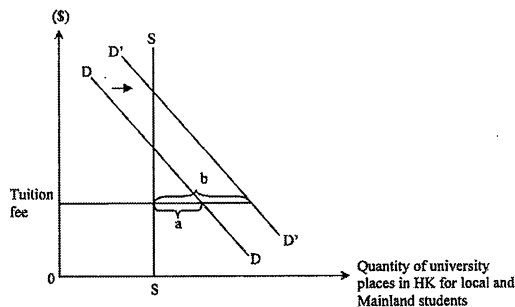
There is an increase in demand (because of the increasing number of tourists).

(1)

The excess supply becomes smaller in the golden week.

(1)

2007/CE/I/8



Indicate in the diagram:

- tuition fee is below the equilibrium price
- a is the original excess demand
- rightward shift of the demand curve
- b is the new excess demand

(1)

(1)

(1)

(1)

Verbal elaboration:

There is an increase in demand (because of the increasing number of mainland students).

(1)

The tuition fee is below the equilibrium level / remains unchanged.

(1)

(resulting in a larger excess demand.)

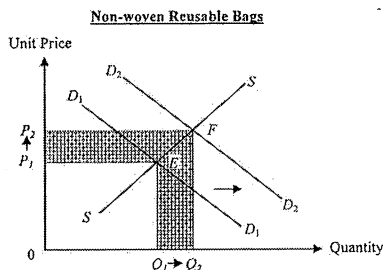
Microeconomics by Topic
6. Demand & Supply (I)

2008/CE/I/1

- number of substitutes: more substitutes are available
 - price of substitutes: a fall in the price of substitutes
 - income of consumers: an increase in income and rice is an inferior good
 - taste of consumers: a change in taste against eating rice
- [Mark the **FIRST TWO** points only.]

(2@, max: 4)

2010/CE/I/1



Indicate in the diagram:

- demand curve shifts to the right
- area $OP_2FQ_2 > \text{area } OP_1EQ_1$

(1)

(1)

Verbal elaboration:

The demand for non-woven reusable bags increases, because they are substitutes of plastic shopping bags.

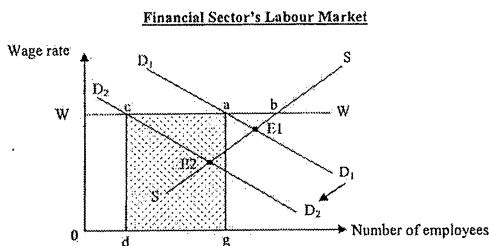
(1)

(1)

Therefore, the total revenue increases.

(1)

2010/CE/II/11(a)



Indicate in the diagram:

- wage rate W above the initial equilibrium wage rate at E_1
- initial excess supply ab
- D_1 shifts to D_2
- cb = the new excess supply
- area $cdga$ = total wage loss

(1)

(1)

(1)

(1)

(1)

Verbal elaboration:

Initially, the wage rate is higher than the initial equilibrium wage rate.

(1)

There is an excess supply of labour / the quantity supplied is greater than the quantity demanded (by ab).

(1)

The demand for labour falls.

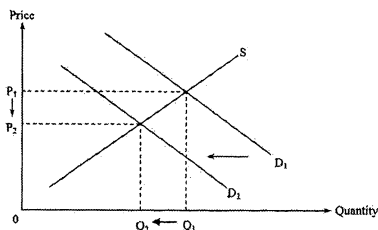
(1)

The new excess supply is cb (which is larger than the initial excess supply ab).

(1)

2012/DSE/II/12(c)

(i)



Indicate in the diagram :

- D shifts to the left

(1)

- P ↓

(1)

Verbal Elaboration :

It would result in a drop in demand for private housing, because

(1)

public housing (living units under HOS) and private housing are substitutes.

(1)

The price of private housing drops.

(1)

(ii) A rise in the mortgage interest rate could cool down the property market, because

(1)

some people fail (or are unwilling) to borrow at a higher interest rate.

(1)

Therefore, the demand for private housing drops.

(1)

2015/DSE/II/4(b)

The supply of large flats would drop, because

(1)

small flats and large flats are in competitive supply.

(1)

2018/DSE/II/13(c)

Nursing, because

(1)

the demand for medical service would rise as a result of the aging population, and

(1)

the demand for nurse is the derived demand for medical service.

(1)

The expected future income of a nurse should thus be higher (than that of someone working in primary education).

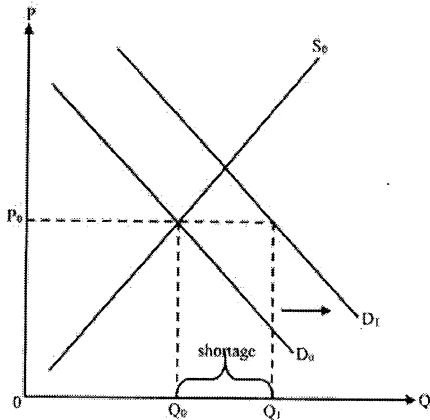
(1)

(c) Verbal elaboration:

Demand for the courses increases. If the price does not adjust fast enough in response to the change in demand, then it would fall short of the equilibrium price and thus result in excess demand or shortage/queues. (3)

Illustrate in the diagram:

- rightward shift in the demand curve (1)
- correct position of shortage (1)



2023/DSE/II/11(a)

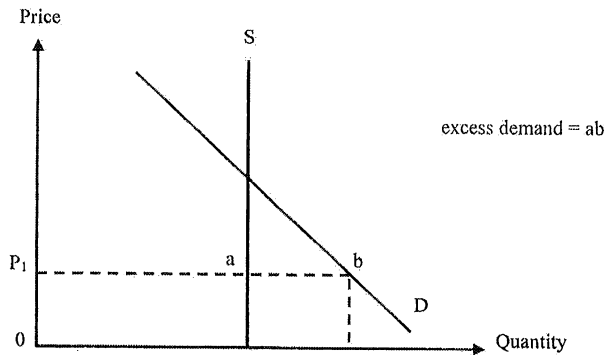
(a) Verbal elaboration:

- As the concert ticket price is set below the equilibrium price, there is an excess demand for concert tickets. (2)

Indicate in the diagram:

- price below equilibrium price (1)
- excess demand (ab) (1)

Figure 5



2024/DSE/II/13(d)

(d) Marks for effective communication (EC: max 2 marks)

Marks	Performance
2	● Support arguments with the source/data and appropriate economic theories. ● Present relevant material. ● Present well-organised and coherent answers without repetition of ideas. ● Use language that expresses ideas clearly and fluently with appropriate use of words/terms/symbols.
1	● Present arguments with some support of the source/data and economic theories. ● Present some irrelevant material. ● Present answers in a less organised way with some repetition. ● Use language that conveys a clear message with some inappropriate use of words/terms/symbols.
0	● Present arguments with no support of the source/data and economic theories. ● Present material unrelated to the gist of the question. ● Present inconsistent arguments. ● Express limited ideas with inappropriate use of words/terms/symbols.

The maximum mark for content is 12 marks.

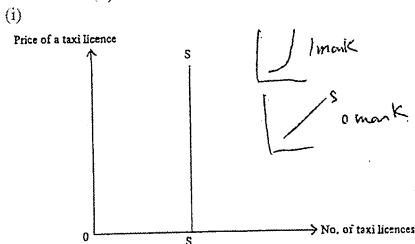
MARKING SCHEME

7.1 ELASTICITIES OF DEMAND & SUPPLY

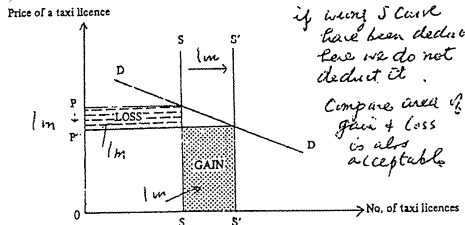
1990/CE/II/05 B	1995/CE/II/07 B	2000/CE/II/10 A	2006/CE/II/09 D (14%)	2010/CE/II/10 D (20%)
1990/CE/II/26 C	1996/CE/II/10 C	2002/CE/II/08 A (44%)	2007/CE/II/09 B (55%)	2012/DSE/II/11 C (45%)
1991/CE/II/19 C	1997/CE/II/08 D	2002/CE/II/10 B (34%)	2007/CE/II/11 D (52%)	2012/DSE/II/12 B (61%)
1991/CE/II/23 C	1997/CE/II/09 C	2003/CE/II/05 C (51%)	2008/CE/II/06 B (24%)	2014/DSE/II/13 B (60%)
1991/CE/II/57 C	1997/CE/II/10 D	2003/CE/II/07 C (55%)	2008/CE/II/10 D (56%)	2015/DSE/II/11 B (47%)
1992/CE/II/28 C	1998/CE/II/06 B	2003/CE/II/09 B (45%)	2009/CE/II/09 C (62%)	2016/DSE/II/15 A (37%)
1993/CE/II/20 D	1998/CE/II/07 C	2003/CE/II/13 D (59%)	2009/CE/II/10 C (26%)	2017/DSE/II/15 B (84%)
1993/CE/II/25 A	2000/CE/II/03 C	2004/CE/II/09 C (51%)	2010/CE/II/07 D (52%)	2018/DSE/II/17 C (63%)
1993/CE/II/28 C	2000/CE/II/05 A	2005/CE/II/12 A (32%)	2010/CE/II/09 C (42%)	2019/DSE/II/11 A
2020/DSE/II/ A	2022/DSE/II/12 D	2023/DSE/II/13 A	2023/DSE/II/14 B	2023/DSE/II/15 B
2024/DSE/II/13 A	2024/DSE/II/14 B	2025/DSE/II/15 B	2025/DSE/II/42 D	

Note: Figures in brackets indicate the percentages of candidates choosing the correct answer

1992/CE/II/1(d)



(ii)



(ii) Indicate in the diagram:

- supply curve shifts to the right
- $P \downarrow$
- correct position of 'gain' and 'loss'
- gain > loss

(1)
(1)
(1)
(1)

Verbal explanation:

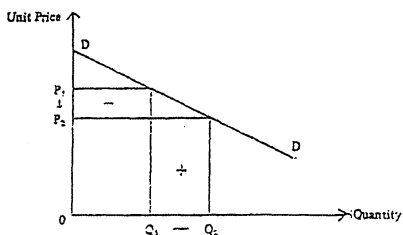
- (I) The price of a taxi licence will decrease, because there is an increase in supply.

(1)
(1)

- (II) If the demand for taxi licences is elastic, the percentage increase in quantity demanded will be larger than the percentage decrease in price.

(2)
(2)

1993/CE/1/1(a)



Indicate in the diagram:

- correct position of 'gain' and 'loss'
- gain > loss

(1)
(1)

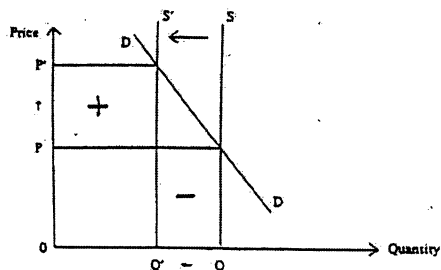
Verbal explanation:

The demand for hamburgers is elastic (because there are substitutes).

The percentage decrease in price is smaller than the percentage increase in quantity demanded.

(2)
(3)

1994/CE/1/1(b)



Indicate in the diagram:

- supply curve shifts to the left
- correct position of 'gain' and 'loss'
- gain > loss

(1)
(1)
(1)

Verbal explanation:

Since the demand is inelastic,

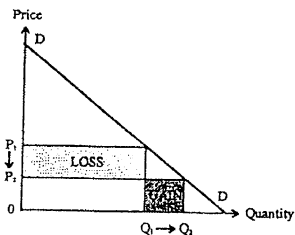
the percentage increase in price is larger than the percentage decrease in quantity sold.

(1)
(3)

Microeconomics by Topic
7. Demand & Supply (II)

1996/CE/1/1(a)

General manager:



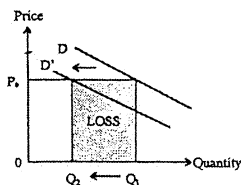
Indicate in the diagram:

- $P \downarrow$ & $Q \uparrow$ (1)
- correct position of 'gain' and 'loss' (1)
- gain < loss (1)

Verbal explanation:

- The demand is inelastic, (1)
- the percentage increase in quantity demanded is smaller than the percentage decrease in price. (2)

Sales manager:



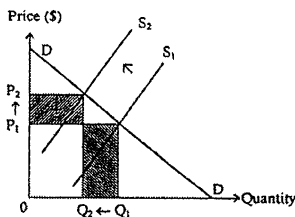
Indicate in the diagram:

- demand curve shifts to the left (1)
- P unchanged & $Q \downarrow$ (1)
- loss (1)

Verbal explanation:

- The demand for this newspaper would decrease (because different newspapers are substitutes for each other). (1)
- The quantity transacted decreases. (1)

1997/CE/1/11(c)



Indicate in the diagram:

- supply curve shifts to the left
- $P \downarrow$ & $Q \uparrow$
- correct position of 'gain' and 'loss'
- gain < loss

(1)
(1)
(1)
(1)

Verbal explanation:

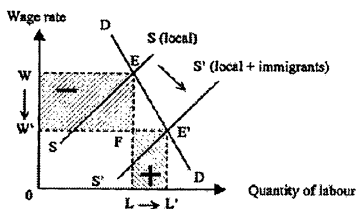
The supply of restaurant services decreases (because of an increase in production cost).

The demand is elastic,

the percentage increase in price is smaller than the percentage decrease in quantity demanded.

(1)
(1)
(3)

2000/CE/1/11(b)



Indicate in the graph:

- S (local people) shifts rightward to S' (local + immigrants)
- correct position of 'gain' and 'loss'
- loss > gain

(1)
(1)
(1)

Verbal explanation:

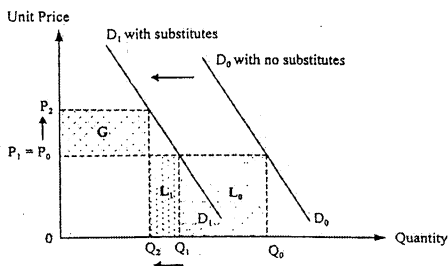
The supply of labour increases.

The demand is inelastic,

the percentage decrease in wage rate is larger than the percentage increase in employment

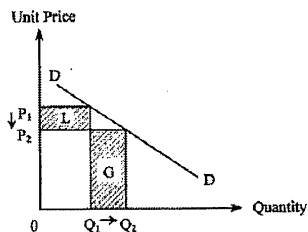
(1)
(1)
(3)

2001/CE/1/2



- (a) Indicate in the graph:
- demand curve shifts to the left (1)
 - fixed postage charge, i.e., $P_1 = P_0$ (1)
 - L_0 = decrease in total revenue (1)
- (b) Indicate in the graph:
- L_1 = new loss in total revenue (1)
 - G = new gain in total revenue (1)

2003/CE/1/9(a)



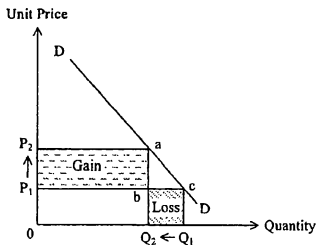
- Indicate in the diagram:
- $P \downarrow$ and $Q \uparrow$ (1)
 - revenue loss < revenue gain (2)

Verbal elaboration:

- The fall in export prices leads to an increase in the export volume. (1)
- As the demand for HK exports is elastic, (1)
- the percentage change in export price is smaller than the percentage change in the export volume. (2)

2005/CE/I/9

(a)



Indicate on the diagram:

- Area abP_1P_2 = Gain; Area bcQ_1Q_2 = Loss

(1)

- Gain > Loss

(1)

Verbal elaboration:

(After the increase in tolls,) the quantity demanded decreases.

(1)

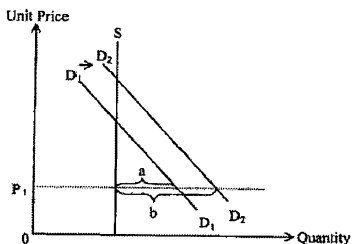
The demand for the tunnel service is inelastic,

(1)

the percentage change in price is larger than the percentage change in quantity demanded.

(2)

(b) (i)



Indicate on the diagram:

- price (P_1) is below the equilibrium price

(1)

- rightward shift of the demand curve

(1)

- excess demand increases from a to b

(1)

Verbal elaboration:

The Cross Harbour Tunnel is a substitute of the Western Harbour Crossing.

(1)

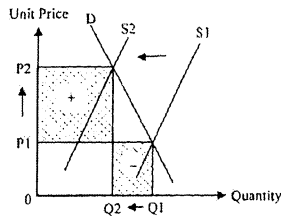
The demand for the Cross Harbour Tunnel's service increases.

(1)

(With a fixed price,) the excess demand would increase.

(1)

2009/CE/I/9(b)



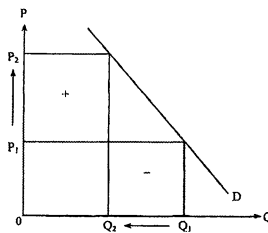
Indicate in the diagram:

- supply curve shifts to the left (1)
- $Q \downarrow$ and $P \uparrow$ (1)
- loss in total revenue < gain in total revenue (2)

Verbal elaboration:

- The supply of textbooks decreases, because the cost of production has increased. (1)
- The price increases and the quantity transacted decreases. (1)
- If the demand is inelastic, (1)
- the percentage change in quantity demanded is smaller than the percentage change in price. (1)

2013/DSE/I/9(a)



Indicate on the diagram:

- correction position of gain and loss (1)
- gain > loss (1)

Verbal elaboration :

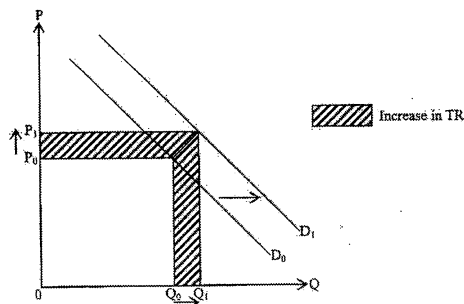
- If the demand for MTR service is inelastic, (1)
- the percentage increase in fare will be higher than the percentage decrease in the number of trips (, so the total expenditure will increase). (2)

2015/DSE/I/11(c)

- Tobacco duty would increase the market price of duty-paid cigarettes and thus reduce its quantity demanded. (1)

- If cigarette consumers are addicted, (1)
- their demand for cigarettes would be inelastic, so (1)
- the percentage decrease in quantity demanded would be smaller than the percentage increase in price. (1)

2017/DSE/I/10(c)



Illustrate in the diagram:

- rightward shift in the demand curve
- higher price and quantity
- higher total revenue

(1)
(1)
(1)

Verbal elaboration:

No, because

there was a rise in demand.

As both the price and the quantity transacted have increased,
total revenue has increased.

(1)
(2)
(2)
(1)
(Max: 7)

2018/DSE/I/5(a)

The supply of medical staff may be inelastic because

they require special training and qualification.

(Even if the government raises their wage rates sharply,) the percentage increase in the quantity supplied of medical staff
would still be smaller than the percentage increase in their wage rates.

(1)
(1)
(1)
(1)

2022/DSE/II/4a

- (a) Imposition of tariff would result in a decrease in the supply of imported goods. (3)

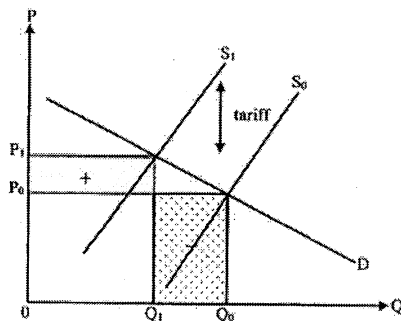
Since the demand for imported goods is elastic, the percentage decrease in quantity
would be bigger than the percentage increase in price.

Total expenditure would therefore drop.

Illustrate in the diagram:

- leftward shift in supply
- correct position of Area (+) and Area (—)
- Area (+) < Area (—)

(1)
(1)
(1)



2023/DSE/II/12aii

- (ii) According to Source B, gasoline is a daily necessity. (1)
Therefore, the demand for gasoline tends to be inelastic. (1)
The percentage decrease in quantity demanded is smaller than the percentage increase in price. (1)
Therefore, total expenditure on gasoline increases as its price increases.

2024/DSE/II/10(c)

- (c) As the demand for flight tickets increases, the demand curve shifts to the right. The equilibrium point moves upward along the same supply curve. Therefore, the percentage change in quantity relative to that in price depends on the price elasticity of supply. (3)
Given that the percentage increase in quantity is smaller than the percentage increase in price, we can conclude that the supply of flight tickets is inelastic.

2025/DSE/II/7

- (a) As each country specialises in producing the good which it can produce at a lower opportunity cost, the total output will increase. (2)
- (b) The opportunity cost to produce 1 unit of computers
in Country A $= 3/5 = 0.6$ units of toys (1)
in Country B $= 4/4 = 1$ unit of toys (1)
Country A would export computers as it has a lower opportunity cost in producing computers than Country B. (1)

7.2 EFFICIENCY & SOCIAL SURPLUS

Microeconomics by Topic
7. Demand & Supply (II)

MARKING SCHEME

1985/AL/I/22 D	1997/AL/I/07 C	2003/AL/I/11 A	2012/AL/I/05 B (45%)	2012/DSE/I/14 C (64%)
1986/AL/I/15 C	1998/AL/I/12 D	2006/AL/I/02 B (71%)	2012/AL/I/07 B (68%)	2014/DSE/I/15 C (54%)
1990/AL/I/02 A	1998/AL/I/24 A	2006/AL/I/23 C (53%)	2012/AL/I/16 D (67%)	2015/DSE/I/22 D (64%)
1991/AL/I/08 D	2000/AL/I/10 A	2008/AL/I/05 C (94%)	PP/DSE/I/12 B	2016/DSE/I/21 C (68%)
1995/AL/I/03 B	2001/AL/I/19 B	2010/AL/I/07 A (85%)	PP/DSE/I/14 A	2016/DSE/I/22 D (61%)
1995/AL/I/06 B	2002/AL/I/10 D	2012/AL/I/01 B (70%)	PP/DSE/I/17 C	2017/DSE/I/12 A (88%)
2019/DSE/I/21 B	2019/DSE/I/22 D	2021/DSE/I/17 A	2021/DSE/I/19 C	2021/DSE/I/20 A
2022/DSE/I/10 C	2022/DSE/I/20 D	2023/DSE/I/20 D	2023/DSE/I/21 C	2023/DSE/I/22 D
2024/DSE/I/19 B	2024/DSE/I/20 C	2024/DSE/I/21 C	2024/DSE/I/22 D	2025/DSE/I/12 B
2025/DSE/I/18 B	2025/DSE/I/21 D	2025/DSE/I/22 B	2025/DSE/I/43 C	

Note: Figures in brackets indicate the percentages of candidates choosing the correct answers.

1988/AL/I/2

The marginal benefit (i.e. value) is the maximum a consumer is willing to pay for the last unit purchased. Errors aside, no maximizing individual will, on a voluntary basis, pay more than what he is willing to pay.

If price is lower than his marginal benefit, to maximise consumer surplus, the consumer will buy more until the price is equal to his marginal benefit. Therefore, at the margin, he will pay the maximum of what he is willing to pay, and price equals marginal benefit.

1989/AL/I/1

Price is a marginal benefit (i.e. the maximum an individual is willing to pay at the margin).

Price serves as a signaling device to direct resource allocation (i.e. allocative function); it also serves as a criterion of determining who wins or loses in competing for goods and services in the market place (i.e. rationing function).

1992/AL/I/6

The market price of the permits will definitely be higher than \$20, because in this market the criteria of drawing lottery will not be used, leaving the market price to measure fully the marginal benefit people place on the permit.

With the permits now going to those who offer the highest prices in the resale market, more goats will likely be killed. This is because the higher permit price implies that either the permit holders are better, more enthusiastic, or more frequent hunters who place a higher marginal benefit on hunting.

(Note: More goats would be killed not because permit holders wished to make compensation for the higher price, but because only better hunters were willing to pay a higher price to obtain the permits.)

1994/AL/I/8

No. The marginal cost of reproducing discs (legal or illegal) is much lower than the selling price. Since illegal copies are sold at a lower price, buyers can enjoy a greater consumer surplus. The marginal benefit is closer to the marginal cost, leading to a more efficient allocation of resources.

Yes. If reproduced discs are sold in the market, in an extreme case, the composers of the songs will not be able to receive any revenue from selling the original discs. There will be no more new songs and original discs in the future (and hence illegal reproduction), and no more consumer surplus and producer surplus in the future.

1997/AL/I/8(a)

When a student is required to pay a high tuition out of his own pocket to buy university education, his own marginal benefit on university education will also be high. He will therefore treasure education more, with the result that he would study harder than when he does not have to pay and put a low marginal benefit on education.

(Note: Whether students are willing to work harder or not depends on the price they have to pay for being lazy, as indicated by the law of demand. The 'price' here is the value of learning forgone.)

2002/AL/I/06(b)

Whoever pays for the purchase of the tunnel, the payment is a cost to society. If the government purchases the tunnel (tax money is a cost to society) and let all cars use it free of charge, there is no way of knowing the total benefit to the users will cover the purchasing cost of the tunnel, even if we ignore the cost of operation. Whether or not here is congestion makes no difference to this answer.

2006/AL/I/5

In a buffet dinner when a consumer eats until his marginal benefit reaches zero, lower than the marginal cost of producing food, we have an inefficient situation.

2006/AL/I/9

It does not mean ladies in mainland China are richer than those in Hong Kong. It only indicates that mainland ladies have a higher maximum willingness to pay for prestigious handbags. Such a higher marginal benefit could be a result of higher wealth or income, or other factors such as a lack of substitutes or consumer preference (e.g. due to culture or advertisement) favourable to these handbags.

Many ladies from China pay such high prices because they do care for famous or prestigious brand names and it is possible that mainland ladies in general have a higher preference for prestige.

2009/AL/I/7(a)

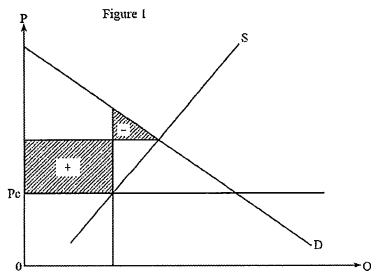
The higher licence fees plus the higher travelling costs indicate that the cost per animal hunted is considerably higher for out-of-state hunters, so in general they would have to be better hunters to equate the marginal gain and marginal cost of hunting.

PP/DSE/I/4

(a) Deadweight loss is the loss in the total social surplus or the potential gains not fully captured when marginal benefit does not equal marginal cost. (2)

(b) (i) Yes.

(1)



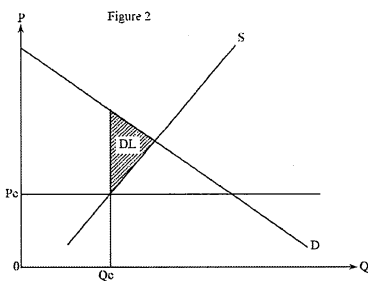
Indicate in the diagram:

- gain of consumer surplus (+)
- loss of consumer surplus (-)

(1)

(1)

(b) (ii)



Indicate in the diagram:

- deadweight loss (DL)
- quantity is Q_e

(1)

(1)

Verbal elaboration:

No, because

when price is set below equilibrium, the quantity transacted will be reduced.

marginal cost of sellers is lower than marginal benefit of consumers / there is deadweight loss.

(1)

(1)

(1)

PP/DSE/I/12(b)(i)

Rationing function: Tickets were distributed to the buyers who were willing to pay at least \$200.

(1)

2015/DSE/I/4(a)

Allocative function: Higher prices of small-sized flats would induce real-estate developers to direct more resources towards the production of small flats.

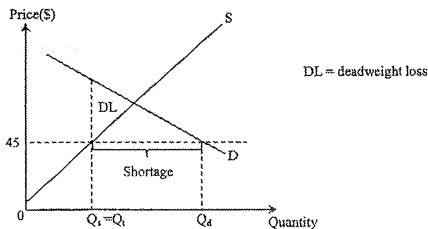
OR

Signal function: Prices help transmit information about consumer demand relative to producer supply.

(2)

2016/DSE/II/11

(b)



Illustrate in the diagram:

- price set below equilibrium (1)
- correct position of Q_s (1)
- correct position of deadweight loss (1)

Verbal elaboration:

The price is set at \$45 which is below equilibrium. (1)

(With the existence of shortage,) the quantity transacted is below equilibrium quantity or there is underproduction. (1)

the marginal benefit is larger than the marginal cost. (1)

(d) Efficiency:

The fee increase would improve efficiency as it results in a price closer to the equilibrium level, thus narrowing the gap between MB and MC and reducing the size of deadweight loss. (2)

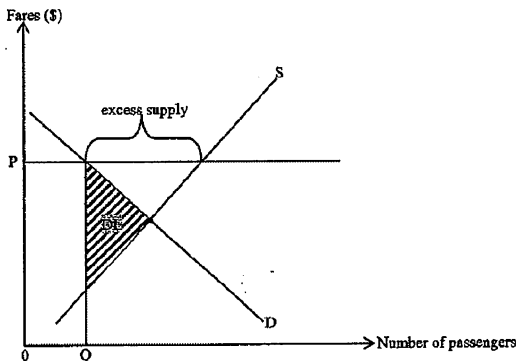
2019/DSE/II/11

- (a) The launch of HSR will increase the geographical mobility of workers because the transportation network improves. (3)

- (b) Verbal elaboration: (2)
Fares are set above the market-clearing (or equilibrium) level. So marginal benefit of train service is larger than marginal cost.

Illustrate in the diagram:

- correct position of P and Q (1)
- correct position of deadweight loss (DL) (1)



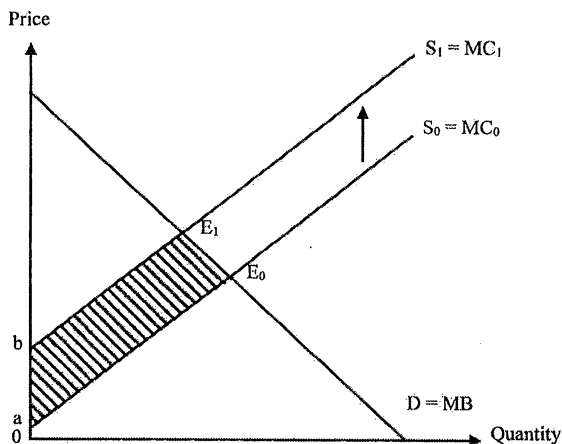
2023/DSE/II/5a-b

(a) Indicate in the diagram:

- upward shift of MC curve (leftward shift of supply curve)
- decrease in total social surplus

(1)

(1)



Decrease in total social surplus = area (abE₁E₀)

(b) Verbal elaboration:

No, efficiency is still attained because at the new equilibrium quantity, marginal benefit is still equal to marginal cost.

(1)

(1)

2023/DSE/II/11b

(b) rationing function: concert tickets are distributed to the highest-valued buyers.

(2)

2025/DSE/II/8

(a) (i) Indicate in the diagram:

- correct position of P_{W+t}
- correct position of Q_1

(1)

(1)

(ii) Indicate in the diagram:

- correct position of Q_M

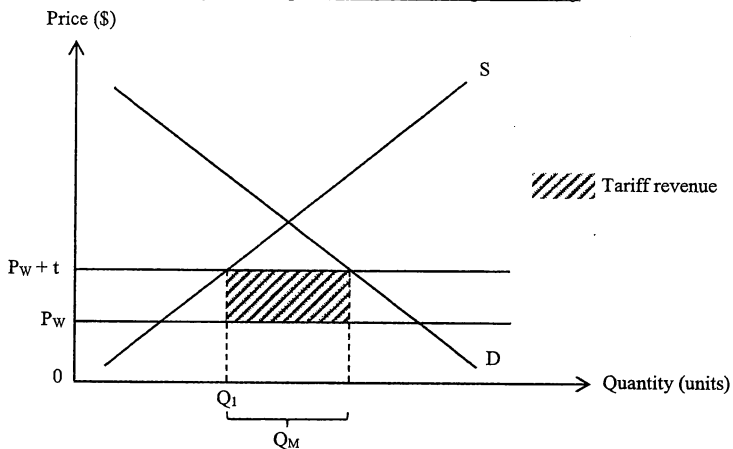
(1)

(iii) Indicate in the diagram:

- correct position of tariff revenue

(1)

Figure 1: Market for agricultural products in the small open economy



- (b) As import volume decreases while the world price remains unchanged, the total import value of agricultural products in the economy decreases. Total import value of goods decreases. Trade balance increases. (3)

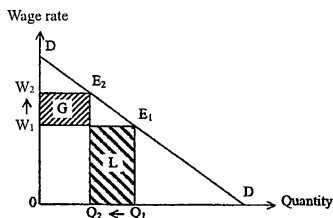
8.1 PRICE CONTROL

MARKING SCHEME

1990/CE/II/04 D	1996/CE/II/08 B	2004/CE/II/12 B (43%)	2009/CE/II/11 A (40%)	2017/DSE/I/18 C (47%)
1992/CE/II/24 D	1996/CE/II/09 B	2004/CE/II/13 C (60%)	2010/CE/II/14 C (44%)	2017/DSE/I/23 D (47%)
1992/CE/II/27 A	1997/CE/II/56 C	2005/CE/II/13 D (49%)	2013/DSE/I/14 C (41%)	2018/DSE/I/18 C (62%)
1993/CE/II/27 B	1998/CE/II/04 D	2007/CE/II/07 C (46%)	2013/DSE/I/15 D (54%)	2020/DSE/I/20 C
1994/CE/II/10 A	1999/CE/II/06 C	2008/CE/II/11 B (52%)	2014/DSE/I/21 A (65%)	2022/DSE/I/16 A
1995/CE/II/09 D	2000/CE/II/06 A	2008/CE/II/12 D (47%)	2015/DSE/I/13 C (57%)	2023/DSE/I/16 D
1995/CE/II/12 A	2002/CE/II/06 A (35%)	2009/CE/II/08 A (50%)	2015/DSE/I/19 C (67%)	2023/DSE/I/17 A
2025/DSE/I/44 C				

Note: Figures in brackets indicate the percentages of candidates choosing the correct answers.

1990/CE/I/1(b)
(i)



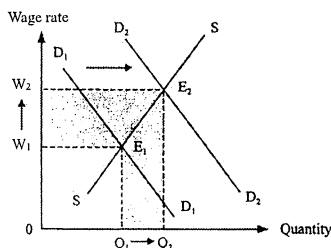
Indicate in the graph:

- $W \uparrow$ & $Q \downarrow$ (1)
- position of 'G' and 'L' (1)
- $G < L$ (1)

Verbal explanation:

- Assume that the demand for Filipino maids is elastic, (2)
- the percentage change in quantity demanded is larger than the percentage change in wage rate. (2)
- In other words, the increase in earnings from the increase in wage rate is smaller than the decrease in earnings from the decrease in quantity demanded. (1)

(ii)



Indicate in the graph:

- demand curve shifts to the right
- $W \uparrow$
- $Q \uparrow$

(1)
(1)
(1)

Verbal explanation:

The demand for local maids will increase, because
Filipino maids and local maids are substitutes.
The equilibrium wage rate will increase.
The equilibrium quantity will increase.

(1)
(1)
(1)
(1)

1994/CE/I/3

Market price = P_0

Quantity transacted = Q_0

The price ceiling is ineffective, because
it is higher than the equilibrium price.

(1)
(1)
(1)
(1)

2006/CE/I/1

(a) No, because

P_A is above the equilibrium price.

(1)
(1)

(b)

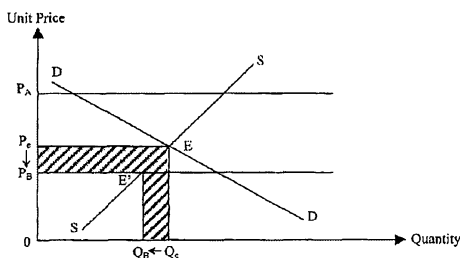


Diagram:

- original total expenditure = area OP_eEQ_e0

- new total expenditure = area $OP_BE'Q_B0$

OR indication of the decrease in total expenditure, e.g., the shared area

(1)
(1)

Verbal Elaboration:

- price $\downarrow$ from P_e to P_B and quantity transacted $\downarrow$ from Q_e to Q_B
- total expenditure of consumers decreases

(1)
(1)

Microeconomics by Topic
8. Market Intervention

2012/DSE/II/5

(a) No, because

there is excess demand / shortage.

other competitive criteria/non-price competition, such as queuing, will emerge.

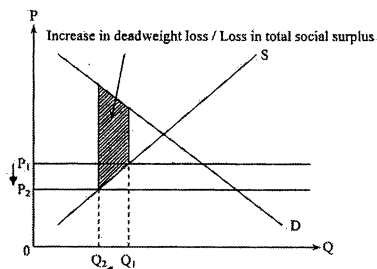
the new competitive criteria do not necessarily depend on income.

(1)
(2) } Max: 3
(2)

(c) No, because

there is greater deadweight loss / total social surplus decreases.

(1)
(1)



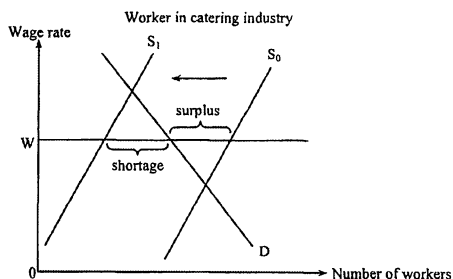
Indicate in the diagram:

- lowering of the price ceiling

- correct position of increase in deadweight loss / loss in total social surplus

(1)
(1)

2013/DSE/II/10(c)



Indicate on the diagram:

- wage rate set above the original equilibrium

- surplus

- magnitude of the leftward shift of the supply curve greater than the surplus

- final shortage

(1)
(1)
(2)
(1)

Verbal elaboration:

An effective price floor will create a surplus of workers.

If the restaurant workers change to work in other industries (for example, turn into security guards),

the supply of workers in this industry drops, and

there may be a shortage for workers at the minimum wage rate.

(1)
(1)
(1)
(1)

2014/DSE/II/3

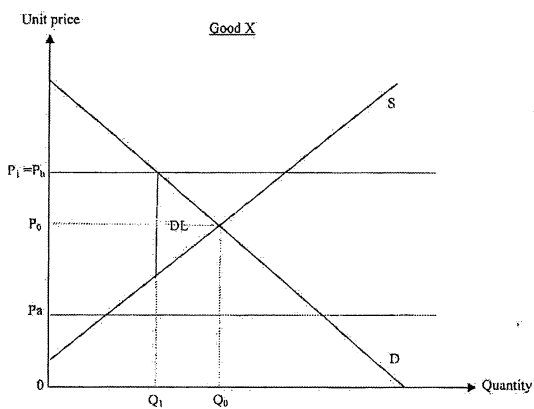


Figure 1

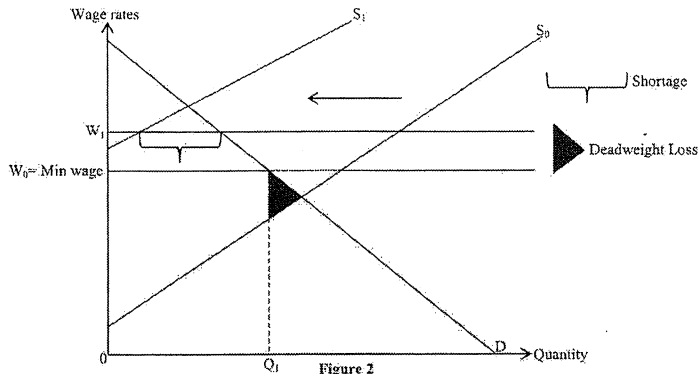
Indicate on the diagram:

- correct P_0 Q_0 (1)
- correct P_1 Q_1 (1)
- correct position of deadweight loss (DL) (1)

Verbal elaboration:

- No, because (1)
- the ineffective price floor (P_a) becomes an effective price floor (P_b) (1)
- at the new quantity (where the horizontal P_b -line intersects the demand. (1)
- marginal benefit (maximum willingness-to-pay) is higher than marginal cost and there exists a deadweight loss. (1)
- (Max: 6)

Foreign Domestic Helpers in Hong Kong



(c) Illustrate in Figure 2:

- correct position of minimum wage
- correct position of Q_1
- correct position of deadweight loss

(1)
(1)
(1)

Verbal elaboration :

At Q_1 marginal benefit is greater than marginal cost.

(1)

(d) Illustrate in Figure 2:

- leftward shift of supply curve
- correct position of shortage at the higher wage rate

(1)
(2)

Verbal elaboration:

Filipino maids working in the Mainland and Filipino domestic helpers in Hong Kong are in competitive supply. (1)
If competition from the Mainland sharply reduces the supply of foreign domestic helpers in Hong Kong, (1)
then shortage of such workers could still arise even at a higher wage rate. (1)

(e) Employers may try to recruit domestic helpers from other countries to work in Hong Kong.

OR

The minimum wage may be raised / abolished by the government.

(1@, max: 1)

2020/DSE/II/10 D

D) Advantage of tariff over quota : The tariff revenue collected must go to government (quota revenue may belong to private firms / importers)

Disadvantage of tariff over quota : It may not be effective in controlling the import level while the local demand increase (the import level can still increase under tariff if demand increase but quota will not)

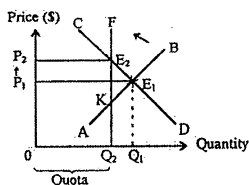
8.2 QUOTA

MARKING SCHEME

1993/CE/II/53 B	1999/CE/II/45 A	2006/CE/II/10 B (52%)	2010/CE/II/13 B (63%)	2017/DSE/I/24 A (69%)
1994/CE/II/12 B	2000/CE/II/49 B	2007/CE/II/08 B (62%)	2012/DSE/I/19 B (56%)	2019/DSE/I/14 A
1994/CE/II/13 D	2005/CE/II/10 C (39%)	2008/CE/II/09 C (54%)	2013/DSE/I/19 B (59%)	2019/DSE/I/15 B
1999/CE/II/07 A	2005/CE/II/48 D (46%)	2009/CE/II/12 B (78%)	2014/DSE/I/23 C (66%)	2019/DSE/I/16 C
2020/DSE/I/16 A	2024/DSE/I/15 C	2025/DSE/I/17 D	2025/DSE/I/45 A	

Note: Figures in brackets indicate the percentages of candidates choosing the correct answers.

1997/CE/I/10(b)



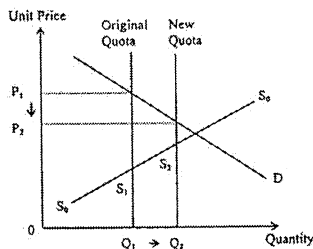
Indicate in the diagram:

- original supply curve AB (1)
- quota $\Rightarrow Q_1 \rightarrow Q_2$ (1)
- new supply curve (AK +) KF / quota quantity (1)
- new market price P_2 (1)

Elaboration:

- The supply of imports decreases, because the effective quota quantity is smaller than the initial equilibrium quantity. (2)
- The market price increases. (1)
- The import volume decreases. (1)

1998/CE/I/2



Indicate in the diagram:

- quota $\uparrow$ (1)
- $P \downarrow$ (1)

Elaboration:

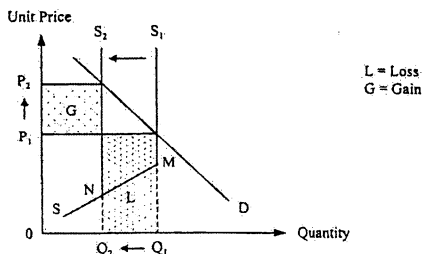
The supply of imports increases.

The market price falls.

2001/CE/I/9(b)

(1)

(1)



Indicate in the graph:

- correct supply curve to show the original quota is effective

- S_1 shifts leftward to S_2

- loss > gain

(1)

(1)

(2)

Verbal explanation:

The supply decreases.

After reduction of quota quantity, the quantity transacted decreases and the price increases.

The condition is that the demand is elastic,

the percentage change in quantity demanded is larger than the percentage change in price.

(1)

(1)

(1)

(2)

2008/CE/I/11(b)

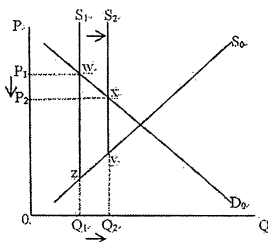
There was quality improvement in Hong Kong-made garments.

(Because one unit of quota must be used for one unit of garment regardless of garment quality, and because the quota has a positive value,) the price of higher quality garment fell relative to lower quality garments under the quota system.

(1)

(1)

2015/DSE/II/3



Indicate on the diagram:

- rightward shift of the supply curve corresponding to an increase in quota

- a reduction in P and a rise in Q

- correct position for the reduction in deadweight loss (area $wxyz$)

(1)

(1)

(1)

Verbal elaboration:

Yes (I agree), because

the increase in effective quota would raise the supply and thus raise the quantity transacted and reduce the price, and

the gap between marginal benefit and marginal cost is reduced as the quantity produced is increased toward its efficient level.

(1)

(1)

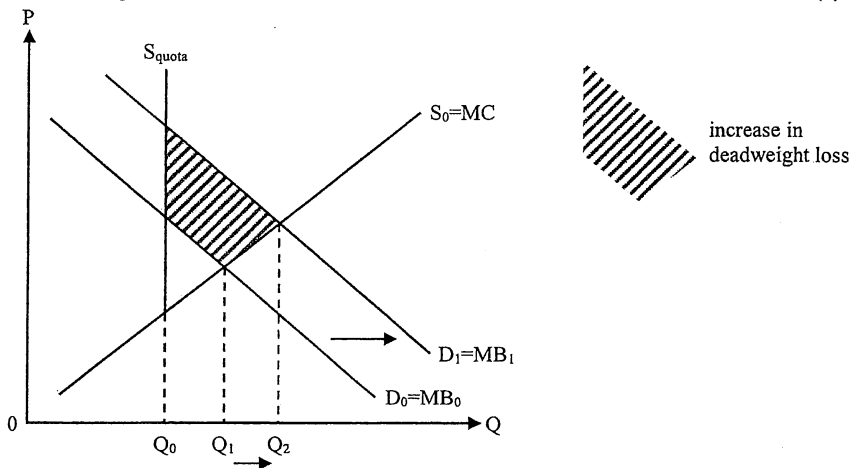
(1)

8.2 QUOTA

2024/DSE/II/3

Verbal elaboration:

- Economic growth implies an increase in income, which would raise the demand for automobiles, given that automobiles are normal goods. (1)
- The gap between marginal benefit and marginal cost increases. / The efficient quantity increases from Q_1 to Q_2 and diverges further from the quantity transacted (Q_0). (1)
- The deadweight loss increases. (1)



Indicate in the diagram:

- Kinked supply curve under quota (1)
- Rightward shift of demand curve (1)
- Increase in deadweight loss (1)

8.3 UNIT TAX & SUBSIDY

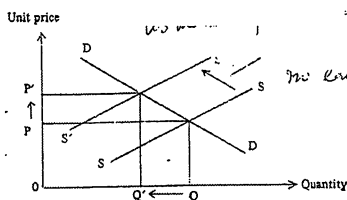
Microeconomics by Topic
8. Market Intervention

MARKING SCHEME

1990/CE/II/51 A	2003/CE/II/06 C (19%)	2010/CE/II/11 C (69%)	1995/CE/II/11 C	2010/CE/II/09 C (42%)
1991/CE/II/58 A	2003/CE/II/11 B (46%)	2010/CE/II/12 A (53%)	2002/CE/II/11 B (58%)	SP/DSE/I/20 D
1992/CE/II/25 A	2003/CE/II/12 D (25%)	2012/DSE/I/16 B (66%)	2004/CE/II/10 A (28%)	2014/DSE/I/16 C (48%)
1993/CE/II/29 B	2003/CE/II/42 A (32%)	2012/DSE/I/17 A (60%)	1990/CE/II/24 C	2015/DSE/I/10 D (32%)
1994/CE/II/08 B	2005/CE/II/09 B (68%)	2013/DSE/I/13 B (63%)	1990/CE/II/10 B	2015/DSE/I/12 B (48%)
1994/CE/II/09 C	2005/CE/II/11 B (47%)	2014/DSE/I/22 B (72%)	1996/CE/II/11 C	2016/DSE/I/19 A (43%)
1996/CE/II/07 A	2006/CE/II/11 B (39%)	2015/DSE/I/16 C (48%)	1999/CE/II/04 D	1994/CE/II/11 B
1998/CE/II/41 C	2006/CE/II/12 C (56%)	2017/DSE/I/16 B (64%)	2000/CE/II/08 B	2020/DSE/I/15 D
1999/CE/II/08 B	2008/CE/II/07 C (63%)	2018/DSE/I/22 A (74%)	2000/CE/II/11 D	2021/DSE/I/15 D
1999/CE/II/09 B	2008/CE/II/08 C (49%)	1991/CE/II/26 D	2001/CE/II/07 A	2021/DSE/I/16 B
2001/CE/II/08 C	2009/CE/II/13 B (47%)	1992/CE/II/26 B	2004/CE/II/11 B (34%)	2022/DSE/I/15 B
2022/DSE/I/21 D	2024/DSE/I/16 D	2024/DSE/I/17 B	2025/DSE/I/20 B	

1991/CE/I/1(b)
(i) Gasoline

Note: Figures in brackets indicate the percentages of candidates choosing the correct answers.



Candidates should indicate on the graph:

- leftward shift of S
- $P \uparrow$
- $Q \downarrow$

(1)
(1)
(1)

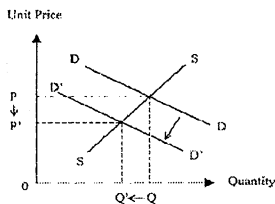
Elaboration:

The supply decreases (due to the gasoline tax).
Therefore, the equilibrium price increases, and
the equilibrium quantity decreases.

(1)
(1)
(1)

Microeconomics by Topic
8. Market Intervention

(ii) New cars



Candidates should indicate on the graph:

- leftward shift of D
- $P \downarrow$
- $Q \downarrow$

(1)
(1)
(1)

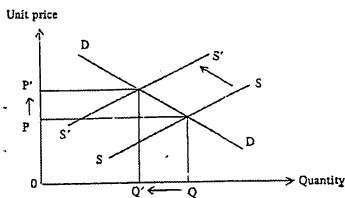
Elaboration:

The demand for new cars decreases, because new cars and gasoline are complements. Therefore, the equilibrium price decreases, and the equilibrium quantity of new cars decreases.

(1)
(1)
(1)
(1)

1992/CE/1/2(b)

(i) Hong-Kong made garments



Candidates should indicate on the graph:

- leftward shift of S
- $P \uparrow$ & $Q \downarrow$

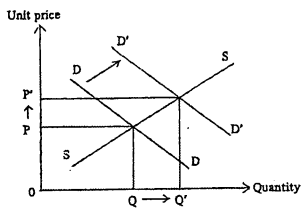
(1)
(1)

Elaboration:

The supply decreases (due to the import tax). Therefore, the equilibrium price increases, and the equilibrium quantity decreases.

(1)
(1)
(1)

(ii) US-made garments



Candidates should indicate on the graph:

- rightward shift of D
- $P \uparrow$ & $Q \uparrow$

(1)

(1)

Elaboration:

The demand increases (due to a rise in the price of substitutes).

(1)

Therefore, the equilibrium price increases, and

(1)

the equilibrium quantity increases.

(1)

1993/CE/1/4(c)

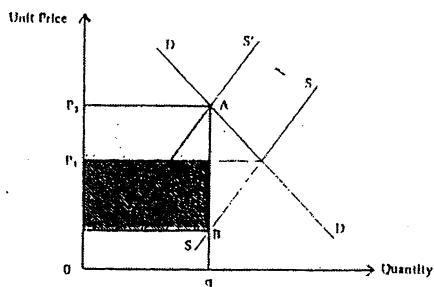


Diagram:

- AB = per unit tax
- correct shift of supply curve
- tax paid by consumers (white rectangle)
- tax paid by suppliers (black rectangle)

(1)

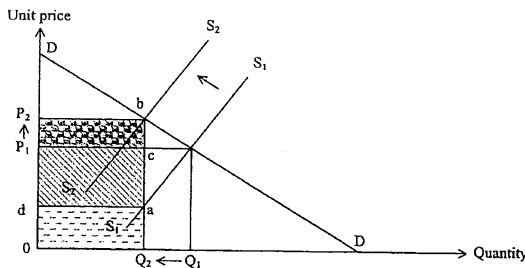
(2)

(2)

2004/CE/1/9(c)

(i) Condition: The elasticity of demand for cars is greater than the elasticity of supply.

(2)



Indicate in the diagram:

- supply curve shifts from S_1 to S_2
- sellers' tax burden = area acP_1d
- buyers' tax burden = area cbP_2P_1
- area $acP_1d >$ area cbP_2P_1

(1)

(1)

(2)

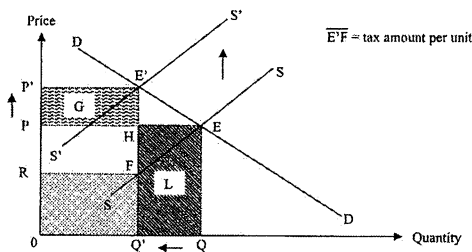
(ii) Indicate in the diagram:

- area $0Q_2ad$ is the net total sales revenue.

(1)

2007/CE/1/9(c)

(i)



Indicate in the diagram:

- upward shift of the supply curve
- a decrease in spending (= G) + increase in spending (= L)
- $G < L$

(1)

(1)

(1)

Verbal elaboration:

There is a decrease in supply.

(1)

The price increases and the quantity decreases.

(1)

The demand is elastic,

(1)

the percentage change in quantity demanded is larger than the percentage change in price.

(1)

(ii) Indicate in the diagram:

- sellers' revenue before tax (= area $0QEP$)
- the revenue net of tax (= area $0Q'FR$)

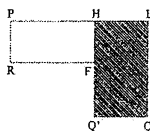
(1)

(1)

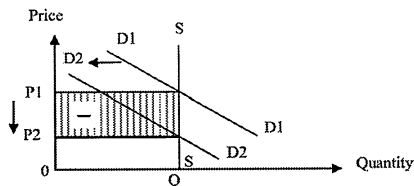
OR

- change in revenue = loss of area $Q'QEPRF$

(2)



2008/CE/1/9(b)(ii)



Indicate in the diagram:

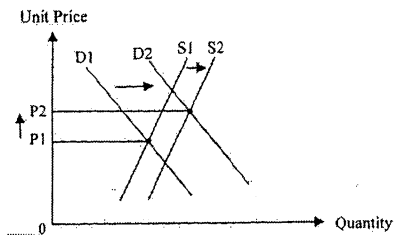
- leftward shift of the demand curve (1)
- market price falls from P_1 to P_2 (1)
- market value decreases from $P_1 \times Q$ to $P_2 \times Q$ (1)

Verbal elaboration:

The demand for stock market investments decreases,
due to lower expected net capital gain. (1)

There is a fall in the price of shares, and
the total market value $[P \times Q]$ falls. (1)

2009/CE/1/11(c)(i)



Indicate in the diagram:

- demand curve shifts to the right (1)
- supply curve shifts to the right (1)
- the new equilibrium price $>$ the original equilibrium price (1)

Verbal elaboration:

The demand for red wine increases (as it is more popular). (1)

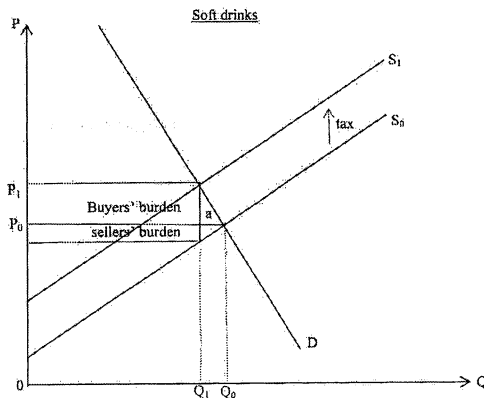
The supply of red wine increases (due to the tax cut). (1)

The increase in demand is larger than the increase in supply. (2)

2014/DSE/II/9

(a) The condition is that the elasticity of demand for the good is smaller than the elasticity of supply.

(1)



Indicate on the diagram:

- parallel upward shift of supply curve due to the per unit tax
- price increases and quantity drops
- correct positions of buyers' tax burden and sellers' tax burden
- buyers' burden > sellers' burden

(1)

(1)

(1)

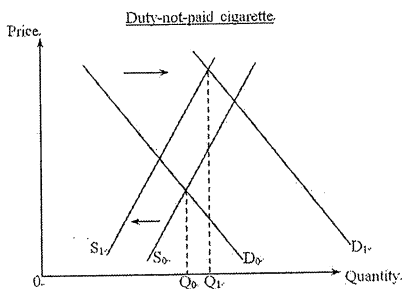
(1)

(b) Indicate on the diagram :

- reduction in consumer surplus: buyers' burden + area a

(1)

2015/DSE/II/11(d)



Indicate on the diagram:

- rightward shift of the demand curve
- leftward shift of the supply curve
- final equilibrium with a higher quantity transacted

(1)

(1)

(1)

Verbal elaboration:

On the one hand, due to the HK Customs' crack-down effort, the supply of duty-not-paid cigarettes would drop.

(1)

On the other hand, as duty-paid cigarettes and duty-not-paid cigarettes are substitutes, tobacco duty would increase the demand for duty-not-paid cigarettes.

(1)

It is thus possible that the total quantity of duty-not-paid cigarettes sold would increase,

(1)

if the increase in demand has a bigger effect on the quantity transacted than the decrease in supply.

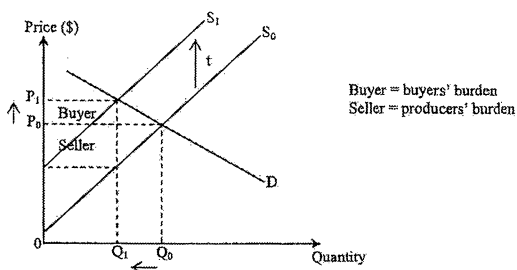
(1)

2016/DSE/II/10(c)

Condition: demand for solar panels is more elastic than its supply.

(2)

Chinese Solar panels sold in European Union



Indicate in the diagram:

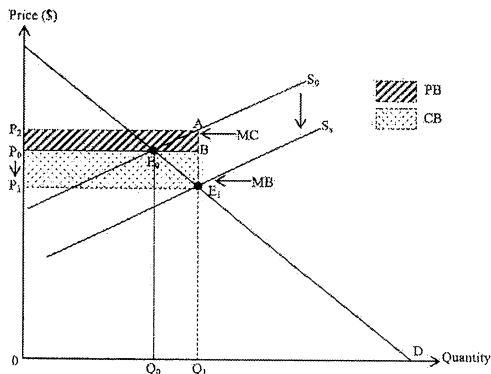
- upward shift of the supply curve
- P increases
- Q decreases
- correct position of buyers' burden and producers' burden
- buyers' burden < producers' burden

(1)
(1)
(1)
(1)
(1)

2017/DSE/II/11

(a) The condition is that the elasticity of demand for education services is lower than the elasticity of supply.

(2)



Illustrate in the diagram:

- parallel downward shift of supply curve due to the per unit subsidy
- correct position of new price (P_1)
- correct position of new quantity (Q_1)
- correct position of producer benefit and consumer benefit
- consumer benefit > producer benefit

(1)
(1)
(1)
(1)
(1)

(b) Illustrate in the diagram:

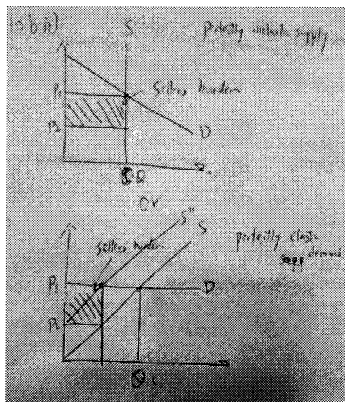
- correct position of marginal benefit
- correct position of marginal cost

(1)
(1)

Verbal elaboration:

At Q_1 marginal cost is larger than marginal benefit.

(1)



B) If the demand is perfectly elastic. Producer will bear all tax burden.

OR

If the supply is perfectly inelastic. Producer will bear all tax burden.

2020/DSE/II/10C

C) Consumer surplus after tariff = abc

Producer surplus before tariff = $df0$

Producer surplus after tariff = $hi0$

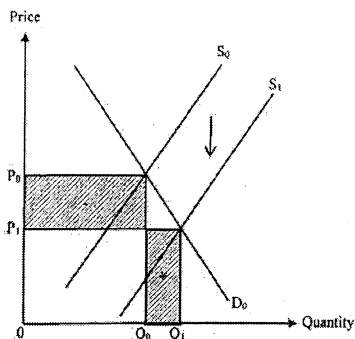
2021/DSE/II/10c

(c) Verbal elaborations:

The provision of subsidy would result in an increase in the supply of masks. If demand is inelastic, the percentage decrease in price would be bigger than the percentage increase in quantity, so that total expenditure on masks would drop. (4)

Illustrate in the diagram:

- downward shift of supply curve (1)
- correct positions of the increase in total expenditure (+) and the decrease in total expenditure (-) (1)
- Area of (+) > Area of (-) (1)



- (d) Wearing mask protects oneself as well as lowering the chance of others being infected. Since the person wearing mask is not compensated by others (whose chances of being infected are reduced), there exists external benefit, with marginal social benefit exceeding marginal social cost. This implies that the number of people wearing mask is below the efficient level i.e. under-consumption of masks. Subsidy can increase the number of people wearing mask towards the efficient level (by lowering the marginal private cost), and thus narrow the gap between marginal social benefit and marginal social cost. (4)

9.1 EXTERNALITIES

MARKING SCHEME

1985/AL/I/02 B	1992/AL/I/23 D	1998/AL/I/13 A	2006/AL/I/30 A (44%)	2013/DSE/I/20 D (69%)
1985/AL/I/23 D	1992/AL/I/24 C	1998/AL/I/19 A	2007/AL/I/30 B (63%)	2014/DSE/I/18 B (93%)
1986/AL/I/06 B	1993/AL/I/08 D	1999/AL/I/19 C	2008/AL/I/28 A (38%)	2014/DSE/I/19 B (81%)
1986/AL/I/25 D	1994/AL/I/08 B	1999/AL/I/27 B	2008/AL/I/30 C (45%)	2015/DSE/I/17 B (45%)
1988/AL/I/17 D	1994/AL/I/15 B	2000/AL/I/19 D	2009/AL/I/28 B (47%)	2018/DSE/I/23 B (73%)
1990/AL/I/25 A	1995/AL/I/23 D	2000/AL/I/20 B	2010/AL/I/28 D (73%)	2020/DSE/I/21 B
1990/AL/I/30 B	1996/AL/I/21 B	2003/AL/I/27 A	2012/AL/I/28 C (23%)	2022/DSE/I/22 B
1991/AL/I/01 C	1997/AL/I/27 C	2005/AL/I/29 C (37%)	SP/DSE/I/18 C	
1992/AL/I/22 D	1997/AL/I/28 B	2006/AL/I/29 A (71%)	2013/DSE/I/18 A (52%)	

Note: Figures in brackets indicate the percentages of candidates choosing the correct answers.

1985/AL/I/7

- (a) More of all goods, including housing, cars and education, is beneficial, but we cannot have more of all goods. If the government subsidizes education, more of it will be produced and consumed. Whether society benefits depends upon the value of the extra education (because of the subsidy) compared to the value of the other goods which the society would have to sacrifice in order to have the extra education (i.e. marginal benefit versus marginal cost). In general, because of the subsidy, the marginal cost of education to society would exceed its marginal benefit.
- (b) Candidates are expected to know that this part of the question concerns the divergence between private and social benefits. If the educated cannot charge a fee for the benefit they generate to the uneducated, then from the society's point of view they will tend to under-invest in education, in the sense that the marginal cost of more education to them will fall short of the marginal social benefit. Subsidizing education by the government will therefore lead to marginal gain to society.

1986/AL/I/10

Level of Output	Marginal Revenue (MR)	Marginal Private Cost (MPC)	Marginal Damage	Marginal Social Cost (MSC)
100	180	130	34	164
200	180	140	32	172
300	180	150	30	180
400	180	160	28	188
500	180	170	26	196
600	180	180	24	204
700	180	200	22	222
800	180	310	20	330

- (a) 600 units, where $MR (\$180) = MPC (\$180)$ and the private gain is maximized (\$150).
- (b) 300 units, where social surplus is maximized (\$24). If the factory owner wants to increase output to, say, 400 units, the marginal profit from this increment is \$20, but the marginal damage to the neighbours is \$28. The neighbours therefore will pay the factory owner \$20 to cut output back to 300 units.
- (c) Again 300 units, where the social surplus is maximum (\$24). If the neighbours seek to reduce the factory's output to, say, 200 units, the marginal reduction in damage is \$30. Since the marginal profit (from the 3rd 100 units) is \$30, the factory owner will be able to pay and compensate neighbours to take the marginal damage.

1988/AL/I/7(a)

If congestion occurs, each user will slow down the speed of others, thereby imposing time cost upon one another without compensating them for this cost. The social time cost generated by one driver is the time cost of himself as well as the time cost he imposes upon all other drivers, whereas the private cost is the time cost to this one driver only.

A toll will reduce the traffic congestion, thereby reducing the likelihood of one user imposing time cost upon others.

2005/AL/I/7(a)

The presence of external effects, such as national pride or glory, may yield values to citizens at large, as most if not all Chinese in some measure enjoy a greater number of gold medals won in the Olympic Games. It would be highly difficult to collect payments for this type of external beneficial effects from all those enjoying the results of the training. (Transaction costs may be mentioned here.) Does national pride worth something to the citizens? Of course. But since payments are not enforced there is no way we can assess actually how much the citizens would be willing to pay.

2007/AL/I/7(a)

Social gains: Non-smokers' exposure to secondhand smoke is harmful to their health, and they are not compensated by the smokers who are responsible for such a negative externality. A ban on smoking can eliminate the existence of this negative externality.

Social costs: Smoking generates marginal benefit (MB) to smokers which exceeds its marginal social cost (MSC) at some levels. To maximize total social surplus (smokers' consumer surplus and tobacco companies' producer surplus), the 'efficient level of smoking' should be that smokers are allowed to smoke until their $M(S)B$ equals MSC. A ban on smoking will lead to under-production and a deadweight loss.

Society will gain from a ban on smoking only if such social gains are larger than the social costs.

It is virtually impossible to show whether the social gains are greater or smaller than the social costs in this case, because there is no guideline to properly estimate these gains and costs. When people are not asked or not required to pay, as they do in the market, any such estimate is not reliable.

2008/AL/I/7(b)

There is a divergence between private and social costs in the sense that the anchor stores generate spillover benefits to the smaller shops.

There is no divergence between these spillovers are all captured by the owners of the shopping centers, and because the anchor stores are capturing their own beneficial spillovers by paying lower rents. On the other hand, the smaller shops, which pay higher rents, are actually buying the spillovers beneficial to them.

2015/DSE/I/12(d)

Negative externality / divergence between private and social costs.

When the public transportation system is overcrowded (i.e., at full capacity), any additional passenger using the system would increase the time cost for other passengers without having to compensate them for this extra cost (implying a divergence between private and social costs). As marginal social cost is now higher than marginal social benefit, the number of passengers using the system would exceed its efficient level. (1)

2016/DSE/I/4

Taking photos may disturb other people in the restaurants/sharing the photos in the social media may disclose the design of dishes. Both possibilities may have an undesirable effect on the revenue/profit of the restaurants, but such effects are not compensated (financially or otherwise) by the photo-takers. Such photo-taking-and sharing behaviour may thus involve negative externality (external cost is involved). (3)

OR

Taking photos and sharing them in social media may have a promotion effect, attracting more customers-bringing higher revenue/profit-to the restaurant, but the photo takers do not receive compensation for such benefits. Such photo-taking-and sharing behaviour may thus involve positive externality (external benefit is involved). (3)

2017/DSE/I/4

(a) Positive externality. Restaurants in remote areas enjoy benefits from increased business due to the game developer's choice of hotspots locations without paying the developer. So external benefit exists and marginal social benefit is greater than marginal social cost. The number of existing hotspots near the restaurants is lower than that required by the efficiency level. (3)

(b) The restaurants may pay the game developer to increase the hotspots nearby. (2)

OR

The game developer may charge the restaurants for placing more hotspots near them. (2)

2019/DSE/II/4

Getting vaccinated protects oneself as well as lowering the chance of others being infected by flu. Since the person receiving vaccination is not compensated by others (whose chances of being infected are reduced), there exists external benefit, with marginal social benefit exceeding marginal social cost. This implies that the number of people getting vaccinated is below the efficient level. Subsidy can increase the quantity of vaccination towards the efficient level and thus narrow the gap between marginal social benefit and marginal social cost. (4)

2022/DSE/II/10a

- (a) The broadcast would attract more customers and thus bring more business to the restaurants. However, in the absence of compensations from these restaurants (beneficiaries) to the shopping malls (benefactors) for broadcasting the Olympic Games, there would exist a positive externality (or external benefit), with the marginal social benefit (MSB) from the broadcast exceeding the marginal private benefit (MPB). (3)

2022/DSE/II/10b

- (2) (b) The difference between MSB and MPB implies under-provision of broadcast. A Market solution may involve the restaurants paying the malls to broadcast more games so that the number of Olympic events broadcast would increase towards its efficient level. (2)

2024/DSE/II/2

- (a) Negative externality may be involved when producers engaged in industrial activities which emit carbon dioxide do not have to pay compensation for the harm inflicted on others. (2)
As a result, equilibrium output determined by equality between marginal private (= social) benefit and marginal private (< social) cost would be higher than the socially efficient level (where marginal social benefit = marginal social cost). Such over-production resulting from the divergence between marginal private cost and marginal social cost implies economic inefficiency (i.e., inefficient allocation of the society's resources in producing such products). (2)
- (b) Payment for carbon credits to engage in production activities that emit carbon dioxide would raise the producers' marginal private cost and thus lower equilibrium output towards the efficient level. At this lower level of output, the gap between marginal private cost and marginal social cost is reduced and economic efficiency is improved. (2)

2025/DSE/II/3

- The tutorial teacher may benefit from an increase in income from advertisement on social media / from his tutorial classes due to an increase in popularity when people view his videos. (2)
If the increase in income equals the benefits enjoyed by the viewers of his videos at the margin, there would be no externality. (1)

9.2 EQUITY

MARKING SCHEME

1985/AL/I/28 A	1993/AL/I/30 C	2007/AL/I/17 D	2012/DSE/I/20 D (82%)	2017/DSE/I/20 A (68%)
1991/AL/I/12 C	1996/AL/I/03 A	SP/DSE/I/19 B	2012/DSE/I/21 D (64%)	2017/DSE/I/21 D (91%)
1991/AL/I/24 D	1998/AL/I/04 C	PP/DSE/I/18 B	2014/DSE/I/20 A (73%)	2017/DSE/I/22 C (73%)
1993/AL/I/26 A	1999/AL/I/26 D	PP/DSE/I/19 D	2015/DSE/I/21 B (63%)	2019/DSE/I/19 C
2020/DSE/I/13 D	2020/DSE/I/19 C	2020/DSE/I/22 D	2020/DSE/I/23 D	2021/DSE/I/21 B
2021/DSE/I/22 D	2022/DSE/I/23 C	2024/DSE/I/23 C	2025/DSE/I/23 D	

PP/DSE/II/11(d)

The STAS increases the income of the needy students' families while the rich families cannot enjoy this benefit. (2)
The STAS will help equalize opportunities as it helps the needy students further their studies. Education can increase future income of the needy students so it may help narrow the income gap over time. (2)

2012/DSE/II/4

(a) Wider (1)

(b) Lower, because (1)
higher income group has a lower post-tax income while the income of lower income group remains unchanged. (2)

2012/DSE/II/10(b)

Equity is a subjective concept that involves value judgments. / Different people use different standards of equity. (1)

Yes, because the \$6 000 results in a higher percentage increase in income of the poor. (2)

OR

No, as it does not help equalize opportunities. Relative to the rich, the poor do not get more resources to improve their productivity / increase future income. (2)

OR

No, as it is not equalizing outcome as not everyone can benefit from the scheme, e.g. non-permanent residents and residents who are below 18 years old are not eligible for the \$6 000. (2)

2013/DSE/II/7

(a) Construction. (1)

This is because the entry to a particular occupation is closely associated with a person's human capital, which tends to be more homogeneous among the construction workers than among finance professionals. (2)

The construction industry often pays their workers by time rates, which are more or less uniform across workers. In addition to basic salaries, compensations to the employees of financing, insurance, real estate and business services often include performance-based commissions, bonuses and profit-sharing, which may result in a greater diversity of income within this industry. (2)

(Max: 3)

- (b) The Gini coefficient, which reflects the degree of income inequality, has increased in the HK economy as a whole. (1)

As shown in Table 1, income inequality in both industries has increased. Other things being equal, this would contribute to the increase in the income inequality in the entire economy. (2)

As shown in Table 2, there is an increase in the proportion of working population in the industry with a larger disparity of income (i.e. financing, insurance, real estate and business services) and a decrease in the industry with a smaller disparity of income (i.e. construction). Other things being equal, this changing pattern of employment distribution across the two industries would contribute to the increase in the income inequality in the entire economy. (2)

(Max: 3)

2013/DSE/II/11

- (a) Objective (1): Yes, as it provides cash subsidies relieving the burden on travelling expenses. (2)

Objective (2): Yes, if the WITSS continues to subsidize the workers on travelling expenses. (2)

OR

Objective (2): No, as employment (as well as unemployment) largely depends on the aggregate demand and supply of the economy (which may fluctuate over time) rather than the WITSS. (2)

- (b) (ii) It helps to reduce the income inequality. It helps equalize outcome as it provides extra money for the lower-income group only / it increases their opportunity of finding a job with higher income because they can travel farther away from their residential area. (3)

2015/DSE/II/8

Proposal 1:

Equalizing opportunities:

Free education for longer years means that the poor are able to receive more education and improve their productivity and hence income-earning power in the future. (2)

Proposal 2:

Equalizing income/outcome:

More welfare payments to the unemployed could help reduce the income inequality between the employed (rich) and the unemployed (poor). (2)

2016/DSE/II/11

- (c) Equalize income/outcome: basic medical service acts like a transfer or subsidy-in-kind, enjoyed mainly by the low-income group; The high-income group may choose not to "queue up" for such medical benefits and turn to private medical service instead. (2)

OR

Equalize opportunities: basic medical service provided to patients with chronic diseases would improve their health, allowing them to continue to work and earn income. Therefore, their income would not be affected by their ability to afford medical service (their background), but rather by their work effort. (2)

- (d) Equity:

After the fee raise, the poor may no longer be able to afford the fee and thus be deprived of the opportunity to enjoy the medical services. Equity may be worsened as the poor cannot enjoy the transfer/subsidy while the rich can still use the service. (2)

OR

After the fee raise, the poor may no longer be able to afford the fee and thus be deprived of the opportunity to enjoy the medical services. The poor may then be deprived of the chance to work and earn income due to their poor health. (2)

2017/DSE/II/11(c)

Yes. Subsidizing education is a measure to equalize opportunities, so the poor will not be deprived of university education due to financial constraint. Education could also raise their future income, thus reducing income inequality over generations. (3)

- (a) Statement A is a positive statement because it is refutable by fact.

(2)

- (b) Equalising income: the districts are with lower average household income so this cash will help narrow the income gap.

Equalising opportunity: the subsidy on education will enhance the chance of the poor to get higher education without being hindered by their backgrounds.

} @2
max: 4

10. (a) The amount of money spent on buying masks constitutes a higher percentage of income for the low-income group than for the high-income group

Marks

(2)

- (b) (i) No, it is not a free good because it is produced from scarce resources.

(2)

- (ii) No, it is not a public good because it is rivalrous in consumption, i.e. it cannot be used concurrently by more-than-one persons.

(2)

OR

No, it is not a public good because it is excludable, i.e. the owner of a mask can exclude other people from accessing or using his/her mask.

- (d) Marks for effective communication (EC: max 2 marks)

Marks	Performance
2	● Support arguments with the source/data and appropriate economic theories. ● Present relevant material. ● Present well-organised and coherent answers without repetition of ideas. ● Use language that expresses ideas clearly and fluently with appropriate use of words/terms/symbols.
1	● Present arguments with some support of the source/data and economic theories. ● Present some irrelevant material. ● Present answers in a less organised way with some repetition. ● Use language that conveys a clear message with some inappropriate use of words/terms/symbols.
0	● Present arguments with no support of the source/data and economic theories. ● Present material unrelated to the gist of the question. ● Present inconsistent arguments. ● Express limited ideas with inappropriate use of words/terms/symbols.

SECTION 1: NATIONAL INCOME

1.1 CIRCULAR FLOWS

MARKING SCHEME

1990/CE/II/35 A	1991/CE/II/42 C	1992/CE/II/44 A	1994/CE/II/31 C	2003/CE/II/25 B (62%)
1990/CE/II/57 D	1991/CE/II/47 C	1993/CE/II/37 B	2002/CE/II/24 B (48%)	2009/CE/II/26 A (68%)

Note: Figures in brackets indicate the percentages of candidates choosing the correct answers.

1.2 EXPENDITURE APPROACH

MARKING SCHEME

1990/CE/II/33 A	1996/CE/II/30 C	2004/CE/II/29 C (36%)	2009/CE/II/28 C (53%)	2013/DSE/I/22 A (60%)
1993/CE/II/39 C	1998/CE/II/30 C	2005/CE/II/27 B (51%)	2010/CE/II/26 B (63%)	2016/DSE/I/23 C (54%)
1993/CE/II/51 A	2001/CE/II/28 C	2006/CE/II/28 C (29%)	2010/CE/II/27 B (63%)	2022/DSE/I/24 A
1996/CE/II/28 D	2003/CE/II/26 C (44%)	2008/CE/II/26 B (71%)	2012/DSE/I/22 B (61%)	2023/DSE/I/25 C
2024/DSE/II/24 B				

Note: Figures in brackets indicate the percentages of candidates choosing the correct answers.

1990/CE/I/3(a)(iii)
 $C + I + G + (X - M)$ (2)
 $= \$ (255 + 109 + 10 + 30 + 218 + 275 - 501 + 86 - 56) \text{ bn.} = \426 bn. (2)
 Expenditure approach (1)

1993/CE/I/4(a)(i)
 Expenditure approach (1)
 $62\% + A\% + 36\% + 88\% - 93\% = 100\%$ (2)
 $\therefore A = 7$ (1)

2002/CE/I/11(a)(i)
 $\$(400 + 180 + 150 + 250 + 500 - 650) \text{ m}$ (2)
 $= \$830 \text{ m}$ (1)

2018/DSE/II/12(c)
 Export of services / net exports (1)

2025/DSE/II/9(Cross Topic)

(a) It is a regressive tax, because (1)
 the same amount of tax needs to be paid when staying in a hotel room of the same (1)
 price regardless of income.
 Therefore, the average tax rate decreases as income increases. (1)

(b) The proposed HAT is an indirect tax. As the tax rate is adjusted from 0% to 3%, the (2)
 indirect tax revenue from the HAT will increase (from zero to some positive
 number).

The re-imposition of the HAT may marginally reduce the profits of the hotels and (2)
 guesthouses, reducing the direct tax revenue from profits tax.

(c) (Complex) division of labour

(1)

Reasons for increasing labour productivity:

- practice makes perfect
- assigning the most suitable person for each task
- saving time in training
- saving time in job rotation
- mechanisation is stimulated

[Mark the FIRST THREE points only.]

} 1@
max: 3

(d) Advantage:

- easier to maintain a stable team of staff as income is more stable
- lower cost of measuring outputs

[Mark the FIRST point only.]

} 1@
max: 1

Disadvantage:

- higher cost of monitoring against shirking
- lower labour productivity due to lower working incentive

[Mark the FIRST point only.]

} 1@
max: 1

(e) No. Some people may have a lower time cost during public holidays than on other days. If the saving in time cost is greater than the additional money cost (due to a higher room rate), the opportunity cost of staying in a hotel in Hong Kong would be lower.

(3)

1.3 VALUE-ADDED APPROACH

MARKING SCHEME

1992/CE/II/13 D	2000/CE/II/35 A	2005/CE/II/28 C (69%)	2010/CE/II/28 A (66%)	2019/DSE/I/26 D
1992/CE/II/46 D	2002/CE/II/26 B (17%)	2007/CE/II/28 B (52%)	2011/CE/II/26 B	2021/DSE/I/23 B
1995/CE/II/59 D	2003/CE/II/27 B (54%)	2007/CE/II/29 A (46%)	2014/DSE/I/24 C (71%)	2022/DSE/I/24 A
1998/CE/II/25 A	2004/CE/II/27 C (75%)	2009/CE/II/27 A (56%)	2017/DSE/I/25 A (40%)	

Note: Figures in brackets indicate the percentages of candidates choosing the correct answers.

1990/CE/I/3(a)(ii)

No, because

the contribution to the national income depends on the valued-added.

for domestic exports of goods, Hong Kong adds value onto the imported raw materials and semi-manufactured goods mainly by providing labour services.

for re-exports, Hong Kong earns income from providing services to the re-export trade.

the income earned from domestically-produced goods for exports can be smaller or greater than that from servicing re-exports.

(3)
(3)
(3)
(2) (max: 7)

1994/CE/I/11(c)

Some vegetables were not current production (i.e. inventory of the previous year).

Some fertilizers used / vegetables sold were not produced by Mr Chan's farm.

(2)
(2)

1997/CE/I/11(a)(ii)

(I) [Output (or value-added) approach:]

$$\$(0.8 - 0.5) + (0.7 - 0.5) + (1.6 - 0.7) \text{ m}$$

$$= \$1.4 \text{ m}$$

(1)
(1)

[Expenditure approach:]

$$\$(0.8 + 1.6 - 1) \text{ m}$$

$$= \$1.4 \text{ m}$$

(1)
(1)

(Mere mention of correct numerical answer \$1.4 m without showing working – max: 1 mark)

(II) [Value-added approach:]

Input costs are deducted at every stage of production.

(2)

[Expenditure approach:]

Only the expenditures on final goods and services are summed up.

(2)

1999/CE/I/9(c)

(i) Value-added approach:

$$\$(230 + 60 - 100) + \$[(350 - 230) + (160 - 60)]$$

$$= \$410$$

(2)
(1)

(ii) Expenditure approach:

$$\$(350 + 160 - 100)$$

$$= \$410$$

(2)
(1)

2001/CE/I/11(a)(i)

$$\begin{aligned} & \$[(90 - 20) + (20 - 2 - 0.5) + 0.5] \text{ m} \\ & = \$88 \text{ m} \end{aligned}$$

(2)

(1)

2003/CE/I/9(b)(i)

Firm A's contribution is the value added by A to the production. Other firms' contribution in the production process should have been deducted.

(2)

Total export value in 2003 may have included part of the inventory in past years which should be deducted from the 2003 figures.

(2)

2004/CE/I/10(b)

The value of outputs of other firms in HK that are bought by this firm for its production.

(2)

2006/CE/I/5

$$\begin{aligned} \text{(a)} \quad & \$ (50\,000 + 60\,000 - 30\,000) - \$5\,000 \\ & = \$75\,000 \end{aligned}$$

(1)

(2)

$$\begin{aligned} \text{(b)} \quad & \$ \{ (60\,000 + 50\,000) - 65\,000 \} + \$ (65\,000 - 30\,000) \\ & = \$80\,000 \end{aligned}$$

(1)

(2)

2014/DSE/II/11(b)

Not all factor inputs used to produce Japanese cars originated from Japan. e.g., some of the parts or raw materials were imported.

(2)

Some Japanese cars sold in 2013 were not produced in the same year. e.g., some of them came from the inventory.

(2)

1.4 ITEMS INCLUDED AND EXCLUDED IN GDP

MARKING SCHEME

1990/CE/II/39 A	1993/CE/II/35 A	1997/CE/II/60 B	2002/CE/II/49 B (52%)	2015/DSE/II/24 C (75%)
1990/CE/II/49 B	1994/CE/II/32 B	1998/CE/II/26 A	2004/CE/II/26 B (71%)	2018/DSE/II/24 A (71%)
1991/CE/II/44 C	1995/CE/II/60 A	1999/CE/II/24 C	2005/CE/II/26 A (77%)	2023/DSE/II/24 B
1992/CE/II/45 D	1996/CE/II/27 B	1999/CE/II/31 A	2006/CE/II/25 D (65%)	
1992/CE/II/47 A	1997/CE/II/24 A	2000/CE/II/33 A	2007/CE/II/26 C (55%)	
1992/CE/II/58 A	1997/CE/II/29 B	2001/CE/II/50 A	2007/CE/II/27 D (59%)	

Note: Figures in brackets indicate the percentages of candidates choosing the correct answers.

1992/CE/I/5(b)

(i) Not included, because (1)
there is no production in return for the money. (2)

(ii) Included, because (1)
it is a return to production in the year. (2)

1993/CE/I/4(b)(i)

No, because (2)
(part of) it is not produced in Hong Kong, and
(part of) it is unreported transaction. (2)

1995/CE/I/11(d)

(i) True, because (1)
it is Mr. Ip's factor income for his production / service rendered. (1)

(ii) False, because (1)
production cost such as the labour wages paid in 1995 are included in the national income for 1995.

OR
the portion completed in 1995 is value-added in 1995.

OR
the portion completed in 1995 is investment in 1995. (2)

1996/CE/I/3(b)

Not included, because
not a consumption expenditure (∵ consumers got their money back) nor an investment expenditure (∵ the products were destroyed)

OR
no value added to GNP (∵ production costs such as cost of materials, wage payments, transportation cost, etc. were offset by loss at the firm level) (2)

- 1996/CE/I/4(a)
GDP at factor cost increased by \$2 000, because (1)
it is a current production.
- OR
the \$2 000 is included in the government expenditure of the current year. (1)
- 1997/CE/I/8
(a) The sewage charges should be included, because (1)
it is a payment for the services produced by the government. (1)
(b) The increased resale value should not be included, because (1)
there is no production of goods or services. (1)
- 1998/CE/I/9(d)
The commission to the broker must be added to Hong Kong's GDP, because (1)
it is an income to the broker / a payment for the broker's service.
OR
it involves provision of service in this current year. (1)
- The payment for shares should not be taken into account, because (1)
it is just a transfer of ownership of existing assets which does not relate to any current production. (1)
- 1999/CE/I/7
(a) Not included, because (1)
charity donation is not a production. (1)
(b) Included, because (1)
it is in exchange for certain service rendered in the current year. (1)
- 2000/CE/I/9(b)(ii)
Smuggled gasoline has no official record in the GDP figures (∵ illegal) (2)
(On the other hand,) sales revenue of petroleum companies ↓ (1)
∴ GDP ↓ (1)
- 2000/CE/I/10(b)(iii)
No, because (1)
the \$5 mn. refers to the value of shares being transferred and share transfer itself is not a production. (2)
- 2001/CE/I/11(a)(ii)
- market value of currently constructed property decreases
- total value of middlemen services (of those property agencies, law firms, etc) decreases
- effect on expenditure components, e.g. consumption ↓, investment ↓
- effect on factor income e.g. wages ↓, profits ↓
(2@, max: 4)
∴ GDP decreases. (1)
(Mere mentioning of the conclusion without elaboration - zero mark)
- 2002/CE/I/10(a)(ii)
Consumers' spending would be greater than \$400 m because there are
- spending on goods not currently produced, e.g., inventories in past year which are sold in this year
- spending on mere transfer of ownership, e.g., secondhand goods, existing stocks and shares, existing properties
- spending on investment items e.g., new houses
- unreported transactions e.g., illegal goods
[Mark the **FIRST THREE** points only.] (2@, max: 6)

2003/CE/I/11(a)
 (i) No, because (1)
 the amount received from the sale demonstrates merely a transfer of ownership which has no current production. (1)

(ii) Yes, because (1)
 it represents the value of services provided by the accounting firms (1)
 which are resident producing units of Hong Kong. (1)

2006/CE/II/10(b)
 (i) $\$(9.37 - 9.07) \times 10\,000$ (1)
 = \$3 000. He gained \$3 000. (1)

No, because (1)
 the capital gain does not involve production. (1)

(ii) Yes, because (1)
 the banking service involved in the transactions is a production. (1)

2008/CE/I/9(a)
 No because (1)
 an increase in the capital value does not involve production. (1)

2009/CE/I/9(c)
 Part of the total spending is on the service provided by the bookstores. The service provided by the bookstores is a current production which should be counted as part of the GDP. (2)

2012/DSE/II/10(a)(i)
 No, because (1)
 there is no corresponding production. (1)

2015/DSE/II/9(d)
 No, because (1)
 it is not sold to the market and thus not counted in production. [Output Approach]

OR
 it is neither counted in consumption nor investment. [Expenditure Approach]

OR
 business losses due to destroyed products (i.e., negative profits) would offset exactly wages paid to workers and other material costs incurred in producing these products (i.e., positive wage and other factor incomes). [Income Approach] (1)

2022/DSE/II/11b

(b) No, it should be counted as transfer payment. (2)

20223DSE/II/10d

(d) No, because (1)
 no current production is involved. (1)

2024/DSE/II/11(c)

- (c) As the barista stays in Hong Kong for only one week, he is not regarded as a Hong Kong resident. (1)
- His income from Ann's coffee shop (a Hong Kong resident producing unit) is therefore regarded as factor income paid abroad which is to be deducted in the calculation of Hong Kong's net factor income from abroad. (1)
- Therefore, Hong Kong's net factor income from abroad would decrease. (1)

1.5 GDP AND GROSS NATIONAL INCOME (GNI)
MARKING SCHEME

1991/CE/II/43 D	1995/CE/II/32 C	2002/CE/II/25 A (29%)	2006/CE/II/29 D (46%)	2010/CE/II/30 A (58%)
1992/CE/II/48 B	1999/CE/II/26 D	2004/CE/II/28 C (53%)	2008/CE/II/27 C (46%)	2011/CE/II/31 A
1993/CE/II/44 B	1999/CE/II/29 B	2006/CE/II/26 A (67%)	2009/CE/II/30 B (70%)	2013/DSE/I/21 A (22%)
2019/DSE/I/25 B	2020/DSE/I/25 A	2022/DSE/I/27 B	2023/DSE/I/26 D	2024/DSE/II/38 A

Note: Figures in brackets indicate the percentages of candidates choosing the correct answers.

1990/CE/I/1(c)

- (i) Yes, because (1)
Filipino maids are resident producing units of Hong Kong. (1)
- (ii) Yes, because (1)
the factor owner is a resident of the Philippines. (1)

2001/CE/I/6

- (a) The answer is 'included' or 'excluded' depending on whether the firm which employ him is a resident producing unit of HK or not. (1)

Explanation of why the firm is or is not a resident producing unit of HK (1)

E.g.1 Excluded:

The foreign airline operates its business in foreign countries.

It is not a resident producing unit of HK.

E.g.2 Included:

A branch of the foreign airline operating in HK employs him.

This branch is a resident producing unit of HK.

- (b) The answer is 'included' or 'excluded' depending on whether the rental income of a resident of HK is earned in HK or not. (1)

Explanation of why the rental income is or is not earned in HK (1)

E.g.1 Included:

The HK company operates its business in HK and the premises are located in HK.

A resident of HK earns that rental income in HK.

E.g.2 Excluded:

The HK company operates its business in HK and the premises are located in a foreign country.

The rental income is earned by a resident of HK but it is an income from abroad.

2002/CE/I/10(d)

- Included in HK's GDP, because (1)
Bank A is a resident producing unit of HK. (1)

- Excluded from HK's GNP, because (1)
Tom is not a resident of HK, and his salary is an income to abroad. (1)

2015/DSE/II/10(a)

- (i) HK's GDP would increase (by \$3 million). (1)
As the course is a current-year production by a resident production unit, its value would be included in GDP. (1)
- (ii) HK's GNP would decrease (by \$5 million), as $\text{change in GNP} = \text{change in GDP} (+\$3 \text{ million}) + \text{change in net income from abroad} (-\$8 \text{ million})$. (1)
As the Brazilian coach is **NOT** a HK resident, the \$8 million he received would be counted as income outflow and enter as a negative item in HK's national income account. (2)

1.6 NOMINAL AND REAL GDP

MARKING SCHEME

1990/CE/II/53 B	1997/CE/II/23 A	2002/CE/II/30 C (32%)	2008/CE/II/32 A (40%)	2017/DSE/I/26 A (52%)
1990/CE/II/56 D	1998/CE/II/23 B	2003/CE/II/28 B (54%)	2009/CE/II/29 A (68%)	2018/DSE/I/25 B (55%)
1991/CE/II/59 D	1998/CE/II/29 A	2003/CE/II/29 D (46%)	2010/CE/II/33 C (37%)	2019/DSE/I/23 A
1993/CE/II/41 C	1999/CE/II/25 A	2005/CE/II/29 B (57%)	2012/DSE/I/23 A (53%)	2020/DSE/I/26 A
1995/CE/II/34 C	2000/CE/II/36 A	2007/CE/II/32 A (63%)	2015/DSE/I/25 D (64%)	2021/DSE/I/24 A
2023/DSE/I/26 B	2024/DSE/II/25 C			

Note: Figures in brackets indicate the percentages of candidates choosing the correct answers.

1990/CE/I/3(a)(iv)

(I) Population in 1988 (2)

(II) GDP deflator in 1988 / Price index in 1988 (2)

1992/CE/I/5

(a) (ii) Real national income = Nominal national income $\times$ (Price index for 1980 $\div$ Price index for 1991)
= \$60 million $\times$ (100 / 150)
= \$40 million (1)
(1)

(iii) Real national income is a better indicator, because (1)
it is free from the effects of the price changes in the year. (2)

1996/CE/I/6

Real wage in 1994 = \$10 000 $\times$ (100 / 120) = \$8 333.3

Real wage in 1995 = \$12 000 $\times$ (100 / 150) = \$8 000

OR

Real wage in 1995 based on 1994

= \$12 000 $\times$ (120 / 150) = \$9 600 < \$10 000

OR

Change in price index: [(150 - 120) / 120] $\times$ 100% = +25%

Change in salary: [(\$12 000 - \$10 000) / \$10 000] $\times$ 100% = +20% (2)

$\therefore$ Real wage decreased. (1)

2000/CE/I/6(b)

For the same amount of money income, real income rose as price level decreased. (3)

2002/CE/II/12(b)(ii)

Not necessary because (1)

if the general price level falls at a higher rate than (at the same rate as) the wage cut, their real income would increase instead (remain unchanged). (4)

(Remark: Mere mentioning of "general price level $\downarrow$ / purchasing power of money $\uparrow$ " - max. 1 mark)

2003/CE/1/5

Per capita real consumption expenditure in 1992

$$= \$2\,500\text{ m} \div 5\text{ m}$$

$$= \$500$$

(2)

Per capita real consumption expenditure in 2002

$$= \$[2\,679\text{ m} \times (100 / 95)] \div 6\text{ m}$$

$$= \$470$$

(2)

$\therefore$ a higher per capita real consumption expenditure in 1992

(1)

OR

Change in per capita nominal consumption expenditure

$$= [\$ (2\,679\text{ m} / 6\text{ m}) - \$ (2\,500\text{ m} \div 5\text{ m})] \div \$ (2\,500\text{ m} \div 5\text{ m})$$

$$= -10.7\%$$

(2)

Change in price index

$$= (95 - 100) \div 100$$

$$= -5\%$$

(2)

$\therefore$ % $\downarrow$ in price index < % $\downarrow$ in per capita nominal consumption expenditure

(1)

Conclusion: a higher general living standard in 1992

(1)

2004/CE/1/5

(a) No, because

(1)

the real GDP (or constant price GDP) for 2002 was higher.

(1)

(b) The general price level was decreasing / deflation / inflation rate was negative.

(2)

2007/CE/1/4(a)

85.3

(1)

2009/CE/1/7(a)

$$\text{Year 2000: } \$16\,000\text{ million} \div 10\text{ million} = \$1\,600$$

(1)

$$\text{Year 2008: } \$24\,840\text{ million} \div (115 \times 100) \div 12\text{ million} = \$1\,800$$

(1)

2009/CE/1/9(c)

Part of the total spending is on the service provided by the bookstores. The service provided by the bookstores is a current production which should be counted as part of the GDP.

(2)

2010/CE/1/7(c)

Not necessarily decreased, because

(1)

the real wage rate will increase if the percentage decrease in the general price level is greater than the percentage decrease in the nominal wage rate.

(2)

2012/DSE/11/11(b)

No, because

(1)

the real income of workers would fall if the percentage increase in price level is greater than the percentage increase in nominal wage.

(2)

2024/DSE/II/5

- (a) $\text{Real GDP} = (\text{nominal GDP} / \text{price level}) \times 100.$ (1)
As nominal GDP falls while price level rises, real GDP must have fallen. (1)

(b) Reasons:

- Population may have dropped by a larger percentage than real GDP.
- Pollution may have become less serious.
- Citizens may have enjoyed more leisure time.
- Any other relevant point

[Mark the FIRST TWO points only.]

} 1@
max: 2

- (c) The tenant would gain, because the purchasing power of the fixed rental paid would decrease when the general price level increased. (2)

OR

The tenant would gain, because the purchasing power of the fixed rental paid would be lower than expected when the actual inflation rate (0.7%) was higher than expected (0%).

1.7 LIMITATIONS OF NATIONAL INCOME STATISTICS

MARKING SCHEME

1993/CE/II/41
C

1999/CE/II/30
C

2002/CE/II/27
D (69%)

2006/CE/II/50
A (38%)

2010/CE/II/29
C (65%)

Note: Figures in brackets indicate the percentages of candidates choosing the correct answers.

1991/CE/I/3

(a) (i) GNP is the total incomes earned by the residents of a country over a period of time (usually 1 year). (4)

(ii) $\$2 \times (6 \times 10) + \$1 \times (12 \times 10)$ (2)
= \$240 (2)

- (b) - same GNP but larger population size will lead to smaller per capita GNP
- relative price level consideration: same nominal GNP but higher price level will lead to lower standard of living
- same per capita GNP but more uneven income distributions may mean lower general living standard
- different GNP composition may mean a higher consumption expenditure in one country (i.e. higher general living standard) but a higher investment expenditure or a higher military expenditure in another (i.e. lower general living standard)
- same GNP but a more serious problem of pollution, traffic congestion, etc. will lead to a lower living standard
- same GNP but more leisure hours will lead to a higher living standard
- same GNP but larger amount of non-market / unreported / illegal goods and services will lead to a higher living standard (3@, max: 9)

[Mark the **FIRST THREE** points only.]

1993/CE/I/4(a)(ii)

- per capita real GDP has increased during the decade
- better quality of goods ($\because$ technological advancement)
- more varieties of goods ($\because$ technological advancement)
- more imports of consumer goods
- better environment protection
- less working hours

(2@, max: 4)

[Mark the **FIRST TWO** points only.]

1994/CE/II/11(d)

In Country A with a lower GDP,

- its per capita GDP may be higher ($\because$ its has a much smaller population) and more likely to live a better life there
- its real GDP may be higher ($\because$ it has a much lower price level)
- its income may be more evenly distributed $\Rightarrow$ more attractive to the very low income people of Country B
- its GDP may have a very different composition e.g. a much larger component of consumption expenditure $\Rightarrow$ more attractive to people who prefer to enjoy life immediately

(3@, max: 6)

[Mark the **FIRST TWO** points only.]

1996/CE/I/10(c)

This is because we do not know:

- the population size / per capita income of both countries.
(If B's population size is three times, say, as that of A, then in general B's living standard is not higher than A.)
 - the income distribution of both countries.
(If B's income is highly unevenly distributed but A's is not, then we cannot conclude that.)
 - other social costs of both countries.
(e.g. If B has a higher crime rate / more serious social instability / etc., then B's living standard may not be generally higher than A.)
 - other social benefits of both countries.
(e.g. If B has less social benefits / less leisure hours / etc., then B's living standard may not be generally higher than A.)
 - what other goods besides rice and clothing these two countries have.
(e.g. If B has less other goods, then the conclusion may be wrong.)
- (2@, max: 6)
- [Mark the **FIRST THREE** points only.]

2005/CE/I/6

- Income distribution: If income is unevenly distributed, the living standard of ordinary people will be overestimated.
 - Composition of output: If a large proportion of total output consists of capital goods, national defense, etc., the living standard will be overestimated.
 - Undesirable effects of production: The problems of pollution or traffic congestion would lower the living standard. Therefore, the living standard would be overestimated without such information.
 - Amount of leisure: Leisure would raise the living standard. Therefore, the living standard would be underestimated without such information.
 - Unreported production activities: The consumption of non-marketed goods / the existence of unreported transactions would improve the living standard. Therefore, the living standard would be underestimated without such information.
- (2@, max: 4)
- [Mark the **FIRST TWO** points only.]

2007/CE/I/5

- Difference in income distribution: Income is less evenly distributed in country X.
 - Difference in the composition of output: In country X, a larger proportion of total output consists of capital goods and national defense.
 - Different undesirable effects of production: There are more problems of pollution or traffic congestion in country X.
 - Different amount of leisure: The people of country X enjoy less leisure.
 - Different amount of non-marketed / unreported production activities: There are fewer non-marketed / unreported goods in country X.
- (2@, max: 4)
- [Mark the **FIRST TWO** points only.]

2009/CE/I/7(b)

(For each of the following points, the case of an increase in the per capita real GDP is used for illustration purposes.)

- Income distribution: If income is unevenly distributed, the corresponding change in the living standard of ordinary people will be overestimated.
 - Composition of output: If a large proportion of total output consists of capital goods, national defense, etc, the corresponding change in the general living standard will be overestimated.
 - Undesirable effects of production: The problems of pollution or traffic congestion would lower the living standard. Therefore, the corresponding change in the general living standard would be overestimated without such information.
 - Amount of leisure: Leisure would raise the living standard. Therefore, the corresponding change in the general living standard would be underestimated without such information.
 - Unreported production activities: The consumption of non-marketed goods / the existence of unreported transactions would improve the living standard. Therefore, the corresponding change in the general living standard would be underestimated without such information.
- (2@, max: 4)
- [Mark the **FIRST TWO** points only.]

2022/DSE/II/3a

(a) Reasons:

- Bigger output may generate (as by-product) more pollution.
- Bigger output may require longer work hours, hence less leisure time for the workers.
- Bigger nominal output may be due to higher inflation without necessarily growth in real output/income.
- Any other relevant point.

} @1
Max:2

[Mark the FIRST TWO points only.]

1.8 UNCLASSIFIED

MARKING SCHEME

2007/CE/I/10(a)

- Some production activities are still undergoing during typhoon.
- Some production activities e.g. emergency services, are generated during typhoon.
- Some production activities disrupted during typhoon can be compensated later.

(2@, max: 4)

[Mark the **FIRST TWO** points only]

2008/CE/I/11(d)

- (i) Part of the decrease in export value would not lead to a corresponding decrease in GDP when, referring to those \$3 billion worth of exports,

- part of it is not locally produced.
 - e.g. some of the production raw materials are imported
 - e.g. part of the value of these exports comes from the value of re-exports
- part of it is not produced in the current year.
 - e.g. some of these exports are second-hand goods
 - e.g. some of these exported goods come from the inventory
- part of those goods not exported have already been produced.
 - e.g. those goods not exported are sold locally

(Any factor, 3)

- (ii) The amount of GDP derived from the spillover effect of the production of the exports will also decrease.

(2)

2016/DSE/II/11(a)

No, because

(1)

public services are usually subsidized - i.e. provided at or below cost, which is in turn lower than market price. Their contribution to GDP (or market value), computed using market price, would thus be higher than \$45 (the subsidized price charged by the clinic).

OR

the public services which are not sold at the market price will be counted using factor cost in the GDP, which is higher than \$45.

(2)

2018/DSE/II/11

The living standard would fall, because the higher proportion of government expenditure or resources / output devoted to national defence would not have much to contribute to the living standard in general.

Overall the citizens would enjoy less goods and services. Since those males drafted for military service are forced to leave their existing jobs, their household income (and consumption) would fall.

(3)

SECTION 2: MACROECONOMIC PROBLEMS

2.1 UNEMPLOYMENT

MARKING SCHEME

1994/CE/II/37 D	1998/CE/II/24 A	2003/CE/II/30 B (44%)	2008/CE/II/30 C (63%)	2017/DSE/II/27 B (76%)
1995/CE/II/37 A	1998/CE/II/28 A	2004/CE/II/33 A (58%)	2009/CE/II/32 C (62%)	2019/DSE/II/24 B
1996/CE/II/33 D	1999/CE/II/32 A	2005/CE/II/32 B (55%)	2010/CE/II/32 A (61%)	2019/DSE/II/39 D
1997/CE/II/26 B	2002/CE/II/29 B (49%)	2006/CE/II/30 C (60%)	2012/DSE/II/24 D (53%)	2020/DSE/II/24 C
1997/CE/II/55 B	2002/CE/II/31 A (15%)	2007/CE/II/30 C (33%)	2016/DSE/II/25 C (48%)	2021/DSE/II/25 D
2022/DSE/II/27 B	2023/DSE/II/27 B	2023/DSE/II/40 D	2024/DSE/II/26 A	

Note: Figures in brackets indicate the percentages of candidates choosing the correct answers.

1994/CE/II/10(a)(i)

- GNP ↓ / living standard ↓
 - skill and experience of the unemployed ↓ / productivity ↓
 - social problems as other losses to society, e.g. higher crime rate, family problems
- [Mark the **FIRST TWO** points only.]

(2@, max: 4)

1995/CE/II/6

The newly graduated job seekers increase the labour force.

Unemployment rate (R) = [Unemployed population (U) ÷ Labour force (L)] × 100%

- if a very high proportion of the new job seekers can find a job, then the unemployment rate will decrease; (2)
- and if quite a number of them cannot find a job, the unemployment rate will increase. (2)

OR

whether the unemployment rate is higher or lower depends on the relative change of the labour force and the unemployed population. (4)

OR

% change in U > % change in L ⇒ R ↑

% change in U < % change in L ⇒ R ↓

(max: 5)

2000/CE/II/7

The no. of unemployed increases at a slower rate than the increase in the labour force. (4)

[Show understanding of "unemployment rate = (no. of unemployed ÷ labour force) × 100%" without other elaboration

- max: 1 mark]

2003/CE/II/6

The unemployment rate would increase, because (1)

an increase in the number of retired persons would not change the unemployed population, (1)

but would reduce the size of labour force / employed population. (1)

Since unemployment rate = unemployed population ÷ labour force

OR

= unemployed population ÷ (employed population + unemployed population), (1)

the unemployment rate would increase.

[Mere mentioning of the formula of unemployment rate - max: 1 mark]

2004/CE/I/7

- productive resources will be left idle / output level will be lower
- political disorder / social unrest
- loss of skills

(2@, max: 4)

[Mark the **FIRST TWO** points only.]

2006/CE/I/9(c)

Unemployment rate = (unemployed population ÷ labour force) × 100% (1)

The unemployed population increased, and (1)

both the labour force and the unemployed population increased at the same rate / percentage. (2)

2007/CE/I/8(b)(i)

Total labour force increases. (1)

Unemployed population increases. (1)

(Given total employment remains unchanged,)

increase in total labour force = increase in unemployed population

OR % increase in the unemployed population > % increase in the labour force (1)

As a result, unemployment rate increases. (1)

2012/DSE/II/6(a)

- loss of output
- loss of human capital
- political unrest

(1@, max: 2)

[Mark the **FIRST TWO** points only.]

2013/DSE/II/11(b)(iii)

The effect on the unemployment rate is uncertain.

The size of the labour force would increase as more people originally outside the labour force are induced to look for jobs.

But the total number of unemployed may also increase as some of these new workers may fail to get a job. The resulting rate of unemployment depends on the percentage increase in the unemployed population relative to that in the labour force. (3)

2015/DSE/II/5

Unemployment rate = (number of unemployed ÷ total labour force) × 100%

Since both total labour force and number of unemployed drop by an equal amount, the percentage decrease in the unemployed population is greater than the percentage decrease in the total labour force, implying a reduction in the unemployment rate. (4)

2016/DSE/II/6

(a) The unemployment rate = $[200\,000 / (3\,800\,000 + 200\,000)] \times 100\% = 5\%$ (2)

- (b)
- loss of output
 - loss of human capital
 - political and social unrest

(1@, max: 3)

[Mark the **FIRST THREE** points only.]

2018/DSE/II/5(b)

Unemployment rate = (number of unemployed people / the total labour force) × 100%

The number of unemployed people will not change while the labour force increases as the retired staff is not in the labour force originally. The unemployment rate will drop. (4)

3a Recession (1 mark)

b. Decrease of inflation rate

Lower RGDP growth rate

Decrease of consumption and investment confidence (2 marks)

c. Loss of human capital

Loss of real output

Social unrest (2 marks)

2022DSE/II/11aii

- (ii) — On the one hand, many companies allowed their employees to work from home. On the other hand, high risks of infection during the pandemic discouraged people from shopping in brick-and-mortar stores and from dining out. The result was a fall in demand for workers in the retail and food services industry.
- Disruption of air travels due to the pandemic reduced demand for accommodations and thus workers in that industry.
- Public administration as well as social and personal services were much less affected by the pandemic and so was the demand for workers in this industry.
- Any other relevant point

@ 2

Max: 2

2023/DSE/II/6

No. Unemployment rate = (unemployed population / labour force) $\times$ 100%. (2)

Despite the decrease in the size of HK's labour force, the unemployment rate would fall (or remain unchanged) if the percentage decrease in the unemployed population was greater than (or equal to) that in the labour force. (2)

2024/DSE/II/12(b)

- (b) Unemployment rate = (unemployed population / labour force) $\times$ 100%. As some unemployed young adults enrol in the full-time training programmes of those football academies, the number of unemployed and the labour force would decrease by the same magnitude. The percentage decrease in the unemployed population would be bigger than that in the labour force. As a result, the unemployment rate decreases. (4)

2.2 THE GENERAL PRICE LEVEL

MARKING SCHEME

1991/CE/II/46 D	1995/CE/II/33 D	2000/CE/II/39 D	2007/CE/II/34 C (59%)	2014/DSE/I/40 C (66%)
1991/CE/II/48 B	1995/CE/II/36 D	2001/CE/II/31 B	2008/CE/II/31 C (69%)	2015/DSE/I/36 A (72%)
1991/CE/II/50 A	1996/CE/II/32 A	2001/CE/II/32 A	2008/CE/II/33 C (71%)	2016/DSE/II/24 D (37%)
1993/CE/II/42 C	1997/CE/II/25 C	2005/CE/II/33 D (51%)	2009/CE/II/33 A (66%)	2016/DSE/II/37 B (65%)
1994/CE/II/20 C	1997/CE/II/27 D	2005/CE/II/34 D (46%)	2009/CE/II/34 B (69%)	2017/DSE/II/34 C (85%)
1994/CE/II/34 B	1997/CE/II/31 B	2006/CE/II/31 D (61%)	2012/DSE/II/32 D (68%)	2022/DSE/II/39 C
1994/CE/II/38 D	1999/CE/II/33 C	2007/CE/II/33 A (55%)	2013/DSE/II/34 C (82%)	2023/DSE/II/38 C
2025/DSE/II/40 C				

Note: Figures in brackets indicate the percentages of candidates choosing the correct answers.

1990/CE/I/3(b)(ii)

- (I) The businessman would gain, because (1)
the value of the loan / interest payments decreases in real terms during inflation. (2)
- (II) The landlord would lose, because (1)
the value of the rent decreases in real terms / the purchasing power of the fixed rent drops in times of inflation. (2)

1993/CE/I/1(e)

- (i) Lose, because (1)
the real value of the receipt will decrease during inflation. (2)
OR
the cost of supplying the lunch package will increase during inflation. (2)
- (ii) Lose, because (1)
the real value of the sum of money and/or the interest income will decrease during inflation. (2)

1998/CE/I/1(a)

- A continuous rise in the general price level (2)

1999/CE/I/1(c)

- (i) Under fixed interest rates, the payment of hire and purchase remains the same.

OR

Under floating interest rates, the fall in nominal interest rate is less than the fall in the general price level. (2)

However, the value of payment to bank for hire purchase will increase since the purchasing power of the payment increase. (2)

- (ii) Under floating interest rates, if the adjustment of the interest rates is fast enough to catch up with the fall in the general price level, there will be no loss nor gain. (2)

2001/CE/I/9(d)

Effect on the general price level

Most foodstuffs, daily necessities and industrial raw materials are imported. (1)

∴ introduction of the tax would lead to a rise in the general price level (1)

Result in inflation?

If the introduction of the tax only results in an once-and-for-all rise in the general price level, then it would not bring forth an inflation. (3)

[Only put down the definition of inflation as 'a persistent or continuous rise in the general price level' without other elaboration - max: 2 marks]

2003/CE/I/9(c)

- (i) Lose, because (1)
the fixed amount of loan repayment and the fixed amount of loan interest payment both have a higher purchasing power than that before unexpected deflation occurs. (2)

- (ii) Gain, because (1)
the fixed pension has a higher purchasing power than that before the unexpected deflation occurs. (2)

2005/CE/I/7

- (a) He would lose, because (1)
he repays a fixed amount of money, the purchasing power of which has increased because of unexpected deflation. (1)

- (b) He would gain, because (1)
he receives a fixed amount of money, the purchasing power of which has increased because of unexpected deflation. (1)

2007/CE/I/4(b)

The table indicates a decrease in the general price level. (2)

- (i) He would gain, because (1)
the fixed amount of pension has a higher purchasing power (in times of unexpected deflation). (1)

- (ii) He would lose, because (1)
the real value of the repayment will increase (during unexpected deflation). (1)

2009/CE/I/5

- (a) Inflation is a continuous rise in the general price level. (2)

- (b) He would gain, because (1)
the real value of his loan repayment will decrease. (2)

2012/DSE/II/11(a)

Persistent increase in the general price level (2)

2014/DSE/II/5

(a) $[(126 - 120) / 120] \times 100\% = 5\%$ (1)

- (b) CPI only includes consumer goods while GDP deflator includes both consumer goods and producer goods.

OR

CPI calculates a fixed basket of goods and the weighting will be revised every 5 years while GDP deflator has variable baskets of goods for each year. (2)

- (c) (i) Kitman would gain, because (1)
the unexpected inflation would lower the real value (purchasing power) of the loan repayment. (1)

2018/DSE/II/10(b)

The foreign domestic helpers will lose under unexpected inflation, as the purchasing power of their fixed wage would drop. (2)

2019/DSE/II/6

- (a) Increased (+1.7%). (1)

- (b) He would gain because when the actual inflation rate (1.7%) was lower than the expected inflation rate (2.3%), the actual real interest rate received (3.3%) was higher than the expected real interest rate (2.7%). OR the purchasing power of the interest received was higher than expected. (3)

- (c) Real wage rates increased because the increase in the nominal wage rate 6.2% was higher than the inflation rate 1.7%. (3)

2.3 BUSINESS CYCLE

MARKING SCHEME

1994/CE/II/35 A	1998/CE/II/27 D	2002/CE/II/28 A (65%)	2008/CE/II/29 A (64%)	2014/DSE/I/39 C (60%)
1994/CE/II/36 D	1999/CE/II/27 D	2004/CE/II/32 A (76%)	2009/CE/II/31 B (91%)	2021/DSE/I/39 D
1995/CE/II/35 A	1999/CE/II/28 D	2005/CE/II/31 A (94%)	2010/CE/II/31 D (89%)	2023/DSE/I/39 A
1996/CE/II/59 D	2000/CE/II/37 D	2006/CE/II/27 A (87%)	2010/CE/II/44 D (65%)	2025/DSE/I/36 A
1997/CE/II/30 A	2001/CE/II/38 C	2007/CE/II/31 D (70%)	2013/DSE/I/33 A (91%)	

Note: Figures in brackets indicate the percentages of candidates choosing the correct answers.

1995/CE/I/7

- Peak / boom / economic prosperity (1)
- high employment rate / low unemployment rate / full employment or close to full employment
 - high consumption level / investment level
 - high production level / national income level
 - price level ↑ / wage rate ↑
 - interest rate ↑

[Mark the **FIRST THREE** points only.] (1@, max: 3)

1995/CE/I/6(a)

- Recession (1)
- negative GDP growth rate / GDP or output decreases (1)
 - increasing unemployment rate (1)
 - falling cost of living (1)

[Mark the **FIRST THREE** points only.]

2008/CE/I/5

- Recovery (1)
- employment increases
 - aggregate demand increases / private consumption expenditure increases / investment increases
 - general price level increases

[Mark the **FIRST TWO** points only.] (1@, max: 2)

2010/CE/I/7

- Recession / contraction (1)
- unemployment (rate) increases
 - aggregate demand decreases / private consumption expenditure decreases / investment decreases
 - inflation rate decreases / deflation / general price level decreases

[Mark the **FIRST TWO** points only.] (1@, max: 2)

2012/DSE/II/6(b)

- falling or even negative real GDP growth
- low / falling inflation or even deflation
- falling business confidence, levels of investment, consumption

[Mark the **FIRST TWO** points only.] (1@, max: 2)

2016/DSE/II/6(c)

- rising real GDP growth rate
- rising inflation / lower deflation
- rising levels of investment / consumption

[Mark the **FIRST TWO** points only.] (1@, max: 2)

2022/DSE/II/11ai

(a) (i) Recession

(1)

Phenomena:

- Falling or even negative real GDP growth
- Low/falling inflation rate or even deflation
- Falling business confidence and levels of investment
- Falling levels of consumption
- Any other relevant point

[Mark the **FIRST** point only.]

} @ 1
max: 1

(a) Recovery

(1)

(b) Economic phenomena:

- rising inflation rate / rising general price level
- falling unemployment rate
- rising consumption/investment
- Any other relevant point

[Mark the FIRST TWO points ONLY.]

} 1@
max: 2

2.4 DISINFLATION, NOMINAL AND REAL INTEREST RATES

MARKING SCHEME

1990/AL/II/11 C	1996/AL/II/21 A	2003/AL/II/21 C	2006/AL/II/21 D (50%)	2011/AL/II/18 D (67%)
1990/AL/II/24 B	1999/AL/II/17 D	2004/AL/II/15 D (71%)	2006/AL/II/23 A (66%)	2012/AL/II/21 A (62%)
1992/AL/II/21 B	1999/AL/II/18 C	2004/AL/II/18 C (53%)	2008/AL/II/21 A (66%)	2014/DSE/II/32 A (57%)
1993/AL/II/29 D	2002/AL/II/22 A (47%)	2005/AL/II/22 C (73%)	2008/AL/II/23 C (33%)	2015/DSE/II/31 B (52%)
1995/AL/II/19 D	2002/AL/II/26 D (39%)	2006/AL/II/08 A (26%)	2010/AL/II/19 B (61%)	2018/DSE/II/34 C (39%)
2019/DSE/II/32 C	2019/DSE/II/37 C	2020/DSE/II/34 C	2024/DSE/II/36 A	2025/DSE/II/35 A

Note: Figures in brackets indicate the percentages of candidates choosing the correct answers.

1994/AL/II/2

Whether an asset will be held depends on its real rate of returns which equals nominal rate minus the inflation rate. The nominal interest returns to cash holding is zero. With high inflation rate, the real rate of return is 'very' negative, which explains why people reduce cash holding.

Though interests are paid on saving deposits, the rate is usually less than the inflation rate. This is particularly so if the inflation rate is high. Thus the real rate of returns to saving balances is still negative. People will switch to other assets with positive real rates.

2003/AL/II/2

- (a) Deflation (or negative inflation) is a sustained fall in the general price level.
Disinflation is a reduction in the inflation rate (i.e., a slower rise in the general price level).
- (b) According to the Fisher equation, the nominal interest rate is equal to the sum of the real interest rate and the expected inflation rate.
Since expected deflation equals the negative of expected inflation, we can express the nominal rate as the real rate *less* expected deflation. Hence, the real rate will exceed the nominal rate when people expect deflation.

2007/AL/II/1(a)

S: Change in the general price level over time.

D: Negative inflation (sustained fall in the price level) vs. reduction in the inflation rate (sustained increase in the price level at a slower rate).

2012/DSE/II/11(a)

- (c) The opportunity cost is the nominal return of the ibonds. (2)
- (d) The ibonds help preserve the purchasing power of the invested sum of money because the interest return of ibonds will increase at the same rate as actual inflation (i.e. the real interest rate on ibonds is virtually fixed by the government). (2)

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- (b) Nominal interest rate(R)= real interest rate (r)+ expected inflation rate (2)

Rearranging terms, we have: expected inflation rate = $R-r$

When people expect deflation, the expected inflation rate would become negative

Thus the real interest rate(r)would be higher than the nominal interest rate(R).

2023/DSE/II/10bi, ii

- (b) (i) Expected real rate of return = $4.5\% - 4.5\% = 0\%$ (1)
Actual real rate of return = $4\% - (-1\%) = 5\%$ (1)

- (ii) Gain, because (1)
the actual real rate of return is higher than the expected real rate of return. (1)

2025/DSE/II/9(Cross Topic)

- (a) It is a regressive tax, because (1)
the same amount of tax needs to be paid when staying in a hotel room of the same (1)
price regardless of income.
Therefore, the average tax rate decreases as income increases. (1)

- (b) The proposed HAT is an indirect tax. As the tax rate is adjusted from 0% to 3%, the (2)
indirect tax revenue from the HAT will increase (from zero to some positive
number).

The re-imposition of the HAT may marginally reduce the profits of the hotels and (2)
guesthouses, reducing the direct tax revenue from profits tax.

- (c) (Complex) division of labour (1)
Reasons for increasing labour productivity:
– practice makes perfect
– assigning the most suitable person for each task
– saving time in training
– saving time in job rotation
– mechanisation is stimulated
[Mark the FIRST THREE points only.]
- } 1@
max: 3

- (d) Advantage:
– easier to maintain a stable team of staff as income is more stable
– lower cost of measuring outputs
[Mark the FIRST point only.]
- } 1@
max: 1

Disadvantage:
– higher cost of monitoring against shirking
– lower labour productivity due to lower working incentive
[Mark the FIRST point only.]

} 1@
max: 1

- (e) No. Some people may have a lower time cost during public holidays than on other (3)
days. If the saving in time cost is greater than the additional money cost (due to a
higher room rate), the opportunity cost of staying in a hotel in Hong Kong would be
lower.

SECTION 3: GOVERNMENT BUDGET

3.1 PRINCIPLES OF TAXATION

MARKING SCHEME

1994/CE/II/46 B	2000/CE/II/40 C	2006/CE/II/41 B (59%)	2009/CE/II/41 C (92%)	2020/DSE/I/39 D
1996/CE/II/40 A	2002/CE/II/39 B (62%)	2007/CE/II/41 B (deleted)	2013/DSE/I/35 A (70%)	2022/DSE/I/40 A
1997/CE/II/47 A	2003/CE/II/44 D (54%)	2009/CE/II/40 D (41%)		

Note: Figures in brackets indicate the percentages of candidates choosing the correct answers.

1990/CE/I/5(a)

Equity –

the rich should pay more than the poor i.e. the tax should be based on the ability to pay (proportional taxation / progressive taxation)

Certainty –

the government should state with certainty the tax obligation of the tax-payer, and not arbitrarily interpret or change it

Convenience –

the payment of tax should be convenient to the tax-payer and the government

Economy –

the cost of collecting and administering the tax should be small in relation to the total tax revenue raised (2@, max: 6)

[Mark the **FIRST THREE** points only.]

1993/CE/I/5(b)(ii)

Equality principle –

property-tax-payers pay taxes in proportion to their rental incomes.

Convenience principle –

the payment of property tax is convenient to the taxpayers.

Certainty principle –

property-tax-payers know their tax obligation clearly.

(3@, max: 6)

[Mark the **FIRST TWO** points only.]

1999/CE/I/10(c)

(i) No, because

(according to the source principle,) the income earned is not derived from H.K.

(1)

(1)

(ii) Airport departure tax.

The certainty principle / convenient principle / economy principle.

(1)

(1)

2002/CE/I/8

(Equality-)

The tax payments should be in proportion to the taxpayers' income.

(Certainty-)

The tax obligation (e.g., the time of payment, the manner of payment and the tax amount) of the taxpayers should be stated with certainty and not be arbitrarily interpreted or changed by the government.

(Convenience-)

The time and methods of the tax payment should be convenient to the taxpayers.

(Economy-)

The administrative costs of tax collection should be as small as possible relative to its yield.

(2@, max: 4)

[Mark the **FIRST TWO** points only.]

2005/CE/I/8

(Equality-)

The tax payments should be in proportion to the taxpayers' income.

(2)

(Certainty-)

The tax obligation (e.g., the time of payment, the manner of payment and the tax amount) of the taxpayers should be stated with certainty and not be arbitrarily interpreted or changed by the government.

(2)

2010/CE/I/6

(Equality / Equity:) the tax payments should be in proportion to the tax-payers' income.

(2)

(Economy:) the administrative costs of tax collection should be as small as possible.

(2)

2014/DSE/II/6(a)

No, because

according to the source principle, income not derived from HK would not be taxed by the HK government.

(1)

(1)

2017/DSE/II/6(b)

Standard tax rate of salaries tax / profits tax / property tax, because

every taxpayer has to pay the tax amount in the same proportion to these different sources (salaries / profits / rental income) of his income.

(2)

[Mark the **FIRST** point only.]

2018/DSE/II/6(b)

Equality: The new tax is no longer proportional, meaning that it is inconsistent with the equality principle.

(2)

Economy: The new tax system may incur a slightly higher administration cost, rendering it less in line with the economy principle.

(2)

[Mark the **FIRST** point only.]

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(a) Principles:

Convenience: Consistent, because online payment can be made easily.

Equality: Inconsistent, because people with higher capital gains are subject to higher tax rates.

Certainty: Consistent, because once the price of the transacted property is fixed and its holding period is recorded, the amount of STCGT can be calculated with certainty.

Economy: Inconsistent, because the cost of developing the new computer system was huge.

@2

max: 4

[Mark the **FIRST TWO** points only.]

3.2 TAXATION IN HONG KONG

MARKING SCHEME

1990/CE/II/31 B	2000/CE/II/42 A	2006/CE/II/40 A (40%)	2012/DSE/I/35 D (69%)	2020/DSE/I/ A
1995/CE/II/44 B	2001/CE/II/37 A	2008/CE/II/40 A (66%)	2015/DSE/I/34 B (20%)	2021/DSE/I/41 C
1997/CE/II/41 C	2002/CE/II/33 A (50%)	2010/CE/II/40 A (59%)	2016/DSE/I/35 A (57%)	2023/DSE/I/35 C
1998/CE/II/40 B	2004/CE/II/44 A (34%)	2012/DSE/I/33 C (56%)	2018/DSE/I/40 A (24%)	2025/DSE/I/41 B

Note: Figures in brackets indicate the percentages of candidates choosing the correct answers.

1991/CE/I/1(a)

- (i) Give 1 mark to a correct example of each of the three sources of government revenue. (3)
Direct taxes: salaries tax, profits tax, property tax
Indirect taxes: rates, motor vehicles first registration tax, tobacco duty, etc.
Other revenue: revenue from land sales, fines & penalties, fees and charges collected from different types of licences, etc.

(Mark only the **FIRST** example of each source.)

- (ii) Increased (2)
- (iii) (I) Sales tax is an indirect tax in consumer expenditure. (2)
∴ the relative share of indirect taxes increases. (2)
- (II) Recession $\Rightarrow$ income $\downarrow \Rightarrow$ direct tax revenue $\downarrow$ (3)
Recession $\Rightarrow$ income $\downarrow \Rightarrow$ consumption $\downarrow \Rightarrow$ indirect tax revenue $\downarrow$ (3)
Recession $\Rightarrow$ other revenue, e.g. revenue from land sales, may decrease. (2)
The relative share of indirect taxes will increase / decrease / remain unchanged, depending on the magnitude of the changes of the three sources of government revenue (2)
(max: 8)

1994/CE/I/7

- (a) (i) Property tax - direct tax (1)
(ii) Rates - indirect tax (1)
- (b) Property tax: the tax burden can't be shifted from the property owner onto the lessee (2)
Rates: the tax burden can be shifted onto the lessee (2)

1997/CE/I/7

- (a) Direct taxes (1)
- (b) Direct taxes (e.g. salaries tax and profits tax) vary with income / are progressive or proportional to incomes (earnings and profits). (2)
∴ When economic conditions change (good), incomes will change (high), direct tax revenue will also change (high). (2)

2003/CE/I/8

- Unemployment $\uparrow$ or salary cuts $\Rightarrow$ salaries tax revenue $\downarrow$
Less profits earned or some firms closed down $\Rightarrow$ profits tax revenue $\downarrow$
Rental income $\downarrow$ or less tenants $\Rightarrow$ property tax revenue $\downarrow$ ((1+1)@, max: 4)
[Mark the **FIRST TWO** points only.]

2004/CE/I/9(a)

- (i) Economic downturn
 $\Rightarrow$ $\uparrow$ expenditure on public assistance, unemployment benefits (1)
 $\Rightarrow$ government expenditure $\uparrow$ (1)
Economic downturn
 $\Rightarrow$ more people become unemployed $\Rightarrow$ salaries $\downarrow$, profits $\downarrow$, rental income $\downarrow$ (1)
 $\Rightarrow$ government revenue $\downarrow$ (1)
- (ii) Ageing population
 $\Rightarrow$ $\uparrow$ demand for social services (1)
 $\Rightarrow$ government expenditure $\uparrow$ (1)
Ageing population
 $\Rightarrow$ declining proportion of the population in the workforce (1)
 $\Rightarrow$ government revenue $\downarrow$ (1)

2005/CE/I/11(b)

Increase in investment will increase employment,
and thus increase salaries tax revenue.

More businesses will increase profit,
and thus increase profits tax revenue.

Increase in demand for office premises will increase rental income,
and thus increase property tax revenue.

(1)
(1)
(1)
(1)
(1)
(1)] (max: 3)

∴ increase in direct tax revenue

(1)

2006/CE/I/11(b)

(i) (I) [Providing that consumption expenditure, profit, employment, salaries will not be greatly adversely affected,]

The tax base will be widened because
most goods and services are untaxed before the introduction of the GST, but will be subject to tax under the GST.

(1)

(1)

(ii) Employment ↑ ⇒ wages ↑

(1)

⇒ salaries tax revenue ↑

(1)

More business / investment ↑ ⇒ profits ↑

(1)

⇒ profits tax revenue ↑

(1)

2007/CE/I/9

(a) The ratio of direct to indirect tax revenue will fall, because
the indirect tax revenue will increase.

(1)

(1)

(b) Economic recession has the following phenomena:

Employment & wages ↓ / profits ↓

⇒ Direct tax revenue ↓ (e.g. profits tax revenue ↓ / salaries tax revenue ↓)

(2)

Private consumption ↓ / transaction volume and price of property and shares ↓

⇒ Indirect tax revenue ↓ (e.g. general sales tax revenue ↓ / revenue from stamp duty ↓)

(2)

However, consumption expenditure being generally less volatile than income may lead to a decrease in the ratio of
direct to indirect tax revenue.

(1)

2014/DSE/II/10(a)

(i) A declining birth rate implies a slower growth in the size of the work force and possibly a shrinking working population
in the future. As fewer workers are there to pay taxes, future tax revenue may be reduced.

(2)

(ii) A rising old-age dependency ratio implies higher percentage of the elderly in the population. The elderly may have
lower (or even zero) income and greater needs for medical services. They may thus require more government support in
terms of social welfare and public health, implying an increase in public expenditure on such programs.

(2)

2015/DSE/II/11(b)

Introduction of new types of indirect taxes, such as general sales tax or value-added tax,

(1)

would make more people or commodities fall into the tax net, thus broadening the tax base.

(1)

2017/DSE/II/6(a)

Fluctuation in consumption is smaller than that of income. Sales-tax revenue generated from consumption would fall
(proportionately) less than income-tax revenue generated from income, thus stabilizing total government revenue, during
economic downturn.

(2)

OR

Introduction of a general sales tax broadens the tax base, so that the tax revenue collected would fluctuate less.

(2)

2018/DSE/II/13(a)(ii)

The fiscal balance may be worsened.

(1)

Government revenue may drop, as the working-age population, which accounts for a major portion of taxpayers, drops. (1)

Government expenditure may increase, as the government spends more on medical services and transfer payments to the growing old-age population. (1)

2019/DSE/II/10

- (a) It is a progressive tax because higher average tax rate is charged for tax payers with higher income.

3.3 CLASSIFICATION AND EFFECTS OF TAXES

MARKING SCHEME

1990/CE/II/47 D	2000/CE/II/43 B	1990/CE/II/59 B	1999/CE/II/42 D	2005/CE/II/42 B (28%)
1990/CE/II/55 B	2002/CE/II/43 D (56%)	1991/CE/II/37 B	1999/CE/II/43 A	2005/CE/II/43 B (44%)
1991/CE/II/37 B	2003/CE/II/38 A (31%)	1991/CE/II/39 B	1999/CE/II/44 C	2006/CE/II/39 B (64%)
1992/CE/II/36 A	2004/CE/II/41 D (69%)	1992/CE/II/43 B	2001/CE/II/35 A	2006/CE/II/42 A (56%)
1993/CE/II/45 C	2004/CE/II/42 D (33%)	1993/CE/II/46 A	2001/CE/II/36 B	2007/CE/II/42 A (78%)
1994/CE/II/47 D	2005/CE/II/39 C (66%)	1993/CE/II/47 B	2001/CE/II/39 D	2007/CE/II/43 B (83%)
1995/CE/II/45 C	2005/CE/II/40 A (35%)	1993/CE/II/48 C	2002/CE/II/40 A (73%)	2008/CE/II/41 B (80%)
1995/CE/II/46 D	2007/CE/II/40 D (70%)	1994/CE/II/48 B	2002/CE/II/42 B (46%)	2008/CE/II/44 B (30%)
1995/CE/II/48 B	2008/CE/II/43 D (61%)	1996/CE/II/43 C	2003/CE/II/39 B (38%)	2009/CE/II/42 C (59%)
1996/CE/II/41 A	2009/CE/II/42 C (59%)	1996/CE/II/45 D	2003/CE/II/40 C (72%)	2009/CE/II/44 C (59%)
1996/CE/II/44 B	2009/CE/II/43 D (34%)	1997/CE/II/45 A	2003/CE/II/41 A (59%)	2010/CE/II/42 D (51%)
1997/CE/II/40 C	2010/CE/II/41 C (74%)	1998/CE/II/37 C	2004/CE/II/41 D (69%)	2013/DSE/I/36 C (80%)
1998/CE/II/39 D	2014/DSE/I/36 D (52%)	1998/CE/II/42 B	2004/CE/II/43 D (63%)	2016/DSE/I/36 C (61%)
2000/CE/II/41 A	2017/DSE/I/35 A (57%)	1999/CE/II/40 B	2005/CE/II/41 B (53%)	2018/DSE/I/35 B (49%)
2020/DSE/I/36 B	2020/DSE/I/38 A	2020/DSE/I/41 C	2023/DSE/I/23 A	2023/DSE/I/36 C

Note: Figures in brackets indicate the percentages of candidates choosing the correct answers.

1990/CE/I/5(b)(i)

Policy A widens the wealth gap (1)
 since the poor pay more in proportion to their income on consumption than the rich (2)
 compared with the rich, the poor pay a larger portion of their income as tax (2)

(max: 4)

Policy B widens the wealth gap (1)

benefits the rich as their tax burden is lowered and income after tax ↑ (2)

no effect on the poor because they are outside the tax net or not paying at the standard rate (2)

(max: 4)

1992/CE/1/5(c)(i)

- raise direct taxes
(disadv: work disincentive / investment disincentive)
 - raise indirect taxes
(disadv: inflationary pressure / more uneven distribution of real income)
 - use past fiscal reserves
(disadv: reduce the reserves for contingent purposes / interest return from the investment of the reserves is forsaken / inflationary pressure)
 - borrowing / issue government bonds
(disadv: the economy has to bear interest cost / burden will be borne by the future generations when the principal is repaid)
 - invite private sector participation
(disadv: reduce government say in the related services)
 - cut other government expenditures
(disadv: less public services in other aspects)
 - raise other sources of government revenue, e.g. privatization / sale of the public assets, increase fees
(disadv: more burden on the users of the public services)
- (3@, max: 9)
- (Mark the **FIRST THREE** points only.)

1993/CE/1/5

- (b) (i) It is a proportional tax, because (1)
a fixed portion of the rental income is taxed (2)
- (d) For:
- increase government revenue
 - discourage speculation in the property market
 - widens the tax base / sources of tax revenue
 - a tax on earning / ability-to-pay
- (2@, max: 2)
- (Mark the **FIRST** point only.)
- Against:
- high administration cost
 - against laissez-faire policy / freedom to buy and sell
 - against simple taxation principle
 - distort resource allocation
 - not fair if only apply to the property market
- (2@, max: 2)
- (Mark the **FIRST** point only.)

1994/CE/1/10(b)

- (i) First progressive (because the tax rate first rises as income rises), but (2)
then proportional (because the tax rate later becomes a flat rate as income rises). (2)
- (ii) (I) - a greater amount of tax allowances (raised from \$119 000 to \$146 000) / more low-income families fall out of the tax net
- families have to earn more than \$483 000 (\$391 500 in 1992/93) before paying tax at the standard rate
- families with an annual income of more than \$119 000 and up to \$483 000 pay less tax./ average tax rate ↓
- less progressivity before reaching the standard rate
- (2@, max: 4)
- (Mark the **FIRST TWO** points only.)
- (ii) (II) The new tax structure, because (2)
(to compare with the old tax structure,) the low-income and sandwich class families earning less than and up to \$483 000 a year pay a lower tax rate, but (2)
those earning more than \$483 000 a year pay the same tax rate. (2)
(Exact figures are **NOT** required in candidates' answers.)
- (iii) Workers would have a higher incentive to work, and (2)
thus earn higher income (i.e. a larger tax base). (1)

1995/CE/I/5

Progressive tax : the tax rate ↑ as income ↑ (2)
e.g., salaries tax / profits tax (fiscal year 2018/19 onwards) (1)
(Mark the **FIRST** example only.)

Proportional tax: the tax rate is constant as income ↑ (2)
e.g., salaries tax for those who pay at the standard rate / property tax (1)
(Mark the **FIRST** example only.)

1996/CE/I/10(b)

Regressive tax, because (1)
the low-income group spends a greater proportion of income on clothing than the high-income group, and therefore (2)
the low-income group pays a larger proportion of income as tax than the high-income group. (2)
(max: 3)

OR

the same amount of tax on clothing represents a greater proportion of income for the low-income group. (3)

1996/CE/I/12(d)

Increased progressivity means the tax burden of taxpayers with higher salary income increased at a faster rate than the low-income group / unemployed. (2)

The provision of unemployment benefits will benefit the low-income group / unemployed but not the high-income group. (1)
∴ More even income distribution. (1)

1997/CE/I/9(d)(ii)

The low-income group spend most of their income on purchasing goods and services for their living, (1)
but the high-income group spend only a small amount of their income for their living. (1)

OR

The low-income group pay a larger portion of their income than the high-income group on the general sales tax. (2)
(max: 2)
⇒ More uneven income distribution (1)

1998/CE/I/5

Proposal A: the tax payment takes a decreasing proportion of taxable income as taxable income rises. (2)
Proposal B: the tax payment takes a increasing proportion of taxable income as taxable income rises. (2)
Proposal C: the tax payment takes the same proportion of taxable income as taxable income rises. (2)
∴ Proposal B is the most effective. (1)

1998/CE/I/7(b)

During inflation, the nominal income of people increases. (1)
With no change in tax allowance, more people are put into the tax net or into higher tax brackets. (2)

1999/CE/I/8

Tax to income ratio

Average tax rates are:
 $\$1\,000 / \$10\,000 = 10\%$; $\$1\,900 / \$20\,000 = 9.5\%$; $\$2\,700 / \$30\,000 = 9\%$ (2)

OR

Marginal tax rates are:
 $\$ (1\,900 - 1\,000) / \$ (20\,000 - 10\,000) = 9\%$; $\$ (2\,700 - 1\,900) / \$ (30\,000 - 20\,000) = 8\%$ (2)

The conclusion is: not a progressive tax, because
the marginal tax rate decreases / the average tax rate decreases / the tax amount increase less proportionately than the taxable income. (3)

2000/CE/I/8

Correct comparison of the amount of money rebated to the high-income group and the low-income group: (3)

e.g. 1 High-income group taxpayers are rebated a larger amount of money than low-income taxpayers.

e.g. 2 The government rebated money to the taxpayers but no money to the poor who were non-taxpayers.

It results in a wider wealth gap. (1)

2000/CE/I/9(a)

(i) A regressive tax is a tax which tax rate decreases as income increases. (2)

Consumers pay the same amount of gasoline tax for purchasing the same amount of gasoline.

⇒ In proportion to individual's income, the richer pay at a lower tax rate while the poorer pay at a higher tax rate.

∴ Gasoline tax is a regressive tax. (3)

(ii) - to reallocate resources: use more public transport but less private transport (2)

- a source of tax revenue (2)

2001/CE/I/7

(a) Indirect tax because the tax burden can be shifted onto others (the buyers). (2)

(b) The poor consumes a larger proportion of their income. (1)

OR

Same amount of tax for the same good paid by people with different incomes (1)

∴ A higher proportion of income of the low-income group is taxed. (1)

∴ The general sales tax increases the income inequality. (1)

(Remark: Mere mentioning of the conclusion without elaboration: zero marks)

2001/CE/I/11(b)

Demand side:

Cost of transaction ↓ ⇒ incentive to buy property ↑ and number of transaction ↑ (1)

Supply side:

Volume / value of business of the construction industry ↑ ⇒ expansion in the production of the construction industry and the derived demand for construction workers ↑ (1)

(i) investment ↑ (1)

(ii) employment opportunities ↑ (1)

(Remark: Mere mentioning of the conclusion without elaboration: zero marks)

2002/CE/I/12(c)

(i) Rates is an indirect tax because (1)

the tax burden can be shifted by the landlord onto the tenants. (1)

(ii) The profits tax is a proportional tax because (1)

its effective tax rate is constant / the tax amount is proportional to the taxable income. (2)

2003/CE/I/10(d)

No, because (1)

as his income increases, the tax rate is not fixed / the tax rate decreases. (2)

2004/CE/I/10(c)

Regressive tax, because (1)

same tax payment relative to different incomes (1)

⇒ decreasing tax rate as income rises (1)

2006/CE/I/11(b)(i)(II)

- The tax is an indirect tax because (1)
the tax burden can be shifted from the producers to the consumers. (1)

2008/CE/I/8

- (a) Regressive tax, because (1)
all red wine buyers pay the same amount of tax for purchasing the same bottle of red wine (1)
higher-income earners pay a less than proportionate amount of tax / the tax rate decreases as the income increases. (2)
- (b) Indirect tax, because (1)
the tax burden can be shifted by the red wine sellers onto the consumers. (1)

2009/CE/I/11

- (a) A reduction in the profits tax rate may boost more business investment and increase profits. (2)
No, because (1)
the increase in government tax revenue resulting from the increase in profit earnings could be big enough to offset the decrease in government tax revenue resulting from the reduction in the tax rate. (1)
- (b) Indirect tax, because (1)
the tax burden can be shifted from the landlords onto the tenants. (1)

2012/DSE/II/4

- (a) Wider (1)
- (b) Lower, because (1)
higher income group has a lower post-tax income while the income of lower income group remains unchanged. (2)

2013/DSE/II/8

Government should increase the income tax rate to reduce the budget deficit

- if the increase in tax revenue due to the rise in tax rate outweighs the reduction in tax payment due to the reduction in the tax base/taxable income (e.g., a fall in the number of tax payers as a result of disincentive effects to labour). (4)
- because increase in tax rate is fairer as income tax is often proportional (or even progressive) and this helps equalize income. (2)
- if there is an inflationary (output) gap as an increase in income tax reduces output and relieves the inflationary pressure. (2)

Government should NOT increase the income tax rate to reduce the budget deficit

- if the increase in tax revenue due to the rise in tax rate falls short of the reduction in tax payment due to the reduction in the tax base/taxable income (e.g., a fall taxable income as a result of reduction in aggregate output). (4)
 - because an increase in income tax rate will result in a greater disincentive to work and more people may quit the job and rely on welfare programmes and this further worsens the budget deficits. (2)
 - as an increase in income tax rate reduces the aggregate output, which may result in unemployment / deflationary (income) gap. (2)
- (max: 4)

2014/DSE/II/6(b)

- Direct tax, because (1)
the tax burden cannot be shifted by these tax-payers to someone else. (1)

2014/DSE/II/7(a)

- negative output effect (fall in GDP) due to a reduction in government expenditure and thus aggregate demand
- negative output effect (fall in GDP) due to a rise in sales/income tax, which would lower consumption/investment demand and thus aggregate demand
- negative (disincentive) effect on labour supply and thus on employment (and output) as a result of higher income taxes

(2@, max: 4)

[Mark the **FIRST TWO** points only.]

2015/DSE/II/11(a)

- (i) The percentage change in direct tax revenue and percentage change in nominal GDP tend to move in the same direction. (1)
 Fluctuation of the former is greater than that of the latter. (1)
- (ii) Growth in the economy (increases in GDP) would be accompanied by growth in salary and profit incomes, which would in turn generate higher revenue from taxes on such incomes. (1)
 Progressive taxation may be one explanation for why direct tax revenue fluctuated more than GDP. (1)

2017/DSE/II/13(b)

- (ii) Proposal A's effect on income distribution:
 Salaries tax payers (higher-income group) would pay more taxes while non-tax-payers (lower-income group) would not be affected. (2)
OR
 High-income tax payers pay more, low-income tax payers pay less. (2)
 The after-tax income inequality would thus be reduced. (1)
 (max: 3)

Proposal B's effect on income distribution:
 There will be no extra effect on current tax payers, so the income inequality is unaffected. (2)
 Reduction of fiscal reserves may result in a tax increase in the future, so future tax payers (mainly the younger portion of the population) may have to pay more taxes, thus *increasing income inequality between generations*. (2)
 (max: 2)

- (iii) Proposal A's effect on other variables:
 Price level would drop as AD drops (while there is no such an effect generated by proposal B). (2)
 Unemployment may increase as AD drops (while there is no such an effect generated by proposal B). (2)
 (max: 2)

Proposal B's effect on other variables:
 A worsening of the government's budgetary position (due to a drain on fiscal reserves) may imply tax increase in the future. (No such effect arises under proposal A.) (2)
 Business confidence may be weakened. Expecting tax increases in the future (due to the problem of fiscal sustainability), which would reduce their after-tax profits, business firms may choose to cut down their investment. (No such effect arises under proposal A.) (2)
 (max: 2)

2018/DSE/II/6(a)

- No, because (1)
 the reduction in the profits-tax rate may induce firms to increase their investment expenditure, so they may end up earning higher profits (tax base). The increase in profits tax revenue due to the increase in profits (tax base) may be greater than the decrease in profit tax revenue due to the reduction in tax rate. (3)

2023/DSE/II/12d

(d) Marks for effective communication (EC: max 2 marks)

Marks	Performance
2	● Support arguments with the source/data and appropriate economic theories. ● Present relevant material. ● Present well-organised and coherent answers without repetition of ideas. ● Use language that expresses ideas clearly and fluently with appropriate use of words/terms/symbols.
1	● Present arguments with some support of the source/data and economic theories. ● Present some irrelevant material. ● Present answers in a less organised way with some repetition. ● Use language that conveys a clear message with some inappropriate use of words/terms/symbols.
0	● Present arguments with no support of the source/data and economic theories. ● Present material unrelated to the gist of the question. ● Present inconsistent arguments. ● Express limited ideas with inappropriate use of words/terms/symbols.

2024/DSE/II/12(d)

- (d) (i) Direct tax, because the tax burden cannot be shifted by the tax-payer (Neymar) to someone else. (2)
- (ii) Proportional tax, because the average tax rate remains unchanged (20%) when the taxable income changes. (2)

2025/DSE/II/9(Cross Topic)

- (a) It is a regressive tax, because (1)
the same amount of tax needs to be paid when staying in a hotel room of the same (1)
price regardless of income. (1)
Therefore, the average tax rate decreases as income increases. (1)
- (b) The proposed HAT is an indirect tax. As the tax rate is adjusted from 0% to 3%, the (2)
indirect tax revenue from the HAT will increase (from zero to some positive
number). (2)
- The re-imposition of the HAT may marginally reduce the profits of the hotels and (2)
guesthouses, reducing the direct tax revenue from profits tax.
- (c) (Complex) division of labour (1)
Reasons for increasing labour productivity:
 - practice makes perfect
 - assigning the most suitable person for each task
 - saving time in training
 - saving time in job rotation
 - mechanisation is stimulated
} 1@
max: 3
[Mark the FIRST THREE points only.]

(d) Advantage:

- easier to maintain a stable team of staff as income is more stable
- lower cost of measuring outputs

[Mark the FIRST point only.]

} 1@
max: 1

Disadvantage:

- higher cost of monitoring against shirking
- lower labour productivity due to lower working incentive

[Mark the FIRST point only.]

} 1@
max: 1

- (e) No. Some people may have a lower time cost during public holidays than on other days. If the saving in time cost is greater than the additional money cost (due to a higher room rate), the opportunity cost of staying in a hotel in Hong Kong would be lower.

(3)

3.4 PUBLIC SECTOR AND FISCAL POLICY

MARKING SCHEME

1990/CE/II/44 D	1993/CE/II/49 A	1996/CE/II/42 C	2002/CE/II/41 A (23%)	2014/DSE/I/25 B (54%)
1991/CE/II/38 C	1994/CE/II/49 C	1997/CE/II/43 A	2008/CE/II/42 A (68%)	2019/DSE/I/34 A
1991/CE/II/40 C	1994/CE/II/50 B	1999/CE/II/41 B	2009/CE/II/45 D (53%)	2022/DSE/I/34 A
1992/CE/II/40 B	1995/CE/II/49 B	2000/CE/II/44 C	2010/CE/II/43 A (59%)	2023/DSE/I/41 A
1992/CE/II/41 D	1996/CE/II/39 C	2000/CE/II/45 B	2010/CE/II/44 D (65%)	

Note: Figures in brackets indicate the percentages of candidates choosing the correct answers.

1993/CE/I/5(e)

- full or high level of employment
- economic growth / rising living standard / per capita GNP ↑
- even or more even distribution of income
- control of inflation / deflation

(Mark the **FIRST THREE** points only.)

(2@, max: 6)

3.5 EFFECT OF A BALANCED BUDGET

MARKING SCHEME

1997/AL/II/04
C

2021/DSE/I/40
A

2016/DSE/I/38
D (deleted)

1993/AL/II/7

If the spending on the airport project is exactly financed by increased taxation of the same amount, the fiscal policy effect (on aggregate demand) is expansionary.

The reason is that if income is not taxed, the private sector will save part of it. If it is taxed, the government will spend all the tax revenue. Therefore the expansionary effect of increased spending is greater than the contractionary effect of increased tax.

(Note: However, taxation, be it a direct tax or an indirect tax, would also have supply-side effects, but these are not considered in the HKAL syllabus.)

2009/AL/II/6(a)

- (i) By increasing disposable income, the cash transfers (Tr) would increase consumption. The aggregate demand would then increase, and thus increase output.
- (ii) If the government simply uses the cash to increase its own expenditure, the aggregate demand would increase by ΔG ($= \Delta Tr$) and thus increase output by a bigger amount. This is because while people will only spend part of the cash transfers on consumption (and save the rest), the government will spend all of it directly.

2016/DSE/II/12

- (a) The relative importance of the gaming industry has been declining because its gross revenue has been falling faster than GDP at current market price. (1)
(1)
- (c) Arguments for:
 - Investment would increase potential output over time, but cash sharing would not have similar long-run effects on the economy. (2)
 - Investment would raise aggregate demand (AD) directly. But the cash-sharing scheme could only raise AD indirectly via an increase in consumption. In case the residents choose not to spend the cash received on consumption, the AD effect would vanish. (2)(max: 3)

Arguments against:

- During economic downturn, a reduction in social welfare due to suspension of the cash-sharing scheme may result in discontent of the citizens and political instability. (2)
 - While people can benefit from the cash-sharing scheme instantly, it may take a longer time for investment to boost their income. (2)
 - Public investment may be inefficient due to high administrative costs or low sensitivity to market / price signals (lack of profit incentives). (2)
- (max: 3)

[Remark: Maximum mark of part (c) is 5 marks]

2024/DSE/II/13(a)

- (a) As shown in Source A, the Government's fiscal balance decreased, going from a fiscal surplus in 2018-19 to a fiscal deficit in 2022-23. (1)
In times of an economic downturn, firms' profits / workers' salaries / property owners' rental income tend(s) to decrease. As a result, the Government's revenue from profits tax / salaries tax / property tax decreases, worsening the fiscal balance. (2)
In times of an economic downturns, the number of unemployed tends to increase. Government expenditure on welfare benefits¹ would increase, reducing the fiscal balance. (2)

max: 4

SECTION 4: MONEY AND BANKING (I)

4.1 MONEY: NATURE AND FUNCTIONS

MARKING SCHEME

1990/CE/II/29 A	1993/CE/II/31 C	1996/CE/II/38 C	2004/CE/II/35 A (75%)	2008/CE/II/36 D (30%)
1990/CE/II/30 C	1993/CE/II/58 C	1997/CE/II/32 B	2004/CE/II/36 B (87%)	2009/CE/II/35 C (72%)
1990/CE/II/40 D	1994/CE/II/41 B	1997/CE/II/38 B (deleted)	2004/CE/II/40 B (49%)	2009/CE/II/36 C (74%)
1990/CE/II/41 C	1994/CE/II/44 C	1997/CE/II/39 D	2005/CE/II/35 C (79%)	2010/CE/II/34 B (85%)
1990/CE/II/58 A	1995/CE/II/38 D	1998/CE/II/32 D	2006/CE/II/33 B (67%)	2010/CE/II/37 D (67%)
1991/CE/II/29 A	1995/CE/II/39 A	1999/CE/II/35 D	2006/CE/II/34 A (60%)	2015/DSE/I/35 C (78%)
1992/CE/II/33 B	1995/CE/II/40 B	2001/CE/II/48 D	2007/CE/II/35 B (83%)	2016/DSE/I/30 C (74%)
1992/CE/II/57 D	1995/CE/II/50 C	2002/CE/II/34 C (82%)	2008/CE/II/34 C (84%)	2019/DSE/I/30 D
1992/CE/II/59 C	1996/CE/II/34 C	2003/CE/II/50 C (43%)	2008/CE/II/35 B (96%)	2021/DSE/I/30 A
2024/DSE/II/31 C	2025/DSE/I/31 D			

Note: Figures in brackets indicate the percentages of candidates choosing the correct answers.

- 1991/CE/I/5(a)

 - a medium of exchange
 - a store of value
 - a standard of deferred payment
 - a unit of account

[Mark the **FIRST THREE** functions only.]

(1@@, max: 3)
- 1993/CE/I/3(d)(ii)

(I) It is not (generally) accepted in exchange. (2)

(II) It cannot provide (or store up the) purchasing power for future use.

OR

It cannot be readily converted into money. (2)
- 1994/CE/I/6

It is **NOT** a standard of deferred payment, because it is (2)

 - not a commonly accepted medium of exchange (2)
 - not of homogeneous quality (2)

OR

It is **NOT** a store of value, because it is (2)

 - not a commonly accepted medium of exchange (2)
 - not durable / unstable value (2)

OR

It is **NOT** a unit of account, because it is (2)

 - not a commonly accepted medium of exchange (2)
 - exchange ratio arbitrary (2)

1995/CE/10(b)(i)
Medium of exchange

(2)

OR

Means of payment

(2)

1996/CE/1/8

Store of value - less purchasing power of the same amount of money

Standard of deferred payment - the real value of money decrease / creditors prefer immediate cash to post-dated cheques

Medium of exchange - the real value of money decreases / the general acceptability of money is adversely affected (in times of a hyperinflation)

(2@, max : 4)

[Mark the **FIRST TWO** points only.]

1997/CE/1/4

(a) (Disagree.)

Salt is used as money, because

(1)

it is generally accepted as the medium of exchange.(2)

(max : 2)

(Remark: Mere mention of the definition of money without saying disagree or giving any other elaboration - No mark.)

(b) - uncasy to maintain a stable quality (loss in value, storage cost)

- may be bulky to carry around a large amount of salt

- may not be scarce enough (cannot control the money supply)

- different kinds of salt may not have uniform quality

(2@, max : 4)

[Mark the **FIRST TWO** points only.]

1998/CE/1/6

Wrong.

Money still exists because a generally accepted medium of exchange is still needed.

(3)

It may be in other forms such as electronic money.

(1)

1999/CE/11(d)

Unit of account -

(1)

it is not a unit of account because the prices of other goods are not expressed in terms of Octopus cards.

(1)

Standard of deferred payment -

(1)

it is not a standard of deferred payment because future payment is not denominated in terms of Octopus cards.

(1)

2000/CE/1/5

Money is a generally accepted medium of exchange.

(2)

(The problem of exchange in the case is the need to search for double coincidence of wants / the act of buying and the act of selling cannot be separated.)

If money is used as a medium of exchange, A can pay money to buy B's vegetables without selling his bananas to B. B can also pay money to buy wood from someone other than A. With money, exchange can take place even if there is no double coincidence of wants / act of buying and the act of selling can be separated.

(2)

2001/CE/I/5

(On divisibility)

Banknotes can be divided into different denominations without a change in value.

Diamond if divided into smaller units will have a fall in total value.

(On homogeneity)

Banknotes with the same face value issued by the same bank share uniform physical characteristics $\Rightarrow$ homogeneous in value

Diamonds do not have uniform physical characteristics $\Rightarrow$ heterogeneous in value

(On homogeneity)

The value of a banknote is more easily recognized (because of uniform physical characteristics).

The value of a diamond is not easily recognized (because diamonds do not have uniform physical characteristics).

(2@, max : 4)

2002/CE/I/11(c)

No because

they are not generally accepted as a medium of exchange.

not generally accepted because they cannot be used to buy other goods and services.

(1)

(1)

(1)

2003/CE/I/7

(a) Medium of exchange

Unit of account

[Mark the **FIRST TWO** points only.]

(1)

(1)

(b) Store of value -

people can store up the purchasing power for future spending on goods and services.

Standard of deferred payment -

future payment is denominated in terms of money.

[Mark the **FIRST TWO** points only.]

(1)

(1)

(1)

(1)

2004/CE/I/9(b)

Store of value / store of wealth, because

the purchasing power / real value of cash continues to decrease in times of inflation (but continues to increase in times of deflation).

(1)

(2)

2004/CE/I/11(c)(ii)

Not generally accepted, because

these coupons cannot be used to buy things in other shops.

(1)

(1)

2005/CE/I/4

Medium of exchange -

money serves as a medium to facilitate people's buying and selling of goods and services.

Unit of account -

money is used to express the prices and value of goods and services.

Store of value -

people can store up the purchasing power for future spending on goods and services.

Standard of deferred payment -

future payment is expressed in terms of money.

[Mark the **FIRST TWO** points only.]

[(1 + 1)@, max: 4]

2006/CE/I/7

(1) unit of account

(2) medium of exchange / means of payment

(3) store of value

(1)

(1)

(1)

2007/CE/I/6

- Accept payment: medium of exchange / means of payment (1)
- HK\$ deposit: store of value / store of wealth (1)

[Mark the **FIRST TWO** points only]

Medium of exchange:

the Hong Kong dollar serves as a medium to facilitate people's buying and selling of goods and services in these shops in Macau. (1)

Store of value:

Hong Kong dollar deposits enable people to store up the purchasing power for future spending on goods and services. (1)

2009/CE/I/9(d)

- (i) Medium of exchange (1)
- Unit of account (1)

[Mark the **FIRST TWO** points only]

- (ii) Not a medium of exchange because these book coupons are not generally accepted by other shops in buying goods and services (1)
- Not a unit of account because the prices of other goods are not expressed in terms of these book coupons (1)

2010/CE/I/5

- (a) Money is defined as a generally accepted medium of exchange. (2)

- (b) (Not generally acceptable)
Pearls are not generally acceptable as a medium of exchange in trading of goods and services.
(Not divisible)
Pearls if divided into smaller units will have a fall in total value.
(Not homogeneous)
Pearls do not have uniform physical characteristics ⇒ Heterogeneous in value / The value of a pearl is not easily recognized. (2@, max: 4)

2012/DSE/II/7

- Gold is more durable while cigarettes may lose their value during transaction.
- Gold is more generally accepted as medium of exchange than cigarettes.
- Gold is more homogenous while cigarettes have different qualities. (2@, max: 4)

[Mark the **FIRST TWO** points only.]

2017/DSE/II/5

Store of value:

As the value of the local currency drops in terms of foreign currency, its purchasing power in terms of the quantity of foreign goods and services would also fall.

Standard of deferred payment:

As the value of the currency becomes highly volatile, the contracting cost of specifying future payments in terms of this currency would increase.

Medium of exchange:

Some multinational firms may refuse to accept the currency if its value fluctuates too much. (2@, max: 4)

[Mark the **FIRST TWO** functions only.]

2018/DSE/II/12(b)

Unit of account: The price of the hotel accommodation service is expressed in terms of RMB. (2)

Medium of exchange: RMB serves as a medium to facilitate people's buying and selling of services in the website. (2)

[Mark the **FIRST TWO** points only.]

2020/DSE/II/5b

- b) During inflation, the real value / purchasing power will decrease of the coupon.

Or

The coupon is not generally acceptable as medium of exchange.

Reasons:

- Not durable, as cigarettes can be easily damaged and cannot be stored for a long time.
- Not homogeneous, as cigarettes do not have uniform physical characteristics.
- Not portable, as people need to carry a large quantity of cigarettes for high-value transactions.
- Not divisible, as cigarettes if divided into smaller units would have a fall in total value.
- Any other relevant point

[Mark the **FIRST TWO** points only.]

Marks

} @2
Max: 4

2022/DSE/II/11d

(d) Reasons:

Consumption vouchers are not generally accepted as medium of exchange, as they (2)

Cannot be used in those shops which have not installed the (SVFs) system.

Consumption vouchers are not a good store of value, as their values would drop to zero (2)

once the deadlines for using the vouchers have passed.

[Mark the **FIRST TWO** points only.]

2024/DSE/II/6

(a) The lack of double coincidence of wants. (1)

(b) (i) Money is anything that is generally accepted as a medium of exchange. (1)

(ii) The use of money allows the acts of buying and selling to be separated, so that exchange no longer requires double coincidence of wants. For example, Villager A may sell his beans for money and use the money to purchase pork from Villager B (who can in turn use the money to buy rice from someone else). (2)

4.2 BANKING SYSTEM AND THE CENTRAL BANK

MARKING SCHEME

1993/CE/II/34 A	2003/CE/II/33 A (57%)	2018/DSE/I/31 C (69%)	2001/CE/II/49 D	2018/DSE/I/33 A (53%)
1994/CE/II/39 B	2005/CE/II/36 A (59%)	1993/CE/II/59 D	2003/CE/II/37 B (46%)	2019/DSE/I/29 A
2000/CE/II/31 A	2007/CE/II/38 B (63%)	1995/CE/II/41 A	2007/CE/II/37 A (56%)	2021/DSE/I/32 C
2001/CE/II/27 A	2010/CE/II/38 B (60%)	1996/CE/II/60 B	2009/CE/II/37 C (42%)	2022/DSE/I/31 A
2002/CE/II/32 A (51%)	2014/DSE/I/31 A (73%)	2000/CE/II/32 C	2012/DSE/I/29 A (64%)	2023/DSE/I/31 B
2024/DSE/II/32 B	<i>Note: Figures in brackets indicate the percentages of candidates choosing the correct answers.</i>			

1992/CE/I/3(c)(i)

- note issuing
- government's banker

(2)
(2)

2001/CE/I/10(a)

Financial intermediaries / to channel people's savings to investments

(2)

2002/CE/I/7

- to issue currency
- to act as the banker of the government
- to act as an advisor to the government
- to supervise / monitor the monetary sector, e.g. private banks
- to act as the lender of last resort
- to administer the clearing house
- to carry out the monetary policy
- to manage the country's foreign exchange reserve

[Mark the **FIRST TWO** points only.]

(1@, max: 2)

4.3 MONEY SUPPLY DEFINITIONS: M1, M2 & M3

MARKING SCHEME

1990/CE/II/32 C	1997/CE/II/35 A	2000/CE/II/27 C	2005/CE/II/38 A (53%)	2013/DSE/I/25 B (41%)
1991/CE/II/34 A	1997/CE/II/36 B	2000/CE/II/28 B	2006/CE/II/35 D (61%)	2015/DSE/I/30 A (70%)
1992/CE/II/32 C	1997/CE/II/37 B	2001/CE/II/25 B	2007/CE/II/39 A (49%)	2016/DSE/I/32 A (76%)
1994/CE/II/43 C	1998/CE/II/35 D	2002/CE/II/33 D (52%)	2008/CE/II/37 A (66%)	2017/DSE/I/33 C (43%)
1995/CE/II/43 A	1998/CE/II/36 D	2003/CE/II/32 A (43%)	2010/CE/II/39 D (48%)	2019/DSE/I/31 B
1996/CE/II/37 A	1999/CE/II/39 B	2004/CE/II/37 A (23%)	2012/DSE/I/28 B (69%)	2020/DSE/I/29 A
2022/DSE/I/41 B	2023/DSE/I/33 B	2024/DSE/II/33 C	2025/DSE/I/32 C	

Note: Figures in brackets indicate the percentages of candidates choosing the correct answers.

1991/CE/I/5(c)

- (i) - demand deposit
- savings deposit
- fixed deposit
- certificates of deposit

[Mark the **FIRST THREE** points only.]

(1@. max: 3)

(ii) - Liquidity:

Is it easily convertible into cash? [Demand deposit is the most liquid.]

- Returns in terms of interest:

How much returns does it yield? [Certificates of deposit offer the highest interest rate.]

- Certainty in interest earned:

Does it involve any risk? [The interest on fixed deposit is certain, and the deposit itself is also protected by the

Deposit Protection Scheme in Hong Kong (, whereas a CD is not).]

(3@. max: 6)

[Mark the **FIRST TWO** points only.]

1992/CE/I/3

- (a) (i) Cash held by non-bank public
Demand deposits

(1)

[Each incorrect component mentioned - deduct 1 mark]

(1)

- (ii) It cannot fulfill the function of money as a medium of exchange.

(3)

- (b) (i) M1 unchanged, because
cash held by the public decreased by \$1 000
demand deposits increased by \$1 000

(1)

OR

they counteract each other

(2)

1993/CE/II/5(c)(i)

M₁: no change;

(1)

M₂: increased (by \$3 million), because

(1)

savings deposits are included in M₂ but not in M₁.

(2)

1995/CE/10(b)(ii)

Not affected, because

both the customer's bank deposit and the shop's bank deposit are included in M2.

(1)

(1)

1996/CE/17

$\Delta M1$ = change in the amount of cash held by the non-bank public + change in the demand deposits with licensed banks

$$= \$ (40 - 50) \text{ mn.} + \$ (30 - 20) \text{ mn.} = 0$$

(2)

$\Delta M2$ = $\Delta M1$ + change in the time and savings deposits with licensed banks + change in the negotiable certificates of deposit issued by licensed banks and held by the non-bank public

$$= 0 + \$ [(40 + 30) - (40 + 40)] \text{ mn.}$$

$$= -\$10 \text{ mn.}$$

(2)

1999/CE/1/5

$M1$ = currency in the hands of the non-bank public + demand deposits

(2)

$M2$ = $M1$ + saving deposits + time deposits held with banks + negotiable certificates of deposit issued by banks held outside the monetary sector.

(2)

2002/CE/1/6

$M1$ increased

(1)

by \$90 000, because

(1)

the legal tender notes and coins held by the public increased.

(1)

$M3$ decreased

(1)

by \$60 000, because

(1)

this amount was no longer included in the cash in public circulation in HK or in the deposits with the monetary sector of HK.

(1)

[Remark: Mere mentioning of the definition of $M1$ and/or $M3$ - NO marks]

2008/CE/1/6

(1) legal tender notes and coins held by the non-bank public

(1)

(2) demand deposits

(1)

(3) savings deposits

(1)

(4) negotiable certificates of deposit

(1)

2020/DSE/1/8

8a) Monetary base = Cash held by public + Reserve of commercial bank = $800 + 1000 = 1800$ (1 mark)

Money Supply = Cash held by public + Total deposit = $800 + 4000 = 4800$ (1 mark)

b) Change of monetary base = change of cash held by public + change of reserve

$$= 0 + (- 700)$$

$$= - 700 \quad (1 \text{ mark})$$

c) New Maximum deposit after the withdrawal = $(1000 - 700) \times \text{maximum banking multiplier} = 300 \times 4 = 1200$

Change of deposit = $1200 - 4000 = - 2800$ (1 mark)

Change of money supply = change of cash held by public + change of deposit

$$= 0 - 2800$$

$$= -2800 \quad (2 \text{ marks})$$

2020/DSE/II/9C

- C) Do not have interest obligation
Do not have redemption pressure
(2 marks)

2025/DSE/II/10(Cross Topic)

(a) (i) Money supply growth rate = $(20\,837.4 - 21\,691.3) / 21\,691.3 \times 100\% = -3.94\%$ (1)

(ii) QTM: $MV = PY$, where M is money supply, V is velocity of circulation of money, P is general price level, Y is real output, assuming V to be constant. (2)

Growth rate of $Y \approx$ growth rate of M – growth rate of P = $-3.94 - 3.6 = -7.54\%$ (1)

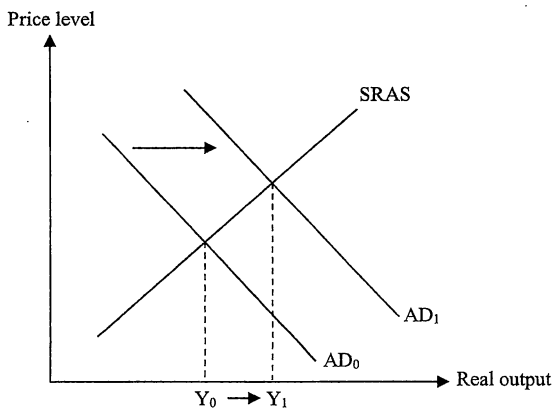
- (b) As discount rate is lowered, the cost for commercial banks to borrow money from the Federal Reserve decreases. (1)
Money supply increases. (1)
Nominal interest rate decreases. (1)

(c) Verbal elaboration:

- As interest rate decreases, investment increases. (1)
- Aggregate demand increases. (1)
- Real output increases. (1)

Indicate in the diagram:

- rightward shift of AD curve (1)
- increase in Y (1)



(d) Argument FOR:

- Increasing expenditure on infrastructure can increase the capital endowment in the future, raising the aggregate output in the long run. (2)

Argument AGAINST:

- Increasing expenditure on infrastructure would raise the US Government's expenses, worsening the fiscal balance. (2)

4.4 DEPOSIT CREATION

MARKING SCHEME

1990/CE/II/45 C	1996/CE/II/36 B	2001/CE/II/26 C	2005/CE/II/37 A (57%)	2010/CE/II/35 B (68%)
1991/CE/II/31 B	1997/CE/II/33 B	2002/CE/II/35 C (84%)	2005/CE/II/50 B (46%)	2010/CE/II/36 C (43%)
1991/CE/II/32 B	1997/CE/II/34 A	2002/CE/II/36 D (56%)	2006/CE/II/36 D (61%)	2012/DSE/I/27 C (71%)
1992/CE/II/34 B	1998/CE/II/33 A	2002/CE/II/37 C (59%)	2006/CE/II/37 B (57%)	2013/DSE/I/26 C (58%)
1992/CE/II/35 D	1998/CE/II/34 D	2002/CE/II/38 A (25%)	2006/CE/II/38 C (37%)	2014/DSE/I/30 D (56%)
1992/CE/II/32 C	1998/CE/II/52 C	2003/CE/II/34 D (17%)	2007/CE/II/36 D (49%)	2016/DSE/I/31 D (65%)
1994/CE/II/40 A	1999/CE/II/36 B	2003/CE/II/35 B (24%)	2008/CE/II/38 B (39%)	2016/DSE/I/33 A (deleted)
1994/CE/II/42 D	1999/CE/II/38 A	2003/CE/II/36 C (16%)	2008/CE/II/39 D (60%)	2017/DSE/I/31 B (36%)
1995/CE/II/42 B	2000/CE/II/29 C	2004/CE/II/38 A (73%)	2009/CE/II/38 A (44%)	2019/DSE/I/28 D
1996/CE/II/35 C	2000/CE/II/30 B	2004/CE/II/39 B (34%)	2009/CE/II/39 D (37%)	2020/DSE/I/33 B
2022/DSE/I/30 D	2022/DSE/I/32 C	2023/DSE/I/34 C	2024/DSE/II/34 B	2025/DSE/II/34 D

Note: Figures in brackets indicate the percentages of candidates choosing the correct answers.

1990/CE/2/(d)(ii)

(I) $\$10\,000 \times (1 - 25\%)$ (2)

= \$7 500 (1)

Possible way: to recall loans / sell investment assets (2)

(II) $\$10\,000 \times (1 / 25\%)$ (2)

= \$40 000 (1)

1991/CE/I/5(b)

(i) $\$200 / \$800 = 25\%$ (2)

(ii) Actual reserves – (Deposits $\times$ Required reserve ratio)

$\$50 - (\$100 \times 25\%) = \$25$ (2)

(iii) Excess reserves $\times$ Banking multiplier

$\$25 \times (1 / 25\%) = \100 (3)

Assumptions:

- no cash drain (2)

- people are willing to borrow (2)

1992/CE/I/3(b)(ii)

- 100% reserves required by the government
 - the bank keeps all the \$1000 as reserves
 - no one borrows from the bank
 - any cash that leaks out of the bank is not redeposited with the banking system
- (Mere listing of points without elaboration - max. 2 marks.)

(3@, max: 6)

[Mark the **FIRST TWO** points only.]

1993/CE/I/5(c)(ii)

$$\begin{aligned} & \$3 \text{ million} \times (1 / 25\%) \\ & = \$12 \text{ million} \end{aligned}$$

(2)

(1)

1994/CE/I/5

$$\begin{aligned} & \text{Maximum amount of deposits to be created} = \$100 \times (1 / 20\%) \\ & = \$500 \end{aligned}$$

(3)

(1)

1995/CE/I/10(a)(iii)

Brief description of the changing process in a correct sequence:

Initial withdrawal of \$M $\Rightarrow$

(Step 1:) Banks lose reserves (i.e. actual reserves < required reserves)

(2)

(Step 2:) Banks call back loans / sell bank assets to the public

(2)

(Step 3:) Further withdrawal of bank deposits to repay the loans / to buy bank assets
The process of bank deposit contraction goes on and on

(2)

(1)

(Step 4:) Ultimately, total bank deposits contract
by $\$M \times (1 / 20\%) = \$5M$

(1)

(2)

(max: 8)

1996/CE/I/11(c)

- (i) No change in total bank deposits, because
it is only a switch from one's bank deposit to another's.

(1)

OR

there may be some deposit contraction at first, but then it would be offset by the deposit expansion later on.

(2)

- (ii) Banks gain excess reserves

(1)

$\Rightarrow$ increase in total bank deposits

(1)

by an amount of $\$5mn \times (1 / 25\%) = \$20mn$

(2)

1997/CE/I/9(c)

(A logical sequence of the following is required.)

An injection of new money (e.g. the initial \$10 b.) to the local banks

(1)

$\Rightarrow$ banks gain excess reserves (\$R)

(1)

and will loan out the excess reserves

(1)

$\Rightarrow$ the loan will be redeposited into the banking system

(1)

But due to the following factor(s), the redeposit amount is smaller than SR:

- initially not all the new money (the \$10b.) are deposited to the local banks
- local banks hold excess reserves
- insufficient demand for loan
- cash drain

(2@, max: 4)

Elaboration on how any one of the above factors would lead to a total bank deposit amount smaller than the maximum amount

(2)

(max: 8)

1998/CE/1/9(a)

(Step 1:) After the bank accepts Mr Wong's \$1 m deposit, it will lend out / invest 75% or \$0.75 m in the first round (2)

(Step 2:) The \$0.75 m loan / investment will be redeposited into the banking system (2)

The process of (Step 1 - Step 2 - Step 1...) will go on and on until (2)

finally the total deposit created = initial deposit $\times$ (1 / reserve ratio)

= $\$1 \times (1 / 25\%) = \4 m (2)

1999/CE/1/6(a)

$\$10\,000 + [\$10\,000 \times (1 - 25\%)]$ (2)

= \$17 500 (2)

2000/CE/1/10(a)

Answer 1

(Assume that the public does not use the \$50 mn cash in hand to repay bank loans when the banks call back loans due to the withdrawals)

(i) $-\$10\text{ mn} \times [1 - (200 / 1\,000)]$ (2)

= $-\$50\text{ mn}$ (1)

(ii) Change in money supply

= Change in cash in hand + Change in bank deposits (1)

= $(+\$10\text{ mn}) + (-\$50\text{ mn})$ (2)

= $-\$40\text{ mn}$ / a contraction of \$40 mn (1)

Answer 2

(Assume that the public considers to make use of the \$50 mn cash in hand to repay bank loans when the banks call back loans due to the withdrawals)

(i) The max. change is the same as the one in Answer 1. However, the actual change depends on how much of the \$50 mn cash in hand is used to repay the bank loans. The min. change is $-\$10\text{ mn}$. (Max: 3)

(ii) Use the same formula in Answer 1. The change in bank deposits varies from a max. decrease of \$50 mn to a min. decrease of \$10 mn, depending on how much of the \$50 mn cash in hand is used to repay the bank loans. The change in cash in hand varies from a max. to $+\$10\text{ mn}$ to a min. of $+\$2\text{ mn}$. Therefore, the final numerical answer varies from $-\$40\text{ mn}$ to $-\$8\text{ mn}$. (Max: 4)

2001/CE/1/10(b)

(i) Actual reserve ratio = $250 / 1\,000$ (1)

= 0.25 (1)

Excess reserve = $(0.25 - 0.2) \times \$1\,000$ (1)

= \$50 mn (1)

(ii) The withdrawal reduces the amount of reserves to \$210 mn. (1)

The new amount of reserves can back up $\$210\text{ mn} \times (1 / 20\%)$ (2)

= \$1 050 mn of deposits (1)

2002/CE/1/10(a)

(i) The required reserve ratio = $(20 / 100) = 0.2$ or multiplier = 5 (1)

Max. amount of deposits created = $\$10\text{ m} \times 5 = \50m (1)

Max. amount of money supply created = Max. amount of deposits created - decrease in cash in public circulation (1)

= $(\$50\text{m} - \$10\text{m}) = \$40\text{ m}$ (2)

(ii) - there is cash leakage

- the banks decide to hold excess reserves

- insufficient demand for loans

(2@, max: 4)

[Mark the **FIRST TWO** points only.]

2003/CE/I/11

- (b) (ii) Some banks cannot lend out their excess reserve as the demand for bank loans ↓
Step 1: loans that can be made by banks would decrease (1)
Step 2: money re-deposited into the banking system would decrease (1)
Correct sequence : Step 1 → Step 2 (1)
Therefore, deposit creation ability ↓ (1)
- (c) Legal reserve ratio = 0.2 (1)
Max. deposit = $\$250 \text{ m} \times (1 / 20\%)$ (1)
= $\$1\,250 \text{ m}$ (2)

2004/CE/I/6(b)

- Initial withdrawal of deposits,
⇒ banks lose reserves / insufficient reserves / actual reserves smaller than required reserves (1)
⇒ banks have to call back loans / sell assets to the public (1)
⇒ further withdrawals of deposits (1)
The process goes on and on. (1)

2005/CE/I/5

- (a) The amount of excess reserve
= $\$[120 - (1\,000 \times 10\%)] \text{ m} = \20 m (1 + 1)
- (b) Maximum possible amount of bank deposits
= $\$120 \text{ m} \times (1 / 10\%) = \$1\,200 \text{ m}$ (1 + 1)

2006/CE/I/6

- (a) 10% (1)
- (b) Yes because (1)
when total deposits reduce to $\$1\,150 (= \$1\,500 - \$350)$,
it only needs a reserve of $\$115 (= \$1\,150 \times 10\%)$ to back up the deposits.
Since the reserve assets only reduce to $\$150 (= \$500 - \$350)$, there is still an excess reserve of $\$35$. (2)

2007/CE/I/7

- (a) $\$200$ (1)
- (b) (i) $-\$100$ (1)
- (ii) $-\$30$ (2)

2008/CE/I/10

- (a) An increase in the reserve ratio
⇒ insufficient reserves / actual reserves smaller than required reserves (1)
⇒ banks have to call back loans / sell assets to the public (1)
⇒ withdrawals of deposits (1)
The process will go on and on (until the actual reserve is equal to the required reserve).* (1)
[*Remark: This point deserves one mark only when the sequence of the above steps is correct]

- (b) New required reserve ratio = $0.2 + 0.05 = 0.25$

Amount of bank deposits that \$1 000 million reserve can back up = $\$1\,000 \text{ million} \times (1 / 25\%)$ (1)
 = \$4 000 million (1)

OR

Change of bank deposits = $-\$250 \text{ million} \times (1 / 25\%)$ (1)
 = $-\$1\,000 \text{ million}$ (1)

Change in MS = Change in Cp + change in Dd / No change in Cp / MS = Cp + Dd (1)
 Change in MS = $-\$1\,000 \text{ million}$ (1)

(MS = money supply;

Cp = cash in the hands of the non-bank public; and

Dd = bank deposits.)

2009/CE/1/6

- (a) The minimum required reserve is $\$1\,000 \times 20\% = \200 (1)
 Excess reserve = \$100 (1)

- (b) Total amount of deposits at maximum = $\$300 \times (1 / 20\%)$ (1)
 = \$1 500 (1)

OR

Amount of extra deposits created = $\$100 \times (1 / 20\%)$ (1)

Therefore, total amount of deposits at maximum = $\$1\,000 + \$500 = \$1\,500$ (1)

2010/CE/1/11(b)

- (i) Banking multiplier = $1 / 40\% = 2.5$ (1)

The max. amount of deposits that can be supported by \$2 000 million reserve at 40% reserve ratio

= $\$2\,000 \text{ mn.} \times (1 / 0.4)$

= \$5 000 mn. (1)

Therefore, the max. change in bank deposits = $\$5\,000 \text{ mn.} - \$8\,000 \text{ mn.} = -\$3\,000 \text{ mn.}$ (1)

- (ii) An increase in the reserve ratio
 ⇒ the actual reserve is smaller than the desired reserve (1)
 ⇒ banks have to call back loans / sell assets to the public (1)
 ⇒ withdrawals of deposits (1)

The process will go on and on (until the actual reserve is equal to the desired reserve). * (1)

[*This 4th mark will be awarded only if the above three marks are already awarded.]

[#If the process starts with “withdrawals of deposits”, then award only a maximum of 2 marks]

1997/AL/11/9

- (a) reserve ratio = 0.25

- (b)

Assets (\$)		Liabilities (\$)	
Reserves	1 000	Deposits	5 500
Loans	4 500		

When the required reserve ratio is 10%, the maximum loan the banking system can extend is \$9 000. Since the maximum demand for loan is \$4 500, the loan extended will be \$4 500. Money supply is \$5 500.

- (c) A government can restrict the maximum amount a banking system can lend, or the maximum money supply. But the actual amount of lending or the actual money supply is also affected by the demand for loan.

1998/AL/II/9

- (a) Money supply = Deposits + amount of currency held by the public
= \$4 000 + \$500 = \$4 500

(b) (i)

Assets (\$)		Liabilities (\$)	
Reserves	1 100	Deposits	4 100
Loans	3 000		

$$\text{Money supply} = \$4\ 100 + \$400 = \$4\ 500$$

(ii)

Assets (\$)		Liabilities (\$)	
Reserves	1 100	Deposits	4 400
Loans	3 300		

$$\text{Money supply} = \$4\ 400 + \$400 = \$4\ 800$$

- (c) It will increase the money supply only if the banks can lend out some or all of their excess reserves.

2017/DSE/I/8

- (a) $\text{RRR} = (\$1\ 000 - \$250) / \$3\ 000 = 0.25$

(1)

- (b) The change in deposits = $\$250 \times (1 / 0.25) = \$1\ 000$

(2)

2019/DSE/II/7

- (a) **Excess reserves** = \$300 million – \$1000 million x 20% = \$100 million (1)

- (b) **Old money supply** = \$1000 million + \$200 million = \$1200 million (4)

$$\text{New money supply} = (\$300 \text{ million} + \$200 \text{ million}) \times 1/20\% + \$0 \text{ million} = \$2500 \text{ million}$$

$$\text{Maximum possible change in the money supply} = \$2500 \text{ million} - \$1200 \text{ million} = \$1300 \text{ million}$$

2021/DSE/II/6

6. (a) Required reserve ratio = $(600 - 300) / 1500 = 20\%$ (1)

- (b) New required reserve ratio = $20 - 5 = 15\%$ (3)

$$\text{New deposit} = 1500 + (600 - 0.15 \times 1500) / 0.15 = 4000 \text{ (\$million)}$$

$$\text{New money supply} = 4000 + 100 = 4100 \text{ (\$million)}^\dagger$$

In real-world practice of monetary policy, adjustment in the required reserve ratio (RRR) is understood as a change in percentage point or basis point (i.e. a level change in RRR) rather than a percentage change in RRR. The following answers are also accepted in case the candidate (mis)understood the 5% reduction in RRR as a percentage change rather than change in percentage point.⁵

$$\text{New required reserve ratio} = 20\% \times (1 - 5\%) = 19\%$$

$$\text{New deposit} = \$3157.89 \text{ million}$$

$$\text{New money supply} = \$3257.89 \text{ million}$$

2022/DSE/IB/8

(a) 25%.

- (b) Original money supply $= 4000 + 1000 = 5000$ (\$million)
- New deposit $= (1000 + 600) \times \frac{1}{25\%} = 6400$ (\$ million)
- New money supply $= 6400 + 400 = 6800$ (\$million)
- Change in Ms $= 6800 - 5000 = 1800$ (\$million)

2023/DSE/II/10i, ii

- (b) (i) Expected real rate of return $= 4.5\% - 4.5\% = 0\%$ (1)
Actual real rate of return $= 4\% - (-1\%) = 5\%$ (1)
- (ii) Gain, because (1)
the actual real rate of return is higher than the expected real rate of return. (1)

2024/DSE/II/8

- (a) The required reserve ratio $= [(\$600 \text{ million} + \$100 \text{ million}) / \$3\,500 \text{ million}] \times 100\%$ (1)
 $= 20\%$
- (b) The new required reserve ratio $= 20\% - 5\% = 15\%$ (1)
The maximum amount of deposits after deposit creation (1)
 $= \$600 \text{ million} / 15\% = \$4\,000 \text{ million}$
The maximum change in deposits (1)
 $= \$4\,000 \text{ million} - \$3\,500 \text{ million}$
 $= \$500 \text{ million}$

SECTION 5: MONEY AND BANKING (II)

5.1 MONETARY BASE (M0) AND MONETARY POLICY

MARKING SCHEME

1994/AL/II/18 A	2005/AL/II/19 C (55%)	1990/AL/II/03 A	2000/AL/II/08 B	2012/DSE/I/30 A (51%)
1995/AL/II/16 C	2005/AL/II/20 D (56%)	1990/AL/II/04 C	2003/AL/II/12 D	2013/DSE/I/28 B (43%)
1995/AL/II/17 D	2007/AL/II/7 A (30%)	1991/AL/II/23 B	2004/AL/II/04 B (74%)	2017/DSE/I/32 A (61%)
1995/AL/II/28 C	2011/AL/II/20 C (66%)	1992/AL/II/04 B	2005/AL/II/17 B (77%)	2017/DSE/I/37 B (66%)
1996/AL/II/19 D	2014/DSE/I/33 A (70%)	1993/AL/II/16 A	2006/AL/II/07 A (84%)	2019/DSE/I/41 B
1999/AL/II/15 C	2015/DSE/I/29 B (43%)	1996/AL/II/18 B	2007/AL/II/10 A (63%)	2020/DSE/I/30 B
2001/AL/II/13 D	2015/DSE/I/32 C (64%)	1997/AL/II/08 A	2010/AL/II/18 B (73%)	2020/DSE/I/32 A
2002/AL/II/21 C (29%)	2018/DSE/I/32 C (40%)	1998/AL/II/17 A	2012/AL/II/14 A (64%)	2021/DSE/I/34 B
2021/DSE/I/35 A	2023/DSE/32 C	2024/DSE/II/28 B	2024/DSE/II/35 D	2024/DSE/II/39 A

1994/AL/II/3

- (a) When the government purchases bonds from the market, it increases the reserves of commercial banks, the lending ability of the latter (through the banking multiplier) and hence the money supply.
- (b) With the reserve ratio at 0.2, 1 million dollars of additional reserves can back up a loan of 5 millions. Thus the money supply increases by an amount of 5 million.

In the calculation, we have assumed that banks are fully loaned up and there is no cash leakage.

1996/AL/II/5(a)

As the public buy government bonds, the reserves in the commercial banks are reduced. Assuming that the banks are fully loaned up, their credit creation power is reduced and less loan can be made. The money supply falls.

2004/AL/II/1(c)

S: Monetary aggregates, both including currency (banknotes and coins) held by the non-bank public.

D: M0 (currency + reserves) vs. M1 / M2 / M3 (currency + deposits).

2007/AL/II/3(a)

The monetary base (or high-powered money) is the sum of currency held by the non-bank public (C) and reserves (R, deposits of commercial banks at the central bank plus vault cash).

The money supply is the sum of currency in circulation (C) and bank deposits (D).

Money supply is often a multiple of base money because, under fractional reserve banking, commercial banks are only required to keep a fraction of their deposits as reserves and are thus able to use their excess reserves to generate further deposits through their loan and portfolio investment activities.

The banking multiplier equals unity when commercial banks keep 100% reserves against their deposits.

2008/AL/II/1(c)

S: interest rates charged on short-term loans to commercial banks;

D: rate set by the central bank at the discount window vs. rate set by the (overnight) interbank loans market.

2012/DSE/II/13

(a) Required reserve ratio = $(\$500 \text{ million} - \$100 \text{ million}) / \$2\,000 \text{ million}$ (1)
= 20% (1)

(b) Maximum possible amount of deposits = $\$500 \text{ million} \times (1 / 0.2)$ (1)
= \$2 500 million (1)

(c) Deposit = $\$700 \text{ million} \times (1 / 0.2) = \$3\,500 \text{ million}$ (2)
Cash in public circulation = 0 (1)
Money supply = Deposit + Cash in public circulation = \$3 500 million (1)

2013/DSE/II/12

(a) Monetary base = $\$400 \text{ million} + \$1\,000 \text{ million} = \$1\,400 \text{ million}$ (2)
Money supply = $\$1\,000 \text{ million} + \$2\,000 \text{ million} = \$3\,000 \text{ million}$ (2)

(b) New money supply = $\$1\,000 \text{ million} + [\$4\,000 \times (1 / 25\%)] = \$2\,600 \text{ million}$ (2)

2014/DSE/II/7(b)

Advantages: An increase in money supply would lower the interest rate and raise investment, leading to an increase in real output in the short run (given upward-sloping aggregate supply). Given idle resources in the economy, more labour (and/or capital) would be used to produce the bigger output, implying an increase in employment or decrease in unemployment. (Contractionary fiscal policy would not result in an increase in GDP and employment.) (2)

Disadvantages: In the long run (with vertical aggregate supply), an increase in money supply would not create any effect on output and employment. But if the fiscal deficits grow so fast that the money supply has to be increased at a higher rate than of real output, high inflation would even arise. (Contractionary fiscal policy would not result in inflation.) (2)

2014/DSE/II/12

(a) Monetary base = cash in public circulation + reserves held by commercial banks (1)
The monetary base increases because the commercial banks have more reserves. (2)

(b) Credit creation process:

When the central bank purchases bonds from commercial banks, more cash is injected to the commercial banks as reserves and there would be excess reserves in the banking system. (1)

The banks would lend out the excess reserves. (1)

And the bank loans will be re-deposited into the banking system. (1)

[The process will go on and on (until the actual reserves are equal to required reserves).]

No. The monetary base will remain unchanged as the sum of cash in public circulation and reserves in the banking system will not be affected by the credit creation process. (2)

2015/DSE/II/6

(a) (i) Legal reserve ratio = $\$1\,000 \text{ million} / \$5\,000 \text{ million} = 0.2$ (1)

(ii) Money supply = $\$150 \text{ million} + \$5\,000 \text{ million} = \$5\,150 \text{ million}$ (2)

(b) New monetary base = $\$150 \text{ million} + \$1\,050 \text{ million} = \$1\,200 \text{ million}$ (2)

(c) New money supply = $\$150 \text{ million} + [\$1\,050 \text{ million} \times (1 / 0.2)] = \$5\,400 \text{ million}$ (3)

2016/DSE/II/8

- (a) Monetary base = \$500 million + \$1 000 million = \$1 500 million (1)
 Money supply = \$500 million + \$4 000 million = \$4 500 million (1)

- (b) Change in monetary base = \$1 500 million - \$1 500 million = 0 (2)

Change in money supply

- = New money supply - Original money supply
 = [\$500 million + \$1 000 million $\times$ (1 / 50%)] - \$4 500 million = -\$2 000 million (2)

OR

- = Change in cash held by non-bank public + Change in deposits
 = 0 + [\$1 000 million $\times$ (1 / 50%) - \$4 000 million] = -\$2 000 million (2)

- (c) (i) Under a fractional reserve banking system, banks are required to keep only a fraction of their deposits as reserves. They can create credit by lending out the remaining fraction in the form of loans to businesses and private individuals, which would eventually find their way back into the banking system in the form of deposits, resulting in a multiple increase in deposits / a banking multiplier larger than unity. (3)

- (ii) If the legal reserve ratio is 100% (RRR = 1), the money supply will be equal to the monetary base.

OR

- If all of the banks decide not to lend out their excess reserves even when $RRR < 1$ (e.g. during financial crisis, when default rates are high), the money supply will be equal to the monetary base. (1)

2018/DSE/II/7(a)

- (i) New M_0 = \$500 million + \$100 million + \$40 million = \$640 million (2)

- (ii) Banks do not hold excess reserves / there is sufficient demand for loans. (1)
 There is no cash leakage. (1)

- (iii) Old M_S = \$2 000 million + \$100 million = \$2 100 million
 $RRR = (\$500 \text{ million} - \$100 \text{ million}) / \$2 000 \text{ million} = 0.2$ (1)
 New M_S = \$100 million + [(\$500 million + \$40 million) $\times$ (1 / 0.2)] = \$2 800 million (2)
 $\therefore M_S$ increases by \$700 million (= \$2 800 million - \$2 100 million). (1)

[Credits would also be awarded to those candidates who reasoned that the public's initial cash holding of \$100 million might also be deposited in to the banking system to yield \$500 million of additional deposits, resulting in an aggregate increase of \$1 200 million of deposits and thus a net increase of \$1 100 million of money supply. As a side note, for the topic of money and banking at the DSE level, cash held by the general public should be treated as fixed (i.e., change of cash in public is assumed to be zero except under other changes).]

2023/DSE/II/12c

- (c) Increase in discount rate raises the cost for commercial banks to borrow money from the Federal Reserve. Monetary base tends to decrease. (1)
 Money supply decreases. (1)
 Interest rates increase.¹ (1)
 Investment decreases. (1)
 Aggregate demand decreases. Price level tends to decrease, relieving the pressure on the rising price level. (1)

2025/DSE/II/10(Cross Topic 5.1 /5.2 /5.3 /6.1)

(a) (i) Money supply growth rate = $(20\,837.4 - 21\,691.3) / 21\,691.3 \times 100\% = -3.94\%$

(1)

(ii) QTM: $MV = PY$, where M is money supply, V is velocity of circulation of money, P is general price level, Y is real output, assuming V to be constant.

(2)

Growth rate of Y $\approx$ growth rate of M – growth rate of P = $-3.94 - 3.6 = -7.54\%$

(1)

(b) As discount rate is lowered, the cost for commercial banks to borrow money from the Federal Reserve decreases.

(1)

Money supply increases.

(1)

Nominal interest rate decreases.

(1)

(c) Verbal elaboration:

– As interest rate decreases, investment increases.

(1)

– Aggregate demand increases.

(1)

– Real output increases.

(1)

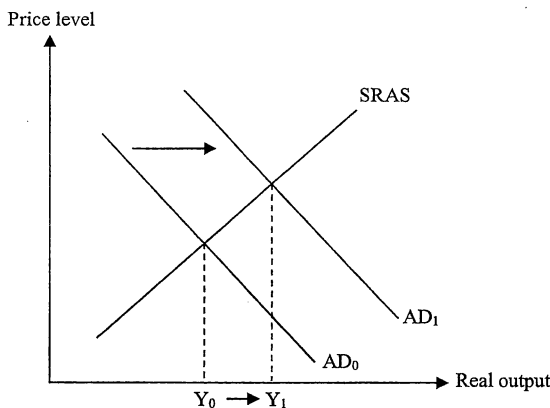
Indicate in the diagram:

– rightward shift of AD curve

(1)

– increase in Y

(1)



(d) Argument FOR:

– Increasing expenditure on infrastructure can increase the capital endowment in the future, raising the aggregate output in the long run.

(2)

Argument AGAINST:

– Increasing expenditure on infrastructure would raise the US Government's expenses, worsening the fiscal balance.

(2)

5.2 MONEY DEMAND

MARKING SCHEME

1990/AL/II/13 D	1995/AL/II/14 B	2005/AL/II/15 D (68%)	2013/DSE/I/27 B (67%)	2017/DSE/I/30 A (69%)
1991/AL/II/15 C	1995/AL/II/15 B	2008/AL/II/20 D (56%)	2015/DSE/I/28 A (55%)	2018/DSE/I/29 C (65%)
1993/AL/II/28 A	2001/AL/II/22 C	2009/AL/II/15 B (65%)	2016/DSE/I/29 A (41%)	2019/DSE/I/33 D
2020/DSE/I/28 D	2020/DSE/I/37 D	2020/DSE/I/31 D	2022/DSE/I/33 B	2022/DSE/35 A
2023/DSE/29 A	2024/DSE/II/27 A	2025/DSE/I/33 B		

Note: Figures in brackets indicate the percentages of candidates choosing the correct answers.

1991/AL/II/1

Consider a model with two assets: money and bonds (or interest-yielding assets).

The opportunity cost of holding money is the highest-valued forgone option (i.e. return to holding other assets).

When the price level is assumed to be fixed, the nominal interest rate is the same as the real interest rate.

2002/AL/II/3(a)

Yes, the interest rate, being the return to holding interest-bearing assets such as bonds, represents the opportunity cost of holding money – just like the (relative) price of a commodity being the amount of an alternative commodity forgone is the opportunity cost of acquiring that commodity.

2003/AL/II/4

- (a) Liquidity is the property of an asset that makes it easily marketable at short notice without loss. Since money is a universally accepted medium of exchange, it is clearly the most liquid asset on earth.
- (b) Money can be regarded as a risk-free asset in a world of stable prices, but not when prices may fluctuate over time. In addition to inflation risk, one can also imagine 'bank run' risks when one is thinking of money not merely as cash but also as bank deposits. (Instability in the purchasing power of money may also arise from exchange rate risk and political or country risk.)
- (c) Risk-averse agents are willing to hold money even though its expected rate of return is lower than that of other assets (such as bonds and stocks) because it is more liquid (less transaction-costly as a medium of exchange, i.e. transactions demand for money) and less risky (as a stock of value, i.e. asset demand for money) than other assets.

2009/AL/II/2

- (c) As implied by the Fisher equation, these two interest rates are related to the rate of expected (rather than actual) inflation. Both interest rates are determined today and to be paid tomorrow with certainty. Depending on the actual realization of the price level tomorrow (which is unknown today), however, the nominal interest may yield higher or lower return in real terms tomorrow (i.e. the actual real interest is not necessarily the same as expected).
- (d) The nominal interest rate, which is the rate of return on "bonds" (i.e., other financial assets forgone if people hold money), can be interpreted as the opportunity cost of holding money (whose rate of return is zero).

2020/DSE/II/9A

9 a) Oligopoly.

Price searcher
Homogeneous or heterogenous product
Difficult to enter the market
Imperfect market information
Interdependent of pricing strategies among firms
Non price competition (mark the first 3 points)

2023/DSE/II/10a

(a) Advantage of holding Silver Bonds over cash:

Purchasing power can be better preserved in times of inflation since the nominal rate of return on Silver Bonds rises with inflation while that of cash remains at zero regardless of inflation rate.

(2)

Disadvantage of holding Silver Bonds over cash:

Liquidity is lower since it takes days for Silver Bonds to be redeemed and converted into cash.

(2)

5.3 THE QUANTITY THEORY OF MONEY (QTM)

MARKING SCHEME

1990/AL/II/14 C	1995/AL/II/18 D	2000/AL/II/22 D	2005/AL/II/21 D (77%)	2014/DSE/I/37 D (56%)
1990/AL/II/26 C	1996/AL/II/14 A	2001/AL/II/23 C	2007/AL/II/11 B (77%)	2015/DSE/I/33 A (58%)
1991/AL/II/14 A	1996/AL/II/20 C	2002/AL/II/19 C (52%)	2008/AL/II/18 D (61%)	2016/DSE/I/34 D (34%)
1991/AL/II/16 A	1997/AL/II/12 B	2002/AL/II/23 B (73%)	2009/AL/II/22 C (71%)	2017/DSE/I/36 C (64%)
1992/AL/II/20 B	1997/AL/II/22 D	2003/AL/II/11 D	2011/AL/II/24 A (59%)	2018/DSE/I/37 B (72%)
1994/AL/II/17 D	1998/AL/II/14 C	2004/AL/II/19 C (44%)	2012/AL/II/16 C (69%)	2019/DSE/I/35 A
1994/AL/II/22 D	1999/AL/II/14 D	2005/AL/II/14 C (74%)	2012/AL/II/20 B (48%)	2021/DSE/I/38 A
1994/AL/II/24 B	2000/AL/II/16 D	2005/AL/II/18 A (56%)	2013/DSE/I/32 B (74%)	2023/DSE/I/37 B
2024/DSE/II/37 D	2025/DSE/I/39 B			

Note: Figures in brackets indicate the percentages of candidates choosing the correct answers.

1990/AL/II/8(a)(ii)

It all depends.

If the growth rate of the money supply approximately equals the growth rate of real output, then the inflationary consequence will be minimal.

However, if the growth rate of the money supply exceeds the growth rate of real output, inflation will occur.

1991/AL/II/2

Explain $MV = PY$ as an identity in accounting. It is only when V and Y are constant that a change in M gives rise to an equi-proportional change in P , the money price level. (M/P) is always a constant --- prediction of the classical QTM.

The classical QTM is used in conjunction with the classical theory, which states that (in the long run) flexible relative prices imply full employment. Y in this context refers to full employment output. The classical QTM requires the (long run) assumption of full employment rather than explaining full employment.

1993/AL/II/4

According to the Quantity Theory of Money, $PY = MV$

where P = price level, Y = output, M = money supply and V = velocity.

Assume V is constant. If the percentage change in income is 5% and the percentage change in the money supply is 10%, the inflation rate is (i.e. P increases by) 5% (= 10% - 5%).

(We cannot predict the rate of inflation if V is not constant.)

1999/AL/II/4

- (a) Inflation is related to monetary factors in the sense that, other things being equal, prices must rise as money supply is increased because we then have more money to spend on the same quantity of goods. [This can also be explained in terms of the quantity theory, assuming constant velocity and output.]
- (b) The quantity equation implies that the inflation rate is equal to the negative of the output growth rate when money supply and velocity are constant. In other words, prices can also rise when we use the same quantity of money to chase after a smaller quantity of goods.

2000/AL/II/8

- (a) Using the equation of exchange, we have $MV = PY$. If velocity is constant, nominal income (PY) must grow at the same rate as the money supply. As both Y and P can change, we cannot separate the impact on P and Y . If we know how Y grows, we can predict the impact of M on P .
- (b) (i) Given velocity is constant, nominal income grows at the same rate as the money supply.
 Nominal income growth rate in year 2 = $(110 - 100) / 100 = 10\%$
 Nominal income growth rate in year 3 = $(121 - 110) / 100 = 10\%$
- (ii) Inflation rate in year 2 = 10%
 Inflation rate in year 3 = $10\% - 3\% = 7\%$
- (c) If the velocity of circulation is constant and real income grows at a constant known rate, then setting the growth of money supply at a specific rate can achieve the target rate of inflation.

2002/AL/II/2(a)

True.

$$QTM: g_P = g_M + g_V - g_Y$$

$g_P > 0$ even when $g_M = 0$ as long as $g_V > g_Y$ (such as when velocity is constant and output is declining).

2007/AL/II/3(b)

The velocity of circulation of money is the number of times an average dollar changes hands – as a medium of exchange to facilitate market transactions – within a certain time period.

Using the Equation of Exchange, we can measure the velocity by the ratio of nominal GDP to money supply.

2010/AL/II/2(c)

MV represents the total amount of circulating money in the economy within a given period of time. PY or GDP represents the total money value of (final) goods and services produced in the economy within the same period. $MV = PY$ because, in a monetary economy, money changes hands whenever buyers use the money they hold (supplied by the central bank) to purchase goods and services from the sellers.

The Equation of Exchange implied that:

$$g_P = g_M + g_V - g_Y$$

which implies in turn 2 possible sources of inflation (Quantity Theory of Money as a theory of inflation):

- growth in the money supply --- i.e., $g_M > 0$, given $g_V = g_Y = 0$ (or given $g_M > g_Y$ and $g_V = 0$); and
- output decline --- i.e., $g_Y < 0$, given $g_M = g_V = 0$.

2012/DSE/II/13(d)

QTM: $MV = PY$, where M = money stock, V = velocity of circulation of money, P = general price level and Y = real output, assuming V constant. (2)

An increase in M leads to an increase in P when Y increases by a smaller proportion.

OR

An increase in M leads to an increase in P by the same percentage when Y is constant. (2)

2018/DSE/II/7(b)

$MV = PY$, where M : money supply, V : velocity of circulation of money, P : price level, and Y : output (2)

Assume V and Y are both constant. Then when M increases, P would increase by the same proportion.

OR

Assume V is constant. Then when the percentage increase in M is higher than that in Y , P would increase as well. (2)

2020/DSE/II/4C

C) M = Nominal Money Supply

V = Velocity of Money

P = General price level.

Y = Real National Income

While both V and Y are assumed to be constant in the long run. Percentage change of Money supply equal to the percentage change of price level. Continuously expansionary monetary policy (increase of M) will lead to persistent increase of general price level (increase of P) (4 marks)

2022/DSE/II/6a

(a) QTM: $MV = PY$, where (4)

M = money supply,

V = velocity of circulation of money, assumed to be constant,

P = price level, and

Y = real output.

% increase in Y being larger than % increase in M would result in a drop in P , i.e. deflation occurs.¹

2023/DSE/DSE/II/12b

(b) Quantity Theory of Money: $MV = PY$ (1)

M = money supply

V = velocity of circulation of money

P = price level

Y = real output

Assuming V to be constant; (1)

as M increases, P increases if the growth rate of M exceeds that of Y . (1)

SECTION 6: AGGREGATE DEMAND AND AGGREGATE SUPPLY

(Note: Despite the fact that the AD-AS model was not explicitly required in the CE and AL syllabuses, in tackling a number of CE/AL questions, the AD-AS framework can still be applied.)

6.1 DEMAND- AND SUPPLY-SIDE FACTORS AFFECTING NATIONAL INCOME

MARKING SCHEME

1990/CE/II/37 C	1993/CE/II/38 A	1997/CE/II/44 D	2001/CE/II/30 D	2008/CE/II/28 D (55%)
1991/CE/II/41 B	1993/CE/II/43 D	1997/CE/II/46 B	2004/CE/II/31 A (31%)	2019/DSE/I/27 A
1991/CE/II/45 B	1993/CE/II/51 A	2000/CE/II/34 C	2005/CE/II/30 C (73%)	2019/DSE/I/38 C
1992/CE/II/49 B	1997/CE/II/22 A	2000/CE/II/38 C	2006/CE/II/32 C (48%)	2020/DSE/I/27 C
2020/DSE/I/35 B	2021/DSE/I/36 B	2022/DSE/I/28 D	2022/DSE/I/37 A	2023/DSE/I/28 A
2023/DSE/I/30 A	2025/DSE/I/37 C			

- 1990/CE/I/3/(b)(i)
 HK\$ depreciate
 ⇒ Export prices ↓ (1)
 ⇒ Export volume ↑ (2)
 ⇒ Boom in export sector (2)
 ⇒ Aggregate demand ↑ ⇒ price level ↑ (2)
 (max: 6)
- 1990/CE/I/05(b)(ii)
 ↑ sales tax on all commodities ⇒ prices of all goods ↑ (2)
 ↓ standard rate ⇒ income after tax of the rich ↑
 ⇒ consumption ↑ ⇒ aggregate demand ↑ ⇒ general price level ↑ (4)
- 1992/CE/I/5(c)
 (ii) a continuous rise in the general price level (2)
- (iii) Construction of the airport
 ⇒ ↑ in aggregate demand for resources / demand for workers (2)
 ⇒ inflation occurs when Hong Kong is near full employment (2)
- 1994/CE/I/10(a)(ii)
Local construction workers:
 - more job opportunities
 - income ↑
Whole economy:
 - consumption expenditure ↑ / more business
 - GNP ↑
 - employment ↑
 - social capital ↑
 - higher economic growth

1995/CE/I/9(c)

- (i) - consumption expenditure
- investment expenditure
- government expenditure
- exports or net export
- imports or net export

(Mark the **FIRST TWO** points only.)

(1@, max: 2)

- (ii) [Note: the answer must match with the answer to (c)(i)]

Consumption expenditure $\uparrow \Rightarrow$ GDP $\uparrow$

(1)

e.g. US residents spent more on food, hotel accommodation, traveling, souvenirs, etc.

(2)

OR

Investment expenditure $\uparrow \Rightarrow$ GDP $\uparrow$

(1)

e.g. more investment in the provision of transport service, accommodation, food supplies, etc.

(2)

OR

Government expenditure $\uparrow \Rightarrow$ GDP $\uparrow$

(1)

e.g. US government expenditure on infrastructure related to the event, expansion / maintenance of stadiums, etc.

(2)

OR

Net exports $\uparrow \Rightarrow$ GDP $\uparrow$

(1)

e.g. more tourists' spending on transport, accommodation, food, clothing, souvenirs, etc.

(2)

(3@, max: 6)

(Mark the **FIRST TWO** points only.)

1996/CE/I/12(c)

Demand-side factor:

- government expenditure (in current year) increases

(2)

Supply-side factor:

- the labour productivity would be raised

(2)

OR

- labour supply would increase because workers originally outside the labour force will be attracted

(2)

(max: 2)

$\therefore$ GDP (both in current year and in future) would increase.

(1)

(Mark the **FIRST** demand-side factor and the **FIRST** supply-side factor only.)

1997/CE/I/9(d)(i)

prices of most goods and services after tax $\uparrow$ / cost of production $\uparrow$ / supply of goods and services $\downarrow$

(1)

$\therefore$ price level $\uparrow$

(1)

2004/CE/I/10

After-tax income (i.e. "disposable income") increased.

(2)

Consumption expenditure would increase.

(2)

$\therefore$ GDP would increase.

2005/CE/I/11(a)

Zero tariff encourages more export of goods.

(2)

Easier market entry encourages more investment.

(2)

Both measures will increase the GDP of Hong Kong.

(1)

2007/CE/I/8(b)(ii)

- a larger labour force

- These top-quality Mainland students have a higher productivity.

- These top-quality Mainland students can increase the productivity of the local employees when they work together / brings knowledge and new ideas to Hong Kong.

(1@, max: 2)

[Mark the **FIRST TWO** points only]

2009/CE/10(a)

- net exports: more net exports (or more export of services) due to more tourists from mainland China
- consumption: more earnings and income will boost consumption expenditure
- investment: more investment (local and from the Mainland), e.g., more hotels and trading firms.
- government expenditure: more government expenditure, e.g., more on infrastructure to facilitate the trade (2@, max: 4)

2019/CE/19(c)

- (i) - more facilities (attractions) will be constructed $\Rightarrow$ increase in investment expenditure (2)
- (ii) - more tourists from mainland China and other places $\Rightarrow$ increase in net exports (export of services)
- OR
- more local visitors $\Rightarrow$ increase in private consumption expenditure (2)

2019/DSE/II/10

(b) Verbal elaboration:

(4)

The middle income group is charged a lower average tax rate, their disposable income rises, consumption and aggregate demand increase, narrowing the deflationary gap.

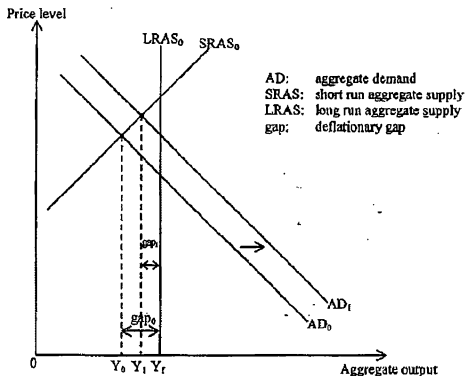
Illustrate in the diagram:

- correct initial deflationary gap
- AD shift rightward ²
- correct new Y with a smaller gap

(1)

(1)

(1)



For: Pros of the monetary policy OR Cons of the fiscal policy

(2)

- The monetary policy increases Y via investment which will help enhance present/future productivity (while the tax policy may not have such an effect).
- The tax policy may worsen the fiscal balance of the government (while the monetary policy will not have such an effect).
- any other relevant point

Against: Pros of the fiscal policy OR Cons of the monetary policy

(2)

- The tax policy may narrow the income gap (while the monetary policy won't have such an effect).
- Bank loans and money supply may not increase even if there is a fall in the required reserve ratio (while the tax policy will not be affected by the action of commercial banks).
- any other relevant point

D) The investment may include importer raw material which is not the contribution of local resident producing unit.

The investment may include the value of inventories which are not produced in current year.

The investment may include the value of intermediate goods (electricity and other raw materials) which are the contribution of other resident producing unit.

(Mark the first 2 answers) (3 marks)

2023/DSE/II/7b

(b) Verbal elaboration:

- As more travellers visit the country, export of services increases. (1)
- Aggregate demand increases. (1)
- Aggregate output increases, narrowing the deflationary (output) gap. (1)

Indicate in the diagram:

- rightward shift of AD curve (1)
- increase in Y, decrease in deflationary (output) gap (1)

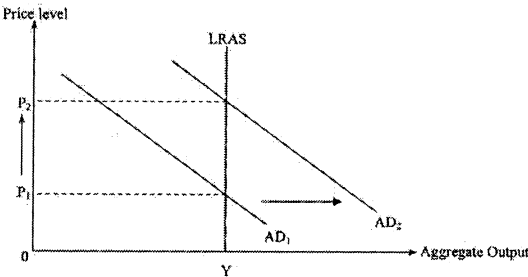
6.2 LONG RUN AND SHORT-RUN EQUILIBRIUM

MARKING SCHEME

1994/AL/II/9 A	2012/DSE/I/34 C (51%)	2014/DSE/I/29 D (69%)	2015/DSE/II/39 B (84%)	2018/DSE/I/26 A (80%)
1997/AL/II/21 D	2012/DSE/I/36 D (52%)	2014/DSE/I/34 C (42%)	2016/DSE/II/26 D (65%)	2018/DSE/I/27 D (76%)
1999/AL/II/4 C	2013/DSE/I/23 D (76%)	2014/DSE/I/35 B (67%)	2016/DSE/II/27 C (57%)	2018/DSE/II/28 B (80%)
2000/AL/II/20 D	2013/DSE/I/24 B (65%)	2014/DSE/I/38 D (75%)	2016/DSE/II/28 B (64%)	2018/DSE/I/38 D (58%)
2001/AL/II/14 A	2013/DSE/I/29 A (56%)	2015/DSE/I/26 A (62%)	2017/DSE/II/28 C (76%)	2018/DSE/I/39 A (65%)
2012/AL/II/30 C (79%)	2013/DSE/I/30 D (59%)	2015/DSE/I/27 D (68%)	2017/DSE/II/29 D (60%)	2020/DSE/I/ C
2012/DSE/II/25 A (72%)	2014/DSE/I/27 A (61%)	2015/DSE/I/37 C (59%)	2017/DSE/I/37 B (66%)	2021/DSE/II/27 D
2012/DSE/I/26 A (73%)	2014/DSE/I/28 C (48%)	2015/DSE/I/38 A (73%)	2017/DSE/II/40 B (45%)	2021/DSE/I/28 C
2022/DSE/I/29 B	2022/DSE/I/36 C	2022/DSE/I/38 A	2024/DSE/II/29 B	2024/DSE/II/30 C
2025/DSE/II/28 D	2025/DSE/I/29 A	2025/DSE/I/30 A		

Note: Figures in brackets indicate the percentages of candidates choosing the correct answers.

- 2012/DSE/II/10(c)
- People’s consumption expenditure increases as their disposable income increases. (2)
- Aggregate demand increases. (1)
- The price level increases, but aggregate output remains unchanged in the long run. (2)



- Indicate in the diagram:
- AD shifts out (1)
- Vertical LRAS (1)
- P increases and Y unchanged (1)

- (a) Wealth effect: As many of the assets in the economy are denominated in nominal values, the higher the price level, the lower the purchasing power of money. This reduces the wealth of the economy (in real terms). As a result, households and firms reduce their purchases of all goods and services and the real output drops.

OR

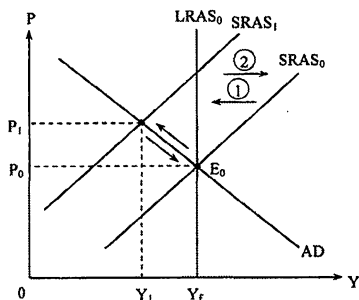
Interest rate effect: As the price level rises, households and firms demand more money to finance their transactions. Given the fixed supply of money, the interest rate would rise. An increase in interest rate would cause decline in investment and consumption and also the real output.

OR

Net exports effect: As the domestic price level rises, foreign-made goods become relatively cheaper so the quantity demanded of imports increases. However, the rise in the domestic price level also means that domestic-made goods are relatively more expensive to foreign buyers so the quantity demanded of exports decreases. When the volumes of exports decrease and of imports increase, net exports will decrease, so as the real output. (3)

- (b) Increase in the price of raw materials results in a rise in factor prices. So the SRAS will decrease. Price level increases and the output will fall in the short run.

The resulting excess supply of labour / deflationary (output) gap will result in a downward pressure on the input prices. In the long run, wages will adjust downward and the SRAS will increase and the price level will fall back to the original level. (5)



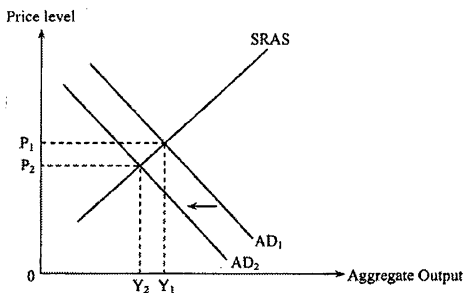
Indicate in the diagram:

- SRAS shifts to the left (1)
- P increases, Y drops (1)
- SRAS shifts back to the original position (1)
- final fall in P, back to the original level (1)

2013/DSE/II/12(c)

An upward adjustment of required reserve ratio reduces the money supply, resulting in an increase in interest rate and a drop in investment expenditure. (3)

The aggregate demand will fall, and so will the price level and output level. (2)



Indicate in the diagram:

- AD shifts to the left
- price level falls
- output level falls

(1)
(1)
(1)

2014/DSE/II/10(b)

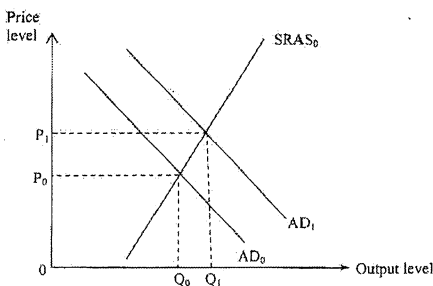
Relaxation of one-child policy would likely result in a rise in the birth rate, implying a bigger work force and a bigger aggregate supply and output in the future. (3)

2014/DSE/II/12(c)

The purchase of government bonds increases money supply, resulting in a drop in the interest rate. (2)

Investment / consumption expenditure increases and aggregate demand increases. (2)

In the short run, the output and price levels increases. (1)



Indicate on the diagram:

- AD curve shifts to the right
- higher price level
- higher output

(1)
(1)
(1)

2015/DSE/II/8

Proposal 1:

More money spent on education would increase government expenditure, leading to an increase in aggregate demand and thus aggregate output.

OR

Free education for longer years would improve labour productivity and thus aggregate output over time.

OR

Provision of free education beyond normal school age would reduce labour supply (by keeping more people at school) and thus aggregate output in the current period. (2)

Proposal 2:

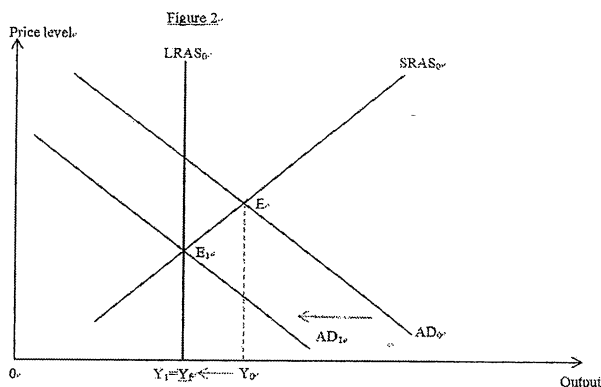
Unemployment benefits would increase the disposable income and consumption of the jobless, and thus aggregate demand, resulting in a rise in aggregate output.

OR

Unemployment benefits may, however, produce disincentive effects on labour supply, resulting in de-skilling of labour and a drop in aggregate output now and in the future. (2)

2015/DSE/II/12

- (b) Export would fall, (1)
 resulting in a drop in aggregate demand, (1)
 so output would fall back to Y_F . (1)



Indicate on the diagram:

- AD curve shifts to the left (1)
- correct position of initial short-run aggregate output Y_0 (1)
- correct position of new short-run aggregate output $Y_1 (=Y_F)$ (1)

- (c) Hong Kong is facing an inflationary gap. (1)
 There is an excess demand in the factor market, (1)
 creating pressure for factor prices to adjust upwards, (1)
 so that costs of production would increase over time and short-run aggregate supply (SRAS) would fall over time (, restoring output to Y_F in the long run). (1)

2016/DSE/II/5

When the price level increases, the adjustment of input prices is imperfect / incomplete (e.g. due to the long term contracts of factor of production), so the real cost of production -- i.e. nominal cost divided by the general price level -- will fall. Firms would thus use more factor inputs to produce larger output. Therefore, the SRAS curve is upward-sloping, i.e. a higher price level would result in a higher output level. (3)

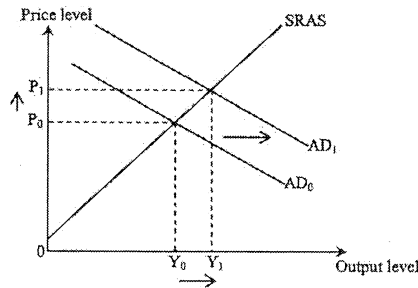
2016/DSE/II/12(b)

The cash-sharing scheme would raise the disposable income of the residents and therefore result in an increase in consumption.

(2)

Aggregate demand would rise, raising price and output levels accordingly.

(2)



Indicate in the diagram:

- rightward shift of AD

(1)

- higher P and Y

(1)

2017/DSE/II/9

(a) Depression / trough

(1)

(b) A reduction in interest rate would result in a rise in investment and/or consumption.

(1)

Aggregate demand would thus increase, and so would aggregate output as well.

(2)

(c) Consistent.

(1)

When interest rate dropped in 2009, GDP growth rate increased and unemployment rate dropped.

(3)

(d) - Subsidizing research and development

- importation of labour

- investment in infrastructures

(1)

[Mark the **FIRST** policy only.]

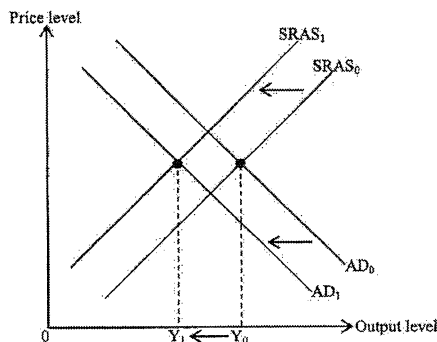
could increase potential output or production capacity of the economy as a whole in the long run, and thus shift LRAS rightwards.

(1)

2017/DSE/II/12(b)

The new visa requirement would increase labour cost for the British firm owners, resulting in a drop in SRAS.
 Political uncertainty would result in a drop in investment from the global investors and thus a fall in AD.
 Aggregate output would thus fall.

(2)
 (2)
 (1)



Indicate in the diagram:

- leftward shift of AD curve
- leftward shift of SRAS curve
- lower output

(1)
 (1)
 (1)

2017/DSE/II/13(b)

Proposal A's effect on aggregate output:

An increase in salaries tax would reduce AD via a drop in disposable (after-tax) income and thus consumption. As a result, aggregate output would fall.

[An increase in salaries tax may also reduce SRAS by creating a disincentive effect on labor.]

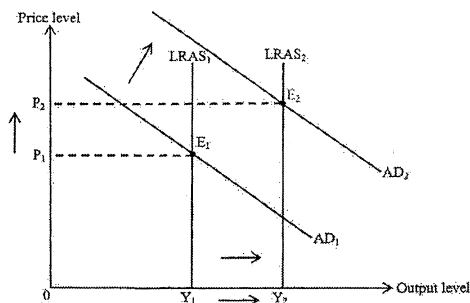
(4)

Proposal B's effect on aggregate output:

There would be no extra effect on aggregate output as a component of AD/AS does not change.

(2)

2018/DSE/II/9



Indicate in the diagram:

- rightward shift of AD (due to more visitors from the Mainland)
- rightward shift of LRAS (due to faster development of the HK economy)
- correct position of P_2 (at E_2), with a bigger horizontal shift in AD than in LRAS

(1)
 (1)

2018/DSE/II/11

The SRAS curve would decrease, as less labour is available for production.

(1)

The AD curve would also decrease, as the UN embargo reduces exports from Country Z.

(1)

As a result, aggregate output would fall.

(1)

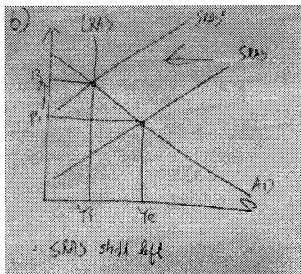
2018/DSE/II/13(a)(i)

Potential output would drop as the working-age population gradually decreases.

(2)

(from 5.36M in 2014 to 4.76M in 2054)

2020/DSE/II/6



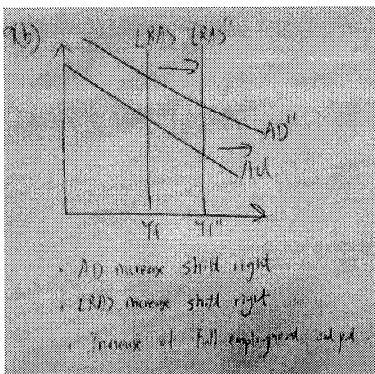
Inflationary gap exist as short run output is larger than full employment output.

Excess demand in labor market will drive up input price. (1 mark)

Production cost of firms increase and SRAS decrease and shift left (1 mark)

Short run output will decrease to full employment output level to close the inflationary gap (1 mark)

2020/DSE/II/9B



B) Infrastructure development will boost Government consumption expenditure (eg. Employment of more workers) and Gross investment expenditure (purchase of more capital good). Both G and I will increase the aggregate demand and the aggregate demand curve will shift rightward. (3 marks)

The construction of infrastructure will increase the quantity of factor endowment of an economy hence the production capacity will increase. LRAS increase and shift rightward. Full employment level will increase. (3 marks)

(a) Verbal elaborations:

Deflationary (output) gap refers to the difference between aggregate output (Y) and the full-employment output (Y_f) when Y is below Y_f . (1)

Illustrate in Figure 2:

– correct position of gap₀ (1)

(b) Verbal elaborations:

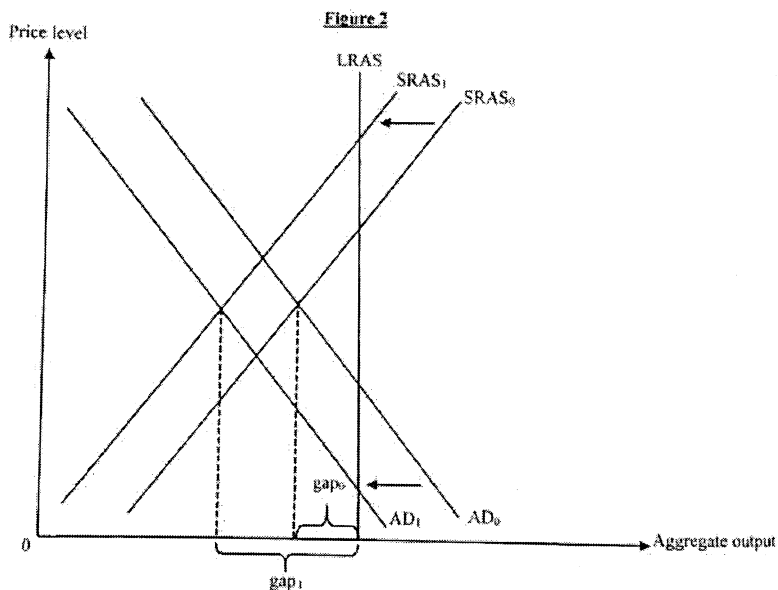
A temporary closure of factories would reduce aggregate supply in the short run. On the other hand, loss of jobs and pessimism would lower consumption (and possibly investment) expenditure and thus reduce aggregate demand. Together, shifts of the two curves would result in a drop in aggregate output (Y). Assuming, for simplicity, no change in long run aggregate supply and thus full-employment output (Y_f), then the fall in Y implies a wider deflationary gap. ⁶ (4)

Illustrate in Figure 2:

– leftward shift of SRAS curve (1)

– leftward shift of AD curve (1)

– correct position of gap₁ (1)

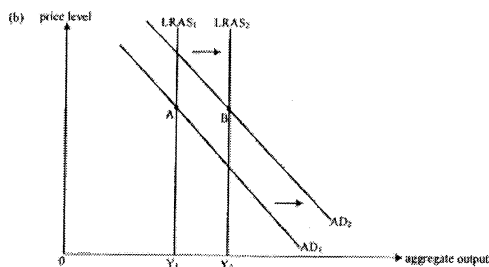


Marks

(4)

- (c) The central bank could use open-market purchase (i.e. buy government bonds from the public) to increase money supply. Then, the interest rate would fall, thus inducing firms to invest more, implying an increase in aggregate demand. As a result, aggregate output (Y) would rise in the short run, but long-run output (Y_f) would remain unchanged, so that the deflationary gap ($= Y - Y_f$) would become narrower.

2022/DSE/II/9b



Illustrate in the diagram:

- arightward shift in the AD curve (1)
- arightward shift in the LRAS curve (1)
- ahigher level of aggregate output (1)

Verbal elaboration:

Better infrastructure would lead to an increase in production capacity and potential Output and thus an increase in long-run aggregate supply. More infrastructure investment would lead to an increase in aggregate demand. Aggregate output would hence increase. (5)

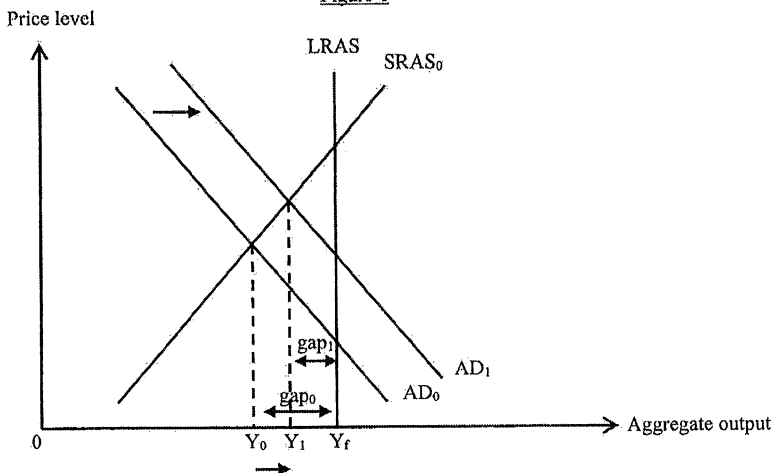
2023/DSE/II/7

(a) Indicate in the diagram:

- correct position of vertical LRAS curve and deflationary (output) gap

(1)

Figure 1



(b) Verbal elaboration:

- As more travellers visit the country, export of services increases.
- Aggregate demand increases.
- Aggregate output increases, narrowing the deflationary (output) gap.

(1)

(1)

(1)

Indicate in the diagram:

- rightward shift of AD curve
- increase in Y , decrease in deflationary (output) gap

(1)

(1)

2023/DSE/II/12c

- (c) Increase in discount rate raises the cost for commercial banks to borrow money from the Federal Reserve. Monetary base tends to decrease. (1)
- Money supply decreases. (1)
- Interest rates increase.¹ (1)
- Investment decreases. (1)
- Aggregate demand decreases. Price level tends to decrease, relieving the pressure on the rising price level. (1)

(a) Reasons:

- Wealth effect: As many of the assets in the economy are denominated in nominal values, the higher the price level, the lower the purchasing power of these monetary assets. This reduces the wealth of the economy (in real terms). As a result, consumption decreases and the aggregate quantity of output demanded drops.
- Interest rate effect: As the price level rises, households and firms demand more money to finance their transactions. Given the fixed supply of money, the interest rate would rise. An increase in interest rate would cause a decline in investment and consumption and thus the aggregate quantity of output demanded.
- Net exports effect: As the domestic price level rises, foreign-made goods become relatively cheaper so the quantity demanded of imports increases. The rise in the domestic price level also means that domestically produced goods are relatively more expensive to foreign buyers, so the quantity demanded of exports decreases. When the volume of exports decreases and that of imports increases, net exports will decrease (if the price elasticities of demand for exports and/or imports are sufficiently high). As net exports are a component of aggregate expenditure, the aggregate quantity of output demanded would drop as net exports decline.

3@
max: 3

[Mark the FIRST point ONLY.]

(b) Verbal elaboration:

The economy is facing a deflationary (output) gap. (1)

There is an excess supply in the factor market. (1)

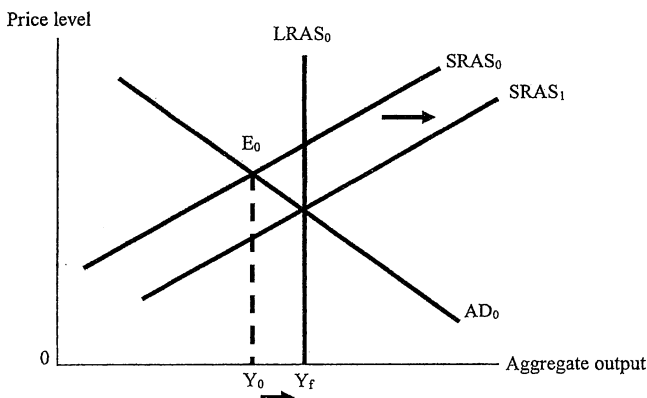
It creates pressure for factor prices to adjust downward. Costs of production would drop. (1)

Short run aggregate supply (SRAS) would increase over time, (1)

thus restoring output level to the full-employment output level, Y_f , in the long run.

Indicate in the diagram:

- Rightward shift of short run aggregate supply curve, reaching Y_f . (1)



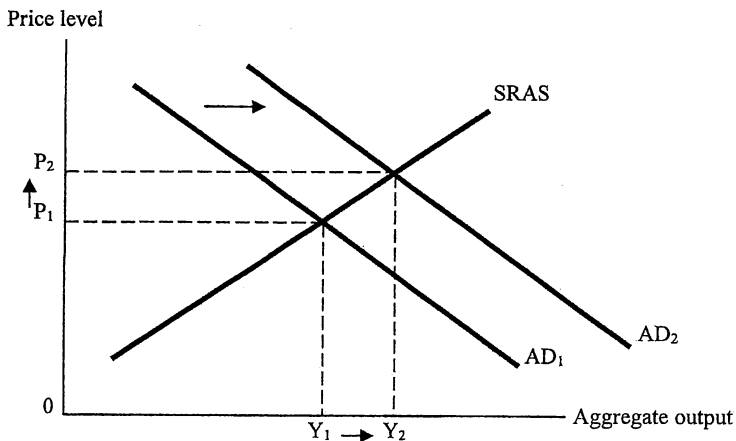
2024/DSE/II/12(a)

(a) Verbal elaboration:

- Due to the construction, gross investment expenditure will increase. (1)
- Aggregate demand will increase. (1)
- Aggregate output will increase in the short run. (1)

Indicate in the diagram:

- rightward shift of aggregate demand curve (1)
- increase in aggregate output (1)



6.3 FISCAL BALANCE AND TRADE BALANCE

MARKING SCHEME

1996/AL/II/26 A	2012/DSE/I/31 C (40%)	2016/DSE/I/39 A (69%)	2017/DSE/I/39 D (51%)	2019/DSE/I/40 A
2003/AL/II/26 C	2013/DSE/I/31 B (45%)	2016/DSE/I/40 C (34%)	2018/DSE/I/36 D (62%)	2021/DSE/I/29 D
2021/DSE/I/37 A	2023/DSE/I/45 C	2025/DSE/I/13 A	2025/DSE/I/14 D	2025/DSE/I/38 D

2002/AL/II/6

- (a) The 3rd explanation, fall in exports, can be a common cause of unemployment and the twin deficits.

$$X \downarrow \Rightarrow AD \downarrow \Rightarrow$$

$Y \downarrow \Rightarrow$ [Effect 1:] unemployment

$Y \downarrow \Rightarrow$ [Effect 2:] tax revenue $\downarrow \Rightarrow$ fiscal deficit

$Y \downarrow \Rightarrow$ [Effect 3:] imports $\downarrow \Rightarrow$ trade deficit

[since the effect of ΔX on NX is bigger than its effect through ΔM , i.e. $X \downarrow > M \downarrow$]

Note: For a given size of the workforce, an increase in unemployment implies a decrease in labour employment, which implies in turn a decrease in output (Y), given the production technology and other factors of production.

- (b) Fiscal contraction has the following consequences:

$$G \downarrow \text{ and } T \uparrow \Rightarrow AD \downarrow \Rightarrow$$

$Y \downarrow \Rightarrow$ [Effect 1:] unemployment

$Y \downarrow \Rightarrow$ [Effect 2:] tax revenue $\downarrow$ but fiscal balance still $\uparrow$

[because the effect of ΔG on $(T - G)$ is bigger than its effect through ΔT , i.e. $G \downarrow > T \downarrow$]

$Y \downarrow \Rightarrow$ [Effect 3:] imports $\downarrow \Rightarrow$ trade balance $\uparrow$

(No,) this single stone can at best kill two birds, but not all three.

2006/AL/II/6(a)(i)

Income taxes as expenditure-reducing policy

Income tax rate $\uparrow$

$\Rightarrow$ disposable income $\downarrow$

$\Rightarrow$ consumption demand and import expenditure

$\Rightarrow$ improvement in trade balance (given exports) $\downarrow$

2023/DSE/II/12d

- (d) Marks for effective communication (EC: max 2 marks)

Marks	Performance
2	● Support arguments with the source/data and appropriate economic theories. ● Present relevant material. ● Present well-organised and coherent answers without repetition of ideas. ● Use language that expresses ideas clearly and fluently with appropriate use of words/terms/symbols.
1	● Present arguments with some support of the source/data and economic theories. ● Present some irrelevant material. ● Present answers in a less organised way with some repetition. ● Use language that conveys a clear message with some inappropriate use of words/terms/symbols.
0	● Present arguments with no support of the source/data and economic theories. ● Present material unrelated to the gist of the question. ● Present inconsistent arguments. ● Express limited ideas with inappropriate use of words/terms/symbols.

SECTION 7: INTERNATIONAL TRADE

7.1 GAINS FROM FREE TRADE

MARKING SCHEME

1990/CE/II/42 A	1996/CE/II/53 B	2001/CE/II/44 D	2006/CE/II/44 B (65%)	2013/DSE/I/38 A (50%)
1990/CE/II/48 D	1997/CE/II/53 B	2002/CE/II/44 A (44%)	2006/CE/II/46 A (54%)	2013/DSE/I/39 A (64%)
1991/CE/II/53 A	1998/CE/II/43 B	2002/CE/II/45 C (44%)	2007/CE/II/44 C (67%)	2014/DSE/I/41 A (59%)
1992/CE/II/51 A	1998/CE/II/44 A	2002/CE/II/47 C (69%)	2007/CE/II/45 D (51%)	2014/DSE/I/44 D (39%)
1992/CE/II/53 C	1998/CE/II/45 B	2003/CE/II/45 C (31%)	2007/CE/II/46 B (62%)	2015/DSE/I/40 A (66%)
1993/CE/II/52 A	1999/CE/II/46 A	2003/CE/II/46 B (24%)	2008/CE/II/45 C (65%)	2015/DSE/I/41 D (45%)
1994/CE/II/52 A	1999/CE/II/48 D	2004/CE/II/45 B (50%)	2009/CE/II/46 C (43)	2016/DSE/I/41 B (60%)
1994/CE/II/53 C	2000/CE/II/46 D	2004/CE/II/46 D (40%)	2010/CE/II/45 A (61)	2017/DSE/I/43 B (65%)
1995/CE/II/51 B	2000/CE/II/47 A	2004/CE/II/47 C (65%)	2010/CE/II/46 A (64)	2018/DSE/I/43 D (38%)
1995/CE/II/52 D	2000/CE/II/48 B	2005/CE/II/45 B (47%)	2010/CE/II/47 B (51)	1992/AL/II/17 A
1996/CE/II/46 D	2001/CE/II/42 A	2005/CE/II/46 A (47%)	2012/DSE/I/37 D (65%)	1996/AL/II/25 A
1996/CE/II/47 D	2001/CE/II/43 A	2006/CE/II/43 C (43%)	2012/DSE/I/38 B (47%)	1997/AL/II/15 D
2019/DSE/I/42 C	2020/DSE/I/42 C	2021/DSE/I/43 C	2022/DSE/I/43 B	2023/DSE/I/42 D
2024/DSE/II/42 B	2024/DSE/II/44 D	2025/DSE/I/15 B	2025/DSE/I/42 D	

1991/CE/I/3/(c)

(i) The opportunity cost of producing 1 unit of food

in Country A: $12 \div 6 = 2$ units of clothing

(2)

in Country B: $20 \div 5 = 4$ units of clothing

(2)

Country A has a comparative advantage over Country B in food production because

A produces food at a relatively lower opportunity cost than B.

(3)

(ii) Increase in food = $(6 \text{ units} \times 20) - [(6 \text{ units} \times 10) + (5 \text{ units} \times 10)]$

(1)

= 10 units

(1)

Increase in clothing = $(20 \text{ units} \times 20) - [(12 \text{ units} \times 10) + (20 \text{ units} \times 10)]$

(1)

= 80 units

(1)

(iii) Terms of trade / exchange ratio of trading the two commodities

(2)

1992/CE/1/2/(a)

- (ii) The opportunity cost of producing 1 unit of rice
in Country A is 0.5 units of clothing whereas that (2)
in Country B is 0.75 units of clothing (2)

Country A produces rice at a lower opportunity cost than Country B (1)

∴ Country A should specialize in rice production (1)

- (iii) The opportunity cost of Country A in producing 18 ($= 2 \times 9$) of rice is 9 ($= 1 \times 9$) units of clothing (2)

⇒ Country A gains 1 unit of clothing from every 18 units of rice sold to Country B (2)

OR

The opportunity cost of Country A in producing 10 ($= 1 \times 10$) units of clothing is 20 ($= 2 \times 10$) units of rice (2)

⇒ Country A saves up 2 units of rice when it buys 10 units of clothing from Country B (2)

- (iv) No, because (1)

the outputs of Country A are doubled but the opportunity cost of producing rice remains the same as in (a)(ii) above

(i.e. 1 unit of rice is exchanged for 0.5 units of clothing). (3)

1993/CE/1/2/(a)

- (i) The opportunity cost of one unit of clothing is 2 units of toys, (1)
while that for toys is 1/2 units of clothing. (1)

No, we can't, because

to find out the comparative advantage, one has to compare the opportunity cost of toy production in the U.S.A with that in China (and see which cost is lower). (3)

- (ii) The principle of comparative advantage: if each country specializes in producing the good in which it has a lower opportunity cost, then the total output will increase. (3)

The opportunity cost of producing clothing (or toys) in China is the same as that in the U.S.A., i.e. 1 unit of clothing is exchanged for 2 units of toys. (2)

⇒ there is no gain from specialization and trade (1)

- (iii) The new opportunity cost of producing one unit of clothing in China is 3/2 units of toys which is smaller than that (2 units) in the U.S.A. (2)

⇒ China should specialize in producing clothing and the U.S.A. should specialize in producing toys. Trade between the two countries can benefit both. (3)

1994/CE/1/9/(c)

Comparative advantage

Unit cost of car production in Japan in terms of apples $= 5/2 (= 2.5)$ units (1)

Unit cost of car production in the U.S.A. in terms of apples $= 4/1 = 4$ units (1)

∴ Japan has a comparative advantage in producing cars. Conversely, the U.S.A. has a comparative advantage in growing apples. (1)

OR

Unit cost of apples growing in Japan in terms of cars $= 2/5 (= 0.4)$ units (1)

Unit cost of apples growing in U.S.A. in terms of cars $= 1/4 (= 0.25)$ units (1)

∴ The U.S.A. has a comparative advantage in growing apples. Conversely, Japan has a comparative advantage in producing cars. (1)

Gain from trade: Japan

Before specialization (within Japan) 2.5 units of apples (1)

After specialization and trade 3 units of apples (1)

Gain $(3 - 2.5 =) 0.5$ units of apples (1)

Units of apples obtained from 1 unit of cars

Gain from trade: U.S.A.

Before specialization (within the USA) 0.25 units of cars (1)

After specialization and trade 0.3333 units of cars (1)

Gain $(0.3333 - 0.25 =) 0.0833$ units of cars (1)

Units of cars obtained from 1 unit of apples

1995/CE/I/8

- (a) The principle of comparative advantage.
If each country specializes in producing the good in which it has a lower opportunity cost, then the total output will increase. (1)

(b) $1.5Y < 1X < 4Y$

OR

$0.25X < 1Y < 0.66X$ (4)

1996/CE/I/10(a)

- (i) Within A, the production cost of 1 unit of clothing is 3 units of rice. (1)
Within B, the production cost of 1 unit of clothing is $3/2$ units of rice. (1)
 $\therefore$ B has a lower cost or a comparative advantage over A in the production of clothing. (1)

- (ii) (I) Under no trade, the unit price of clothing in A is 3 units of rice forgone. When A trades with B, the import price of 1 unit of clothing is 2 units of rice exported to B.
 $\therefore$ A can save 1 unit of rice per unit of clothing imported from B.

OR

Similarly, the unit price of rice in A before trade is $1/3$ units of clothing. With trade, the export price of 1 unit of rice is $1/2$ units of clothing imported.

$\therefore$ A can gain $1/6$ units of clothing from exporting 1 unit of rice. (3)

- (II) Including the transport cost, the (total) cost of 1 unit of clothing imported from B is 3 units of rice. This equals to the unit cost of production of clothing in A.

$\therefore$ there is no gain for A to trade with B. (3)

1997/CE/I/10(a)

- (i) Same exchange ratio of food and clothing before trade, which is 1 unit of food (F) for 2 units of clothing (C) / same opportunity cost in both countries: $1F = 2C$ (1)+ (1)
 $\Rightarrow$ no such terms of trade (1)

- (ii) (I) In Country A, the cost of 1 unit of food is $(20/5 = 4)$ units of clothing. (1)

$\therefore$ In comparison,

[either by showing the cost of 1 unit of food in Country B is 2 units of clothing or explicitly shown in (a)(i)]
Country B has a lower opportunity cost in producing food. (1)

$\therefore$ Country B has the comparative advantage in food production. (1)

(II)

	Food (units)	Clothing (units)
after trade	-1	+3
before trade	-1	+2
gain from trade		+1

(2)

$\therefore$ 1 unit of clothing is gained. / 0.5 units of food is saved.

(1)

1998/CE/I/10(a)

(i) The opportunity cost of producing 1 unit of computer in

Hong Kong is 10 units of toy (1)

Mainland China is 2.5 units of toy (1)

Taiwan is 5 units of toy (1)

Taiwan has the comparative advantage in producing computers (because her opportunity cost is the lowest among the three economies). (1)

(ii) HK's opportunity cost ($1C = 10T$, $C = \text{clothing}$, $T = \text{toy}$) is the same as the international price ($1C = 10T$) (2)

$\therefore$ HK will not gain. (1)

Mainland China's opportunity cost of producing toys (cost of producing $1T = 0.04C$) is lower than her export price of toys ($1T = 0.1C$) (2)

OR

Mainland China's opportunity cost of producing computers (cost of producing $1C = 25T$) is higher than her import price of computers ($1C = 10T$)

$\therefore$ Mainland China will gain. (1)

1999/CE/I/9(a)

France's opportunity cost of producing one unit of cars is 2 ($= 4/2$) units of red wine. (1)

Japan's opportunity cost of producing one unit of cars is $1/2$ units of red wine.

OR

France's import price of one unit of cars is 1.5 ($= 3/2$) units of red wine. (1)

$\therefore$ France should import cars from Japan. (1)

France's gain = 2 units of red wine - 1.5 units of red wine = 0.5 units of red wine (3)

2000/CE/I/11(a)

(i) Country A can produce $(1\,200 \div 0.4)$ units of wine or $(900 \div 0.6)$ units of clothing.

$\therefore$ A's opportunity cost of wine production is $(900 \div 0.6) / (1\,200 \div 0.4) = 0.5$ units of clothing. (2)

Country B can produce $(400 \div 0.4)$ units of wine or $(2\,400 \div 0.6)$ units of clothing.

$\therefore$ B's opportunity cost of wine production is $(2\,400 \div 0.6) / (400 \div 0.4) = 4$ units of clothing. (2)

A has a lower opportunity cost / comparative advantage in wine production. (1)

$\Rightarrow$ A exports wine. (1)

(ii) No country will gain when the transportation cost per unit of wine $\geq$ total gain of trade per unit of wine / difference in the two domestic opportunity costs. (2)

$\therefore$ min. transportation cost = $[4 - 0.5] = 3.5$ units of clothing (2)

2001/CE/I/9(a)

(i) The opportunity cost of producing 1 unit of garments

Hong Kong: $2 \div 4 = 0.5$ units of computers (1)

USA: $5 \div 4 = 1.25$ units of computers (1)

Hong Kong has a lower opportunity cost in producing garments / comparative advantage in garment production. (1)

$\therefore$ Hong Kong will export garments. (1)

(ii) The range of T.O.T. with transportation is:

1 unit of computers $<$ 1 unit of garments $<$ 1.25 units of computers (3)

2002/CE/I/9

- (a) No, we cannot because (1)
the amount of resources of both countries are not given. (2)
- (b) Yes, we can because (1)
in Country A, the cost of 1 unit of toys is $5/15 = 1/3$ units of computers (1)
in Country B, the cost of 1 unit of toys is $5/10 = 1/2$ units of computers (1)
∴ Country A has a lower cost in producing toys. (1)

2003/CE/I/10(c)

- (i) The opportunity cost of producing 1 unit of paper (1)
in Country A: $10 \div 10 = 1$ unit of cars (1)
in Country B: $30 \div 15 = 2$ units of cars (1)
Country A has a comparative advantage in producing paper, because (1)
its opportunity cost is lower. (1)
Total gain from trading 1 unit of paper for Countries A and B is $[2 - 1 =] 1$ unit of cars (1)
- (ii) No, because (1)
(given that the opportunity cost of 1 unit of paper in Country A is 1 unit of cars while Country X's selling price of 1 unit of paper is 0.5 units of cars)
the lowest possible price which Country A can offer is higher than the price offered by Country X.
- OR**
Country B's gain from importing paper from Country X is greater than the maximum possible gain from importing paper from Country A. (2)

2004/CE/I/11(a)

- (i) No, because (1)
with the same amount of resources, Economy A produces a smaller amount of output than HK. / Economy A has a lower productivity than HK in producing clothing. (1)
- Comparison of output: 10 units vs 12 units **OR** 20 units vs 24 units (1)
- (ii) The opportunity cost of producing one unit of car in Economy A is 2 units of clothing. (1)
The opportunity cost of producing one unit of car in HK is 4 units of clothing. (1)
i.e. lower opportunity cost in Economy A. (1)
- ∴ Economy A has a comparative advantage in producing cars. (1)
- To trade at 5 units of clothing to 2 units of cars, Economy A gains $(5/2 - 2 =) 0.5$ units of clothing from each car exported. (2)

2005/CE/I/10

- (a) (i) Country B. This is because (1)
(with the same amount of resources,) the quantity of output of Country B is larger than that of Country A. (1)
- (ii) - different education and training
- different degree of technological advancement
- different degree of division of labour
- different working environment
- different labour incentives (1@, max: 3)
- [Mark the **FIRST THREE** points only.]

- (b) (i) Principle of comparative advantage: If each country specializes in the production of the good it has a lower opportunity cost, then the total output of all countries will increase. (2)

The opportunity cost of producing 1 unit of watches

in Country A: $5 \div 10 = 0.5$ units of TV sets. (1)

in Country B: $15 \div 20 = 0.75$ units of TV sets. (1)

(i.e., the opportunity cost of producing watches in Country A is lower than that in Country B.)

Therefore, Country A should produce and export watches. (1)

- (ii) (I) The possible range of mutually beneficial terms of trade is
 $0.5 \text{ units of TV sets} < 1 \text{ unit of watches} < 0.75 \text{ units of TV sets}$ (2)

(II) (A country has to compare the terms of trade with its own opportunity cost of production of the good in order to determine whether it should export or import the good.)

If the export price of Country A (the watch exporting country) is lower than its own opportunity cost of production which is 0.5 units of TV sets, Country A would lose from the trade.

OR

If the import price of Country B (the watch importing country) is higher than its own opportunity cost of production which is 0.75 units of TV sets, Country B would lose from the trade. (4)

2006/CE/1/8

- (a) The opportunity cost in the production of 1 unit of clothing

in Country A is $200 \div 100 = 2$ units of computers (1)

in Country B is $450 \div 150 = 3$ units of computers (1)

Therefore, Country A has the comparative advantage in producing clothing. (1)

- (b) Country A gains $2.5 - 2$ (1)
 $= 0.5$ units of computers from 1 unit of export. (1)

2007/CE/1/10(b)

- (i) $(10 - 5)$ units of vegetables (1)
 $= 5$ units of vegetables (2)

- (ii) $5 \text{ units of vegetables} < 1 \text{ unit of jewels} < 9 \text{ units of vegetables}$
OR
 $1/9 \text{ units of jewels} < 1 \text{ unit of vegetables} < 1/5 \text{ units of jewels}$ (3)

2008/CE/1/7

- (a) No because (1)
they have the same amount of output with the same amount of resources. (2)

- (b) The opportunity cost of producing food

in Country A $= 50 \div 20 = 2.5$ units of machines (1)

in Country B $= 60 \div 40 = 1.5$ units of machines (1)

Yes because (1)
the opportunity cost of producing food in Country B is lower. (1)

2009/CE/1/10(b)

- (i) The opportunity cost of producing one unit of computers

in Mainland China: $90 \div 50 = 1.8$ units of toys (1)

in Taiwan: $250 \div 200 = 1.25$ units of toys (1)

Taiwan enjoys a comparative advantage in producing computers, because she has a lower opportunity cost in producing computers. (1)

- (ii) Gain of Taiwan from exporting a unit of computers $= (1.5 - 1.25 - 0.2)$ (1)
 $= 0.05$ units of toys (2)

2010/CE/I/10(b)

- (i) If each economy specializes to produce the good in which it enjoys a comparative advantage in production / lower opportunity cost of production, the total output of all economies will increase. (2)
- (ii) In HK: the opportunity cost of producing 1 unit of medicine is 2.5 units of watches.
In Japan: the opportunity cost of producing 1 unit of medicine is 2 units of watches.

The possible range of mutually beneficial terms of trade is
2 units of watches < 1 unit of medicine < 2.5 units of watches (4)

2012/DSE/II/9

- (a) In country A, opportunity cost of producing 1S = 0.2W (1)
In country B, opportunity cost of producing 1S = 0.25W (1)
- (b) If each country specializes in producing the good in which the country has a lower opportunity cost, the total output will increase. (2)
Country A would export shirts. (1)
- (c) $0.2W < 1S < 0.25W$ OR $4S < 1W < 5S$ (2)

2013/DSE/II/5

- (a) In Country A, the opportunity cost of producing 1 smartphone (S) = 0.75 Food (F)
In Country B, the opportunity cost of producing 1 smartphone (S) = 0.5 Food (F)
Country B has a comparative advantage in producing smartphones. (3)
- (b) Yes, because the potential gain from trade $(0.75F - 0.5F = 0.25F)$ per unit of smartphones is greater than the transportation cost (0.2F). (3)

2014/DSE/II/8

- (a) In Country A, the opportunity cost in producing 1 unit of watches (W) = 4 units of Paper (P).
In Country B, the opportunity cost in producing 1 unit of watches (W) = 2 units of Paper (P).
Country B will specialize in producing watches and export watches because the opportunity cost of producing watches is lower there than in Country A. (3)
- (b) The terms of trade is $1W = 2.5P$. Per unit of watches imported, Country A will gain 1.5 P (= $4P - 2.5P$). (3)

2015/DSE/II/7

- (a) Uncertain. Since the amount of resources owned by the two countries is not given, their productivities (i.e., output-input ratios) cannot be determined and compared. (2)
- (b) In Country A, the opportunity cost in producing 1 unit of clothing (C) in terms of rice (R) = $15/3 = 5$ units of rice (R).
In Country B, the opportunity cost in producing 1 unit of clothing (C) in terms of rice (R) = $8/4 = 2$ units of rice (R).
Therefore Country A (B) enjoys comparative advantage in producing rice (clothing). (2)

Per unit gain from trade for Country B = $4R - 2R = 2R$
Total gain from trade = $(4R - 2R) \times 3 = 6R$

OR

In other words, Country B would be the clothing-exporting country, producing 4C, exporting 3C in exchange for 12R and keeping 1C for domestic consumption.
Without trade, it could have produced 1 C and 6R on its own. Its total gain from trade is thus 6R (= $12R - 6R$). (2)

2016/DSE/II/7

- (a) In Country A, the opportunity cost of producing 1C = $6/2 = 3T$
 The terms of trade: 1C will exchange for $(11/2 =) 5.5T$
 As the opportunity cost of producing 1C is lower than the terms of trade, Country A will produce and export C. (3)
- (b) The gain from trade = $5.5 - 3 = 2 = 0.5T$ per unit of C (3)

2017/DSE/II/7

- (a) (i) Uncertain. Without knowing the total amount of resources of both countries, the productivity (output per unit of resource) for food and computer cannot be determined. (2)
- (ii) Total producible outputs in the two countries when they utilize all of their resources in producing either food or computer:

	Food (units)	Computer (units)
Country A	200	100
Country B	200	200

- The opportunity cost of producing 1 unit of computer in Country A:
 $= 200/100 = 2$ units of food (1)
- The opportunity cost of producing 1 unit of computer in Country B:
 $= 200/200 = 1$ unit of food (1)
- Country B has a lower opportunity cost in producing computers, so it will produce and export computers. (1)
- (b) Terms of Trade (TOT) = $7/4 = 1.75 F$ per unit of C (1)
 Per unit gain for Country B = $(1.75 - 1)F = 0.75F$ (1)
 Total gain = $0.75 \times 16 = 12F$ (1)

2018/DSE/II/8

- (a) The opportunity cost of producing 1T in China = 5C
 The opportunity cost of producing 1T in Japan = 2C (2)
- (b) If each country specialises in producing the good in which the country has lower opportunity cost, then total world output would increase. Japan would export toys, as it has a lower cost of producing toys than China. (3)
- (c) Range of TOT: $2.05C < 1T < 4.95C$ (2)

2019/DSE/II/8

- (a) (i) Yes. (2)
When Country B has the same resource as Country A, the output of cars would be 200 units which is higher than that of country A.

- (ii) Yes. (3)
The opportunity cost of producing 1C in Country A = 1T
The opportunity cost of producing 1C in Country B = 0.6T
Country B has a comparative advantage in producing cars.

7A) Country B. The production cost of toy is lower than country A. (0.5 car lower than 1 car)

B. Terms of trade 5 toys for 4 cars = 1 toy for 0.8 car. (1 mark)

Exporting country gain = Terms of trade - exporting country production cost - transportation cost.
 = 0.8 car - 0.5 car - 0.1 car = 0.2 car. (2 marks)

In Country A, the opportunity cost of producing 1W = $10/4 = 2.5$ R

In Country B, the opportunity cost of producing 1W = $6/3 = 2$ R

Range of terms of trade: $2R < 1W < 2.5R$

OR

In Country A, the opportunity cost of producing 1R = $4/10 = 0.4$ W

In Country B, the opportunity cost of producing 1R = $3/6 = 0.5$ W

Range of terms of trade: $0.4W < 1R < 0.5W$

- (a) The opportunity cost of producing 1R in Country E = 2T (4)

The opportunity cost of producing 1R in Country F = 3T

Since Country E has a lower opportunity cost in producing rice, Country E exports rice and Country F imports rice.

Gains from trade for Country E = $2.2T - 2T = 0.2T$

Gains from trade for Country F = $3T - 2.2T = 0.8T$

- (b) Here, Country F has no absolute advantage in producing either good, and yet it has (2)

Comparative advantage in producing toys. As shown in (a), Country F can still enjoy gains from trade (=0.8T per 1R) by exporting toys in exchange for rice from Country E.

- (a) (i) No, because with the same amount of resources, Country A can only produce the same number of computers as Country B. (2)

(ii) Yes.

Opportunity cost of producing 1 unit of computers:

in Country A = $500/800 = 0.625$ units of rice (1)

in Country B = $300/400 = 0.75$ units of rice (1)

As Country A can produce computers at a lower opportunity cost than Country (1)

B, Country A has a comparative advantage in producing computers.

- (b) 0.625 units of rice < 1 unit of computers < 0.7 units of rice (2)

- (a) Absolute advantage cannot be determined (1)
 as the actual amounts of resources in the two countries are not known. (1)

- (b) Outputs when each country allocates all its resources to producing either bread or clothing:

	Bread (units)		Clothing (units)
Country X	$100 / 0.25 = 400$	OR	$75 / 0.75 = 100$
Country Y	$80 / 0.25 = 320$	OR	$240 / 0.75 = 320$

- Hence, opportunity cost to produce 1 unit of bread (2)
 in Country X = $100/400 = 0.25$ units of clothing
 in Country Y = $320/320 = 1$ unit of clothing

As Country X has a lower cost in producing bread, it is the bread-exporting country.

- The total gain from trade from exporting 30 units of bread (1)
 $= 30 \times (7/10 - 0.25)$ (1)
 $= 13.5$ units of clothing

7.2 TRADE BARRIERS

MARKING SCHEME

1990/AL/II/22 A	1998/AL/II/29 B	2004/AL/II/25 B (63%)	2009/AL/II/26 A (45%)	2015/DSE/I/45 D (68%)
1994/AL/II/30 B	1999/AL/II/21 D	2004/AL/II/26 B (67%)	2010/AL/II/27 A (62%)	2016/DSE/I/42 C (56%)
1996/AL/II/29 B	2000/AL/II/26 A	2007/AL/II/25 A (70%)	2012/AL/II/24 B (63%)	2017/DSE/I/42 C (57%)
1998/AL/II/28 D	2001/AL/II/11 B	2009/AL/II/25 B (77%)	2014/DSE/I/45 D (64%)	2018/DSE/I/41 D (43%)
2019/DSE/I/43 A	2020/DSE/I/43 A	2021/DSE/I/44 B	2022/DSE/I/44 D	2023/DSE/I/43 D
2024/DSE/I/43 C	2025/DSE/I/18 B	2025/DSE/I/43 C		

Note: Figures in brackets indicate the percentages of candidates choosing the correct answers.

1993/AL/II/5

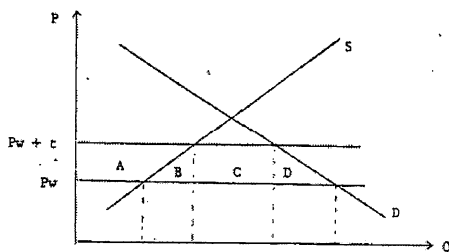


Diagram 1 : Tariff

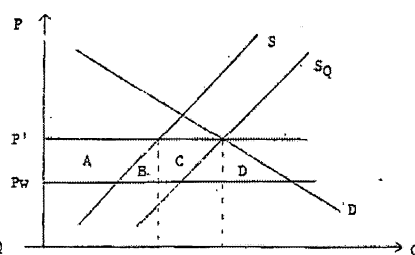


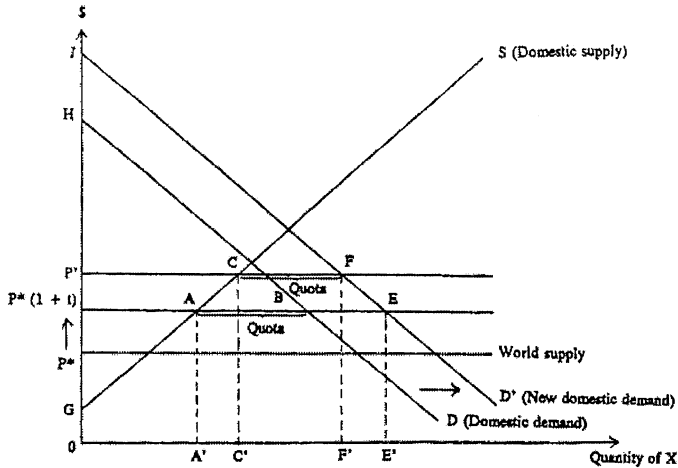
Diagram 2 : Quota

Diagram 1 summarizes the effect of a tariff:

- Area A: increase in domestic producers' surplus
- Area A + B + C + D: decrease in consumers' surplus
- Area C: tariff revenue accruing to the government
- Area B and D: deadweight loss to society

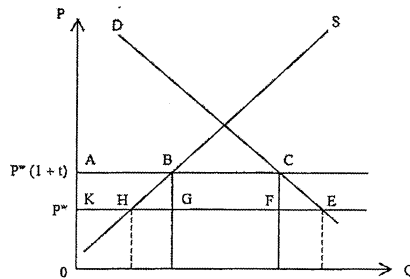
In diagram 2, a quota results in the supply curve S_Q . Areas A, [A + B + C + D], [B and D] have the same interpretation as before. C is no longer tariff revenue. C may be captured by the importers, foreign exporters or the government depending on how the import quotas are allocated.

1995/AL/II/8



- (a) Suppose the country wishes to restrict the imports of X to AB. This can be achieved by imposing a quota of AB or a tariff of t .
- (b) After the demand grows, under the original tariff rate, the price remains to be $p^*(1+t)$ and the import becomes AE. Under the original quota, the price becomes P' [higher than $p^*(1+t)$] and the import is CF ($= AB$) [smaller than AE]. The consumer surplus under the tariff system is $p^*(1+t)EJ$ [which is larger than] $P'FJ$ under the quota system.

2001/AL/II/4



In the absence of tariffs, the price in the domestic economy is P^* . After imposing tariffs, domestic price is raised to $P^*(1+t)$. Producer surplus is increased by ABHK. Consumer surplus is reduced by ACEK. Tariff revenue collected by the government is BCFG. Loss in welfare is given by BGH + CEF.

2011/AL/II/1(d)

S: examples of trade protection

D: direct vs. indirect effects on price of imports.

OR

indirect vs. direct effects on quantity of imports.

Marks

- (a) Inferior good, as people with less income during recession choose to consume more junk food.

(2)

(b) Illustrate in Figure 1

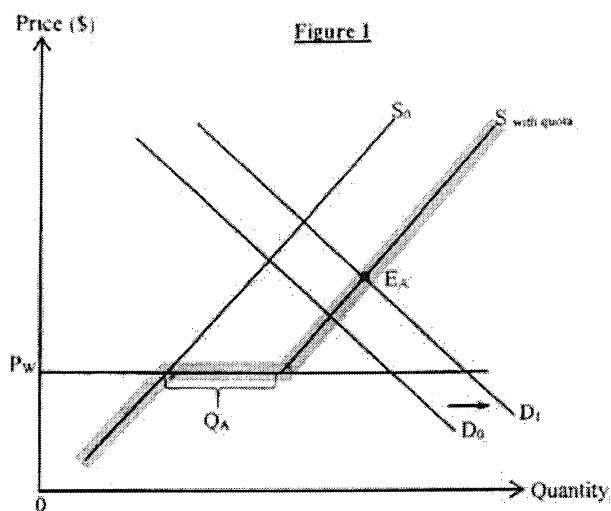
- correct position of the quota (Q_A)
- rightward shift of demand curve
- a new kinked supply curve ($S_{\text{with quota}}$)
- correct position of the final equilibrium point (E_A)

(1)

(1)

(1)

(1)



SECTION 8: INTERNATIONAL FINANCE

8.1 BALANCE OF PAYMENTS

MARKING SCHEME

1990/CE/II/36 B	1994/CE/II/56 D	1998/CE/II/46 D	2004/CE/II/49 B (43%)	2009/CE/II/47 B (34%)
1990/CE/II/46 B	1995/CE/II/54 D	1998/CE/II/53 C	2004/CE/II/50 B (63%)	2010/CE/II/48 A (42%)
1990/CE/II/60 A	1995/CE/II/56 C	1999/CE/II/47 A	2005/CE/II/44 D (56%)	2010/CE/II/49 A (51%)
1991/CE/II/49 C	1995/CE/II/57 D	1999/CE/II/53 A	2005/CE/II/47 A (63%)	2012/DSE/I/40 A (52%)
1991/CE/II/50 D	1995/CE/II/58 D	1999/CE/II/54 A	2006/CE/II/45 C (23%)	2014/DSE/I/43 D (38%)
1991/CE/II/60 C	1996/CE/II/48 C	2000/CE/II/50 C	2006/CE/II/48 D (68%)	2015/DSE/I/43 D (54%)
1992/CE/II/52 B	1996/CE/II/49 C (deleted)	2000/CE/II/51 A	2007/CE/II/48 C (69%)	2015/DSE/I/44 D (19%)
1992/CE/II/60 B	1996/CE/II/50 A	2001/CE/II/41 B (deleted)	2007/CE/II/50 C (57%)	2016/DSE/I/43 B (48%)
1993/CE/II/40 D	1996/CE/II/51 B	2001/CE/II/45 D	2008/CE/II/46 B (57%)	2017/DSE/I/41 B (48%)
1993/CE/II/54 B	1997/CE/II/50 C	2002/CE/II/48 A (31%)	2008/CE/II/47 A (70%)	2018/DSE/I/42 X (deleted)
1994/CE/II/51 B	1997/CE/II/51 D	2003/CE/II/43 B (67%)	2008/CE/II/48 C (47%)	2018/DSE/I/44 A (46%)
1994/CE/II/55 A	1997/CE/II/52 C	2003/CE/II/48 C (15%)	2008/CE/II/49 B (40%)	2019/DSE/I/44 C
2021/DSE/I/42 D	2022/DSE/I/42 A	2023/DSE/I/44 C	2024/DSE/II/40 B	2024/DSE/II/41 D

2025/DSE/I/44

C

1990/CE/I/3(a)(i)

Domestic exports of goods + Re-exports of goods - Imports of goods

(1)

= \$(218 + 275 - 501) bn.

= -\$8 bn.

(2)

1991/CE/II/4(b)

(i) Yes, because

(1)

it is trading of physical goods.

(1)

It worsens Hong Kong's balance of payments. / This item represents a payment in HK's b.o.p. account.

(1)

(ii) No, because

(1)

it is trading of service.

(1)

It favours Hong Kong's balance of payments. / This item represents a receipt in HK's b.o.p. account.

(1)

1992/CE/I/4(c)

	<u>Balance of invisible trade</u>	<u>Balance of payments</u>	
(i)	no effect	unfavourable	(2)
(ii)	favourable	favourable	(2)

1993/CE/I/2(b)(i)

Re-exports increase, because
more raw materials and machine to China via Hong Kong. (1)

OR

more Chinese exports via Hong Kong. (2)

1993/CE/I/4(a)

(ii) Yes, because (1)
the ratios of both exports and imports to total GDP increased (2)

(iii) Trade deficit in 1980 (1)
Trade surplus in 1990 (1)

Reasons:

- re-exports ↑ e.g. expansion of China trade
 - exports ↑ due to China factor, etc.
 - expansion of tertiary sector service related to trade, e.g. bank service, insurance service, tourism
 - expansion of infrastructure service related to trade, e.g. container port, airport, communication network (2@, max: 4)
- [Mark the **FIRST TWO** points only.]

1994/CE/I/8

It occurs when the country's capital account and other items in the current account altogether has a surplus and the surplus is greater than the trade deficit. (4)

1995/CE/I/9(b)

(i) Souvenirs imported from HK $\Rightarrow$ the US current account / trade account was unfavourably affected --- A (2)
Souvenirs bought by tourists $\Rightarrow$ the US current account / invisible trade account was favourably affected --- B (2)

A effect < B effect $\Rightarrow$ favourable to the US balance of payments (1)

(ii) It affected the US capital account favourably. (1)
 $\Rightarrow$ favourable to the US b.o.p. (1)

1996/CE/I/9

(a) $\$(24 - 35 + 17 - 19)$ mn. = $-\$13$ mn. (1)

(b) No, because (1)
there is a balance of payments deficit of \$28 mn. / the credit entry in the foreign exchange & gold account represents a decrease in reserves. (2)

1997/CE/I/6

If the initial export value of goods is significantly higher than the initial import value of goods and the difference is more than compensate for the difference in their individual growth rates, then there is a case of visible trade surplus. (3)

1997/CE/I/11(a)(i)

(I) No, because (1)
there is no export of goods but an import of \$1m. (1)

(II) No, because (1)
total payments (= \$1 m.) > total receipts (\$0.6 m.) (1)

1998/CE/I/8

- No, because (1)
no information on other items, such as the invisible trade account (, net current transfers, net factor income from abroad,)
and the capital account. (2)

1998/CE/I/11(c)

(ii) Current account

OR

Visible trade account / Invisible trade account (2)

- (iii) The fall in the local demand for Japanese cars will improve HK's visible trade balance. (2)
The fall in the re-export demand for Japanese cars will worsen HK's invisible trade balance. (2)
The combined effect on HK's balance of payments is uncertain, depending on the relative strength of the above two forces. (1)

1999/CE/I/6(b)

- Not included, because (1)
this \$10 000 is not the value of a service. (1)

2001/CE/I/8

- (a) (i) Goods (1)
Services (1)
(ii) E.g. interest (other examples are: dividend, rental income, wages) earned from / paid to foreign countries (2)
(b) The capital and financial account (2)

2004/CE/I/8

- Re-exports = visible trade surplus + import of goods - domestic exports
= \$(40 + 30 - 10) m (1)
= \$60 m (2)

2006/CE/I/11(a)

- (i) Visible trade balance = value of exports of goods - value of imports of goods (1)
[Most goods in HK are imported.] In Hong Kong, an increase in private consumption usually means an increase in value of imports of goods. (1)
∴ Visible trade balance most likely worsens. (1)
[other things being equal or assuming no change in value of exports of goods.]
(ii) The condition is:
Increase in the value of exports of goods < increase in the value of imports of goods (2)

2008/CE/I/11(c)

- Visible trade balance = value of exported goods - value of imported goods (1)
No because (1)
there may also be a fall in the value of exported goods (1)
which is larger than the fall in the value of imported goods. (1)

- 2009/CE/I/8
- (a) P is services. (1)
- (b) Q Account is Capital and Financial Account. (1)
- (c) (i) Visible trade balance = -130 billion dollars (1)
- (ii) Re-export of goods = 2 323 billion dollars (1)

2010/CE/I/9(d)

Imported souvenirs will decrease (the visible trade balance and hence) the balance of HK's current account. (1)

Souvenirs sold to inbound tourists will increase (the invisible trade balance of) HK's current account. (2)

The decrease in the current account balance is smaller than the increase in the current account balance. (1)

There is value added to the souvenirs by local service providers. (1)

- 2013/DSE/II/6
- (a) Balance of current account = $\$[35 - 50 + 47 - 27]$ million = \$5 million (1)
- (b) No, because (1)
- the BOP balance is -\$10 million, which implies a BOP deficit. (2)

2015/DSE/II/12(a)

A drop in the export of services to mainland tourists (1)

would reduce HK's current account surplus (or enlarge its deficit). (1)

2016/DSE/II/13(b)(i)

Proposal A on current account:

An increase in income tax would lower disposable income (1)

and result in a drop in consumption, AD, as well as output level. (1)

Import would fall, implying (1)

a bigger surplus or smaller deficit in the current account balance. (1)

Proposal B on current account:

A fee charged on departing travellers would result in an increase in the cost of using the airport. (1)

Travelling industry would be adversely affected, reducing the number of both incoming and outgoing travellers. (1)

Both import and export of services would drop, (1)

causing an ambiguous change in the current account balance. (1)

2017/DSE/II/12(c)

Invisible trade: the increase in cost of travel would result in a fall in the number of British citizens visiting the EU and hence the import of services.

Factor income: with fewer British workers working in the EU, there would be a drop in total wage payment to such workers.

Unilateral transfer: with fewer workers working in the EU, there would be a drop in their remittance back to Britain. (2@, max: 4)

2018/DSE/II/11

Balance of payments would be worsened, because (1)

export of goods would fall because of the embargo; (1)

export of services would also fall as fewer tourists would want to visit Country Z because of the threat of war; and (1)

foreign investors would withdraw their investments in Country Z, so capital may flow out of the country. (1)

10 a1) The balance of current account in 2017 = $-808 + 255 = -533$

The balance of current account in 2018 = $-891 + 269 = -622$

According to data of source A, the current account worsen (increase in deficit) by 89.

Ii) Not necessarily. As the other two components of current account (income and current transfer) are not given in data, there is no conclusion can be drawn about the change of current account.

2022/DSE/II/4c

(c) Other effects of devaluation:

- Export of services may rise, as more tourists are attracted to come and spend
Because devaluation makes local goods and services cheaper in terms of foreign
currency.
- It may attract capital inflows or foreign investment if the cost of investment in
terms of foreign currency decreases as a result of devaluation.
- Any other relevant point.

@ 2
max:2

[Mark the FIRST point only.]

2024/DSE/II/11(d)

- (d) As more Mainland visitors consume at Ann's coffee shop, the value of HK's exports of services increases. (1)
- As Ann buys more coffee beans from Brazil, the value of HK's imports of goods increases. (1)
- As there is value-added by Ann's coffee shop, the increase in the value of HK's exports of services (due to the increase in the sales revenue of Ann's coffee shop) would be greater than the increase in the value of HK's imports of goods (due to the increase in the purchase cost of Brazilian coffee beans by Ann's coffee shop). As a result, the current account balance of HK's balance of payments increases. (1)

8.2 EXCHANGE RATES

MARKING SCHEME

1990/CE/II/38 B	1998/CE/II/48 B	2010/CE/II/50 D (70%)	2018/DSE/II/45 D (57%)	2006/CE/II/47 C (63%)
1990/CE/II/43 A	2003/CE/II/47 B (67%)	2012/DSE/I/39 C (37%)	1993/CE/II/33 C	2009/CE/II/50 D (29%)
1991/CE/II/51 B	2004/CE/II/48 C (66%)	2013/DSE/I/37 D (52%)	1993/CE/II/36 D	2013/DSE/I/40 C (70%)
1993/CE/II/55 A	2005/CE/II/49 C (52%)	2014/DSE/I/42 B (59%)	1996/CE/II/52 D	2015/DSE/I/42 C (62%)
1993/CE/II/57 D	2007/CE/II/49 D (49%)	2016/DSE/I/44 A (52%)	1999/CE/II/49 C	2017/DSE/I/45 D (40%)
1995/CE/II/55 B	2008/CE/II/50 A (37%)	2016/DSE/I/45 D (56%)	2000/CE/II/52 A	2019/DSE/I/45 C
1997/CE/II/54 B	2009/CE/II/49 B (84%)	2017/DSE/I/44 A (69%)	2001/CE/II/46 D	2020/DSE/I/44 A
2020/DSE/II/45 B	2021/DSE/I/45 A	2022/DSE/I/45 D	2024/DSE/II/45 C	2025/DSE/II/45 A

1992/CE/II/2(c)

Depreciation of HK\$

⇒ import prices/cost of imported raw materials (in HK\$)↑

(2)

⇒ unit cost of production ↑

(2)

1998/CE/II/11(b)

Price of Japanese cars (quoted in HK\$) falls.

(1)

⇒ an increase in the quantity demanded for Japanese cars

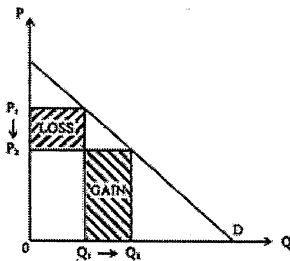
(1)

Elastic demand means the % change in quantity demanded is larger than the % change in price.

(2)

⇒ sales revenue increases (i.e. revenue loss < revenue gain)

(1)



Indicate in the diagram:

- P ↓ and Qd ↑

(1)

- Correct position of 'Gain' and 'Loss'

(1)

1999/CE/1/9(b)

The devaluation $\Rightarrow$ price (in HK\$) of Japanese cars $\downarrow$ ($\Rightarrow$ quantity of Japanese cars demanded $\uparrow$)

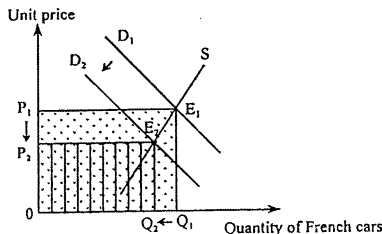
Japanese cars and French cars are substitutes

$\therefore$ demand for French cars $\downarrow$

$\Rightarrow$ both the equilibrium price and quantity of French cars $\downarrow$

$\Rightarrow$ sales revenue of French cars $\downarrow$ (from $0P_1E_1Q_1$ to $0P_2E_2Q_2$)

(1)
(1)
(1)
(1)+(1)
(1)



Indicate in the diagram:

- D shifts to the left
- P $\downarrow$ and Q $\downarrow$
- TR $\downarrow$ (area $P_1E_1Q_1Q_2E_2P_2$)

(1)
(1)
(1)

1999/CE/1/10(d)

HK\$ appreciates against these Asian currencies.

$\Rightarrow$ these Asian countries' exports become relatively cheaper than HK exports in the US market.

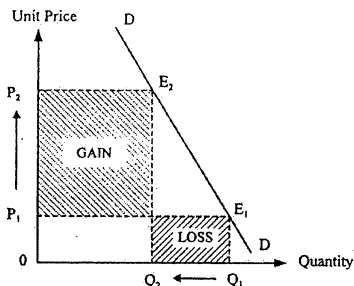
$\Rightarrow$ as there are (cheaper) substitutes, demand for HK exports $\downarrow$

$\Rightarrow$ derived demand for labour in HK's export sector $\downarrow$, so HK's unemployment is very likely to increase.

(1)
(1)
(2)
(2)

(Note: In the context of AD-AS, you may argue that a decrease for HK's exports will lead to a decrease in Hong Kong's aggregate demand, which would result in a (larger) deflationary gap. It represents the existence of (more) unemployed workers in the factor market.)

2002/CE/1/12(a)(ii)



Indicate in the graph:

- selling price $\uparrow$ and quantity demanded $\downarrow$
- gain > loss

(1)
(2)

Verbal explanation:

The selling price increases (due to the depreciation of the HK\$ against the British Pound).

The quantity demanded decreases.

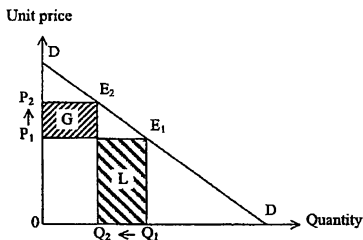
The percentage change in quantity demanded is smaller than the percentage change in price.

Total revenue increases.

(1)
(1)
(1)
(1)

2004/CE/I/11(b)

- (i) Increase, because
HK\$ depreciates against the Euro dollar under the linked exchange rate system. (1)
- (ii) (The rise in price leads to) a fall in quantity demanded. (1)
Elastic demand for imports (1)
% change in $P <$ % change in Q_d (2)

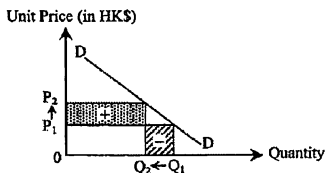


Indicate in the diagram:

- gain in total import value (G) < loss in total import value (L) (2)

2006/CE/I/10

- (a) (i) Increased, because
Euro appreciated against the Hong Kong dollar. (1)
- (ii) With an increase in import price in terms of HK\$, the quantity of imported cars decreased. (1)
If demand is inelastic, % change (increase) in $P >$ % change (decrease) in Q_d . (2)



Indicate in the diagram:

- $P \uparrow$ and $Q_d \downarrow$ (1)
- Correct position of 'Increase' and 'Decrease' in total expenditure (1)
- 'Increase' > 'Decrease' (1)

- (iii) In terms of £, the total value of imported cars from Europe decreased, because
the volume of imports decreased while the price of imported cars in terms of £ remained unchanged. (1)

- (b) (i) $\$(9.37 - 9.07) \times 10\,000$ (1)
= \$3 000. He gained \$3 000. (1)

- No, because (1)
the capital gain does not involve production. (1)

2008/CE/I/11(a)

Import price increases as a result of revaluation and the import quantity decreases.

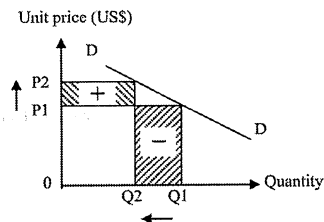
Condition: elastic demand

% change in price < % change in quantity demanded

(1)

(1)

(2)



Indicate in the diagram:

- price increases and quantity decreases

- correct position of 'increase in import value' and 'decrease in import value'

- area of 'increase in import value' < area of 'decrease in import value'

(1)

(1)

(1)

2010/CE/I/8

(a) increased

(1)

(b) uncertain

(1)

(c) increased

(1)

2012/DSE/II/8

No, because

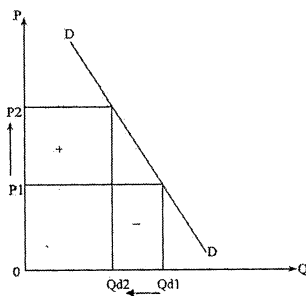
if the demand for import is inelastic, the import value will not decrease

as the percentage increase in price will be greater than percentage decrease in quantity demanded.

(1)

(1)

(2)



Indicate in the diagram:

- $P \uparrow$ and $Q_d \downarrow$

- 'gain in import value' > 'loss in import value'

(1)

(1)

2014/DSE/II/11(a)

Japanese Yen depreciated against Hong Kong dollar.

(1)

As the price of Japanese automobiles does not change, the automobiles become cheaper in terms of HKD and Hong Kong importers will buy more Japanese automobiles.

(1)

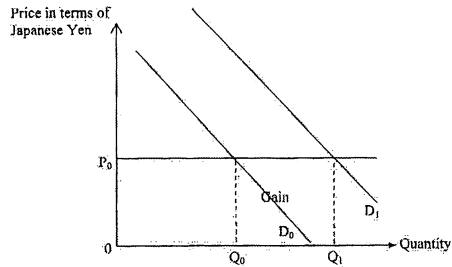
The demand for Japanese automobiles by HK people increases.

(1)

Quantity transacted increases.

(1)

(Therefore, total export value of Japanese automobiles increases.)



Indicate on the diagram:

- demand curve shifts to the right
- a higher quantity transacted
- gain in total value

(1)

(1)

(1)

2018/DSE/II/12(d)

During 2015-16, RMB depreciated against USD. Under HK's linked exchange rate system, the value of HKD would follow that of USD, so RMB depreciated against HKD as well.

Depreciation of RMB implies that goods in HK became more expensive in terms of RMB. Therefore fewer mainland tourists would come to HK (, and those who came would spend less in HK).

As mainland visitors were the major source of tourists in HK, a significant drop in "total tourism expenditure associated to inbound tourism" would result.

(6)

(c) Marks award for effective communication (EC: max 2 marks)

Marks	Performance
2	● Support arguments with the source/data and appropriate economic theories. ● Present relevant material. ● Present well-organised and coherent answers without repetition of ideas. ● Use language that expresses ideas clearly and fluently with appropriate use of words/terms/symbols.
1	● Present arguments with some support of the source/data and economic theories. ● Present some irrelevant material. ● Present answers in a less organised way with some repetition. ● Use language that conveys a clear message with some inappropriate use of words/terms/symbols.
0	● Present arguments with no support of the source/data and economic theories. ● Present material unrelated to the gist of the question. ● Present inconsistent arguments. ● Express limited ideas with inappropriate use of words/terms/symbols.

The maximum marks for content is 11 marks. Answers may include the following:

Effects on aggregate output:

- Re-exports drops, AD drops, and aggregate output drops.
- The stock prices drop and people's wealth drops, consumption drops, and aggregate output drops.
- Some trading firms may consider moving away from Hong Kong to other countries in order to escape from the tariffs/ explore new markets. SRAS drops and aggregate output drops in the future.

Effects on tax revenue:

- Profits tax revenue will drop because some logistic/trading firms suffer and profits drop.
- Income tax revenue will drop as aggregate output and the average household income fall.
- Revenue from stamp duty may drop as the stock market crash may lower the volume of trade in the stock market.

Effects on other variables:

- Unemployment rate may increase as production drops and demand for labour is reduced as firms move away from Hong Kong. The unemployed population increases while the labour force remains unchanged.
- HKD may appreciate against RMB as Chinese government may lower the exchange rate of RMB to lessen the effect of the tariff on exported goods/ there may be large capital outflow from China.
- Government spendings may increase as the government may need to grant subsidy or loans to the affected firms and/or provide unemployment benefits to the jobless.

2021/DSE/II/5

(a) (i) No, because the salary received by Mr Chan was derived from production by a non-resident producing unit (the UK football club) outside Hong Kong. (2)

(ii) Yes. As Mr Chan is a Hong Kong resident, the salary he received would be counted as factor income from abroad. (3)

(b) USD, because the exchange rate of USD to HKD, thus the purchasing power of USD in Hong Kong, is more stable under the linked exchange rate. In other words, USD involves lower exchange-rate risk for Mr Chan. (2)

OR

GBP, because GBP is the legal tender in the UK, so he can use GBP directly to buy goods and services during his stay in the UK.

2022/DSE/II/4b

(b) Devaluation of local currency (e.g., Country B) against its trading partner (e.g., Country C) would drive up the price of imported goods in terms of local currency, thus reducing its quantity demanded by local people (i.e., import volume). If import demand is elastic, then total import expenditure/value would fall (like in the tariff case analysed in part (a)). (2)

2023/DSE/II/9

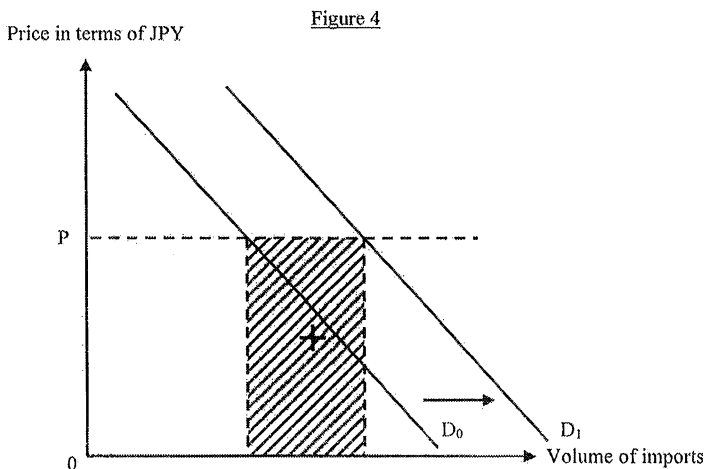
(a) Figure 2 shows that JPY depreciated against USD. Under the Linked Exchange Rate System, JPY would also depreciate against HKD. (1)

(b) Verbal elaboration:

- Import price in terms of HKD decreases. (1)
- At any given price in terms of JPY, Hong Kong will import more from Japan, i.e. Hong Kong's demand for Japanese imports increases. Import volume increases. Given that the price in terms of JPY remains unchanged, total import value in terms of JPY increases, as shown in the data. (1)

Indicate in diagram:

- rightward shift of D curve (1)
- correct position of increase in total import value (+) (1)



2024/DSE/II/13(b, d)

- (b) Source B shows that the USD appreciated against the RMB during the period. Under the Linked Exchange Rate System of Hong Kong, HKD would also appreciate against RMB. (1)
 As HKD appreciated against RMB, prices of goods and services in the Mainland dropped in terms of HKD (assuming that their prices in RMB remained unchanged). The money cost for Hong Kong residents to visit the Mainland would drop. (1)
 As a result, the number of Hong Kong residents visiting the Mainland would increase. (1)

(d) Marks for effective communication (EC: max 2 marks)

Marks	Performance
2	● Support arguments with the source/data and appropriate economic theories. ● Present relevant material. ● Present well-organised and coherent answers without repetition of ideas. ● Use language that expresses ideas clearly and fluently with appropriate use of words/terms/symbols.
1	● Present arguments with some support of the source/data and economic theories. ● Present some irrelevant material. ● Present answers in a less organised way with some repetition. ● Use language that conveys a clear message with some inappropriate use of words/terms/symbols.
0	● Present arguments with no support of the source/data and economic theories. ● Present material unrelated to the gist of the question. ● Present inconsistent arguments. ● Express limited ideas with inappropriate use of words/terms/symbols.

The maximum mark for content is 12 marks.

SECTION 9: EXTENSION OF TRADE THEORY AND ECONOMIC DEVELOPMENT

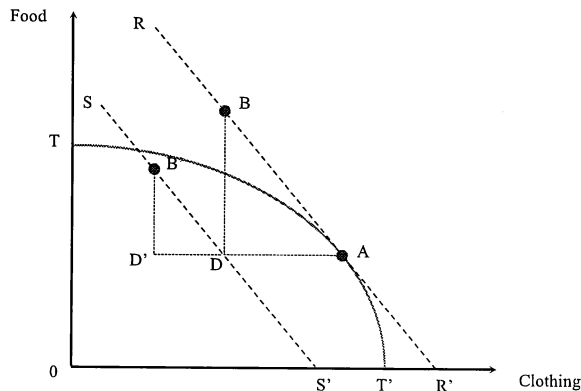
9.1 EXTENSION OF TRADE THEORY

MARKING SCHEME

1990/AL/II/20 B	1997/AL/II/17 C	2001/AL/II/26 C	2003/AL/II/25 B	2008/AL/II/24 C (72%)
1991/AL/II/04 C	1998/AL/II/18 B	2002/AL/II/24 A (43%)	2005/AL/II/27 D (36%)	2009/AL/II/23 A (85%)
1992/AL/II/29 C	1999/AL/II/19 B	2002/AL/II/27 C (44%)	2006/AL/II/26 C (46%)	2009/AL/II/24 C (84%)
1995/AL/II/21 B	2001/AL/II/20 D	2002/AL/II/28 D (35%)	2006/AL/II/27 C (62%)	2010/AL/II/24 B (36%)
1996/AL/II/24 C	2001/AL/II/21 D	2003/AL/II/24 A	2007/AL/II/24 A (54%)	2011/AL/II/25 A (57%)

Note: Figures in brackets indicate the percentages of candidates choosing the correct answers.

1991/AL/II/3

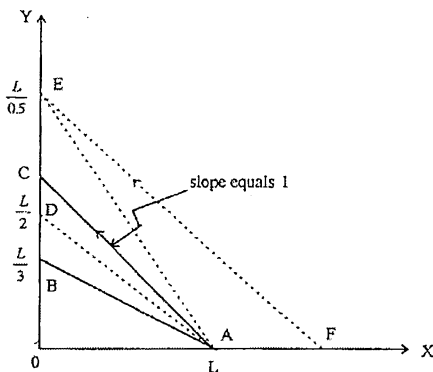


Destruction of DA units of clothing shifts the consumption possibility frontier from RR' to SS'. This is a parallel shift because of the price taking assumption.

A new equilibrium occurs at B'. D'D is the new amount of exports, in exchange for B'D' (the new amount of imports).

1997/AL/II/8

- (a) AB is the production frontier. Country A will export Good X.
- (b) Before technical progress, AB is the production frontier. After trade, AC is the consumption frontier and the country will specialize in the production of Good X and point A is the production point.
- (i) When the labour requirement in the production of Y is 2, Country A still has its comparative advantage in X and will export X. The production frontier becomes AD. After trade, Country A still produces at point A. AC remains to be the consumption frontier. There will be no change in consumption. Welfare of Country A will not change.
- (ii) When the labour requirement in the production of Y is 0.5, Country A has its comparative advantage in Y and will export Y. The production frontier becomes AE. After trade, Country A will produce at point E. EF becomes its consumption frontier. Welfare will increase as the consumption frontier has shifted out (relative to AC).



2012/DSE/II/15

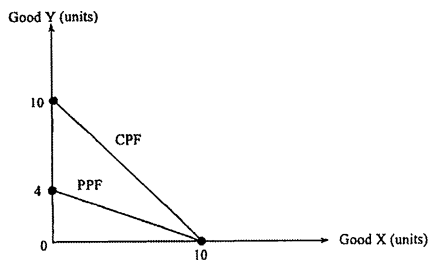
- (a) (i) In country A, the opportunity cost of producing $1X = 1Y$ while in country B, the opportunity cost of producing $1X = 0.667Y$. (1)
- Country B has a lower opportunity cost in producing good X, therefore Country B has a comparative advantage in producing good X. (1)
- (ii) Yes, because their costs of producing goods X and Y are different. (1)
- (b) Country A will gain $(1Y - 0.8Y =) 0.2Y$ (2)
- and Country B will gain $(0.8Y - 0.667Y =) 0.133Y$ (2)
- per unit of X traded respectively. (2)

2013/DSE/II/14(a)

- (i) Country A will import Good Y because
the domestic production cost of Good Y is higher than the terms of trade.

(1)
(1)

(ii)



Indicate on the diagram:

- correct position of PPF
- correct position of CPF

(1)
(1)

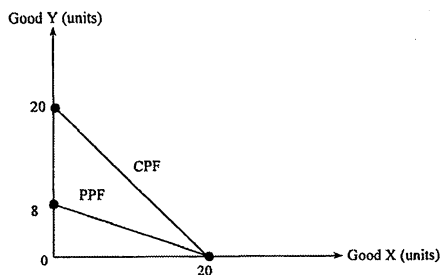
- (iii) Country A will still import Good Y because
the domestic production cost of Good Y is still higher than the terms of trade.

(1)
(1)

Country A will be better off because

the CPF shifts outward / Country A can consume more of Good X and Good Y.

(1)
(1)



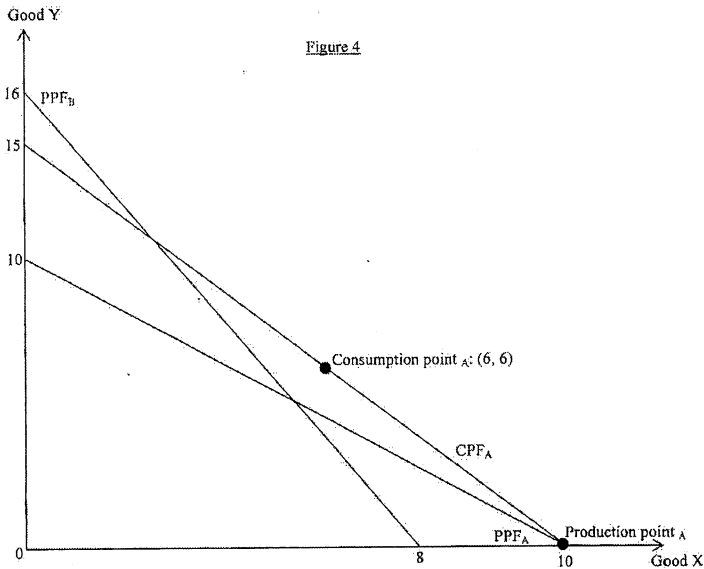
Indicate on the diagram:

- correct position of PPF
- correct position of CPF

(1)
(1)

2014/DSE/II/14

- (a) The marginal (and average) costs of production for Good X and Good Y are constant (and do not increase with the quantities of X or Y). So the opportunity cost of X in terms of Y, which is the slope of the PPF, is constant. (2)
- (b) (i) Country B can produce 8 units of Good X by using all of its resources. Given the amount of resources owned by Country A doubles that of Country B, Country A can only produce 5 ($= 10/2$) units of X by using the same amount of resources as B. So Country A does not have absolute advantage in producing X. (3)
- (ii) In Country A, the opportunity cost of producing one unit of X is $(10/10 =) 1Y$. In Country B, the opportunity cost of producing one unit of X is $(16/8 =) 2Y$. So Country A has comparative advantage in producing Good X. (2)
- (c)

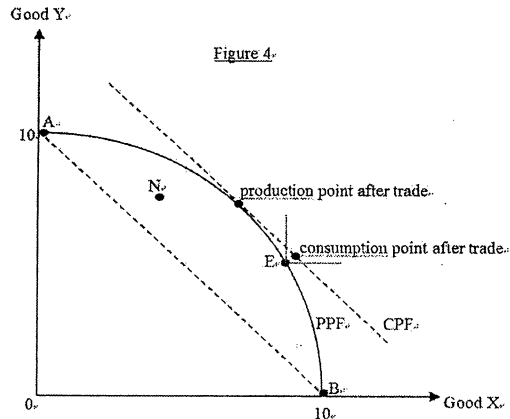


Indicate on the diagram:

- (i) production point of Country A: (10, 0) (1)
- (ii) CPF of Country A: a straight line with X-intercept equals 10 and Y-intercept equals 15 (2)
- (iii) consumption point of Country A: (6, 6) (2)

2015/DSE/II/14

- (a) No,
because there will be idle resources in the country / the country is not maximizing its production. (1)
(1)
- (b) No, because
without trade, the domestic cost of good X trade in country A ($= 2.5Y$) exceeds the world price of X ($= 1Y$) (1)
OR
opportunity cost of 1X = slope of the tangent of PPF at E $>$ slope of AB (2)
(2)
- (c)



Indicate on the diagram:

- correct position of production point after trade (1)
- tangency between CPF and PPF (1)
- CPF is parallel to AB (1)

- (d) Yes, because (1)
the country can consume beyond its PPF (which is also its CPF without trade) / the country can consume more of X and Y after trade. (1)

Indicate on the diagram:

- correct position of consumption point after trade (1)

2016/DSE/II/15

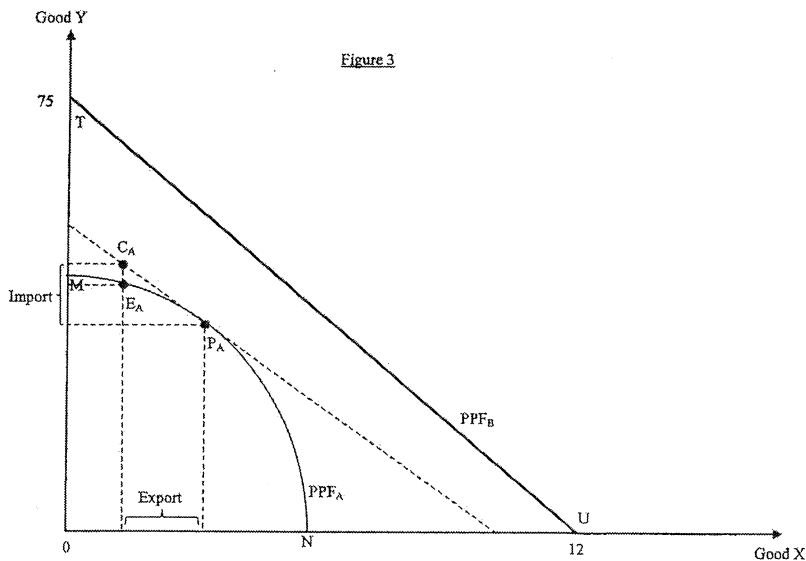
(a) The marginal costs of producing of both X and Y are increasing in Country A. (1)

(b) Country A's opportunity cost of producing $1X = 4Y$ (1)

Country B's opportunity cost of producing $1X = 6.25Y$ (1)

Country A has a lower cost in producing X, and thus it has a comparative advantage in producing X. (1)

(c)



Indicate in the diagram:

(i) The correct position of P_A (1)

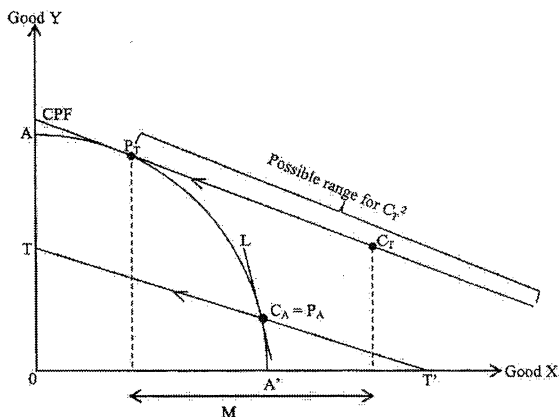
(ii) The correct position of C_A (1)

(iii) The correct volume of export (1)

The correct volume of import (1)

(d) No, because (1)

the opportunity cost of producing X (and Y) will be unaffected by this technological change. (1)



- (a) (i) Without trade, Country A can only consume its own production. (1)

Illustrated in the diagram:

- correct position of $C_A = P_A$ (1)

- (ii) Slope of L is the marginal cost of Good X. (1)

L is steeper than TT' . (1)

That means marginal cost of producing Good X is higher than the world price of Good X. (1)

So Country A did not have a comparative advantage in producing Good X. (1)

(max: 3)

Illustrated in the diagram:

- correct position of the straight line (L) (1)

- (b) (i) Country A would produce at P_T where $TOT =$ domestic marginal cost of production. (1)

Illustrated in the diagram:

- correct position of P_T (1)

- correct position of CPF (1)

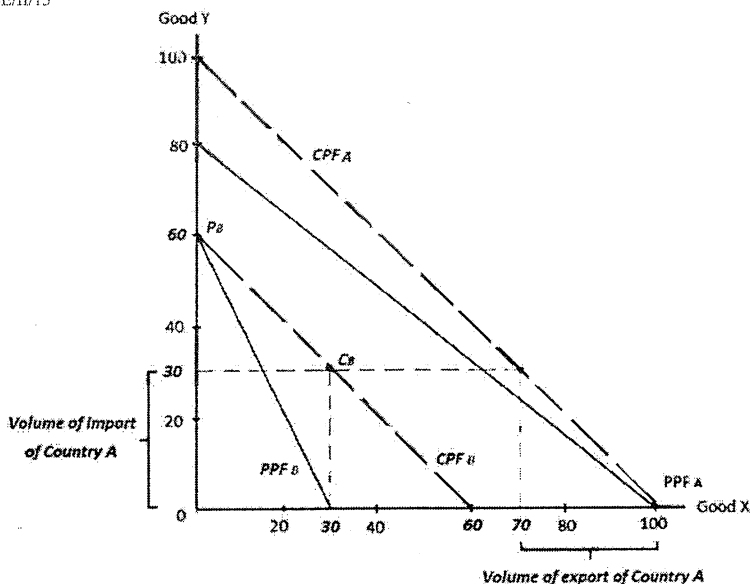
- (ii) Country A can consume at a point which is unattainable from local production before trade. (1)

Illustrated in the diagram:

- correct position of C_T (1)

- (ii) Illustrated in the diagram:

- correct position of M (1)



- (a) Illustrated in the diagram:
 - correct position of production possibility frontier of Country B (PPF_B) (1)
- (b) Country A has absolute advantage in producing Good X, because with the same amount of resources (30 units), Country A can produce a larger quantity of Good X (60 units) than Country B (30 units) (1)
OR
 with the same amount of resources (50 units), Country A can produce a larger quantity of Good X (100 units) than Country B (50 units).
OR
 Country A can produce 1 unit of X with fewer units of resources (0.5 units of resources) than Country B (1 unit of resources). (2)
- (c) Illustrated in the diagram:
 (i) - correct position of production point of Country B (P_B) (1)
 (ii) - correct position of consumption possibility frontier of Country B (CPF_B) (1)
 - correct position of consumption point of Country B after trade (C_B) (1)
 (iii) - correct position of consumption possibility frontier of Country A (CPF_A) (1)
 - correct position of volume of import of Country A (1)
 - correct position of volume of export of Country A (1)

9.2 ECONOMIC DEVELOPMENT

MARKING SCHEME

PP/DSE/II/16(b)

- (i) - Longer life expectancy
- Higher literacy rate

- (ii) - Income is less evenly distributed.
- A larger proportion of total output consists of national defence and capital goods.
- There are more problems of pollution.
- The people enjoy less leisure.
- There are fewer non-marketed / unreported goods.

(2@, max: 4)

[Mark the **FIRST TWO** points only.]

2012/DSE/II/15(d)

- (i) HDI measures the level of economic and social development of a country in 3 dimensions: health (measured by life expectancy at birth), education (measured by mean years of schooling and expected years of schooling) and living standard (measured by Gross National Income per capita). (2)

- (ii) More production, business activities and employment opportunities in the import-export industry lead to higher per capita GDP.

OR

Foreign health product or technology imported lead to better health

(2)

2013/DSE/II/14(b)

- (i) Hong Kong.

HK: GNI per capita rank - HDI rank = -4

Given HDI rank = 13, GNI per capita rank = 9

Japan: GNI per capita rank - HDI rank = 11

Given HDI rank = 12, GNI per capita rank = 23

(2)

- (ii) Japanese have longer years of schooling on average. (1)

Japanese have a longer life expectancy on average. (1)

- (iii) Any example of policy that can enhance the education level or health condition of Hong Kong people. E.g., increase government subsidy on tertiary education. (2)

2014/DSE/II/14(d)

- tax allowance for further education (or education subsidy)
- teaching grants -- to universities for more degree programmes
- bigger immigration quota for experts and professionals

[Mark the **FIRST TWO** points only.]

2015/DSE/II/14(e)

- (i) GDP may increase because of trade-induced transfer of superior technology from its trading partners, resulting in an increase in productivity.

OR

Exchange in health technology may increase life expectancy.

(1)

- (ii) - Workers in import-substitute industries may suffer unemployment
- Income distribution may become more uneven
- Pollution may increase (due to specialization in producing more "polluted" goods)

(2@, max: 4)

[Mark the **FIRST TWO** points only.]

2016/DSE/II/15(e)

- (i) - lower per-capita GDP
 - lower human capital / labour productivity
 - lower marginal productivity of capital (due to complementarity of labour and capital)
 - slower research and development
 (2@, max: 4)
 [Mark the **FIRST TWO** points only.]
- (ii) - importation of high-skilled labour
 - scholarships to attract higher-quality overseas students
 - relaxation of immigration policy to target the smarter brains
 - introduction of return migration policy
 (2@, max: 2)
 [Mark the **FIRST** point only.]

2017/DSE/II/15(c)

- (i) Enhanced economic growth (as shown by higher levels or positive rates of growth) of per-capita real GDP because (1)
 external trade could increase aggregate demand (through exports) and hence real GDP. (1)
- (ii) No change in economic development (as reflected by the same level of HDI) because (1)
 more people may leave school earlier in order to take part in trading activities, thus decreasing the average number of years of schooling, while GDP rises as a result of trade. (1)
OR
 pollution (as a by-product) may become more serious as production is increased after trade / new disease or virus brought in by foreigners may shorten the life expectancy at birth of the local people. (1)

2018/DSE/II/15(d)

- (i) Uncertain, because (1)
 per capita GNI would fall while life expectancy may rise. (2)
OR
 No, because (1)
 the effect on life expectancy would become significant only in the long term, so that the (immediate) GNI effect should dominate in the current period. (2)
- (ii) Uncertain, because (1)
 while per capita GNI may increase, average year of schooling and/or average life expectancy may drop. (2)

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- (b) Income inequality improved in 2020. (1)
- Policies:
- Increase in government transfer payments to the low-income groups.
 - Increased progressivity of the income tax system.
 - Any other relevant point.
- [Mark the **FIRST** point only.]
- } @ 1
max:1